Estimation and Inference in Boundary Discontinuity Designs: Location-Based Methods Supplemental Appendix

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Abstract

This supplemental appendix presents more general theoretical results encompassing those reported in the paper, their theoretical proofs, and other technical results. It also provides a discussion of implementation details, and additional numerical results.

Keywords: regression discontinuity, treatment effects estimation, causal inference.

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SA-1 Introduction

This supplemental appendix considers four generalizations of the setup in the paper.

1. The score $\mathbf{X}_i$ is d-dimensional with $d \geq 2$ and support $\mathscr{X} \subseteq \mathbb{R}^d$.
2. The boundary region $\mathscr{B}$ is a substantively more general low-dimensional manifold with "effective dimension" $d-1$, and hence not necessarily piecewise linear with finitely many line segments.
3. Derivative estimation and inference for all parameters of interest.
4. Different bandwidths $\mathbf{h} = (h_1, \ldots, h_d)^\top$ are allowed for each dimension of the score variable.

In addition, this supplement presents several other technical and methodological results not covered in the main text due to space limitations.

The paper presents results for the special case $d = 2$, $\mathscr{B}$ piecewise linear with finitely many line segments, $h = h_1 = h_2$, and $\boldsymbol{\nu} = \mathbf{0}$. The assumptions in the main paper are more specialized primitive conditions; after imposing these restrictions on the supplemental results, the mapping between the paper and this supplemental appendix is as follows.

- Theorem SA-1 $\Longrightarrow$ Theorem 1.
- Theorem SA-2 $\Longrightarrow$ Theorem 2.
- Theorems SA-3 and SA-8 $\Longrightarrow$ Theorem 3.
- Theorem SA-9 $\Longrightarrow$ Theorem 4.
- Theorem SA-10 $\Longrightarrow$ Theorem 5.
- Theorem SA-12 $\Longrightarrow$ Theorem 6.
- Theorem SA-13 $\Longrightarrow$ Theorem 7.
- Theorem SA-14 $\Longrightarrow$ Theorem 8.
- Theorems SA-15 and SA-18 $\Longrightarrow$ Theorem 9.
- Theorem SA-19 $\Longrightarrow$ Theorem 10.
- Theorem SA-20 $\Longrightarrow$ Theorem 11.
- Theorem SA-22 $\Longrightarrow$ Theorem 12.

The generalizations considered in this supplemental appendix are conceptually straightforward, but technically more involved. Following the structure of the paper, we first consider sharp designs. Section SA-6 generalizes the results to fuzzy designs. Section SA-7 presents a novel Gaussian strong approximation result that may be of independent interest.

SA-1.1 Notation and Definitions

For textbook references on empirical process, see van der Vaart and Wellner [1996], Dudley [2014], and Giné and Nickl [2016]. For textbook references on geometric measure theory, see Simon et al. [1984], Federer [2014], and Folland [2002].

(i) *Multi-index Notations.* For a multi-index $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_d) \in \mathbb{N}_0^d$, let $|\boldsymbol{\nu}| = \sum_{l=1}^d \nu_l$, $\boldsymbol{\nu}! = \nu_1! \cdots \nu_d!$, and $\mathbf{u}^{\boldsymbol{\nu}} = \prod_{l=1}^d u_l^{\nu_l}$ for $\mathbf{u} = (u_1, \ldots, u_d) \in \mathbb{R}^d$. For any integer $p \geq 0$, let $\mathbf{r}_p(\mathbf{u}) = (\mathbf{u}^{\boldsymbol{\alpha}} : |\boldsymbol{\alpha}| \leq p)^\top \in \mathbb{R}^{\mathfrak{p}_p}$ collect all monomials of total degree at most p in a fixed order, where $\mathfrak{p}_p = \frac{(d+p)!}{d!p!}$. Define $\mathbf{e}_{\boldsymbol{\nu}}$ to be the $\mathfrak{p}_p$-dimensional unit vector such that $\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{r}_p(\mathbf{u}) = \mathbf{u}^{\boldsymbol{\nu}}$ for all $\mathbf{u} \in \mathbb{R}^d$ and $|\boldsymbol{\nu}| \leq p$. In addition, let $f^{(\boldsymbol{\nu})}(\mathbf{x}) = \partial^{\nu_1} \cdots \partial^{\nu_d} f(\mathbf{x})$.

(ii) *Norms.* For a vector $\mathbf{a} \in \mathbb{R}^k$, $\|\mathbf{a}\| = (\sum_{i=1}^k \mathbf{a}_i^2)^{1/2}$, $\|\mathbf{a}\|_\infty = \max_{1 \leq i \leq k} |\mathbf{a}_i|$. For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\|\mathbf{A}\|_p = \sup_{\|\mathbf{x}\|_p = 1} \|\mathbf{A}\mathbf{x}\|_p$, $p \in \mathbb{N} \cup \{\infty\}$, $\lambda_{\min}(\mathbf{A})$ denotes its minimum eigenvalue, and $|\mathbf{A}| = |\det(\mathbf{A})|$ denotes the absolute value of its determinant. For a function f on a metric space (S, d), $\|f\|_\infty = \sup_{s \in S} |f(s)|$. For a probability measure Q on $(\mathcal{S}, \mathscr{S})$ and $p \geq 1$, define $\|f\|_{Q,p} = (\int_{\mathcal{S}} |f|^p \, \mathrm{d}Q)^{1/p}$, and $Q(f) = \int f \, \mathrm{d}Q$. For a set $E \subseteq \mathbb{R}^d$, denote by $\mathfrak{m}(E)$ the Lebesgue measure of E.

(iii) *Empirical Process.* We use standard empirical process notations: $\mathbb{E}_n[g(\mathbf{v}_i)] = \frac{1}{n} \sum_{i=1}^n g(\mathbf{v}_i)$ and $\mathbb{G}_n[g(\mathbf{v}_i)] = \frac{1}{\sqrt{n}} \sum_{i=1}^n (g(\mathbf{v}_i) - \mathbb{E}[g(\mathbf{v}_i)])$. Let $(\mathcal{S}, d)$ be a semi-metric space. The covering number $N(\mathcal{S}, d, \varepsilon)$ is the minimal number of balls $B_s(\varepsilon) = \{t : d(t, s) < \varepsilon\}$ needed to cover $\mathcal{S}$. A $\mathbb{P}$-*Brownian bridge* is a mean-zero Gaussian random function $\mathbb{G}_{\mathbb{P}}(f), f \in L_2(\mathcal{X}, \mathbb{P})$ with covariance $\mathbb{E}[\mathbb{G}_{\mathbb{P}}(f)\mathbb{G}_{\mathbb{P}}(g)] = \mathbb{P}(fg) - \mathbb{P}(f)\mathbb{P}(g)$, for $f, g \in L_2(\mathcal{X}, \mathbb{P})$. A class $\mathscr{F} \subseteq L_2(\mathcal{X}, \mathbb{P})$ is $\mathbb{P}$-*pregaussian* if there is a version of $\mathbb{P}$-Brownian bridge $\mathbb{G}_{\mathbb{P}}$ such that $\mathbb{G}_{\mathbb{P}} \in C(\mathscr{F}; \rho_{\mathbb{P}})$ almost surely, where $\rho_{\mathbb{P}}$ is the semi-metric on $L_2(\mathcal{X}, \mathbb{P})$ is defined by $\rho_{\mathbb{P}}(f, g) = (\|f - g\|_{\mathbb{P},2}^2 - (\int f \, \mathrm{d}\mathbb{P} - \int g \, \mathrm{d}\mathbb{P})^2)^{1/2}$, for $f, g \in L_2(\mathcal{X}, \mathbb{P})$.

(iv) *Geometric Measure Theory.* For a set $A \subseteq \mathcal{X}$, the *De Giorgi perimeter of A related to $\mathcal{X}$* is $\mathscr{L}(A) = \mathtt{TV}_{\{\mathbb{1}_A\}, \mathcal{X}}$. For $d \in \mathbb{N}$ and $0 \leq m \leq d$, the m-dimensional Hausdorff (outer) measure is $\mathfrak{H}^m(A) = \lim_{\delta \downarrow 0} \mathfrak{H}_\delta^m(A)$, $A \subseteq \mathbb{R}^d$, where for each $\delta > 0$, $\mathfrak{H}_\delta^m(A)$ is defined by taking $\mathfrak{H}_\delta^m(\emptyset) = 0$, and for any non-empty $A \subseteq \mathbb{R}^d$, $\mathfrak{H}_\delta^m(A) = \frac{\pi^{m/2}}{\Gamma(m/2+1)} \inf \sum_{j=1}^\infty (\operatorname{diam}(C_j)/2)^m$, and the infimum is taken over all countable collections $C_1, C_2, \cdots$ of subsets of $\mathbb{R}^d$ such that $\operatorname{diam}(C_j) < \delta$ and $A \subseteq \cup_{j=1}^\infty C_j$. Integration against $\mathfrak{H}^m$ is defined via Carathéodory's Theorem following the classical measure-theoretic literature. The Hausdorff dimension $\dim_{\mathfrak{H}}(A)$ of A is defined by $\dim_{\mathfrak{H}}(A) = \inf\{t \geq 0 : \mathfrak{H}^t(A) = 0\}$. A set $A \subseteq \mathbb{R}^d$ is said to be k-rectifiable if A is of Hausdorff dimension k, and there exist a countable collection $\{f_i\}$ of continuously differentiable maps $f_i : \mathbb{R}^k \to \mathbb{R}^d$ such that $\mathfrak{H}^k(A \setminus \cup_{i=0}^\infty f_i(\mathbb{R}^k)) = 0$. B is a *rectifiable curve* if there exists a Lipschitz continuous function $\gamma : [0,1] \to \mathbb{R}^d$ such that $B = \gamma([0,1])$. We define the curve length function of B to be $\mathfrak{L}(B) = \sup_{\pi \in \Pi} s(\pi, \gamma)$, where $\Pi = \{(t_0, t_1, \ldots, t_N) : N \in \mathbb{N}, 0 \leq t_0 < t_1 < \ldots \leq t_N \leq 1\}$ and $s(\pi, \gamma) = \sum_{i=0}^{N-1} \|\gamma(t_i) - \gamma(t_{i+1})\|_2$ for $\pi = (t_0, t_1, \ldots, t_N)$.

(v) *Bounds and Asymptotics.* For real sequences, we write $a_n = o(b_n)$ if $\limsup \frac{|a_n|}{|b_n|} = 0$, and $a_n \lesssim b_n$ if there exist constants C and $N > 0$ such that $n > N$ implies $|a_n| \leq C|b_n|$. For sequences of random variables, we write $a_n = o_{\mathbb{P}}(b_n)$ if $\operatorname{plim}_{n \to \infty} \frac{|a_n|}{|b_n|} = 0$, and $|a_n| \lesssim_{\mathbb{P}} |b_n|$ if $\limsup_{M \to \infty} \limsup_{n \to \infty} \mathbb{P}[|\frac{a_n}{b_n}| \geq M] = 0$.

Let $\Phi(\cdot)$ be the cumulative distribution function of a standard univariate Gaussian random variable. For a bandwidth vector $\mathbf{h} = (h_1, \ldots, h_d)^\top \in \mathbb{R}_{++}^d$, let $\mathbf{H} = \operatorname{diag}(\mathbf{h})$, $\underline{h} = \min_{1 \leq l \leq d} h_l$, $\bar{h} = \max_{1 \leq l \leq d} h_l$, and

$K_{\mathbf{H}}(\mathbf{u}) = |\mathbf{H}|^{-1}K(\mathbf{H}^{-1}\mathbf{u})$. All limits are taken such that $\bar{h} \to 0$, $\bar{h}/\underline{h} \lesssim 1$, and the bandwidths vanish as $n \to \infty$. Most of our results hold with fixed but small enough bandwidths. We do not make this distinction explicit to avoid overly-complex statements and technical discussions.

SA-2 Sharp Design Setup

We partition $\mathscr{X}$ into two areas, $\mathscr{A}_t \subset \mathbb{R}^d$ with $t \in \{0,1\}$, which represent the control and treatment regions, respectively. That is, $\mathscr{X} = \mathscr{A}_0 \cup \mathscr{A}_1$, where $\mathscr{A}_0$ and $\mathscr{A}_1$ are two disjoint regions in $\mathbb{R}^d$. The observed outcome is $Y_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)Y_i(0) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)Y_i(1)$. The "boundary" is $\mathscr{B} = \mathrm{bd}(\mathscr{A}_0) \cap \mathrm{bd}(\mathscr{A}_1) \subseteq \mathbb{R}^d$, where $\mathrm{bd}(\cdot)$ denotes the topological boundary of a set, and has effective dimension $d-1$. We assume that $\mathscr{B}$ belongs to $\mathscr{A}_1$, that is, $\mathscr{B} \subseteq \mathscr{A}_1$ and $\mathscr{B} \cap \mathscr{A}_0 = \emptyset$.

SA-2.1 Assumptions

Assumption 1 generalizes as follows.

Assumption SA-1 (Data Generating Process)**.** *Let $t \in \{0,1\}$.*

(i) $(Y_1(t), \mathbf{X}_1^\top)^\top, \ldots, (Y_n(t), \mathbf{X}_n^\top)^\top$ *are independent and identically distributed random vectors.*

(ii) *The distribution of $\mathbf{X}_i$ has compact support $\mathscr{X} \subseteq \mathbb{R}^d$ with nonempty interior and a Lebesgue density $f_X(\mathbf{x})$ that is continuous and bounded away from zero on $\mathscr{X}$.*

(iii) $\mu_t(\mathbf{x}) = \mathbb{E}[Y_i(t)|\mathbf{X}_i = \mathbf{x}]$ *has a $(p+1)$-times continuously differentiable extension to an open neighborhood of $\mathscr{X}$.*

(iv) $\sigma_t^2(\mathbf{x}) = \mathbb{V}[Y_i(t)|\mathbf{X}_i = \mathbf{x}]$ *is bounded away from zero and continuous on $\mathscr{X}$.*

(v) $\sup_{\mathbf{x}\in\mathscr{X}} \mathbb{E}[|Y_i(t)|^{2+v}|\mathbf{X}_i = \mathbf{x}] < \infty$ *for some $v \geq 2$.*

Assumption 2 generalizes as follows.

Assumption SA-2 (Boundary and Kernel)**.**

(i) $\mathscr{B} \subset \mathrm{int}(\mathscr{X})$, *and $\mathscr{B}$ is compact $(d-1)$-rectifiable, with $\mathfrak{H}^{d-1}(\mathscr{B})$ positive and finite.*

(ii) $K : \mathbb{R}^d \to [0,\infty)$ *is compactly supported and Lipschitz continuous, or $K(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in [-1,1]^d)$.*

(iii) *For each $t \in \{0,1\}$, there exists a Lebesgue-measurable set $U \subseteq \mathbb{R}^d$ with $0 < \mathfrak{m}(U) < \infty$, such that $K(\mathbf{u}) \geq \kappa > 0$ for all $\mathbf{u} \in U$, $\lambda_{\min}(\int_U \mathbf{r}_p(\mathbf{z})\mathbf{r}_p(\mathbf{z})^\top \mathrm{d}\mathbf{z}) > 0$, and*

$$\liminf_{\bar{h}\downarrow 0} \inf_{\mathbf{x}\in\mathscr{B}} \int_U K(\mathbf{u})\mathbb{1}(\mathbf{x} + \mathbf{H}\mathbf{u} \in \mathscr{A}_t)\,\mathrm{d}\mathbf{u} \gtrsim 1,$$

uniformly over diagonal bandwidth matrices satisfying $\bar{h}/\underline{h} \lesssim 1$.

For $d = 2$, Assumption SA-2(i) holds whenever $\mathscr{B}$ is a compact rectifiable curve with positive and finite length. For the special case in the main paper, where $\mathscr{B}$ is piecewise linear with finitely many line segments and K is positive in a neighborhood of the origin, Assumption SA-2(iii) also holds: locally, each side of the boundary contains a half-plane or polygonal cone of positive aperture, and the finite number of line segments gives a uniform lower bound over $\mathscr{B}$. More generally, Assumption SA-2(iii) is a local nondegeneracy condition linking the geometry of $\mathscr{B}$ to the support and shape of the kernel K. The density $f_X(\mathbf{x})$ does not appear explicitly in this condition because Assumption SA-1(ii) already requires it to be bounded away from zero on $\mathscr{X}$.

More precisely, Assumption SA-2(iii) rules out boundary-kernel configurations in which, after local rescaling by $\mathbf{H}$, one side of the boundary receives too little kernel-weighted mass to guarantee the local polynomial regression is well-defined uniformly over $\mathscr{B}$. Thus, the condition ensures that the local design is sufficiently rich on each side of the boundary. Lemma SA-1 shows that Assumptions SA-1(ii) and SA-2 imply that the population Gram matrix is full rank for $\bar{h}$ small enough, uniformly in $\mathbf{x} \in \mathscr{B}$. This guarantees that the treatment effect estimators are well-defined in large samples under the conditions imposed. Assumption SA-2(iii) is stated uniformly over $\mathscr{B}$ in anticipation of our uniform estimation and inference results, but it can be relaxed to hold only pointwise in $\mathbf{x} \in \mathscr{B}$ for pointwise results.

Assumption 3 generalizes as follows.

Assumption SA-3 (Weight Function and Boundary)**.** *Let* $w : \mathscr{B} \to (0, \infty)$ *satisfy* $\sup_{\mathbf{x}\in\mathscr{B}} w(\mathbf{x}) < \infty$, $\inf_{\mathbf{x}\in\mathscr{B}} w(\mathbf{x}) > 0$, *and* $\int_{\mathscr{B}} w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) < \infty$.

SA-2.2 Parameters

The multidimensional generalizations of the three causal parameters studied in the paper are:

1. *Boundary Average Treatment Effect Curve* (BATEC):

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0)|\mathbf{X}_i = \mathbf{x}], \qquad \mathbf{x} \in \mathscr{B}.$$

2. *Weighted Boundary Average Treatment Effect* (WBATE):

$$\tau_{\texttt{WBATE}} = \frac{\int_{\mathscr{B}} \tau(\mathbf{x}) w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}{\int_{\mathscr{B}} w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}.$$

3. *Largest Boundary Average Treatment Effect* (LBATE):

$$\tau_{\texttt{LBATE}} = \sup_{\mathbf{x}\in\mathscr{B}} \tau(\mathbf{x}).$$

More generally, this supplemental appendix considers the derivatives of the BATEC parameter

$$\tau^{(\boldsymbol{\nu})}(\mathbf{x}) = \mu_1^{(\boldsymbol{\nu})}(\mathbf{x}) - \mu_0^{(\boldsymbol{\nu})}(\mathbf{x}), \qquad \mathbf{x} \in \mathscr{B},$$

and the corresponding weighted and largest derivative parameters, respectively given by

$$\tau^{(\boldsymbol{\nu})}_{\texttt{WBATE}} = \frac{\int_{\mathscr{B}} \tau^{(\boldsymbol{\nu})}(\mathbf{x}) w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}{\int_{\mathscr{B}} w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})} \qquad \text{and} \qquad \tau^{(\boldsymbol{\nu})}_{\texttt{LBATE}} = \sup_{\mathbf{x}\in\mathscr{B}} \tau^{(\boldsymbol{\nu})}(\mathbf{x}),$$

where $|\boldsymbol{\nu}| \leq p$.

SA-2.3 Estimators and Other Related Quantities

The estimator of the (derivative of the) BATEC parameters

$$\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) = \widehat{\mu}_1^{(\boldsymbol{\nu})}(\mathbf{x}) - \widehat{\mu}_0^{(\boldsymbol{\nu})}(\mathbf{x}) \qquad \mathbf{x} \in \mathscr{B},$$

where $\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\beta}}_t(\mathbf{x})$ for $t \in \{0,1\}$ with

$$\widehat{\boldsymbol{\beta}}_t(\mathbf{x}) = \underset{\boldsymbol{\beta} \in \mathbb{R}^{\mathfrak{p}_p}}{\arg\min}\, \mathbb{E}_n\big[\big(Y_i - \mathbf{r}_p(\mathbf{X}_i - \mathbf{x})^\top \boldsymbol{\beta}\big)^2 K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big], \qquad \mathbf{x} \in \mathscr{B}, \tag{SA-1}$$

with $K_{\mathbf{H}}(\mathbf{u}) = |\mathbf{H}|^{-1} K(\mathbf{H}^{-1}\mathbf{u})$ for a d-variate kernel function $K(\cdot)$ and a diagonal bandwidth matrix $\mathbf{H} = \operatorname{diag}(\mathbf{h})$ with $\mathbf{h} = (h_1, \ldots, h_d)^\top \in \mathbb{R}^d_{++}$. Thus,

$$|\mathbf{H}| = \prod_{l=1}^{d} h_l \qquad \text{and} \qquad \mathbf{H}^{-1} = \operatorname{diag}(h_1^{-1}, \ldots, h_d^{-1}).$$

Furthermore, if all bandwidths are proportional to each other, that is, $h_l = c_l h$ for fixed constants $c_l > 0$, $l = 1, \ldots, d$, and common bandwidth $h > 0$, then $\mathbf{H} = h \operatorname{diag}(\mathbf{c})$ with $\mathbf{c} = (c_1, \ldots, c_d)^\top$, and hence

$$|\mathbf{H}| = h^d \prod_{l=1}^{d} c_l, \qquad \text{and} \qquad \mathbf{H}^{-1} = h^{-1} \operatorname{diag}(c_1^{-1}, \ldots, c_d^{-1}).$$

Therefore, $|\mathbf{H}| \asymp h^d$ and $\mathbf{H}^{-1} \asymp h^{-1}\mathbf{I}_d$.

Under the assumptions imposed,

$$\widehat{\boldsymbol{\beta}}_t(\mathbf{x}) = \mathbf{S}^{-1} \widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} \mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) Y_i \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big], \qquad \mathbf{S} = \operatorname{diag}(\mathbf{r}_p(\mathbf{h})),$$

and

$$\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}} = \mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

where its population analog is

$$\boldsymbol{\Gamma}_{t,\mathbf{x}} = \mathbb{E}\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big].$$

For use in the proofs, let $\boldsymbol{\beta}_t(\mathbf{x})$ denote the population Taylor coefficient vector satisfying

$$\boldsymbol{\beta}_t(\mathbf{x})^\top \mathbf{r}_p(\mathbf{u}) = \sum_{|\boldsymbol{\omega}| \le p} \frac{\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!} \mathbf{u}^{\boldsymbol{\omega}}, \qquad \mathbf{u} \in \mathbb{R}^d.$$

Note that $\big\|\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\big\|_2 = \big\|\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\big\|_\infty = \mathbf{h}^{-\boldsymbol{\nu}}$. In addition, define

$$\mathbf{Q}_{t,\mathbf{x}} = \mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t) \varepsilon_i\big],$$

where $\varepsilon_i = Y_i - [\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\mu_0(\mathbf{X}_i) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\mu_1(\mathbf{X}_i)] = Y_i - \mathbb{E}[Y_i|\mathbf{X}_i]$.

For $\mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}$ and $t \in \{0,1\}$, we introduce the following population quantities

$$\Omega^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_2} = \Omega^{(\boldsymbol{\nu})}_{0,\mathbf{x}_1,\mathbf{x}_2} + \Omega^{(\boldsymbol{\nu})}_{1,\mathbf{x}_1,\mathbf{x}_2}, \qquad \Omega^{(\boldsymbol{\nu})}_{t,\mathbf{x}_1,\mathbf{x}_2} = \frac{(\boldsymbol{\nu}!)^2}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} \mathbf{e}_{\boldsymbol{\nu}}^\top \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}_1} \boldsymbol{\Sigma}_{t,\mathbf{x}_1,\mathbf{x}_2} \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}_2} \mathbf{e}_{\boldsymbol{\nu}},$$

$$\boldsymbol{\Sigma}_{t,\mathbf{x}_1,\mathbf{x}_2} = |\mathbf{H}|\mathbb{E}\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_1))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_2))^\top K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_1) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_2)\sigma_t^2(\mathbf{X}_i)\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

and their sample analogs

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_2} = \widehat{\Omega}^{(\boldsymbol{\nu})}_{0,\mathbf{x}_1,\mathbf{x}_2} + \widehat{\Omega}^{(\boldsymbol{\nu})}_{1,\mathbf{x}_1,\mathbf{x}_2}, \qquad \widehat{\Omega}^{(\boldsymbol{\nu})}_{t,\mathbf{x}_1,\mathbf{x}_2} = \frac{(\boldsymbol{\nu}!)^2}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\Gamma}}^{-1}_{t,\mathbf{x}_1} \widehat{\boldsymbol{\Sigma}}_{t,\mathbf{x}_1,\mathbf{x}_2} \widehat{\boldsymbol{\Gamma}}^{-1}_{t,\mathbf{x}_2} \mathbf{e}_{\boldsymbol{\nu}},$$

$$\widehat{\boldsymbol{\Sigma}}_{t,\mathbf{x}_1,\mathbf{x}_2} = |\mathbf{H}|\mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_1))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_2))^\top K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_1) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_2)\widehat{\varepsilon}_i(\mathbf{x}_1)\widehat{\varepsilon}_i(\mathbf{x}_2)\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

where $\widehat{\varepsilon}_i(\mathbf{x}) = Y_i - \mathbf{r}_p(\mathbf{X}_i - \mathbf{x})^\top[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\widehat{\boldsymbol{\beta}}_0(\mathbf{x}) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\widehat{\boldsymbol{\beta}}_1(\mathbf{x})]$.

Finally, we introduce more notation for the leading bias and variance of the estimator of the (derivative of the) BATEC parameter. For $\mathbf{x} \in \mathscr{B}$ and $t \in \{0,1\}$, define

$$B^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \sum_{|\boldsymbol{\omega}|=p+1} \frac{\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!}\mathbf{h}^{\boldsymbol{\omega}-\boldsymbol{\nu}}\mathbb{E}\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^{\boldsymbol{\omega}} K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

where $B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}) = B^{(\boldsymbol{\nu})}_{1,\mathbf{x}}(\mathbf{h}) - B^{(\boldsymbol{\nu})}_{0,\mathbf{x}}(\mathbf{h})$. When $\mathbf{h} = \mathbf{c}h$ for a fixed vector $\mathbf{c}$ and h is a common bandwidth, $B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}) = h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{c})$ for some $\widetilde{B}^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{c})$. Also, define

$$V^{(\boldsymbol{\nu})}_{\mathbf{x}} = V^{(\boldsymbol{\nu})}_{0,\mathbf{x}} + V^{(\boldsymbol{\nu})}_{1,\mathbf{x}},$$

with

$$V^{(\boldsymbol{\nu})}_{t,\mathbf{x}} = (\boldsymbol{\nu}!)^2 \mathbf{e}_{\boldsymbol{\nu}}^\top \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \boldsymbol{\Sigma}_{t,\mathbf{x},\mathbf{x}} \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \mathbf{e}_{\boldsymbol{\nu}} = n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}\Omega^{(\boldsymbol{\nu})}_{t,\mathbf{x},\mathbf{x}}.$$

The corresponding empirical analogs of these leading terms, leaving the unknown derivatives explicit, are

$$\widehat{B}^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\Gamma}}^{-1}_{t,\mathbf{x}} \sum_{|\boldsymbol{\omega}|=p+1} \frac{\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!}\mathbf{h}^{\boldsymbol{\omega}-\boldsymbol{\nu}}\mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^{\boldsymbol{\omega}} K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

and $\widehat{B}^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}) = \widehat{B}^{(\boldsymbol{\nu})}_{1,\mathbf{x}}(\mathbf{h}) - \widehat{B}^{(\boldsymbol{\nu})}_{0,\mathbf{x}}(\mathbf{h})$. Also, $\widehat{V}^{(\boldsymbol{\nu})}_{\mathbf{x}} = \widehat{V}^{(\boldsymbol{\nu})}_{0,\mathbf{x}} + \widehat{V}^{(\boldsymbol{\nu})}_{1,\mathbf{x}}$ with $\widehat{V}^{(\boldsymbol{\nu})}_{t,\mathbf{x}} = n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}\widehat{\Omega}^{(\boldsymbol{\nu})}_{t,\mathbf{x},\mathbf{x}}$.

SA-2.4 Preliminary Lemmas

Let $\mathbf{X} = (\mathbf{X}_1^\top, \cdots, \mathbf{X}_n^\top)$, and recall that $t \in \{0,1\}$.

Lemma SA-1 (Invertibility). *Suppose Assumptions SA-1(i,ii) and SA-2 hold. Then,*

$$\liminf_{\bar{h}\to 0} \inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\boldsymbol{\Gamma}_{t,\mathbf{x}}) > 0.$$

Lemma SA-2 (Gram)**.** *Suppose Assumptions SA-1(i,ii) and SA-2 hold. If* $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$*, then*

$$\sup_{\mathbf{x}\in\mathscr{B}} \big\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}} - \mathbf{\Gamma}_{t,\mathbf{x}}\big\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}, \qquad \sup_{\mathbf{x}\in\mathscr{B}} \big\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} - \mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\big\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}.$$

If, in addition, $\bar{h} = o(1)$*, then* $\inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}) \gtrsim_{\mathbb{P}} 1$, $\sup_{\mathbf{x}\in\mathscr{B}} \|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} 1$*, and* $\sup_{\mathbf{x}\in\mathscr{B}} \|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}\| \lesssim_{\mathbb{P}} 1$.

Lemma SA-3 (Linear Approximation)**.** *Suppose Assumptions SA-1(i,ii,iv,v) and SA-2 hold. If* $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$*, then*

$$\sup_{\mathbf{x}\in\mathscr{B}} \big\|\mathbf{Q}_{t,\mathbf{x}}\big\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}.$$

If, in addition, $\bar{h} = o(1)$*, then*

$$\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x}) - \mathbb{E}\big[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big] - \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\mathbf{Q}_{t,\mathbf{x}}\big| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} \Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}\Bigg).$$

Lemma SA-4 (Covariance)**.** *Suppose Assumptions SA-1 and SA-2 hold. If* $\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \big\|\widehat{\mathbf{\Sigma}}_{t,\mathbf{x}_1,\mathbf{x}_2} - \mathbf{\Sigma}_{t,\mathbf{x}_1,\mathbf{x}_2}\big\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1},$$

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \big|\widehat{\Omega}_{\mathbf{x}_1,\mathbf{x}_2}^{(\boldsymbol{\nu})} - \Omega_{\mathbf{x}_1,\mathbf{x}_2}^{(\boldsymbol{\nu})}\big| \lesssim_{\mathbb{P}} (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1} \Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg),$$

and

$$\sup_{\mathbf{x}\in\mathscr{B}} \big|(\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2} - (\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2}\big| \lesssim_{\mathbb{P}} \sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} \Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg).$$

Lemma SA-5 (Bias)**.** *Suppose Assumptions SA-1(i,ii,iii) and SA-2 hold. If* $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\sup_{\mathbf{x}\in\mathscr{B}} \big|\mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}] - \mu_t^{(\boldsymbol{\nu})}(\mathbf{x})\big| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}},$$

implying

$$\sup_{\mathbf{x}\in\mathscr{B}} \big|\mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}] - \mu_t^{(\boldsymbol{\nu})}(\mathbf{x}) - \widehat{B}_{t,\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\big| = o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}),$$

with $\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{B}_{t,\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h}) - B_{t,\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\big| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}$*, and hence* $\sup_{\mathbf{x}\in\mathscr{B}} |B_{t,\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})| \lesssim \bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}$.

SA-3 Boundary Average Treatment Effect Curve

SA-3.1 Point Estimation and MSE Expansions

Theorem SA-1 (Convergence Rates: Sharp BATEC)**.** *Suppose Assumptions SA-1 and SA-2 hold. If* $\log(1/\bar{h})/(n|\mathbf{H}|) = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\left|\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}(\mathbf{x})\right| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\Big(\frac{1}{\sqrt{n|\mathbf{H}|}} + \frac{1}{n^{\frac{1+v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Big)$$

for $\mathbf{x} \in \mathscr{B}$*, and*

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}(\mathbf{x})\right| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Big).$$

The conditional mean-squared error (MSE) is

$$\mathrm{MSE}^{(\boldsymbol{\nu})}(\mathbf{x}) = \mathbb{E}\big[(\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}(\mathbf{x}))^2\big|\mathbf{X}\big]$$

for $\mathbf{x} \in \mathscr{B}$, and the conditional integrated MSE (IMSE) is

$$\mathrm{IMSE}^{(\boldsymbol{\nu})} = \int_{\mathscr{B}} \mathrm{MSE}^{(\boldsymbol{\nu})}(\mathbf{x}) w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}),$$

where $w(\mathbf{x})$ satisfies Assumption SA-3.

Theorem SA-2 (MSE Expansions: Sharp BATEC)**.** *Suppose Assumptions SA-1, SA-2 and SA-3 hold. If* $\log(1/\bar{h})/(n^{\frac{v}{2+v}}|\mathbf{H}|) = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\mathrm{MSE}^{(\boldsymbol{\nu})}(\mathbf{x}) = \big(B_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\big)^2 + \frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} V_{\mathbf{x}}^{(\boldsymbol{\nu})} + o_{\mathbb{P}}\big(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}} + (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\big)$$

for $\mathbf{x} \in \mathscr{B}$*, and*

$$\mathrm{IMSE}^{(\boldsymbol{\nu})} = \int_{\mathscr{B}} \Big[\big(B_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\big)^2 + \frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} V_{\mathbf{x}}^{(\boldsymbol{\nu})}\Big] w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) + o_{\mathbb{P}}\big(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}} + (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\big).$$

Theorem SA-2 can be used to develop feasible bandwidth selectors for diagonal bandwidth matrices by minimizing the displayed criterion over $\mathbf{h} \in \mathbb{R}^d_{++}$. If $\mathbf{h} = \mathbf{c}h$ for a fixed vector $\mathbf{c}$ and a common bandwidth h, the problem reduces to scalar bandwidth selection. If $\widetilde{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{c}) \neq 0$, the asymptotic MSE-optimal common scale is

$$h_{\mathrm{MSE},\boldsymbol{\nu}}(\mathbf{x}) = \left(\frac{(d + 2|\boldsymbol{\nu}|)V_{\mathbf{x}}^{(\boldsymbol{\nu})}}{(2p + 2 - 2|\boldsymbol{\nu}|)(\widetilde{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{c}))^2}\frac{1}{n}\right)^{\frac{1}{2p+d+2}}$$

for $\mathbf{x} \in \mathscr{B}$. Similarly, if $\int_{\mathscr{B}}(\widetilde{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{c}))^2 w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) \neq 0$, the asymptotic IMSE-optimal common scale is

$$h_{\mathrm{IMSE},\boldsymbol{\nu}} = \left(\frac{(d + 2|\boldsymbol{\nu}|)\int_{\mathscr{B}} V_{\mathbf{x}}^{(\boldsymbol{\nu})} w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}{(2p + 2 - 2|\boldsymbol{\nu}|)\int_{\mathscr{B}}(\widetilde{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{c}))^2 w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}\frac{1}{n}\right)^{\frac{1}{2p+d+2}}.$$

In practice, the unknown bias and variance quantities can be replaced with (consistent) estimators thereof. For example, the unknown functions $\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})$ can be estimated using higher-order local polynomial estimators in $\widehat{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})$, and $\widehat{V}_{\mathbf{x}}^{(\boldsymbol{\nu})} = \widehat{V}_{0,\mathbf{x}}^{(\boldsymbol{\nu})} + \widehat{V}_{1,\mathbf{x}}^{(\boldsymbol{\nu})}$ with $\widehat{V}_{t,\mathbf{x}}^{(\boldsymbol{\nu})} = (\boldsymbol{\nu}!)^2 \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} \widehat{\boldsymbol{\Sigma}}_{t,\mathbf{x},\mathbf{x}} \widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} \mathbf{e}_{\boldsymbol{\nu}}$, which corresponds to a standard variance estimator (which is also used for asymptotic inference as discussed below).

Finally, some pointwise statements admit weaker side conditions when the empirical-process classes are restricted to a fixed boundary point. We state the stronger uniform side conditions throughout this section to simplify the exposition.

SA-3.2 Distributional Approximation and Inference

Let $\mathbf{V} = ((Y_1, \mathbf{X}_1^\top), \cdots, (Y_n, \mathbf{X}_n^\top))^\top$, and recall that $t \in \{0,1\}$. For $|\boldsymbol{\nu}| \le p$, define the feasible t-statistic

$$\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) = \frac{\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}(\mathbf{x})}{\sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}}}, \qquad \mathbf{x} \in \mathscr{B}.$$

The associated $100(1-\alpha)\%$ confidence interval estimator is

$$\widehat{\mathrm{I}}_{\alpha}^{(\boldsymbol{\nu})}(\mathbf{x}) = \Big[\ \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \mathscr{q}_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}}\ ,\ \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) + \mathscr{q}_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}}\ \Big],$$

where $\mathscr{q}_\alpha$ denotes an appropriate quantile depending on the desired confidence level $\alpha \in (0,1)$, and coverage objective (pointwise versus uniform over $\mathscr{B}$).

Theorem SA-3 (Confidence Intervals: Sharp BATEC)**.** *Suppose Assumptions SA-1 and SA-2 hold. If $n|\mathbf{H}| \to \infty$ and $n|\mathbf{H}|\bar{h}^{2(p+1)} = o(1)$, then*

$$\sup_{u \in \mathbb{R}} \big|\mathbb{P}\big(\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) \le u\big) - \Phi(u)\big| = o(1), \qquad \mathbf{x} \in \mathscr{B},$$

and

$$\mathbb{P}\big(\tau^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}_{\alpha}^{(\boldsymbol{\nu})}(\mathbf{x})\big) = 1 - \alpha + o(1), \qquad \mathbf{x} \in \mathscr{B},$$

provided that $\mathscr{q}_\alpha = \inf\{c > 0 : \mathbb{P}(|\widehat{Z}| \ge c|\mathbf{V}) \le \alpha\}$ with $\widehat{Z}|\mathbf{V} \sim \mathsf{Normal}(0,1)$.

For uniform inference, we rely on a new strong approximation result established in Section SA-7. First, we simplify the statistic $\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}$, which is not a sum of independent random variables. Let

$$\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) = \mathbb{E}_n\Big[(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2} \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\big[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\boldsymbol{\Gamma}_{1,\mathbf{x}}^{-1} - \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\boldsymbol{\Gamma}_{0,\mathbf{x}}^{-1}\big]\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\varepsilon_i\Big],$$

where recall that $\varepsilon_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\mu_t(\mathbf{X}_i) = Y_i - \mathbb{E}[Y_i|\mathbf{X}_i]$.

Theorem SA-4 (Stochastic Linearization: Sharp BATEC)**.** *Suppose Assumptions SA-1 and SA-2 hold. If $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$ and $\bar{h} = o(1)$, then*

$$\sup_{\mathbf{x} \in \mathscr{B}} \big|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) - \overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \sqrt{\log(1/\bar{h})}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|}\Big).$$

We can now exploit the linear structure of $(\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$, that is, an average of i.n.i.d. random vectors.

Define the following functions indexed by $\mathbf{x} \in \mathscr{B}$:

$$g_{\mathbf{x}}(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in \mathscr{A}_1)\mathscr{K}_1^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) - \mathbb{1}(\mathbf{u} \in \mathscr{A}_0)\mathscr{K}_0^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}), \qquad \mathbf{u} \in \mathscr{X},$$

and

$$\mathscr{K}_t^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) = n^{-1/2}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^{\top}\mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))K_{\mathbf{H}}(\mathbf{u}-\mathbf{x}), \qquad \mathbf{u} \in \mathscr{X}, \quad t \in \{0,1\}.$$

Define the associated class of functions $\mathscr{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathscr{B}\}$ and $\mathscr{R} = \{\mathrm{Id}\}$, where $\mathrm{Id}(x) = x$, for all $x \in \mathbb{R}$. Then, the *residual-based empirical process* is

$$R_n(g, r) = n^{-1/2}\sum_{i=1}^{n}\Big[g(\mathbf{X}_i)r(Y_i) - g(\mathbf{X}_i)\mathbb{E}[r(Y_i)|\mathbf{X}_i]\Big], \qquad g \in \mathscr{G}, r \in \mathscr{R},$$

and therefore

$$\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) = R_n(g_{\mathbf{x}}, \mathrm{Id}), \qquad \mathbf{x} \in \mathscr{B}.$$

Theorem SA-23 gives a new Gaussian strong approximation that can be applied to our current setup. Specifically, our new theorem allows polynomial moment bounds on the conditional distribution of $Y_i|\mathbf{X}_i$.

Theorem SA-5 (Gaussian Strong Approximation: $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$)**.** *Suppose Assumptions SA-1 and SA-2 hold, and $\mathscr{X} = \prod_{l=1}^{d}[a_l, b_l]$, with $-\infty < a_l < b_l < \infty$ for $l = 1, \ldots, d$. Suppose also that there exists a constant $C > 0$ such that for $t \in \{0,1\}$ and for any $\mathbf{x} \in \mathscr{B}$, the De Giorgi perimeter of the set $E_{t,\mathbf{x}} = \{\mathbf{y} \in \mathscr{A}_t : \mathbf{H}^{-1}(\mathbf{y}-\mathbf{x}) \in \mathrm{supp}(K)\}$ satisfies $\mathscr{L}(E_{t,\mathbf{x}}) \leq C\bar{h}^{d-1}$. For the associated classes $\mathscr{G}$ and $\mathscr{R} = \{\mathrm{Id}\}$ defined above, suppose the singleton regularity condition in part (2) of Theorem SA-23 holds. If $\liminf_{n\to\infty}\log\bar{h}/\log n > -\infty$ and $n|\mathbf{H}| \to \infty$, then (on a possibly enlarged probability space) there exists a mean-zero Gaussian process $Z^{(\boldsymbol{\nu})}$ indexed by $\mathscr{B}$ with almost surely continuous sample path such that*

$$\mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}}\big|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) - Z^{(\boldsymbol{\nu})}(\mathbf{x})\big|\Big] \lesssim (\log n)^{\frac{3}{2}}\left(\frac{1}{n|\mathbf{H}|}\right)^{\frac{1}{2d+2}\cdot\frac{v}{v+2}} + \log(n)\left(\frac{1}{n^{\frac{v}{2+v}}|\mathbf{H}|}\right)^{\frac{1}{2}},$$

where $\lesssim$ is up to a universal constant, and $Z^{(\boldsymbol{\nu})}$ has the same covariance structure as $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$; that is, $\mathbb{C}\mathrm{ov}[\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}_2)] = \mathbb{C}\mathrm{ov}[Z^{(\boldsymbol{\nu})}(\mathbf{x}_1), Z^{(\boldsymbol{\nu})}(\mathbf{x}_2)]$ for all $\mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}$.

The rectangular support condition in Theorem SA-5 is imposed only for the process-level strong approximation. It permits an affine rescaling of the score support to $[0,1]^d$; combined with the bounded and bounded-away-from-zero density in Assumption SA-1(ii), $\mathbb{P}_X$ can then be used as the surrogate measure and the Rosenblatt transform provides the normalizing transformation with bounded constants. See Cattaneo and Yu [2025, Section 3.1] for details. This normalization simplifies the verification of that stronger coupling and is not used in the pointwise, aggregate, or supremum-based confidence-band results. Because the local-polynomial estimators use only shrinking neighborhoods of $\mathscr{B} \subset \mathrm{int}(\mathscr{X})$, the condition is not practically restrictive for the process-level result in settings where the relevant score support can be represented by a compact rectangle after rescaling.

The De Giorgi perimeter condition in Theorem SA-5 is used only to verify the total variation requirement of the residual empirical process strong approximation theorem (Theorem SA-23) for the class of boundary-indexed local-polynomial weighting functions. Even when the kernel is smooth, the factors $\mathbb{1}(\mathbf{u} \in \mathscr{A}_t)$

introduce discontinuities along the assignment boundary. The localized perimeter bound controls the total variation of these discontinuities uniformly over $\mathbf{x} \in \mathscr{B}$, because the relevant jumps occur on the sets $E_{t,\mathbf{x}} = \{\mathbf{y} \in \mathscr{A}_t : \mathbf{H}^{-1}(\mathbf{y} - \mathbf{x}) \in \operatorname{supp}(K)\}$. The separate singleton regularity condition concerns the truncated conditional means introduced by the polynomial-moment truncation step. Informally, these additional technical restrictions avoid an overly "wiggly" assignment boundary and irregular truncation-induced conditional means.

The following lemma shows that the simpler boundary assumptions imposed in the paper are sufficient to verify the De Giorgi perimeter condition for standard compactly supported kernels.

Lemma SA-6 (Localized Perimeter: Piecewise-Linear Boundary). *Suppose $d = 2$, $\mathbf{H} = h\mathbf{I}_2$, $\mathscr{B}$ is piecewise linear with finitely many line segments, $\mathscr{B}$ lies in the interior of $\mathscr{X}$, and $\operatorname{supp}(K)$ is bounded. If either $K(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in [-1,1]^2)$ or $\operatorname{supp}(K)$ has finite De Giorgi perimeter, then, for $t \in \{0,1\}$,*

$$\sup_{\mathbf{x}\in\mathscr{B}} \mathscr{L}\big(\{\mathbf{y} \in \mathscr{A}_t : h^{-1}(\mathbf{y} - \mathbf{x}) \in \operatorname{supp}(K)\}\big) \lesssim h.$$

Theorem SA-5 can be used to construct confidence bands for $(\tau^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$. Let $(\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$ be a (conditionally on $\mathbf{V}$) mean-zero Gaussian process with feasible (conditional) covariance function

$$\mathbb{C}\mathrm{ov}\big[\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}_2)\big|\mathbf{V}\big] = \frac{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_1}\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_2,\mathbf{x}_2}}}, \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}.$$

Theorem SA-6 (Confidence Bands: Sharp BATEC; process approximation). *Suppose the assumptions and conditions in Theorem SA-5 hold. If $\liminf_{n\to\infty} \log \bar{h}/\log n > -\infty$, $(\log n)^3/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$ and $n|\mathbf{H}|\bar{h}^{2(p+1)}\log n = o(1)$, then*

$$\sup_{u\in\mathbb{R}} \Big|\mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \leq u\Big) - \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \leq u\Big|\mathbf{V}\Big)\Big| = o_{\mathbb{P}}(1)$$

and

$$\mathbb{P}\Big[\tau^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_\alpha(\mathbf{x}), \text{ for all } \mathbf{x} \in \mathscr{B}\Big] = 1 - \alpha + o(1),$$

provided that $\mathscr{q}_\alpha = \inf\big\{c > 0 : \mathbb{P}\big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \geq c\big|\mathbf{V}\big) \leq \alpha\big\}$.

Theorem SA-5 gives a strong approximation of the entire process $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$ by a Gaussian process $Z^{(\boldsymbol{\nu})}$. This process-level coupling is stronger than what is needed for pointwise normal approximation, and also stronger than the supremum approximation needed for confidence bands. Indeed, as demonstrated in the following theorem, approximating only the supremum of the process using the results in Chernozhukov et al. [2014b] removes the need for the De Giorgi perimeter condition, and allows for a wider class of boundaries $\mathscr{B}$.

Theorem SA-7 (Gaussian Approximation for $\sup_{\mathbf{x}\in\mathscr{B}} |\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|$). *Suppose Assumptions SA-1 and SA-2 hold. Then for n large enough, on a possibly enriched probability space, there exists a Gaussian process $Z^{(\boldsymbol{\nu})}(\mathbf{x})$, $\mathbf{x} \in \mathscr{B}$, with almost sure continuous trajectories such that*

$$\mathbb{C}\mathrm{ov}[Z^{(\boldsymbol{\nu})}(\mathbf{x}_1), Z^{(\boldsymbol{\nu})}(\mathbf{x}_2)] = \mathbb{C}\mathrm{ov}[\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}_2)], \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B},$$

and

$$\mathbb{P}\Big[\big|\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|-\sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|\big|>\frac{\log n}{(n|\mathbf{H}|)^{1/6}}+\frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}}+\frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}}|\mathbf{H}|)^{1/2}}\Big]\lesssim\frac{1}{\log n}+\frac{\log n}{n}.$$

Using this result, we obtain the following theorem for confidence bands that does not require the De Giorgi perimeter condition.

Theorem SA-8 (Confidence Bands: Sharp BATEC; suprema approximation)**.** *Suppose Assumptions SA-1 and SA-2 hold. If* $\liminf_{n\to\infty}\log\bar{h}/\log n>-\infty$, $(\log n)^9/(n^{v/(2+v)}|\mathbf{H}|)=o(1)$, *and* $\bar{h}^{2(p+1)}n|\mathbf{H}|\log n=o(1)$*, then*

$$\sup_{u\in\mathbb{R}}\Big|\mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big|\leq u\Big)-\mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big|\leq u\Big|\mathbf{V}\Big)\Big|=o_{\mathbb{P}}(1)$$

and

$$\mathbb{P}\Big[\tau^{(\boldsymbol{\nu})}(\mathbf{x})\in\widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\alpha}(\mathbf{x}),\ \textit{for all }\mathbf{x}\in\mathscr{B}\Big]=1-\alpha+o(1),$$

provided that $\mathscr{q}_\alpha=\inf\big\{c>0:\mathbb{P}\big(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big|\geq c\big|\mathbf{V}\big)\leq\alpha\big\}$.

SA-3.3 Covariate Adjustment

Following Calonico et al. [2019] and Calonico et al. [2026], we can incorporate covariate adjustment into our framework. To conserve space, we only give a brief description of the basic idea. Suppose that $\mathbf{Z}_1,\ldots,\mathbf{Z}_n$ are pre-intervention covariates of dimension $d_Z\geq 1$.

For efficiency improvements, the covariate-adjusted BATEC estimator is

$$\widetilde{\tau}(\mathbf{x})=\mathbf{e}_{\mathfrak{p}_p+1}^\top\widetilde{\boldsymbol{\gamma}}(\mathbf{x})$$

with

$$\widetilde{\boldsymbol{\gamma}}(\mathbf{x})=\operatorname*{arg\,min}_{\boldsymbol{\gamma}\in\mathbb{R}^{2\mathfrak{p}_p+d_Z}}\mathbb{E}_n\big[\big(Y_i-\widetilde{\mathbf{r}}_p(\mathbf{X}_i-\mathbf{x},T_i,\mathbf{Z}_i)^\top\boldsymbol{\gamma}\big)^2K_h(\mathbf{X}_i-\mathbf{x})\big],$$

where $T_i=\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_1)$, and

$$\widetilde{\mathbf{r}}_p(\mathbf{X}_i-\mathbf{x},T_i,\mathbf{Z}_i)=[\mathbf{r}_p(\mathbf{X}_i-\mathbf{x})^\top,T_i\cdot\mathbf{r}_p(\mathbf{X}_i-\mathbf{x})^\top,\mathbf{Z}_i^\top]^\top$$

contains the full polynomial basis function for each treatment group but the pre-intervention covariates are not interacted with the treatment indicator.

For heterogeneity analysis, the covariate-adjusted BATEC estimator is

$$\check{\kappa}(\mathbf{x},\mathbf{z})=\mathbf{e}_{\mathfrak{p}_p+1}^\top\check{\boldsymbol{\gamma}}(\mathbf{x})+\mathbf{z}^\top\left[\mathbf{0}_{d_Z\times(2+d_Z)\mathfrak{p}_p}\quad\mathbf{I}_{d_Z}\quad\mathbf{0}_{d_Z\times(\mathfrak{p}_p-1)d_Z}\right]\check{\boldsymbol{\gamma}}(\mathbf{x})$$

with

$$\check{\boldsymbol{\gamma}}(\mathbf{x})=\operatorname*{arg\,min}_{\boldsymbol{\gamma}\in\mathbb{R}^{2\mathfrak{p}_p(d_Z+1)}}\mathbb{E}_n\big[\big(Y_i-\check{\mathbf{r}}_p(\mathbf{X}_i-\mathbf{x},T_i,\mathbf{Z}_i)^\top\boldsymbol{\gamma}\big)^2K_h(\mathbf{X}_i-\mathbf{x})\big],$$

where $\mathbf{I}_d$ denotes the $(d \times d)$ identity matrix, $\mathbf{0}_{d_1 \times d_2}$ denotes the $(d_1 \times d_2)$ matrix of zeros, $\mathbf{z}$ takes values on the support of $\mathbf{Z}_i$, and

$$\check{\mathbf{r}}_p(\mathbf{X}_i - \mathbf{x}, T_i, \mathbf{Z}_i) = [\mathbf{r}_p(\mathbf{X}_i - \mathbf{x})^\top, T_i \cdot \mathbf{r}_p(\mathbf{X}_i - \mathbf{x})^\top, (\mathbf{r}_p(\mathbf{X}_i - \mathbf{x}) \otimes \mathbf{Z}_i)^\top, T_i \cdot (\mathbf{r}_p(\mathbf{X}_i - \mathbf{x}) \otimes \mathbf{Z}_i)^\top]^\top$$

contains the full interaction between the polynomial basis function, the treatment assignment indicator, and the pre-intervention covariates.

Employing the theoretical results in the supplemental appendix, it is not difficult to develop valid pointwise and uniform estimation and inference results for these covariate-adjusted BATEC estimators, transformations thereof, and their fuzzy counterparts. We plan to formalize these results in future research.

SA-4 Weighted Boundary Average Treatment Effect

Without loss of generality, we set $\int_{\mathscr{B}} w(\mathbf{b})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) = 1$. For $|\boldsymbol{\nu}| \leq p$, the sharp (derivative) WBATE parameter is

$$\tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \int_{\mathscr{B}} \tau^{(\boldsymbol{\nu})}(\mathbf{b}) w(\mathbf{b})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}),$$

where the weight function satisfies Assumption SA-3. The original WBATE parameter is the special case $\boldsymbol{\nu} = \mathbf{0}$.

The sharp (derivative) WBATE estimator is

$$\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \int_{\mathscr{B}} \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) w(\mathbf{b})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}),$$

Our first lemma in this section studies the conditional bias of $\tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}$. Let

$$B^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}(\mathbf{h}) = B^{(\boldsymbol{\nu})}_{1,\mathtt{WBATE}}(\mathbf{h}) - B^{(\boldsymbol{\nu})}_{0,\mathtt{WBATE}}(\mathbf{h}), \qquad B^{(\boldsymbol{\nu})}_{t,\mathtt{WBATE}}(\mathbf{h}) = \int_{\mathscr{B}} B^{(\boldsymbol{\nu})}_{t,\mathbf{b}}(\mathbf{h}) w(\mathbf{b})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}),$$

for $t \in \{0, 1\}$.

Lemma SA-7 (Bias: Sharp WBATE)**.** *Suppose Assumption SA-1(i)-(iii), SA-2 and SA-3 hold. If* $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\mathbb{E}[\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] - \tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = B^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}(\mathbf{h}) + o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).$$

If, in addition, $\mathbf{h} = \mathbf{c}h$ *for a fixed vector* $\mathbf{c}$ *and common bandwidth* h*, then* $B^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}(\mathbf{h}) = h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}(\mathbf{c}) + o(h^{p+1-|\boldsymbol{\nu}|})$.

The next lemma studies the conditional variance of $\tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}$, and a plug-in estimator thereof. Let

$$\Omega^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \Omega^{(\boldsymbol{\nu})}_{1,\mathtt{WBATE}} + \Omega^{(\boldsymbol{\nu})}_{0,\mathtt{WBATE}}, \qquad \Omega^{(\boldsymbol{\nu})}_{t,\mathtt{WBATE}} = \int_{\mathscr{B}}\int_{\mathscr{B}} \Omega^{(\boldsymbol{\nu})}_{t,\mathbf{b}_1,\mathbf{b}_2} w(\mathbf{b}_1) w(\mathbf{b}_2)\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2)$$

and impose the aggregate variance nondegeneracy condition

$$\Omega^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} \gtrsim \frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}. \tag{SA-2}$$

This scalar condition rules out cancellation of the local covariance kernel after integrating the weighted boundary process; it is automatically distinct from the pointwise variance lower bound when derivative aggregates are considered. For the aggregate variance and inference results below, we also use the local boundary-measure regularity condition

$$\sup_{\mathbf{x}\in\mathscr{B}} \mathfrak{H}^{d-1}(\mathscr{B}\cap B(\mathbf{x},r)) \leq C_{\mathscr{B}} r^{d-1}, \qquad 0<r\leq r_0, \tag{SA-3}$$

for some constants $C_{\mathscr{B}}<\infty$ and $r_0>0$. In the proofs below, write $\nu_{\mathscr{B},k}$ for the k-fold product measure $(\mathfrak{H}^{d-1})^k$ on $\mathscr{B}^k$. Also let

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \widehat{\Omega}^{(\boldsymbol{\nu})}_{1,\mathtt{WBATE}} + \widehat{\Omega}^{(\boldsymbol{\nu})}_{0,\mathtt{WBATE}}, \qquad \widehat{\Omega}^{(\boldsymbol{\nu})}_{t,\mathtt{WBATE}} = \int_{\mathscr{B}}\int_{\mathscr{B}} \widehat{\Omega}^{(\boldsymbol{\nu})}_{t,\mathbf{b}_1,\mathbf{b}_2} w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2),$$

for $t\in\{0,1\}$.

Lemma SA-8 (Variance: Sharp WBATE)**.** *Suppose Assumptions SA-1, SA-2 and SA-3 hold, and* (SA-2) *and* (SA-3) *hold. If* $\frac{\log(1/\bar{h})}{n|\mathbf{H}|}=o(1)$ *and* $\bar{h}=o(1)$*, then*

$$\mathbb{V}[\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] = \Omega^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big),$$

where

$$\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} \lesssim \Omega^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} \lesssim \frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}.$$

Furthermore,

$$\mathbb{V}[\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] = \widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big).$$

Theorem SA-9 (MSE Expansion: Sharp WBATE)**.** *Suppose Assumptions SA-1, SA-2 and SA-3 hold, and* (SA-2) *and* (SA-3) *hold. If* $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|)=o(1)$ *and* $\bar{h}=o(1)$*, then*

$$\mathbb{E}[(\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}-\tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}})^2|\mathbf{X}] = \Omega^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} + \big(B^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}(\mathbf{h})\big)^2 + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big) + o_{\mathbb{P}}(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}}).$$

MSE-optimal diagonal bandwidth selection follows directly from Theorem SA-9; scalar bandwidth selectors are obtained as the proportional-bandwidth special case $h_j=c_jh$.

For inference, we consider the feasible t-statistics

$$\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \frac{\widehat{\tau}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}-\tau^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}}}.$$

Theorem SA-10 (Asymptotic Normality: Sharp WBATE)**.** *Suppose Assumptions SA-1, SA-2 and SA-3 hold, and* (SA-2) *and* (SA-3) *hold. If* $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|)=o(1)$ *and* $n|\mathbf{H}|\bar{h}^{2p+2}/\bar{h}^{d-1}=o(1)$*, then*

$$\sup_{u\in\mathbb{R}}\big|\mathbb{P}(\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}\leq u)-\Phi(u)\big| = o(1).$$

Condition (SA-3) controls the amount of boundary measure in bandwidth-sized neighborhoods, which is what gives the $\bar{h}^{d-1}$ effective integration factor in aggregate variances. The piecewise-linear boundary and scalar bandwidth assumptions used in the main paper imply this condition.

SA-5 Largest Boundary Average Treatment Effect

For $|\boldsymbol{\nu}| \leq p$, we consider the sharp (derivative) LBATE estimand and estimator

$$\tau^{(\boldsymbol{\nu})}_{\texttt{LBATE}} = \sup_{\mathbf{b}\in\mathscr{B}} \tau^{(\boldsymbol{\nu})}(\mathbf{b}) \qquad \text{and} \qquad \widehat{\tau}^{(\boldsymbol{\nu})}_{\texttt{LBATE}} = \sup_{\mathbf{b}\in\mathscr{B}} \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}).$$

The original LBATE parameter is the special case $\boldsymbol{\nu} = \mathbf{0}$.

Theorem SA-11 (Convergence Rate: Sharp LBATE)**.** *Suppose Assumptions SA-1 and SA-2 hold. If* $\log(1/\bar{h})/(n|\mathbf{H}|) = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\big|\widehat{\tau}^{(\boldsymbol{\nu})}_{\texttt{LBATE}} - \tau^{(\boldsymbol{\nu})}_{\texttt{LBATE}}\big| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Big).$$

Recall from Theorem SA-8 that $(\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$ is a (conditionally on $\mathbf{V}$) mean-zero Gaussian process with feasible (conditional) covariance function

$$\mathbb{C}\mathrm{ov}\Big[\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}_2)\Big|\mathbf{V}\Big] = \frac{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_1,\mathbf{x}_1}\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x}_2,\mathbf{x}_2}}}, \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}.$$

Consider the confidence interval

$$\widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\alpha,\texttt{LBATE}} = \Bigg[\sup_{\mathbf{b}\in\mathscr{B}}\Big(\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}}}\Big), \ \sup_{\mathbf{b}\in\mathscr{B}}\Big(\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}}}\Big)\Bigg],$$

where $\mathscr{q}_\alpha = \inf\big\{c > 0 : \mathbb{P}\big(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \geq c\big|\mathbf{V}\big) \leq \alpha\big\}$.

Theorem SA-12 (Confidence Interval: Sharp LBATE)**.** *Suppose the assumptions and conditions in Theorem SA-8 hold. Then*

$$\mathbb{P}\Big[\tau^{(\boldsymbol{\nu})}_{\texttt{LBATE}} \in \widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\alpha,\texttt{LBATE}}\Big] \geq 1 - \alpha + o(1).$$

SA-6 Imperfect Compliance

We extend our previous results to settings with imperfect treatment compliance, where treatment assignment and treatment status may not be equal for some units. See, e.g., Hernán and Robins [2020] for a textbook introduction.

We generalize our notation and assumptions to explicitly account for imperfect treatment compliance. Let

$$W_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\cdot W_i(0) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\cdot W_i(1)$$

be the observed treatment status, where $W_i(t)$ denotes the potential treatment status under treatment assignment $t \in \{0,1\}$ for each unit. Accordingly, the observed outcome now is

$$Y_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0) \cdot Y_i(0, W_i(0)) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1) \cdot Y_i(1, W_i(1)),$$

where the potential outcomes are now functions of two arguments: $Y_i(t,r)$ denotes the potential outcome for unit i when this unit is assigned to treatment $t \in \{0,1\}$ and takes treatment status $r \in \{0,1\}$.

The usual (fuzzy) estimands and estimators in settings with imperfect treatment compliance are, respectively,

$$\zeta(\mathbf{x}) = \frac{\tau_Y(\mathbf{x})}{\tau_W(\mathbf{x})} \qquad \text{and} \qquad \widehat{\zeta}(\mathbf{x}) = \frac{\widehat{\tau}_Y(\mathbf{x})}{\widehat{\tau}_W(\mathbf{x})},$$

where, for each $\mathbf{x} \in \mathscr{B}$,

$$\tau_Y(\mathbf{x}) = \mathbb{E}[Y_i(1, W_i(1)) - Y_i(0, W_i(0)) | \mathbf{X}_i = \mathbf{x}] \qquad \text{and} \qquad \tau_W(\mathbf{x}) = \mathbb{E}[W_i(1) - W_i(0) | \mathbf{X}_i = \mathbf{x}],$$

and

$$\widehat{\tau}_Y(\mathbf{x}) = \mathbf{e}_{\mathbf{0}}^\top \widehat{\boldsymbol{\beta}}_{Y,1}(\mathbf{x}) - \mathbf{e}_{\mathbf{0}}^\top \widehat{\boldsymbol{\beta}}_{Y,0}(\mathbf{x}) \qquad \text{and} \qquad \widehat{\tau}_W(\mathbf{x}) = \mathbf{e}_{\mathbf{0}}^\top \widehat{\boldsymbol{\beta}}_{W,1}(\mathbf{x}) - \mathbf{e}_{\mathbf{0}}^\top \widehat{\boldsymbol{\beta}}_{W,0}(\mathbf{x})$$

with $\widehat{\boldsymbol{\beta}}_{A,t}(\mathbf{x})$ denoting the local polynomial fit (SA-1) when using outcome variable $A \in \{Y, W\}$ on side $t \in \{0,1\}$.

To recycle the notation and assumptions from the sharp BD design setup, we redefine $Y_i(t) = Y_i(t, W_i(t))$ for each $t \in \{0,1\}$, which enables us to reuse Assumption SA-1 in the fuzzy BD design setting. We also impose the following analogous assumption for the reduced-form potential outcome/status pairs.

Assumption SA-4 (Fuzzy DGP)**.** *Let* $t \in \{0,1\}$.

(i) $(Y_1(t), W_1(t), \mathbf{X}_1^\top)^\top, \ldots, (Y_n(t), W_n(t), \mathbf{X}_n^\top)^\top$ *are independent and identically distributed random vectors.*

(ii) $\mu_{W,t}(\mathbf{x}) = \mathbb{E}[W_i(t) | \mathbf{X}_i = \mathbf{x}]$ *has a* $(p+1)$*-times continuously differentiable extension to an open neighborhood of* $\mathscr{X}$*.*

(iii) $\boldsymbol{\Upsilon}_t(\mathbf{x}) = \mathbb{C}\mathrm{ov}[(Y_i(t), W_i(t))^\top | \mathbf{X}_i = \mathbf{x}]$ *has entries that are continuous on* $\mathscr{X}$*.*

More generally, for $|\boldsymbol{\nu}| \le p$, we consider Wald-type ratios of derivative discontinuities of the form

$$\zeta^{(\boldsymbol{\nu})}(\mathbf{x}) = \frac{\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})} \qquad \text{and} \qquad \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) = \frac{\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})}.$$

For $\boldsymbol{\nu} = \mathbf{0}$ this is the usual fuzzy Wald curve. For $|\boldsymbol{\nu}| > 0$, it is a ratio of derivative jumps, not the derivative of the level Wald ratio. Here, for each $\mathbf{x} \in \mathscr{B}$,

$$\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x}) = \mu_{Y,1}^{(\boldsymbol{\nu})}(\mathbf{x}) - \mu_{Y,0}^{(\boldsymbol{\nu})}(\mathbf{x}) \qquad \text{and} \qquad \tau_W^{(\boldsymbol{\nu})}(\mathbf{x}) = \mu_{W,1}^{(\boldsymbol{\nu})}(\mathbf{x}) - \mu_{W,0}^{(\boldsymbol{\nu})}(\mathbf{x}),$$

and

$$\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x}) = \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\beta}}_{Y,1}(\mathbf{x}) - \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\beta}}_{Y,0}(\mathbf{x}) \qquad \text{and} \qquad \widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x}) = \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\beta}}_{W,1}(\mathbf{x}) - \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \widehat{\boldsymbol{\beta}}_{W,0}(\mathbf{x})$$

with $\widehat{\boldsymbol{\beta}}_{A,t}(\mathbf{x})$ denoting the local polynomial fit (SA-1) when using outcome variable $A \in \{Y, W\}$ on side $t \in \{0,1\}$. That is, under the assumptions imposed,

$$\widehat{\boldsymbol{\beta}}_{A,t}(\mathbf{x}) = \mathbf{S}^{-1}\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1}\mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})A_i\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big], \qquad A \in \{Y, W\},$$

corresponds to the local polynomial regression fit when using either Y_i or W_i as outcome. We also define

$$\mathbf{Q}_{A,t,\mathbf{x}} = \mathbb{E}_n\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\varepsilon_{A,i}\big],$$

where $\varepsilon_{A,i} = A_i - [\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\mu_{A,0}(\mathbf{X}_i) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\mu_{A,1}(\mathbf{X}_i)] = A_i - \mathbb{E}[A_i|\mathbf{X}_i]$, where $\mu_{A,t}(\mathbf{X}_i) = \mathbb{E}[A_i(t)|\mathbf{X}_i]$ and $A \in \{Y, W\}$. Let $\boldsymbol{\beta}_{A,t}(\mathbf{x})$ denote the corresponding population Taylor coefficient vector satisfying

$$\boldsymbol{\beta}_{A,t}(\mathbf{x})^\top\mathbf{r}_p(\mathbf{u}) = \sum_{|\boldsymbol{\omega}|\leq p}\frac{\mu_{A,t}^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!}\mathbf{u}^{\boldsymbol{\omega}}, \qquad \mathbf{u} \in \mathbb{R}^d.$$

Finally, for $A_1, A_2 \in \{Y, W\}$, $\mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}$ and $t \in \{0,1\}$, we introduce the following population quantities

$$\Omega^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}_1,\mathbf{x}_2} = \Omega^{(\boldsymbol{\nu})}_{A_1,A_2,0,\mathbf{x}_1,\mathbf{x}_2} + \Omega^{(\boldsymbol{\nu})}_{A_1,A_2,1,\mathbf{x}_1,\mathbf{x}_2}, \qquad \Omega^{(\boldsymbol{\nu})}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2} = \frac{(\boldsymbol{\nu}!)^2}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\mathbf{e}_{\boldsymbol{\nu}}^\top\boldsymbol{\Gamma}_{t,\mathbf{x}_1}^{-1}\boldsymbol{\Sigma}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2}\boldsymbol{\Gamma}_{t,\mathbf{x}_2}^{-1}\mathbf{e}_{\boldsymbol{\nu}},$$

where

$$\begin{aligned}\boldsymbol{\Sigma}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2} = |\mathbf{H}|\mathbb{E}\big[&\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_1))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_2))^\top \\ &\cdot K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_1)K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_2)\sigma^2_{A_1,A_2,t}(\mathbf{X}_i)\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],\end{aligned}$$

with $\sigma^2_{A_1,A_2,t}(\mathbf{X}_i) = \mathbb{C}\mathrm{ov}(A_1(t), A_2(t)|\mathbf{X}_i)$, and their sample analogs

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}_1,\mathbf{x}_2} = \widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,0,\mathbf{x}_1,\mathbf{x}_2} + \widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,1,\mathbf{x}_1,\mathbf{x}_2}, \qquad \widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2} = \frac{(\boldsymbol{\nu}!)^2}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\mathbf{e}_{\boldsymbol{\nu}}^\top\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}_1}^{-1}\widehat{\boldsymbol{\Sigma}}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2}\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}_2}^{-1}\mathbf{e}_{\boldsymbol{\nu}},$$

where

$$\begin{aligned}\widehat{\boldsymbol{\Sigma}}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2} = |\mathbf{H}|\mathbb{E}_n\big[&\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_1))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_2))^\top \\ &\cdot K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_1)K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_2)\widehat{\varepsilon}_{A_1,i}(\mathbf{x}_1)\widehat{\varepsilon}_{A_2,i}(\mathbf{x}_2)\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],\end{aligned}$$

where $\widehat{\varepsilon}_{A,i}(\mathbf{x}) = A_i - \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top\mathbf{S}[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\widehat{\boldsymbol{\beta}}_{A,0}(\mathbf{x}) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\widehat{\boldsymbol{\beta}}_{A,1}(\mathbf{x})]$, with $A \in \{Y, W\}$.

SA-6.1 Second-order Linearization

The results in this section will follow from the following exact second-order linearization:

$$\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) - \zeta^{(\boldsymbol{\nu})}(\mathbf{x}) = \mathfrak{w}(\mathbf{x})^\top\mathfrak{T}(\mathbf{x}) + \mathfrak{R}_{\mathtt{F}}(\mathbf{x})$$

with

$$\mathfrak{w}(\mathbf{x}) = \Big[\frac{1}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})}, -\frac{\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})^2}\Big]^\top \qquad \text{and} \qquad \mathfrak{T}(\mathbf{x}) = \Big[\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau_Y^{(\boldsymbol{\nu})}(\mathbf{x}), \widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\Big]^\top,$$

and second-order error

$$\mathfrak{R}_{\mathtt{F}}(\mathbf{x})=\frac{\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})^2\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})}\big(\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\big)^2-\frac{1}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})}\big(\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})\big)\big(\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\big).$$

For example, under the assumptions imposed, if $\inf_{\mathbf{x}\in\mathscr{B}}|\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})|>0$, $\log(1/\bar{h})/(n^{\frac{1+v}{2+v}}|\mathbf{H}|)=o(1)$, $\bar{h}=o(1)$, and

$$\mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}+\bar{h}^{p+1}\Big)=o(1),$$

then two applications of Theorem SA-1 (taking as outcomes Y and W) imply

$$\begin{aligned}\sup_{\mathbf{x}\in\mathscr{B}}|\mathfrak{R}_{\mathtt{F}}(\mathbf{x})|&\lesssim_{\mathbb{P}}\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})|^2+\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})|\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})-\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})|\\&\lesssim_{\mathbb{P}}\mathbf{h}^{-2\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}+\bar{h}^{p+1}\Big)^2,\end{aligned}\tag{SA-4}$$

thereby demonstrating that the second-order term $\mathfrak{R}_{\mathtt{F}}(\mathbf{x})$ is (uniformly) higher-order relative to the leading term $\mathfrak{w}(\mathbf{x})^\top\mathfrak{T}(\mathbf{x})$.

SA-6.2 Fuzzy BATEC

SA-6.2.1 Point Estimation and AMSE Expansions

Theorem SA-13 (Convergence Rates: Fuzzy BATEC)**.** *Suppose Assumptions SA-1, SA-2, and SA-4 hold, and* $\inf_{\mathbf{x}\in\mathscr{B}}|\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})|>0$*. If* $\log(1/\bar{h})/(n^{\frac{1+v}{2+v}}|\mathbf{H}|)=o(1)$*,* $\bar{h}=o(1)$*, and*

$$\mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}+\bar{h}^{p+1}\Big)=o(1),$$

then

$$\big|\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x})-\zeta^{(\boldsymbol{\nu})}(\mathbf{x})\big|\lesssim_{\mathbb{P}}\mathbf{h}^{-\boldsymbol{\nu}}\Big(\frac{1}{\sqrt{n|\mathbf{H}|}}+\frac{1}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}+\bar{h}^{p+1}\Big)$$

for $\mathbf{x}\in\mathscr{B}$*, and*

$$\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x})-\zeta^{(\boldsymbol{\nu})}(\mathbf{x})\big|\lesssim_{\mathbb{P}}\mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}+\bar{h}^{p+1}\Big).$$

The approximate (conditional) mean-squared error (AMSE) is

$$\mathrm{AMSE}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})=\mathbb{E}\big[(\mathfrak{w}(\mathbf{x})^\top\mathfrak{T}(\mathbf{x}))^2\big|\mathbf{X}\big]=\mathfrak{w}(\mathbf{x})^\top\mathbb{E}\big[\mathfrak{T}(\mathbf{x})\mathfrak{T}(\mathbf{x})^\top\big|\mathbf{X}\big]\mathfrak{w}(\mathbf{x})$$

for $\mathbf{x}\in\mathscr{B}$, and the approximate (conditional) integrated MSE (AIMSE) is

$$\mathrm{AIMSE}_{\mathtt{F}}^{(\boldsymbol{\nu})}=\int_{\mathscr{B}}\mathrm{AMSE}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})w(\mathbf{x})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}),$$

where $w(\mathbf{x})$ satisfies Assumption SA-3. These constructions disregard the higher-order term $\mathfrak{R}_{\mathtt{F}}(\mathbf{x})$.

To state the AMSE expansions with imperfect compliance, we generalize the notation introduced previously for the leading bias and variance. For the bias, letting $A \in \{Y, W\}$, we define

$$B^{(\boldsymbol{\nu})}_{A,\mathbf{x}}(\mathbf{h}) = B^{(\boldsymbol{\nu})}_{A,1,\mathbf{x}}(\mathbf{h}) - B^{(\boldsymbol{\nu})}_{A,0,\mathbf{x}}(\mathbf{h}),$$

$$B^{(\boldsymbol{\nu})}_{A,t,\mathbf{x}}(\mathbf{h}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \sum_{|\boldsymbol{\omega}|=p+1} \frac{\mu^{(\boldsymbol{\omega})}_{A,t}(\mathbf{x})}{\boldsymbol{\omega}!} \mathbf{h}^{\boldsymbol{\omega}-\boldsymbol{\nu}} \mathbb{E}\big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^{\boldsymbol{\omega}} K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\big],$$

where recall that $\mu_{A,t}(\mathbf{x}) = \mathbb{E}[A_i(t)|\mathbf{X}_i = \mathbf{x}]$. Therefore, the leading bias takes the form:

$$B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h}) = \mathfrak{w}(\mathbf{x})^\top \mathbf{B}^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}), \qquad \mathbf{B}^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}) = \big(B^{(\boldsymbol{\nu})}_{Y,\mathbf{x}}(\mathbf{h}), B^{(\boldsymbol{\nu})}_{W,\mathbf{x}}(\mathbf{h})\big)^\top. \tag{SA-5}$$

When $\mathbf{h} = \mathbf{c}h$ for a fixed vector $\mathbf{c}$ and h is a common bandwidth, $B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h}) = h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c})$ for some $\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c})$.

For the variance, letting $A_1, A_2 \in \{Y, W\}$, we define

$$V^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}} = V^{(\boldsymbol{\nu})}_{A_1,A_2,0,\mathbf{x}} + V^{(\boldsymbol{\nu})}_{A_1,A_2,1,\mathbf{x}},$$

where

$$V^{(\boldsymbol{\nu})}_{A_1,A_2,t,\mathbf{x}} = (\boldsymbol{\nu}!)^2 \mathbf{e}_{\boldsymbol{\nu}}^\top \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \boldsymbol{\Sigma}_{A_1,A_2,t,\mathbf{x},\mathbf{x}} \boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}} \mathbf{e}_{\boldsymbol{\nu}} = n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}} \Omega^{(\boldsymbol{\nu})}_{A_1,A_2,t,\mathbf{x},\mathbf{x}}.$$

Therefore, the leading variance takes the form:

$$V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} = \mathfrak{w}(\mathbf{x})^\top \mathbf{C}^{(\boldsymbol{\nu})}_{\mathbf{x}} \mathfrak{w}(\mathbf{x}), \qquad \mathbf{C}^{(\boldsymbol{\nu})}_{\mathbf{x}} = \begin{pmatrix} V^{(\boldsymbol{\nu})}_{Y,Y,\mathbf{x}} & V^{(\boldsymbol{\nu})}_{Y,W,\mathbf{x}} \\ V^{(\boldsymbol{\nu})}_{W,Y,\mathbf{x}} & V^{(\boldsymbol{\nu})}_{W,W,\mathbf{x}} \end{pmatrix}.$$

For inference, we impose $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$ directly. This scalar nondegeneracy condition applies to the linearized fuzzy Wald score and does not require the reduced-form conditional covariance matrix of $(Y_i(t), W_i(t))$ to be uniformly positive definite.

Theorem SA-14 (AMSE Expansions: Fuzzy BATEC)**.** *Suppose Assumptions SA-1, SA-2, SA-3 and SA-4 hold, and $\inf_{\mathbf{x}\in\mathscr{B}} |\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$. If $\log(1/\bar{h})/(n^{\frac{v}{2+v}}|\mathbf{H}|) = o(1)$ and $\bar{h} = o(1)$, then*

$$\mathrm{AMSE}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) = \big(B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h})\big)^2 + \frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} + o_{\mathbb{P}}\big(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}} + (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\big)$$

for $\mathbf{x} \in \mathscr{B}$, and

$$\mathrm{AIMSE}^{(\boldsymbol{\nu})}_{\mathtt{F}} = \int_{\mathscr{B}} \Big[\big(B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h})\big)^2 + \frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}\Big] w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) + o_{\mathbb{P}}\big(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}} + (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\big).$$

Theorem SA-14 can be used to develop feasible bandwidth selectors for diagonal bandwidth matrices by minimizing the displayed criterion over $\mathbf{h} \in \mathbb{R}^d_{++}$. If $\mathbf{h} = \mathbf{c}h$ for a fixed vector $\mathbf{c}$ and a common bandwidth h, the problem reduces to scalar bandwidth selection. If $\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c}) \neq 0$, the approximate MSE-optimal common scale is

$$h_{\mathtt{F},\mathrm{AMSE},\boldsymbol{\nu}}(\mathbf{x}) = \left(\frac{(d+2|\boldsymbol{\nu}|)V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}}{(2p+2-2|\boldsymbol{\nu}|)(\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c}))^2}\frac{1}{n}\right)^{\frac{1}{2p+d+2}}$$

for $\mathbf{x} \in \mathscr{B}$. Similarly, if $\int_{\mathscr{B}}(\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c}))^2 w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) \neq 0$, the approximate IMSE-optimal common scale is

$$h_{\mathtt{F},\mathrm{AIMSE},\boldsymbol{\nu}} = \left(\frac{(d+2|\boldsymbol{\nu}|)\int_{\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}{(2p+2-2|\boldsymbol{\nu}|)\int_{\mathscr{B}}(\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{c}))^2 w(\mathbf{x})\, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})}\frac{1}{n}\right)^{\frac{1}{2p+d+2}}.$$

In practice, the unknown bias and variance quantities can be replaced with (consistent) estimators thereof, as discussed before.

SA-6.2.2 Distributional Approximation and Inference

Let $\mathbf{V} = ((Y_1, W_1, \mathbf{X}_1^\top), \cdots, (Y_n, W_n, \mathbf{X}_n^\top))^\top$, and recall that $t \in \{0,1\}$. For $|\boldsymbol{\nu}| \leq p$, define the feasible t-statistic

$$\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) = \frac{\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) - \zeta^{(\boldsymbol{\nu})}(\mathbf{x})}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}}}, \qquad \mathbf{x} \in \mathscr{B},$$

where we define

$$\widehat{\mathfrak{w}}(\mathbf{x}) = \Big[\frac{1}{\widehat{\tau}^{(\boldsymbol{\nu})}_W(\mathbf{x})}, -\frac{\widehat{\tau}^{(\boldsymbol{\nu})}_Y(\mathbf{x})}{\big(\widehat{\tau}^{(\boldsymbol{\nu})}_W(\mathbf{x})\big)^2}\Big]^\top,$$

and

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} = \widehat{\mathfrak{w}}(\mathbf{x}_1)^\top \widehat{\boldsymbol{\Theta}}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\widehat{\mathfrak{w}}(\mathbf{x}_2), \qquad \widehat{\boldsymbol{\Theta}}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} = \begin{pmatrix} \widehat{\Omega}^{(\boldsymbol{\nu})}_{Y,Y,\mathbf{x}_1,\mathbf{x}_2} & \widehat{\Omega}^{(\boldsymbol{\nu})}_{Y,W,\mathbf{x}_1,\mathbf{x}_2} \\ \widehat{\Omega}^{(\boldsymbol{\nu})}_{W,Y,\mathbf{x}_1,\mathbf{x}_2} & \widehat{\Omega}^{(\boldsymbol{\nu})}_{W,W,\mathbf{x}_1,\mathbf{x}_2}\end{pmatrix}, \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}.$$

We also define the population version of the denominator by

$$\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} = \mathfrak{w}(\mathbf{x}_1)^\top \boldsymbol{\Theta}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\mathfrak{w}(\mathbf{x}_2), \qquad \boldsymbol{\Theta}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} = \begin{pmatrix} \Omega^{(\boldsymbol{\nu})}_{Y,Y,\mathbf{x}_1,\mathbf{x}_2} & \Omega^{(\boldsymbol{\nu})}_{Y,W,\mathbf{x}_1,\mathbf{x}_2} \\ \Omega^{(\boldsymbol{\nu})}_{W,Y,\mathbf{x}_1,\mathbf{x}_2} & \Omega^{(\boldsymbol{\nu})}_{W,W,\mathbf{x}_1,\mathbf{x}_2}\end{pmatrix}, \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}.$$

Lemma SA-9 (Covariance: Fuzzy BATEC)**.** *Suppose Assumptions SA-1, SA-2 and SA-4 hold,* $\inf_{\mathbf{x}\in\mathscr{B}} |\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$*, and* $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$*. If* $\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} = o(1)$*,* $\bar{h} = o(1)$*, and*

$$\mathbf{h}^{-\boldsymbol{\nu}}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\right) = o(1),$$

then

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \big|\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} - \Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\big| \lesssim_{\mathbb{P}} \frac{\mathbf{h}^{-\boldsymbol{\nu}}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\right),$$

$$\sup_{\mathbf{x}\in\mathscr{B}} \big|(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}})^{-1/2} - (\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}})^{-1/2}\big| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg).$$

The associated $100(1-\alpha)\%$ confidence interval estimator is

$$\widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\mathtt{F},\alpha}(\mathbf{x}) = \Bigg[\ \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) - \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}}\ ,\ \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) + \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}}\ \Bigg],$$

where $\mathscr{q}_\alpha$ denotes an appropriate quantile depending on the desired confidence level $\alpha \in (0,1)$, and coverage objective (pointwise versus uniform over $\mathscr{B}$).

The following theorem establishes pointwise asymptotic normality and validity of confidence intervals.

Theorem SA-15 (Confidence Intervals: Fuzzy BATEC)**.** *For a fixed boundary point* $\mathbf{x} \in \mathscr{B}$*, suppose Assumptions SA-1, SA-2 and SA-4 hold,* $|\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$*, and* $V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$*. If* $n|\mathbf{H}| \to \infty$*,* $n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}} \to \infty$*, and* $n|\mathbf{H}|\bar{h}^{2(p+1)} \to 0$*, then*

$$\sup_{u\in\mathbb{R}}\Big|\mathbb{P}\big(\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) \le u\big) - \Phi(u)\Big| = o(1),$$

and

$$\mathbb{P}\big(\zeta^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\mathtt{F},\alpha}(\mathbf{x})\big) = 1 - \alpha + o(1),$$

provided that $\mathscr{q}_\alpha = \inf\{c > 0 : \mathbb{P}(|\widehat{Z}| \ge c|\mathbf{V}) \le \alpha\}$ *with* $\widehat{Z}|\mathbf{V} \sim \mathsf{Normal}(0,1)$*.*

For uniform inference, we rely on the strong approximation result established in Section SA-7. Since $\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}$ is not directly a sum of independent terms, we introduce the linearized statistic

$$\begin{aligned}\overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) = \mathbb{E}_n\Big[&(\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\big[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\boldsymbol{\Gamma}^{-1}_{1,\mathbf{x}} - \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\boldsymbol{\Gamma}^{-1}_{0,\mathbf{x}}\big]\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\\ &\cdot K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathfrak{w}(\mathbf{x})^\top\boldsymbol{\varepsilon}_i\Big],\end{aligned}$$

where $\boldsymbol{\varepsilon}_i = (\varepsilon_{Y,i}, \varepsilon_{W,i})^\top = (Y_i - \mathbb{E}[Y_i|\mathbf{X}_i], W_i - \mathbb{E}[W_i|\mathbf{X}_i])^\top$ is the vector of regression errors.

The following theorem quantifies the approximation error of the stochastic linearization:

Theorem SA-16 (Stochastic Linearization: Fuzzy BATEC)**.** *Suppose Assumptions SA-1, SA-2 and SA-4 hold,* $\inf_{\mathbf{x}\in\mathscr{B}}|\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$*, and* $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$*. If* $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$*,* $\bar{h} = o(1)$*, and*

$$\mathbf{h}^{-\boldsymbol{\nu}}\Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg) = o(1),$$

then

$$\sup_{\mathbf{x}\in\mathscr{B}}\Big|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) - \overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})\Big| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \mathbf{h}^{-\boldsymbol{\nu}}\ \sqrt{\log(1/\bar{h})}\Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg).$$

Next, we present a coupling result for the supremum of the fuzzy linearized T-statistics. Unlike Theorem SA-5, this result uses a Gaussian approximation for suprema and therefore requires only VC-type entropy and moment bounds, not the total-variation/perimeter condition used for the process-level coupling.

Theorem SA-17 (Gaussian Approximation for $\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})|$)**.** *Suppose Assumptions SA-1, SA-2 and SA-4 hold, $\inf_{\mathbf{x}\in\mathscr{B}}|\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, and $\inf_{\mathbf{x}\in\mathscr{B}} V_{\mathtt{F},\mathbf{x}}^{(\boldsymbol{\nu})} > 0$. Then for n large enough, on a possibly enriched probability space, there exists a Gaussian process $Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}), \mathbf{x}\in\mathscr{B}$ with almost sure continuous trajectories such that*

$$\mathbb{C}\mathrm{ov}[Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_1), Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_2)] = \mathbb{C}\mathrm{ov}[\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_2)], \qquad \mathbf{x}_1,\mathbf{x}_2\in\mathscr{B},$$

and

$$\mathbb{P}\Big[\big|\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| - \sup_{\mathbf{x}\in\mathscr{B}}|Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})|\big| > \frac{\log n}{(n|\mathbf{H}|)^{1/6}} + \frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}}|\mathbf{H}|)^{1/2}}\Big] \lesssim \frac{1}{\log n} + \frac{\log n}{n}.$$

To construct a feasible confidence band for $(\zeta^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x}\in\mathscr{B})$, let $(\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) : \mathbf{x}\in\mathscr{B})$ be a mean-zero Gaussian process conditional on $\mathbf{V}$. Its feasible conditional covariance function is defined as

$$\mathbb{C}\mathrm{ov}\left[\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_1), \widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}_2) \mid \mathbf{V}\right] = \frac{\widehat{\Omega}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}^{(\boldsymbol{\nu})}}{\sqrt{\widehat{\Omega}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_1}^{(\boldsymbol{\nu})}\widehat{\Omega}_{\mathtt{F},\mathbf{x}_2,\mathbf{x}_2}^{(\boldsymbol{\nu})}}}, \quad \mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}.$$

Theorem SA-18 (Confidence Bands: Fuzzy BATEC; suprema approximation)**.** *Suppose Assumptions SA-1, SA-2, and SA-4 hold, $\inf_{\mathbf{x}\in\mathscr{B}}|\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, and $\inf_{\mathbf{x}\in\mathscr{B}} V_{\mathtt{F},\mathbf{x}}^{(\boldsymbol{\nu})} > 0$. If $\liminf_{n\to\infty}\log\bar{h}/\log n > -\infty$, $(\log n)^9/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$, $\bar{h}^{p+1}\sqrt{n|\mathbf{H}|\log n} = o(1)$, and*

$$\mathbf{h}^{-\boldsymbol{\nu}}\sqrt{\log(1/\bar{h})}\left(\sqrt{\log(1/\bar{h})/(n|\mathbf{H}|)} + \log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|) + \bar{h}^{p+1}\right) = o(1),$$
$$\log^2(n)\mathbf{h}^{-\boldsymbol{\nu}}\left(\sqrt{\frac{\log n}{n|\mathbf{H}|}} + \frac{\log n}{n^{v/(2+v)}|\mathbf{H}|} + \bar{h}^{p+1}\right) = o(1),$$

then

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u\right) - \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u \,\middle|\, \mathbf{V}\right)\right| = o_{\mathbb{P}}(1)$$

and

$$\mathbb{P}\left(\zeta^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}_{\mathtt{F},\alpha}^{(\boldsymbol{\nu})}(\mathbf{x}) \textit{ for all } \mathbf{x}\in\mathscr{B}\right) = 1-\alpha+o(1),$$

provided that $\mathfrak{q}_\alpha = \inf\left\{c>0 : \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \ge c \,\middle|\, \mathbf{V}\right) \le \alpha\right\}$.

SA-6.3 Fuzzy WBATE

We now consider WBATE summaries of these fuzzy ratios of derivative discontinuities. For $|\boldsymbol{\nu}|\le p$, the parameter is

$$\zeta_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = \int_{\mathscr{B}} \zeta^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}),$$

where the weight function $w(\cdot)$ satisfies Assumption SA-3. Without loss of generality, we assume that $\int_{\mathscr{B}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) = 1$. The corresponding plug-in estimator is

$$\widehat{\zeta}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = \int_{\mathscr{B}} \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}).$$

The original fuzzy WBATE parameter is the special case $\boldsymbol{\nu} = \mathbf{0}$.

Our AMSE results for the fuzzy WBATE estimator leverage the leading linear component from the second-order linearization established in Section SA-6.1,

$$\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} = \int_{\mathscr{B}} \mathfrak{w}(\mathbf{b})^\top \mathfrak{T}(\mathbf{b}) w(\mathbf{b}) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}).$$

The corresponding approximate MSE is

$$\mathrm{AMSE}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} = \mathbb{E}\big[\big(\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}\big)^2 \big| \mathbf{X}\big].$$

The leading bias is defined as the integral of the pointwise leading bias of the fuzzy BATEC estimator:

$$B^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}(\mathbf{h}) = \int_{\mathscr{B}} B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}}(\mathbf{h}) \; w(\mathbf{b}) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}),$$

where $B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}}(\mathbf{h})$ is given in Equation (SA-5).

Lemma SA-10 (Bias: Fuzzy WBATE Linearization)**.** *Suppose Assumptions SA-1, SA-2, SA-3 and SA-4 hold, and* $\inf_{\mathbf{x}\in\mathscr{B}} |\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$. *If* $\frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|} = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\mathbb{E}[\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} | \mathbf{X}] = B^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}(\mathbf{h}) + o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).$$

If, in addition, $\mathbf{h} = \mathbf{c}h$ *for a fixed vector* $\mathbf{c}$ *and common bandwidth* h*, then* $B^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}(\mathbf{h}) = h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}(\mathbf{c})$.

The conditional variance of this linearized fuzzy WBATE component involves the integration of the cross-covariance structure of the reduced-form processes. Let

$$\Omega^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} = \int_{\mathscr{B}} \int_{\mathscr{B}} \Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2} w(\mathbf{b}_1) w(\mathbf{b}_2) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2),$$

and impose the aggregate variance nondegeneracy condition

$$\Omega^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} \gtrsim \frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}. \tag{SA-6}$$

This condition rules out cancellation after weighting and integrating the linearized fuzzy Wald process. Let its plug-in estimator be

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} = \int_{\mathscr{B}} \int_{\mathscr{B}} \widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2} w(\mathbf{b}_1) w(\mathbf{b}_2) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1) \, \mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2).$$

Lemma SA-11 (Variance: Fuzzy WBATE Linearization)**.** *Suppose the assumptions and conditions of Lemma SA-9 and Assumption SA-3 hold, and* (SA-6) *and* (SA-3) *hold. Then*

$$\mathbb{V}[\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} | \mathbf{X}] = \Omega^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big),$$

where

$$\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} \lesssim \Omega^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} \lesssim \frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}.$$

In addition,

$$\mathbb{V}[\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] = \widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big).$$

Theorem SA-19 (AMSE Expansion: Fuzzy WBATE)**.** *Suppose the assumptions of Lemma SA-11 hold, including (SA-3). If* $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$ *and* $\bar{h} = o(1)$*, then*

$$\mathrm{AMSE}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} = \Omega^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} + \big(B^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}(\mathbf{h})\big)^2 + o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} + \bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}}\Big).$$

For inference, we define the feasible t-statistic for the aggregate effect:

$$\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} = \frac{\widehat{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}} - \zeta^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}}}}.$$

Theorem SA-20 (Asymptotic Normality: Fuzzy WBATE)**.** *Suppose Assumptions SA-1, SA-2, SA-3 and SA-4 hold,* $\inf_{\mathbf{x}\in\mathscr{B}} |\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$*,* $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$*, and (SA-6) and (SA-3) hold. If* $\log(1/\bar{h})/(n^{v/(2+v)}|\mathbf{H}|) = o(1)$*,* $n|\mathbf{H}|\bar{h}^{2p+2}/\bar{h}^{d-1} = o(1)$*, and*

$$\frac{\mathbf{h}^{-2\boldsymbol{\nu}}\log^2(1/\bar{h})}{n|\mathbf{H}|\bar{h}^{d-1}} = o(1), \qquad \frac{\mathbf{h}^{-2\boldsymbol{\nu}}\log^4(1/\bar{h})}{n^{\frac{4v}{2+v}-1}|\mathbf{H}|^3\bar{h}^{d-1}} = o(1),$$

then

$$\sup_{u\in\mathbb{R}}\big|\mathbb{P}(\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F,WBATE}} \le u) - \Phi(u)\big| = o(1).$$

The local boundary-measure condition in (SA-3) is used here for the same reason as in the sharp WBATE results: it controls the effective size of the boundary pairs that can have nonzero local-kernel covariance. The more specialized piecewise-linear boundary conditions in the main paper imply this condition.

SA-6.4 Fuzzy LBATE

For $|\boldsymbol{\nu}| \le p$, we consider the LBATE summary of the fuzzy ratio of derivative discontinuities, with estimand and estimator

$$\zeta^{(\boldsymbol{\nu})}_{\mathtt{LBATE}} = \sup_{\mathbf{b}\in\mathscr{B}} \zeta^{(\boldsymbol{\nu})}(\mathbf{b}) \qquad \text{and} \qquad \widehat{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{LBATE}} = \sup_{\mathbf{b}\in\mathscr{B}} \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}).$$

The original fuzzy LBATE parameter is the special case $\boldsymbol{\nu} = \mathbf{0}$.

Theorem SA-21 (Convergence Rate: Fuzzy LBATE)**.** *Suppose the assumptions and conditions of Theorem SA-13 hold. Then*

$$\big|\widehat{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{LBATE}} - \zeta^{(\boldsymbol{\nu})}_{\mathtt{LBATE}}\big| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Big).$$

Recall from Theorem SA-18 that $(\widehat{Z}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$ is a (conditionally on $\mathbf{V}$) mean-zero Gaussian process with feasible (conditional) covariance function

$$\mathbb{C}\mathrm{ov}\Big[\widehat{Z}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}_1), \widehat{Z}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}_2)\Big|\mathbf{V}\Big] = \frac{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_1}\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_2,\mathbf{x}_2}}}, \qquad \mathbf{x}_1, \mathbf{x}_2 \in \mathscr{B}.$$

Consider the confidence interval

$$\widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\alpha,\mathtt{F},\mathtt{LBATE}} = \bigg[\sup_{\mathbf{b}\in\mathscr{B}}\Big(\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b},\mathbf{b}}}\Big), \sup_{\mathbf{b}\in\mathscr{B}}\Big(\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b},\mathbf{b}}}\Big)\bigg],$$

where $\mathscr{q}_\alpha = \inf\big\{c > 0 : \mathbb{P}\big(\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{Z}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})\big| \geq c\big|\mathbf{V}\big) \leq \alpha\big\}$.

Theorem SA-22 (Confidence Interval: Fuzzy LBATE)**.** *Suppose the assumptions and conditions in Theorem SA-18 hold. Then*

$$\mathbb{P}\Big[\zeta^{(\boldsymbol{\nu})}_{\mathtt{LBATE}} \in \widehat{\mathrm{I}}^{(\boldsymbol{\nu})}_{\alpha,\mathtt{F},\mathtt{LBATE}}\Big] \geq 1 - \alpha + o(1).$$

SA-7 Gaussian Strong Approximation

We present a Gaussian strong approximation theorem, which is the key technical tool behind Theorem SA-5. The theorem builds on and generalizes the results in Cattaneo and Yu [2025]. Consider the *residual-based empirical process* given by

$$R_n[g, r] = \frac{1}{\sqrt{n}}\sum_{i=1}^{n}\Big[g(\mathbf{x}_i)r(y_i) - \mathbb{E}[g(\mathbf{x}_i)r(y_i)|\mathbf{x}_i]\Big], \qquad g \in \mathscr{G}, r \in \mathscr{R},$$

where $\mathscr{G}$ and $\mathscr{R}$ are classes of functions satisfying certain regularity conditions.

SA-7.1 Definitions for Function Spaces

Let $\mathscr{F}$ be a class of measurable functions from a probability space $(\mathbb{R}^q, \mathscr{B}(\mathbb{R}^q), \mathbb{P})$ to $\mathbb{R}$. We introduce several definitions that capture properties of $\mathscr{F}$.

(i) $\mathscr{F}$ is pointwise measurable if it contains a countable subset $\mathscr{G}$ such that for any $f \in \mathscr{F}$, there exists a sequence $(g_m : m \geq 1) \subseteq \mathscr{G}$ such that $\lim_{m\to\infty} g_m(\mathbf{u}) = f(\mathbf{u})$ for all $\mathbf{u} \in \mathbb{R}^q$.

(ii) Let $\mathrm{supp}(\mathscr{F}) = \cup_{f\in\mathscr{F}}\mathrm{supp}(f)$. A probability measure $\mathbb{Q}_{\mathscr{F}}$ on $(\mathbb{R}^q, \mathscr{B}(\mathbb{R}^q))$ is a surrogate measure for $\mathbb{P}$ with respect to $\mathscr{F}$ if

  (i) $\mathbb{Q}_{\mathscr{F}}$ agrees with $\mathbb{P}$ on $\mathrm{supp}(\mathbb{P}) \cap \mathrm{supp}(\mathscr{F})$.

  (ii) $\mathbb{Q}_{\mathscr{F}}(\mathrm{supp}(\mathscr{F}) \setminus \mathrm{supp}(\mathbb{P})) = 0$.

  Let $\mathscr{Q}_{\mathscr{F}} = \mathrm{supp}(\mathbb{Q}_{\mathscr{F}})$.

(iii) For $q = 1$ and an interval $\mathscr{I} \subseteq \mathbb{R}$, the pointwise total variation of $\mathscr{F}$ over $\mathscr{I}$ is

$$\mathtt{pTV}_{\mathscr{F},\mathscr{I}} = \sup_{f\in\mathscr{F}}\sup_{P\geq 1}\sup_{\mathscr{P}_P\in\mathscr{I}}\sum_{i=1}^{P-1}|f(a_{i+1}) - f(a_i)|,$$

where $\mathscr{P}_P = \{(a_1, \ldots, a_P) : a_1 \leq \cdots \leq a_P\}$ denotes the collection of all partitions of $\mathscr{I}$.

(iv) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the total variation of $\mathscr{F}$ over $\mathscr{C}$ is

$$\mathtt{TV}_{\mathscr{F},\mathscr{C}} = \inf_{\mathscr{U} \in \mathscr{O}(\mathscr{C})} \sup_{f \in \mathscr{F}} \sup_{\phi \in \mathscr{D}_q(\mathscr{U})} \int_{\mathbb{R}^q} f(\mathbf{u}) \operatorname{div}(\phi)(\mathbf{u}) \, \mathrm{d}\mathbf{u} / \| \|\phi\|_2 \|_\infty,$$

where $\mathscr{O}(\mathscr{C})$ denotes the collection of all open sets that contains $\mathscr{C}$, and $\mathscr{D}_q(\mathscr{U})$ denotes the space of infinitely differentiable functions from $\mathbb{R}^q$ to $\mathbb{R}^q$ with compact support contained in $\mathscr{U}$.

(v) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the local total variation constant of $\mathscr{F}$ over $\mathscr{C}$, is a positive number $\mathtt{K}_{\mathscr{F},\mathscr{C}}$ such that for any cube $\mathscr{D} \subseteq \mathbb{R}^q$ with edges of length ℓ parallel to the coordinate axes,

$$\mathtt{TV}_{\mathscr{F},\mathscr{D}\cap\mathscr{C}} \leq \mathtt{K}_{\mathscr{F},\mathscr{C}} \ell^{q-1}.$$

(vi) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the envelopes of $\mathscr{F}$ over $\mathscr{C}$ are

$$\mathtt{M}_{\mathscr{F},\mathscr{C}} = \sup_{\mathbf{u} \in \mathscr{C}} M_{\mathscr{F},\mathscr{C}}(\mathbf{u}), \qquad M_{\mathscr{F},\mathscr{C}}(\mathbf{u}) = \sup_{f \in \mathscr{F}} |f(\mathbf{u})|, \qquad \mathbf{u} \in \mathscr{C}.$$

(vii) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the Lipschitz constant of $\mathscr{F}$ over $\mathscr{C}$ is

$$\mathtt{L}_{\mathscr{F},\mathscr{C}} = \sup_{f \in \mathscr{F}} \sup_{\mathbf{u}_1, \mathbf{u}_2 \in \mathscr{C}} \frac{|f(\mathbf{u}_1) - f(\mathbf{u}_2)|}{\|\mathbf{u}_1 - \mathbf{u}_2\|_\infty}.$$

(viii) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the L_1 bound of $\mathscr{F}$ over $\mathscr{C}$ is

$$\mathtt{E}_{\mathscr{F},\mathscr{C}} = \sup_{f \in \mathscr{F}} \int_{\mathscr{C}} |f| \, \mathrm{d}\mathbb{P}.$$

(ix) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the uniform covering number of $\mathscr{F}$ with envelope $M_{\mathscr{F},\mathscr{C}}$ over $\mathscr{C}$ is

$$\mathtt{N}_{\mathscr{F},\mathscr{C}}(\delta, M_{\mathscr{F},\mathscr{C}}) = \sup_{\mu} N(\mathscr{F}, \|\cdot\|_{\mu,2}, \delta \|M_{\mathscr{F},\mathscr{C}}\|_{\mu,2}), \qquad \delta \in (0, \infty),$$

where the supremum is taken over all finite discrete measures on $(\mathscr{C}, \mathscr{B}(\mathscr{C}))$. We assume that $M_{\mathscr{F},\mathscr{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathscr{C}$.

(x) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, the uniform entropy integral of $\mathscr{F}$ with envelope $M_{\mathscr{F},\mathscr{C}}$ over $\mathscr{C}$ is

$$J_{\mathscr{C}}(\delta, \mathscr{F}, M_{\mathscr{F},\mathscr{C}}) = \int_0^\delta \sqrt{1 + \log \mathtt{N}_{\mathscr{F},\mathscr{C}}(\varepsilon, M_{\mathscr{F},\mathscr{C}})} \, \mathrm{d}\varepsilon,$$

where it is assumed that $M_{\mathscr{F},\mathscr{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathscr{C}$.

(xi) For a non-empty $\mathscr{C} \subseteq \mathbb{R}^q$, $\mathscr{F}$ is a VC-type class with envelope $M_{\mathscr{F},\mathscr{C}}$ over $\mathscr{C}$ if (i) $M_{\mathscr{F},\mathscr{C}}$ is measurable and $M_{\mathscr{F},\mathscr{C}}(\mathbf{u})$ is finite for every $\mathbf{u} \in \mathscr{C}$, and (ii) there exist $\mathtt{c}_{\mathscr{F},\mathscr{C}} > 0$ and $\mathtt{d}_{\mathscr{F},\mathscr{C}} > 0$ such that

$$\mathtt{N}_{\mathscr{F},\mathscr{C}}(\varepsilon, M_{\mathscr{F},\mathscr{C}}) \leq \mathtt{c}_{\mathscr{F},\mathscr{C}} \varepsilon^{-\mathtt{d}_{\mathscr{F},\mathscr{C}}}, \qquad \varepsilon \in (0, 1).$$

If a surrogate measure $\mathbb{Q}_{\mathscr{F}}$ for $\mathbb{P}$ with respect to $\mathscr{F}$ has been assumed, and it is clear from the context, we drop the dependence on $\mathscr{C} = \mathcal{Q}_{\mathscr{F}}$ for all quantities in the previous definitions. That is, to save notation, we set $\mathtt{TV}_{\mathscr{F}} = \mathtt{TV}_{\mathscr{F},\mathcal{Q}_{\mathscr{F}}}$, $\mathtt{K}_{\mathscr{F}} = \mathtt{K}_{\mathscr{F},\mathcal{Q}_{\mathscr{F}}}$, $\mathtt{M}_{\mathscr{F}} = \mathtt{M}_{\mathscr{F},\mathcal{Q}_{\mathscr{F}}}$, $M_{\mathscr{F}}(\mathbf{u}) = M_{\mathscr{F},\mathcal{Q}_{\mathscr{F}}}(\mathbf{u})$, $\mathtt{L}_{\mathscr{F}} = \mathtt{L}_{\mathscr{F},\mathcal{Q}_{\mathscr{F}}}$, and so on, whenever there is no confusion.

SA-7.2 Residual-based Empirical Process

The following theorem generalizes Cattaneo and Yu [2025, Theorem 2] by requiring only bounded polynomial moments for y_i conditional on $\mathbf{x}_i$.

Theorem SA-23 (Strong Approximation for Residual-based Empirical Processes)**.** *Suppose $(\mathbf{z}_i = (\mathbf{x}_i, y_i) : 1 \leq i \leq n)$ are i.i.d. random vectors taking values in $(\mathbb{R}^{d+1}, \mathscr{B}(\mathbb{R}^{d+1}))$ with common law $\mathbb{P}_Z$, where $\mathbf{x}_i$ has distribution $\mathbb{P}_X$ supported on $\mathscr{X} \subseteq \mathbb{R}^d$, y_i has distribution $\mathbb{P}_Y$ supported on $\mathscr{Y} \subseteq \mathbb{R}$, $\sup_{\mathbf{x}\in\mathscr{X}} \mathbb{E}[|y_i|^{2+v}|\mathbf{x}_i = \mathbf{x}] \leq C_y < \infty$ for some $v > 0$, and the following conditions hold:*

- (i) *$\mathscr{G}$ is a real-valued pointwise measurable class of functions on $(\mathbb{R}^d, \mathscr{B}(\mathbb{R}^d), \mathbb{P}_X)$.*
- (ii) *There exists a surrogate measure $\mathbb{Q}_{\mathscr{G}}$ for $\mathbb{P}_X$ with respect to $\mathscr{G}$ such that $\mathbb{Q}_{\mathscr{G}} = \mathfrak{m} \circ \phi_{\mathscr{G}}$, where the normalizing transformation $\phi_{\mathscr{G}} : \mathcal{Q}_{\mathscr{G}} \to [0,1]^d$ is a diffeomorphism.*
- (iii) *$\mathscr{G}$ is a VC-type class with envelope $\mathtt{M}_{\mathscr{G}}$ over $\mathcal{Q}_{\mathscr{G}}$ with $\mathtt{c}_{\mathscr{G}} \geq e$ and $\mathtt{d}_{\mathscr{G}} \geq 1$.*
- (iv) *$\mathscr{R}$ is a real-valued pointwise measurable class of functions on $(\mathbb{R}, \mathsf{Borel}(\mathbb{R}), \mathbb{P}_Y)$.*
- (v) *$\mathscr{R}$ is a VC-type class with envelope $M_{\mathscr{R},\mathscr{Y}}$ over $\mathscr{Y}$ with $\mathtt{c}_{\mathscr{R},\mathscr{Y}} \geq e$ and $\mathtt{d}_{\mathscr{R},\mathscr{Y}} \geq 1$, where $M_{\mathscr{R},\mathscr{Y}}(y) + \mathtt{pTV}_{\mathscr{R},(-|y|,|y|)} \leq \mathtt{v}(1 + |y|)$ for all $y \in \mathscr{Y}$, for some $\mathtt{v} > 0$.*
- (vi) *There exists a constant $\mathtt{k}$ such that $|\log_2 \mathtt{E}_{\mathscr{G}}| + |\log_2 \mathtt{TV}| + |\log_2 \mathtt{M}_{\mathscr{G}}| \leq \mathtt{k} \log_2 n$, where the constant $\mathtt{TV} = \max\{\mathtt{TV}_{\mathscr{G}}, \mathtt{TV}_{\mathscr{G}\times\mathscr{U}_{\mathscr{R}},\mathcal{Q}_{\mathscr{G}}}\}$ with $\mathscr{U}_{\mathscr{R}} = \{\theta(\cdot, r, \tau) : r \in \mathscr{R}, \tau \in (0,\infty]\}$, and $\theta(\mathbf{x}, r, \tau) = \mathbb{E}[r(y_i)\mathbb{1}(|y_i| \leq \tau)|\mathbf{x}_i = \mathbf{x}]$ for $\mathbf{x} \in \mathscr{X}$.*

Define the residual-based empirical process

$$R_n(g, r) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} g(\mathbf{x}_i)(r(y_i) - \mathbb{E}[r(y_i)|\mathbf{x}_i]), \qquad g \in \mathscr{G}, r \in \mathscr{R}.$$

Then (1) on a possibly enlarged probability space, there exists a sequence of mean-zero Gaussian processes $(Z_n^R(g, r) : g \in \mathscr{G}, r \in \mathscr{R})$ with almost sure continuous trajectories such that:

- *$\mathbb{E}[R_n(g_1, r_1)R_n(g_2, r_2)] = \mathbb{E}[Z_n^R(g_1, r_1)Z_n^R(g_2, r_2)]$ for all $(g_1, r_1), (g_2, r_2) \in \mathscr{G} \times \mathscr{R}$, and*
- *$\mathbb{E}\big[\big\|R_n - Z_n^R\big\|_{\mathscr{G}\times\mathscr{R}}\big] \leq C\mathtt{v}\mathtt{d} \log(\mathtt{c}n)\rho_n$,*

with

$$\rho_n = \sqrt{\mathtt{d}\log(\mathtt{c}n)}\, \mathtt{r}_n^{\frac{v}{v+2}} (\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}})^{\frac{2}{v+2}} + \mathtt{M}_{\mathscr{G}} n^{-\frac{v/2}{2+v}} + \frac{\mathtt{M}_{\mathscr{G}}}{\sqrt{n}} \Big(\frac{\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}}{\mathtt{r}_n}\Big)^{\frac{2}{v+2}},$$

where C is a positive constant that depends only on v and C_y, $\mathtt{c} = \mathtt{c}_{\mathscr{G}} \mathtt{c}_{\mathscr{R},\mathscr{Y}} \mathtt{k}$, $\mathtt{d} = \mathtt{d}_{\mathscr{G}} + \mathtt{d}_{\mathscr{R},\mathscr{Y}} + \mathtt{k}$, and

$$\mathtt{r}_n = \min\Big\{ \frac{(\mathtt{c}_1^d \mathtt{M}_{\mathscr{G}}^{d+1} \mathtt{TV}^d \mathtt{E}_{\mathscr{G}})^{1/(2d+2)}}{n^{1/(2d+2)}}, \frac{(\mathtt{c}_1^{d/2} \mathtt{c}_2^{d/2} \mathtt{M}_{\mathscr{G}} \mathtt{TV}^{d/2} \mathtt{E}_{\mathscr{G}} \mathtt{L}^{d/2})^{1/(d+2)}}{n^{1/(d+2)}} \Big\},$$

$$\mathtt{c}_1 = d \sup_{\mathbf{x} \in \mathcal{Q}_{\mathscr{G}}} \prod_{j=1}^{d-1} \sigma_j(\nabla \phi_{\mathscr{G}}(\mathbf{x})), \qquad \mathtt{c}_2 = \sup_{\mathbf{x} \in \mathcal{Q}_{\mathscr{G}}} \frac{1}{\sigma_d(\nabla \phi_{\mathscr{G}}(\mathbf{x}))}, \qquad \mathtt{L} = \max\{\mathtt{L}_{\mathscr{G}}, \mathtt{L}_{\mathscr{G} \times \mathscr{U}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}}\};$$

and (2) if $\mathscr{R}$ is a singleton and the truncated conditional-mean class $\mathscr{U}_{\mathscr{R}}$ satisfies

$$\mathtt{TV}_{\mathscr{G} \times \mathscr{U}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}} \lesssim \mathtt{TV}_{\mathscr{G} \times \mathscr{V}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}}, \qquad \mathtt{L}_{\mathscr{G} \times \mathscr{U}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}} \lesssim \mathtt{L}_{\mathscr{G} \times \mathscr{V}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}},$$

then we can replace $\mathtt{TV}$ and $\mathtt{L}$ in the previous conditions and statements by $\mathtt{TV}_{sing} = \max\{\mathtt{TV}_{\mathscr{G}}, \mathtt{TV}_{\mathscr{G} \times \mathscr{V}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}}\}$, and $\mathtt{L}_{sing} = \max\{\mathtt{L}_{\mathscr{G}}, \mathtt{L}_{\mathscr{G} \times \mathscr{V}_{\mathscr{R}}, \mathcal{Q}_{\mathscr{G}}}\}$, respectively, with $\mathscr{V}_{\mathscr{R}} = \{\theta(\cdot, r) : r \in \mathscr{R}\}$, and $\theta(\mathbf{x}, r) = \mathbb{E}[r(y_i)|\mathbf{x}_i = \mathbf{x}]$ for $\mathbf{x} \in \mathcal{X}$.

Remark SA-1. *The class $\mathscr{U}_{\mathscr{R}}$ comprises truncated conditional means at all truncation levels. Its Lipschitz and total-variation constants can be bounded, for example, if $f(y \mid \mathbf{x})$ is Lipschitz in $\mathbf{x}$ uniformly over $(\mathbf{x}, y)$ in the support of $(\mathbf{x}_i, y_i)$. When $\mathscr{R}$ is a singleton, regularity of the untruncated conditional mean class $\mathscr{V}_{\mathscr{R}}$ is enough whenever the truncated conditional means inherit the same bounds, as stated in part (2) of the theorem; this is automatic under bounded support for the residual component and holds under standard smooth conditional-density conditions.*

SA-7.3 Proof of Theorem SA-23

We use a truncation argument. Let $\kappa_n \geq 1$ be the level of truncation. For each $r \in \mathscr{R}$, define

$$\tilde{r}(y) = r(y) \mathbb{1}(|y| \leq \kappa_n), \qquad y \in \mathbb{R},$$

and define the class $\tilde{\mathscr{R}} = \{\tilde{r} : r \in \mathscr{R}\}$. Suppose Z_n^R is a mean-zero Gaussian process indexed by $\mathscr{G} \times \mathscr{R} \cup \mathscr{G} \times \tilde{\mathscr{R}}$, whose construction is given below. Then

$$R_n(g, r) - Z_n^R(g, r) = \big[R_n(g, \tilde{r}) - Z_n^R(g, \tilde{r})\big] + \big[R_n(g, r) - R_n(g, \tilde{r})\big] + \big[Z_n^R(g, \tilde{r}) - Z_n^R(g, r)\big].$$

Part 1: Strong approximation for truncated residual empirical process.

The truncation class $\tilde{\mathscr{R}}$ is pointwise measurable whenever $\mathscr{R}$ is, and it is VC-type with envelope $M_{\tilde{\mathscr{R}},\mathscr{Y}} = M_{\mathscr{R},\mathscr{Y}} \mathbb{1}(|\cdot| \leq \kappa_n)$ over $\mathscr{Y}$ and constants $\mathtt{c}_{\mathscr{R},\mathscr{Y}}$ and $\mathtt{d}_{\mathscr{R},\mathscr{Y}}$ up to universal factors. Moreover, the envelope and pointwise total variation satisfy

$$M_{\tilde{\mathscr{R}},\mathscr{Y}}(y) + \mathtt{pTV}_{\tilde{\mathscr{R}},(-|y|,|y|)} \lesssim \mathtt{v} \kappa_n, \qquad y \in \mathscr{Y},$$

because truncating r only adds jumps at $\pm \kappa_n$. Therefore Cattaneo and Yu [2025, Theorem 2], applied with $\alpha = 0$ and with the bounded-outcome constant proportional to $\mathtt{v}\kappa_n$, implies on a possibly enlarged probability space that there exists a sequence of mean-zero Gaussian processes $(Z_n^R(g, r) : (g, r) \in \mathscr{G} \times \tilde{\mathscr{R}})$ with almost sure continuous trajectories on $(\mathscr{G} \times \tilde{\mathscr{R}}, \rho_{\mathbb{P}})$ such that $\mathbb{E}[R_n(g_1, r_1) R_n(g_2, r_2)] = \mathbb{E}[Z_n^R(g_1, r_1) Z_n^R(g_2, r_2)]$ for all $(g_1,r_1),(g_2,r_2)\in\mathscr{G}\times\tilde{\mathscr{R}}$, and, after integrating the tail bound in that theorem,

$$\begin{aligned}
&\mathbb{E}[\|R_n(g,\tilde r)-Z_n^R(g,\tilde r)\|_{\mathscr{G}\times\mathscr{R}}]\\
&\leq C_1\mathtt{v}\kappa_n\Bigg(\sqrt{d}\min\Big\{\frac{(\mathtt{c}_1^d\mathtt{M}_{\mathscr{G}}^{d+1}\mathtt{TV}^d\mathtt{E}_{\mathscr{G}})^{\frac{1}{2d+2}}}{n^{1/(2d+2)}},\frac{(\mathtt{c}_1^{\frac{d}{2}}\mathtt{c}_2^{\frac{d}{2}}\mathtt{M}_{\mathscr{G}}\mathtt{TV}^{\frac{d}{2}}\mathtt{E}_{\mathscr{G}}\mathtt{L}^{\frac{d}{2}})^{\frac{1}{d+2}}}{n^{1/(d+2)}}\Big\}(\mathtt{d}\log(\mathtt{c}n))^{3/2}+\frac{\mathtt{d}\log(\mathtt{c}n)}{\sqrt{n}}\mathtt{M}_{\mathscr{G}}\Bigg)\\
&=C_1\mathtt{v}\kappa_n\Bigg(\sqrt{d}\mathtt{r}_n(\mathtt{d}\log(\mathtt{c}n))^{\frac{3}{2}}+\frac{\mathtt{d}\log(\mathtt{c}n)}{\sqrt{n}}\mathtt{M}_{\mathscr{G}}\Bigg),
\end{aligned}$$

where C_1 is a positive universal constant. The total-variation constant $\mathtt{TV}$ used in the theorem controls the truncated class because

$$\mathscr{V}_{\tilde{\mathscr{R}}}=\{\theta(\cdot,r,\kappa_n):r\in\mathscr{R}\}\subseteq\mathscr{U}_{\mathscr{R}}.$$

Thus $\mathtt{TV}=\max\{\mathtt{TV}_{\mathscr{G}},\mathtt{TV}_{\mathscr{G}\times\mathscr{U}_{\mathscr{R}},\mathcal{Q}_{\mathscr{G}}}\}$ is an upper bound for $\max\{\mathtt{TV}_{\mathscr{G}},\mathtt{TV}_{\mathscr{G}\times\mathscr{V}_{\tilde{\mathscr{R}}},\mathcal{Q}_{\mathscr{G}}}\}$. The same argument applies to $\mathtt{L}$.

In the special case that $\mathscr{R}=\{r_*\}$ is a singleton, take $\tilde y_i=r_*(y_i)\mathbb{1}(|y_i|\leq\kappa_n)/(\mathtt{v}\kappa_n)$. Then $\mathbb{E}[\exp(|\tilde y_i|)]\leq 2$, and $\tilde y_i$ is supported on $\tilde{\mathscr{Y}}=[-1,1]$. Moreover,

$$\frac{1}{\mathtt{v}\kappa_n}R_n(g,\tilde r_*)=\frac{1}{\sqrt{n}}\sum_{i=1}^n g(\mathbf{x}_i)(\tilde y_i-\mathbb{E}[\tilde y_i|\mathbf{x}_i]),\qquad g\in\mathscr{G}.$$

The right-hand side is a residual empirical process based on the sample $(\mathbf{x}_i,\tilde y_i)$, $1\leq i\leq n$, indexed by $\mathscr{G}\times\{\mathrm{Id}\}$, where $\mathrm{Id}:\mathbb{R}\to\mathbb{R}$ is the identity function. Under the singleton domination condition in part (2) of the theorem, applying Cattaneo and Yu [2025, Theorem 2] with $\alpha=0$ to this empirical process gives the same upper bound with $\mathtt{TV}$ and $\mathtt{L}$ replaced by $\mathtt{TV}_{\mathrm{sing}}$ and $\mathtt{L}_{\mathrm{sing}}$.

Part 2: Truncation error for the empirical process — $\|R_n(g,r)-R_n(g,\widetilde r)\|_{\mathscr{G}\times\mathscr{R}}$

Consider the class of differences due to truncation, $\Delta\mathscr{R}=\{r-\tilde r:r\in\mathscr{R}\}$. The assumptions imply $\mathscr{G}\times\Delta\mathscr{R}$ is VC-type in the sense that for all $0<\varepsilon<1$,

$$\sup_{\mathcal{Q}}N(\mathscr{G}\times\Delta\mathscr{R},\|\cdot\|_{\mathcal{Q},2},\varepsilon\|\mathtt{M}_{\mathscr{G}}(M_{\mathscr{R},\mathscr{Y}}-M_{\tilde{\mathscr{R}},\mathscr{Y}})\|_{\mathcal{Q},2})\leq\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}}(\varepsilon^2/4)^{-\mathtt{d}_{\mathscr{G}}-\mathtt{d}_{\mathscr{R},\mathscr{Y}}},$$

where the supremum is over all finite discrete measures on $\mathbb{R}^{d+1}$, and $M_{\tilde{\mathscr{R}},\mathscr{Y}}(y)=M_{\mathscr{R},\mathscr{Y}}(y)\mathbb{1}(|y|\leq\kappa_n)$. The function $\mathtt{M}_{\mathscr{G}}(M_{\mathscr{R},\mathscr{Y}}-M_{\tilde{\mathscr{R}},\mathscr{Y}})$ is an envelope for $\mathscr{G}\times\Delta\mathscr{R}$, since all functions in $\Delta\mathscr{R}$ vanish on $[-\kappa_n,\kappa_n]$. Denote $\mathbf{X}=(\mathbf{x}_i)_{1\leq i\leq n}$. The linear envelope condition on $\mathscr{R}$ and the conditional $(2+v)$ moment bound on y_i imply

$$\begin{aligned}
\mathbb{E}\Big[\max_{1\leq i\leq n}\mathtt{M}_{\mathscr{G}}^2(M_{\mathscr{R},\mathscr{Y}}(y_i)-M_{\tilde{\mathscr{R}},\mathscr{Y}}(y_i))^2\Big|\mathbf{X}\Big]^{\frac{1}{2}}&\lesssim\mathtt{M}_{\mathscr{G}}\mathbb{E}\Big[\big(\max_{1\leq i\leq n}M_{\mathscr{R},\mathscr{Y}}(y_i)\big)^2\Big|\mathbf{X}\Big]^{\frac{1}{2}}\lesssim\mathtt{M}_{\mathscr{G}}n^{\frac{1}{2+v}},\\
\sup_{(g,r)\in\mathscr{G}\times\mathscr{R}}\mathbb{E}[g(\mathbf{x}_i)^2r(y_i)^2\mathbb{1}(|y_i|\geq\kappa_n)]^{\frac{1}{2}}&\lesssim\sup_{(g,r)\in\mathscr{G}\times\mathscr{R}}\mathbb{E}\left[g(\mathbf{x}_i)^2\mathbb{E}[|r(y_i)|^{2+v}|\mathbf{x}_i]^{\frac{2}{2+v}}\mathbb{P}(|y_i|\geq\kappa_n|\mathbf{x}_i)^{\frac{v}{2+v}}\right]^{\frac{1}{2}}\\
&\lesssim\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}\kappa_n^{-v}}.
\end{aligned}$$

By Jensen's inequality, we also have

$$\mathbb{E}\Big[\max_{1\le i\le n}\mathtt{M}_{\mathscr{G}}^2(\mathbb{E}[M_{\mathscr{R},\mathscr{Y}}(y_i)-M_{\tilde{\mathscr{R}},\mathscr{Y}}(y_i)|\mathbf{x}_i])^2\Big|\mathbf{X}\Big]^{\frac{1}{2}}\lesssim \mathtt{M}_{\mathscr{G}}n^{\frac{1}{2+v}},$$
$$\sup_{(g,r)\in\mathscr{G}\times\mathscr{R}}\mathbb{E}[g(\mathbf{x}_i)^2\mathbb{E}[r(y_i)-\tilde{r}(y_i)|\mathbf{x}_i]^2]^{\frac{1}{2}}\lesssim\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}\kappa_n^{-v}},$$
$$\mathbb{E}[\mathtt{M}_{\mathscr{G}}^2(M_{\mathscr{R},\mathscr{Y}}(y_i)-M_{\tilde{\mathscr{R}},\mathscr{Y}}(y_i))^2]^{1/2}\lesssim\mathtt{M}_{\mathscr{G}}\kappa_n^{-v/2}.$$

Denote $A=(\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}})^{\frac{1}{2\mathtt{d}_{\mathscr{G}}+2\mathtt{d}_{\mathscr{R},\mathscr{Y}}}}/4$ and $D=2\mathtt{d}_{\mathscr{G}}+2\mathtt{d}_{\mathscr{R},\mathscr{Y}}$. Chernozhukov et al. [2014b, Corollary 5.1] gives

$$\begin{aligned}\mathbb{E}\left[\|R_n(g,r)-R_n(g,\widetilde{r})\|_{\mathscr{G}\times\mathscr{R}}\right]&\lesssim\mathbb{E}\left[\sup_{g\in\mathscr{G}}\sup_{h\in\Delta\mathscr{R}}\frac{1}{\sqrt{n}}\sum_{i=1}^n g(\mathbf{x}_i)(h(y_i)-\mathbb{E}[h(y_i)|\mathbf{x}_i])\right]\\&\lesssim\sqrt{D\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}\kappa_n^{-v}\log(A\sqrt{\mathtt{M}_{\mathscr{G}}/\mathtt{E}_{\mathscr{G}}})}+\frac{D\mathtt{M}_{\mathscr{G}}n^{\frac{1}{2+v}}}{\sqrt{n}}\log(A\sqrt{\mathtt{M}_{\mathscr{G}}/\mathtt{E}_{\mathscr{G}}})\\&\lesssim\sqrt{D\log(A\sqrt{\mathtt{M}_{\mathscr{G}}/\mathtt{E}_{\mathscr{G}}})}\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}\kappa_n^{-v/2}+\frac{D\log(A\sqrt{\mathtt{M}_{\mathscr{G}}/\mathtt{E}_{\mathscr{G}}})\mathtt{M}_{\mathscr{G}}}{\sqrt{n^{\frac{v}{2+v}}}}.\end{aligned}$$

Part 3: Truncation error for the Gaussian process — $\|Z_n^R(g,r)-Z_n^R(g,\tilde{r})\|_{\mathscr{G}\times\mathscr{R}}$

The class $\mathscr{G}\times\tilde{\mathscr{R}}\cup\mathscr{G}\times\mathscr{R}$ is VC-type with respect to envelope function $2\mathtt{M}_{\mathscr{G}}\mathtt{M}_{\mathscr{R},\mathscr{Y}}$ in the sense that for all $0<\varepsilon<1$,

$$\sup_Q N(\mathscr{G}\times\mathscr{R}\cup\mathscr{G}\times\tilde{\mathscr{R}},\|\cdot\|_{Q,2},2\varepsilon\|\mathtt{M}_{\mathscr{G}}\mathtt{M}_{\mathscr{R},\mathscr{Y}}\|_{Q,2})\le\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R}}(\varepsilon^2/4)^{-\mathtt{d}_{\mathscr{G}}-\mathtt{d}_{\mathscr{R}}},$$

where the supremum is over all finite discrete measures on $\mathbb{R}^{d+1}$. Here $\mathtt{c}_{\mathscr{R}}$ and $\mathtt{d}_{\mathscr{R}}$ are VC-type constants for $\mathscr{R}\cup\tilde{\mathscr{R}}$; truncation by the fixed indicator $\mathbb{1}(|y|\le\kappa_n)$ preserves the entropy exponent, so these constants can be chosen with $\mathtt{c}_{\mathscr{R}}\ge\mathtt{c}_{\mathscr{R},\mathscr{Y}}$ and $\mathtt{d}_{\mathscr{R}}=\mathtt{d}_{\mathscr{R},\mathscr{Y}}$ up to universal constants. Hence $\mathscr{G}\times\tilde{\mathscr{R}}\cup\mathscr{G}\times\mathscr{R}$ is pre-Gaussian, and on some probability space there exists a mean-zero Gaussian process $\bar{Z}_n^R$ indexed by $\mathscr{F}=\mathscr{G}\times\tilde{\mathscr{R}}\cup\mathscr{G}\times\mathscr{R}$ with the same covariance structure as R_n and almost surely continuous paths with respect to the metric ρ, given by

$$\rho((g_1,r_1),(g_2,r_2))=\mathbb{E}[(\bar{Z}_n^R(g_1,r_1)-\bar{Z}_n^R(g_2,r_2))^2]^{\frac{1}{2}}=\mathbb{E}[(R_n(g_1,r_1)-R_n(g_2,r_2))^2]^{\frac{1}{2}},$$

for $(g_1,r_1),(g_2,r_2)\in\mathscr{F}$. Recall the definition of $\Delta\mathscr{R}$ in Part 2. The increment diameter between each original function and its truncated counterpart satisfies

$$\sigma\equiv\sup_{g\in\mathscr{G},r\in\mathscr{R}}\rho((g,r),(g,\tilde{r}))\le\sup_{g\in\mathscr{G},h\in\Delta\mathscr{R}}\mathbb{E}[g(\mathbf{x}_i)^2h(y_i)^2]^{1/2}\lesssim\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}\kappa_n^{-v}}.$$

The entropy bound above also implies, for all $0<\varepsilon<1$,

$$N(\mathscr{G}\times\mathscr{R}\cup\mathscr{G}\times\tilde{\mathscr{R}},\rho,2\varepsilon\|\mathtt{M}_{\mathscr{G}}M_{\mathscr{R},\mathscr{Y}}\|_{\mathbb{P},2})\le\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}}(\varepsilon^2/4)^{-\mathtt{d}_{\mathscr{G}}-\mathtt{d}_{\mathscr{R},\mathscr{Y}}}.$$

Denote $A = (\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}})^{\frac{1}{2\mathtt{d}_{\mathscr{G}}+2\mathtt{d}_{\mathscr{R},\mathscr{Y}}}}/4$ and $D = 2\mathtt{d}_{\mathscr{G}} + 2\mathtt{d}_{\mathscr{R},\mathscr{Y}}$. By van der Vaart and Wellner [1996, Corollary 2.2.8], choosing any $(g_0, r_0) \in \mathscr{G} \times \mathscr{R}$,

$$\begin{aligned}
\mathbb{E}\Big[\|\bar{Z}_n^R(g,r) - \bar{Z}_n^R(g,\tilde{r})\|_{\mathscr{G}\times\mathscr{R}}\Big] &\lesssim \mathbb{E}\left[\left|\bar{Z}_n^R(g_0,r_0) - \bar{Z}_n^R(g_0,\tilde{r}_0)\right|\right] + \int_0^\sigma \sqrt{\log\left(\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R}}\left(\frac{\mathtt{M}_{\mathscr{G}}}{\varepsilon}\right)^{\mathtt{d}_{\mathscr{G}}+\mathtt{d}_{\mathscr{R}}}\right)}\,\mathrm{d}\varepsilon \\
&\leq \sqrt{D\log(A\sqrt{\mathtt{M}_{\mathscr{G}}/\mathtt{E}_{\mathscr{G}}})}\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}\kappa_n^{-v/2} \\
&\lesssim \sqrt{(\mathtt{d}_{\mathscr{G}} + \mathtt{d}_{\mathscr{R},\mathscr{Y}})\log(\mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}}\mathtt{k}n)}\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}\kappa_n^{-v/2}.
\end{aligned}$$

Since the restriction of $(\bar{Z}_n^R(g,r) : (g,r) \in \mathscr{F})$ to $\mathscr{G} \times \tilde{\mathscr{R}}$ has the same distribution as the Gaussian process constructed in Part 1, the Vorob'ev–Berkes–Philipp theorem [Dudley, 2014, Theorem 1.31] allows us to construct $\bar{Z}_n^R$ on the same probability space as $(\mathbf{x}_i, y_i)_{1\leq i\leq n}$ and Z_n^R, with $\bar{Z}_n^R$ and Z_n^R coinciding on $\mathscr{G} \times \tilde{\mathscr{R}}$. By an abuse of notation, call this extended Gaussian process Z_n^R.

Part 4: Putting Together

Combining the previous three parts, choose κ_n so that

$$\mathtt{r}_n\kappa_n \asymp \sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}\kappa_n^{-v/2}, \qquad \text{that is} \qquad \kappa_n \asymp \left(\frac{\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}}{\mathtt{r}_n}\right)^{\frac{2}{v+2}}.$$

Then

$$\begin{aligned}
\mathbb{E}\big[\|R_n - Z_n^R\|_{\mathscr{G}\times\mathscr{R}}\big] &\lesssim \mathtt{v}(\mathtt{d}\log(\mathtt{c}n))^{3/2}\mathtt{r}_n^{\frac{v}{v+2}}(\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}})^{\frac{2}{v+2}} + \mathtt{v}\mathtt{d}\log(\mathtt{c}n)\mathtt{M}_{\mathscr{G}}n^{-\frac{v/2}{2+v}} \\
&\qquad + \mathtt{v}\mathtt{d}\log(\mathtt{c}n)\mathtt{M}_{\mathscr{G}}n^{-1/2}\left(\frac{\sqrt{\mathtt{M}_{\mathscr{G}}\mathtt{E}_{\mathscr{G}}}}{\mathtt{r}_n}\right)^{\frac{2}{v+2}} \\
&\lesssim \mathtt{v}\mathtt{d}\log(\mathtt{c}n)\rho_n,
\end{aligned}$$

where $\mathtt{d} = \mathtt{d}_{\mathscr{G}} + \mathtt{d}_{\mathscr{R},\mathscr{Y}} + \mathtt{k}$, and $\mathtt{c} = \mathtt{c}_{\mathscr{G}}\mathtt{c}_{\mathscr{R},\mathscr{Y}}\mathtt{k}$. This proves the stated bound. □

SA-8 Proofs

SA-8.1 Proof of Lemma SA-1

Because $\mathscr{B}$ is compact and $\mathscr{B} \subset \mathrm{int}(\mathscr{X})$, compact support of K implies that $\mathbf{x} + \mathbf{H}\,\mathrm{supp}(K) \subseteq \mathscr{X}$ uniformly over $\mathbf{x} \in \mathscr{B}$ for all sufficiently small $\bar{h}$. Assumption SA-1(ii) and uniform continuity of f_X on the compact set $\mathscr{X}$ imply

$$\begin{aligned}
\mathbf{\Gamma}_{t,\mathbf{x}} &= \mathbb{E}\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big] \\
&= \int_{\mathscr{A}_t} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u} - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u} - \mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{u} - \mathbf{x})f_X(\mathbf{u})\,\mathrm{d}\mathbf{u} \\
&= f_X(\mathbf{x})\int_{\mathscr{A}_t} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u} - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u} - \mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{u} - \mathbf{x})\,\mathrm{d}\mathbf{u} + o(1),
\end{aligned}$$

where in the last line we have used $\sup_{\mathbf{u}\in\mathbf{x}+\mathbf{H}\,\mathrm{supp}(K)} |f_X(\mathbf{u}) - f_X(\mathbf{x})| = o(1)$ uniformly over $\mathbf{x} \in \mathscr{B}$ and $\int_{\mathscr{A}_t}(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))^{\mathbf{v}}K_{\mathbf{H}}(\mathbf{u}-\mathbf{x})\,\mathrm{d}\mathbf{u} = O(1)$ for any multi-index $\mathbf{v}$ from a standard change of variable argument.

I. Polynomial Representation of Minimum Eigenvalue

For each bandwidth matrix $\mathbf{H}$, define

$$\mathbf{A}_{t,\mathbf{x}}(\mathbf{H}) = \int_{\mathscr{A}_t} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))^\top K_{\mathbf{H}}(\mathbf{u}-\mathbf{x})\,\mathrm{d}\mathbf{u}.$$

The argument below establishes a uniform positive lower bound for $\lambda_{\min}(\mathbf{A}_{t,\mathbf{x}}(\mathbf{H}))$ for all sufficiently small $\bar{h}$; no pointwise limit of $\mathbf{A}_{t,\mathbf{x}}(\mathbf{H})$ is needed. A change of variable gives

$$\mathbf{A}_{t,\mathbf{x}}(\mathbf{H}) = \int \mathbf{r}_p(\mathbf{z})\mathbf{r}_p(\mathbf{z})^\top K(\mathbf{z})\mathbb{1}(\mathbf{x}+\mathbf{H}\mathbf{z}\in\mathscr{A}_t)\,\mathrm{d}\mathbf{z}.$$

Let $\mathbf{a}\in\mathbb{R}^{\mathfrak{p}_p}$, where $\mathfrak{p}_p = \frac{(d+p)!}{d!p!}$. Then the equivalent representation of minimum eigenvalue gives

$$\begin{aligned}\lambda_{\min}(\mathbf{A}_{t,\mathbf{x}}(\mathbf{H})) &= \min_{\|\mathbf{a}\|=1}\int(\mathbf{a}^\top\mathbf{r}_p(\mathbf{z}))^2K(\mathbf{z})\mathbb{1}(\mathbf{x}+\mathbf{H}\mathbf{z}\in\mathscr{A}_t)\,\mathrm{d}\mathbf{z}\\ &\geq \kappa\min_{\|\mathbf{a}\|=1}\int_U(\mathbf{a}^\top\mathbf{r}_p(\mathbf{z}))^2\mathbb{1}(\mathbf{x}+\mathbf{H}\mathbf{z}\in\mathscr{A}_t)\,\mathrm{d}\mathbf{z}, \qquad \text{(SA-7)}\end{aligned}$$

where in the last line we have used $K(\mathbf{u})\geq\kappa$ for all $\mathbf{u}\in U$.

II. Mass Retaining Ratio in Treatment/Control Region

Denote $E_{\mathbf{H}}(\mathbf{x},t) = \{\mathbf{z}\in U : \mathbf{x}+\mathbf{H}\mathbf{z}\in\mathscr{A}_t\}$. Assumption SA-2 (ii) implies there is some upper bound $\Lambda>0$ of $K(\cdot)$. Hence for $c_0 = 1/2\ \liminf_{\bar{h}\downarrow 0}\inf_{\mathbf{x}\in\mathscr{B}}\int_U K(\mathbf{u})\mathbb{1}(\mathbf{x}+\mathbf{H}\mathbf{u}\in\mathscr{A}_t)\,\mathrm{d}\mathbf{u}$, we have

$$\Lambda\mathfrak{m}(E_{\mathbf{H}}(\mathbf{x},t)) \geq \int_U K(\mathbf{u})\mathbb{1}(\mathbf{x}+\mathbf{H}\mathbf{u}\in\mathscr{A}_t)\,\mathrm{d}\mathbf{u}\geq c_0$$

for small enough $\bar{h}$, which implies

$$\mathfrak{m}(E_{\mathbf{H}}(\mathbf{x},t))\geq\alpha\mathfrak{m}(U), \qquad \alpha = \frac{c_0}{\Lambda\mathfrak{m}(U)}. \qquad \text{(SA-8)}$$

III. L_2 Integral of Polynomials in Full Versus Treatment/Control Regions

Consider $S = \{f\in\mathscr{P}_p : \int_U f(\mathbf{u})^2\,\mathrm{d}\mathbf{u} = 1\}$, where $\mathscr{P}_p$ is the collection of all polynomials in $\mathbf{u}$ with total degree at most p. Let $(\phi_j, 1\leq j\leq\mathfrak{p}_p)$ be an orthonormal basis of $(\mathscr{P}_p, \|\cdot\|_{L_2})$. Then $T(\mathbf{a}) = \sum_{j=1}^{\mathfrak{p}_p} a_j\phi_j$ is an isometry. Since $T(S) = \{\mathbf{a}\in\mathbb{R}^{\mathfrak{p}_p} : \|\mathbf{a}\| = 1\}$ is compact, S is also compact in $(\mathscr{P}_p, \|\cdot\|_{L_2})$. Since $\mathscr{P}_p$ is $\mathfrak{p}_p$-dimensional, equivalence of norms implies that S is also compact in $(\mathscr{P}_p, \|\cdot\|_{L_\infty})$. Now consider

$$\Phi_q(\varepsilon) = \mathfrak{m}(\{\mathbf{u}\in U : |q(\mathbf{u})|<\varepsilon\}), \qquad q\in S, \varepsilon>0,$$

and

$$\psi(q) = \sup\Big\{\varepsilon>0 : \Phi_q(\varepsilon)\leq\frac{\alpha}{2}\mathfrak{m}(U)\Big\}.$$

Since $\int_U q^2 = 1$ and q is polynomial, $\lim_{\varepsilon\downarrow 0}\Phi_q(\varepsilon) = 0$ and $\Phi_q(\|q\|_\infty) = \mathfrak{m}(U)$. Continuity and Lipschitzness of $q\in S$ imply $\psi(q)>0$ for all $q\in S$.

Next, we want to show ψ is lower-semicontinuous on $(\mathscr{P}_p, \|\cdot\|_{L_\infty})$. Suppose $q_n \to q$ uniformly on U. For every $\varepsilon_0 \in (0, \psi(q))$, there exists $\eta > 0$ such that $\Phi_q(\varepsilon_0) \le \frac{\alpha}{2}\mathfrak{m}(U) - \eta$. Continuity of polynomials and the fact that level sets of nonzero polynomials have zero Lebesgue measure imply $\mathbb{1}_{\{|q_n| < \varepsilon_0\}}(\cdot) \to \mathbb{1}_{\{|q| < \varepsilon_0\}}(\cdot)$ almost surely. By Dominated Convergence Theorem, $\Phi_{q_n}(\varepsilon_0) \to \Phi_q(\varepsilon_0)$. Hence for large enough n, $\Phi_{q_n}(\varepsilon_0) \le \frac{\alpha}{2}\mathfrak{m}(U)$, which implies $\varepsilon_0 \le \psi(q_n)$. This implies $\liminf_{n\to\infty} \psi(q_n) \ge \varepsilon_0$. Since ε_0 is arbitrary in $(0, \psi(q))$, we have $\liminf_{n\to\infty} \psi(q_n) \ge \psi(q)$.

Compactness of S and lower-semicontinuity of ψ implies ψ attains its minimum on S. Since $\psi(q) > 0$ for all $q \in S$, we know $\varepsilon_* = \inf_{q \in S} \psi(q) > 0$. Then for every $q \in S$,

$$\begin{aligned}
\int_{E_{\mathbf{H}}(\mathbf{x},t)} q^2 &\ge \varepsilon_*^2\, \mathfrak{m}\Big(E_{\mathbf{H}}(\mathbf{x},t) \setminus \{|q| \le \varepsilon_*\}\Big) \\
&\ge \varepsilon_*^2 \Big(\mathfrak{m}(E_{\mathbf{H}}(\mathbf{x},t)) - \mathfrak{m}(\{|q| \le \varepsilon_*\})\Big) \\
&\ge \varepsilon_*^2\, \frac{\alpha}{2} \mathfrak{m}(U).
\end{aligned}$$

Scaling q from S gives

$$\int_{E_{\mathbf{H}}(\mathbf{x},t)} q^2 \ge \varepsilon_*^2\, \frac{\alpha}{2} \int_U q^2, \qquad q \in \mathscr{P}_p. \tag{SA-9}$$

IV. Lower Bound of Minimum Eigenvalue

Equations (SA-7), (SA-8) and (SA-9) together give for small enough $\bar{h}$,

$$\begin{aligned}
\inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\mathbf{A}_{t,\mathbf{x}}(\mathbf{H})) &\ge \kappa \inf_{\mathbf{x}\in\mathscr{B}} \min_{\|\mathbf{a}\|=1} \int_{E_{\mathbf{H}}(\mathbf{x},t)} (\mathbf{a}^\top \mathbf{r}_p(\mathbf{z}))^2 \,\mathrm{d}\mathbf{z}, \\
&\ge \kappa\varepsilon_*^2\, \frac{\alpha}{2} \min_{\|\mathbf{a}\|=1} \int_U (\mathbf{a}^\top \mathbf{r}_p(\mathbf{z}))^2 \,\mathrm{d}\mathbf{z} \\
&\ge \kappa\varepsilon_*^2\, \frac{\alpha}{2} \lambda_{\min}\Big(\int_U \mathbf{r}_p(\mathbf{z})\mathbf{r}_p(\mathbf{z})^\top \,\mathrm{d}\mathbf{z}\Big),
\end{aligned}$$

which implies $\liminf_{\bar{h}\to 0} \inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\mathbf{A}_{t,\mathbf{x}}(\mathbf{H})) > 0$. Since Assumption SA-1(ii) gives $\inf_{\mathbf{x}\in\mathscr{X}} f_X(\mathbf{x}) > 0$ and the preceding expansion of $\mathbf{\Gamma}_{t,\mathbf{x}}$ is uniform over $\mathbf{x} \in \mathscr{B}$, it follows that

$$\liminf_{\bar{h}\to 0} \inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}}) \ge \inf_{\mathbf{x}\in\mathscr{X}} f_X(\mathbf{x}) \liminf_{\bar{h}\to 0} \inf_{\mathbf{x}\in\mathscr{B}} \lambda_{\min}(\mathbf{A}_{t,\mathbf{x}}(\mathbf{H})) > 0.$$

SA-8.2 Proof of Lemma SA-2

Since $\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}$ is a finite dimensional matrix, it suffices to show the stated rate of convergence for each entry. Let $\mathbf{v}$ be a multi-index such that $|\mathbf{v}| \le 2p$. Define

$$g_n(\xi, \mathbf{x}) = (\mathbf{H}^{-1}(\xi - \mathbf{x}))^{\mathbf{v}} K_{\mathbf{H}}(\xi - \mathbf{x}) \mathbb{1}(\xi \in \mathscr{A}_t), \qquad \xi \in \mathscr{X}, \mathbf{x} \in \mathscr{B}.$$

and $\mathscr{F} = \{g_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathscr{B}\}$. We will show $\mathscr{F}$ is a VC-type class. In order to do this, we study the following quantities.

Constant Envelope Function. We assume K is continuous and has compact support, or $K = \mathbb{1}(\cdot \in [-1,1]^d)$. Hence there exists a constant C_1 such that for all $l \in \mathscr{F}$ and all $\xi \in \mathscr{X}$, $|l(\xi)| \le C_1 |\mathbf{H}|^{-1} = F$.

Diameter of $\mathscr{F}$ in L_2. $\sup_{l\in\mathscr{F}}\|l\|_{\mathbb{P},2}=\sup_{\mathbf{x}\in\mathscr{B}}(\int_{\mathbf{H}^{-1}(\mathscr{A}_t-\mathbf{x})}|\mathbf{H}|^{-1}\mathbf{y}^{2\mathbf{v}}K(\mathbf{y})^2 f_X(\mathbf{x}+\mathbf{H}\mathbf{y})\,\mathrm{d}\mathbf{y})^{1/2}\le C_2|\mathbf{H}|^{-1/2}$ for some constant C_2. We can take C_1 large enough so that $\sigma=C_2|\mathbf{H}|^{-1/2}\le F=C_1|\mathbf{H}|^{-1}$.

Ratio. For some constant C_3, $\delta=\frac{\sigma}{F}=C_3\sqrt{|\mathbf{H}|}$.

Covering Numbers. Case 1: K is Lipschitz. Let $\mathbf{x},\mathbf{x}'\in\mathscr{B}$. Then, for generic evaluation points $\mathbf{x}=(x_1,\dots,x_d)^\top$ and $\mathbf{x}'=(x_1',\dots,x_d')^\top$,

$$\begin{aligned}\sup_{\xi\in\mathscr{X}}|g_n(\xi,\mathbf{x})-g_n(\xi,\mathbf{x}')|\le{}&\left|(\mathbf{H}^{-1}(\xi-\mathbf{x}))^{\mathbf{v}}-(\mathbf{H}^{-1}(\xi-\mathbf{x}'))^{\mathbf{v}}\right|K_{\mathbf{H}}(\xi-\mathbf{x})\\&+|(\mathbf{H}^{-1}(\xi-\mathbf{x}'))^{\mathbf{v}}|\left|K_{\mathbf{H}}(\xi-\mathbf{x})-K_{\mathbf{H}}(\xi-\mathbf{x}')\right|\\\lesssim{}&|\mathbf{H}|^{-1}\underline{h}^{-1}\|\mathbf{x}-\mathbf{x}'\|_\infty,\end{aligned}$$

since we have assumed that K has compact support and is Lipschitz continuous. Hence, for any $\varepsilon\in(0,1]$ and for any finitely supported measure Q and metric $\|\cdot\|_{Q,2}$ based on $L_2(Q)$,

$$N(\mathscr{F},\|\cdot\|_{Q,2},\varepsilon\|F\|_{Q,2})\le N(\mathscr{X},\|\cdot\|_\infty,\varepsilon\|F\|_{Q,2}|\mathbf{H}|\underline{h})\overset{(i)}{\lesssim}\left(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon\|F\|_{Q,2}|\mathbf{H}|\underline{h}}\right)^d\lesssim\left(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon\underline{h}}\right)^d,$$

where in (i) we used compactness of $\mathscr{X}$ and the fact that $\varepsilon\|F\|_{Q,2}|\mathbf{H}|\underline{h}\lesssim\varepsilon\underline{h}\lesssim 1$. Hence, $\mathscr{F}$ forms a VC-type class, and taking $A_1=\operatorname{diam}(\mathscr{X})/\underline{h}$ and $A_2=d$, $\sup_Q N(\mathscr{F},\|\cdot\|_{Q,2},\varepsilon\|F\|_{Q,2})\lesssim(A_1/\varepsilon)^{A_2}$, $\varepsilon\in(0,1]$, and where the supremum is over all finite discrete measures.

Case 2: $K=\mathbb{1}(\cdot\in[-1,1]^d)$. Consider

$$m_n(\xi,\mathbf{x})=(\mathbf{H}^{-1}(\xi-\mathbf{x}))^{\mathbf{v}}|\mathbf{H}|^{-1}\mathbb{1}(\xi\in\mathscr{A}_t),\qquad \xi,\mathbf{x}\in\mathscr{X},$$

$\mathscr{M}=\{m_n(\cdot,\mathbf{x}):\mathbf{x}\in\mathscr{B}\}$ and the constant envelope function $M=C_4|\mathbf{H}|^{-1}\underline{h}^{-|\mathbf{v}|}$, for some constant C_4 only depending on diameter of $\mathscr{X}$. The same argument as before shows that for any discrete measure Q, we have

$$N(\mathscr{M},\|\cdot\|_{Q,2},\varepsilon\|M\|_{Q,2})\le N(\mathscr{X},\|\cdot\|_\infty,\varepsilon\|M\|_{Q,2}|\mathbf{H}|\underline{h}^{|\mathbf{v}|+1})\lesssim\Big(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon\|M\|_{Q,2}|\mathbf{H}|\underline{h}^{|\mathbf{v}|+1}}\Big)^d\lesssim\Big(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon\underline{h}}\Big)^d.$$

The class $\mathscr{G}=\{\mathbb{1}(\mathbf{H}^{-1}(\cdot-\mathbf{x})\in[-1,1]^d):\mathbf{x}\in\mathscr{B}\}$ has VC dimension no greater than $2d$ [van der Vaart and Wellner, 1996, Example 2.6.1], and by van der Vaart and Wellner [1996, Theorem 2.6.4], for any discrete measure Q, $N(\mathscr{G},\|\cdot\|_{Q,2},\varepsilon)\le 2d(4e)^{2d}\varepsilon^{-4d}$, $0<\varepsilon\le 1$. The class $\mathscr{F}$ is contained in the product class $\mathscr{M}\cdot\mathscr{G}$. Since the envelope ratio satisfies $\|F\|_{Q,2}/\|M\|_{Q,2}\gtrsim\underline{h}^{|\mathbf{v}|}$ uniformly in Q, the standard product-covering inequality gives, for a constant $c>0$,

$$\begin{aligned}N(\mathscr{F},\|\cdot\|_{Q,2},\varepsilon\|F\|_{Q,2})&\le N(\mathscr{M},\|\cdot\|_{Q,2},c\varepsilon\underline{h}^{|\mathbf{v}|}\|M\|_{Q,2})N(\mathscr{G},\|\cdot\|_{Q,2},c\varepsilon\underline{h}^{|\mathbf{v}|})\\&\lesssim\left(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon\underline{h}^{|\mathbf{v}|+1}}\right)^d(\varepsilon\underline{h}^{|\mathbf{v}|})^{-4d}\lesssim\varepsilon^{-5d}\underline{h}^{-d(5|\mathbf{v}|+1)}.\end{aligned}$$

Hence, taking $A_1=C\underline{h}^{-(|\mathbf{v}|+1)}$ and $A_2=5d$ for a large enough constant C, $\sup_Q N(\mathscr{F},\|\cdot\|_{Q,2},\varepsilon\|F\|_{Q,2})\lesssim(A_1/\varepsilon)^{A_2}$, $\varepsilon\in(0,1]$, where the supremum is over all finite discrete measures.

Maximal Inequality. By Corollary 5.1 in Chernozhukov et al. [2014b] for the empirical process on class $\mathscr{F}$,

$$\mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{E}_n[g_n(\mathbf{X}_i,\mathbf{x})]-\mathbb{E}[g_n(\mathbf{X}_i,\mathbf{x})]\big|\Big]\lesssim\frac{\sigma}{\sqrt{n}}\sqrt{A_2\log(A_1/\delta)}+\frac{\|F\|_{\mathbb{P},2}A_2\log(A_1/\delta)}{n}$$
$$\lesssim\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n|\mathbf{H}|},$$

where $A_1, A_2, \sigma, F, \delta$ are all given previously. Assuming $\frac{\log(1/\bar{h})}{n|\mathbf{H}|}\to 0$, we conclude that $\sup_{\mathbf{x}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}-\mathbf{\Gamma}_{t,\mathbf{x}}\|\lesssim_{\mathbb{P}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}$. By Weyl's Theorem, $\sup_{\mathbf{x}\in\mathscr{B}}|\lambda_{\min}(\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}})-\lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}})|\leq\sup_{\mathbf{x}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}-\mathbf{\Gamma}_{t,\mathbf{x}}\|\lesssim_{\mathbb{P}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}$. Lemma SA-1 gives $\lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}})\gtrsim 1$ uniformly over $\mathbf{x}\in\mathscr{B}$. Therefore, $\inf_{\mathbf{x}\in\mathscr{B}}\lambda_{\min}(\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}})\gtrsim_{\mathbb{P}} 1$, $\sup_{\mathbf{x}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}\|\lesssim_{\mathbb{P}} 1$, and $\sup_{\mathbf{x}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}\|\lesssim_{\mathbb{P}} 1$. Hence $\sup_{\mathbf{x}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}-\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\|\leq\sup_{\mathbf{x}\in\mathscr{B}}\|\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\|\|\mathbf{\Gamma}_{t,\mathbf{x}}-\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}\|\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}\|\lesssim_{\mathbb{P}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}$.

SA-8.3 Proof of Lemma SA-3

The proof is similar to the proof of Lemma SA-2. Let $\mathbf{v}$ be a multi-index such that $0\leq|\mathbf{v}|\leq p$. Let

$$g_n(\xi,\mathbf{x})=(\mathbf{H}^{-1}(\xi-\mathbf{x}))^{\mathbf{v}}K_{\mathbf{H}}(\xi-\mathbf{x})\mathbb{1}(\xi\in\mathscr{A}_t),\qquad \xi,\mathbf{x}\in\mathscr{X}.$$

Define the class of functions $\mathscr{F}=\{(\xi,u)\in\mathscr{X}\times\mathbb{R}\mapsto g_n(\xi,\mathbf{x})u:\mathbf{x}\in\mathscr{B}\}$.

Envelope Function. Since K is continuous on its compact support, there exists a constant $C_1>0$ such that $|g_n(\xi,\mathbf{x})u|\leq C_1|\mathbf{H}|^{-1}|u|$, for $\xi,\mathbf{x}\in\mathscr{X}$ and $u\in\mathbb{R}$. We define the envelope function $F(\xi,u)=C_1|\mathbf{H}|^{-1}|u|$, for $\xi\in\mathscr{X}$ and $u\in\mathbb{R}$. Moreover, by Assumption SA-1(v), let $M=\max_{1\leq i\leq n}F(\mathbf{X}_i,\varepsilon_i)$, then

$$\mathbb{E}[M^2]^{1/2}\lesssim|\mathbf{H}|^{-1}\mathbb{E}\big[\max_{1\leq i\leq n}|\varepsilon_i|^2\big]^{1/2}\lesssim|\mathbf{H}|^{-1}\mathbb{E}\big[\max_{1\leq i\leq n}|\varepsilon_i|^{2+v}\big]^{1/(2+v)}\lesssim n^{1/(2+v)}|\mathbf{H}|^{-1}.$$

Diameter of $\mathscr{F}$ in L_2. Recall we denote $\varepsilon_i=Y_i-\mathbb{E}[Y_i|\mathbf{X}_i]$, then

$$\sup_{l\in\mathscr{F}}\mathbb{E}[l(\mathbf{X}_i,\varepsilon_i)^2]^{1/2}\leq\sup_{\xi\in\mathscr{X}}\mathbb{E}[\varepsilon_i^2|\mathbf{X}_i=\xi]^{1/2}\sup_{\xi\in\mathscr{X}}\mathbb{E}[g_n(\mathbf{X}_i,\xi)^2]^{1/2}\leq C_3|\mathbf{H}|^{-1/2}=\sigma.$$

Ratio. We set $\delta=\frac{\sigma}{\|F\|_{\mathbb{P},2}}\lesssim|\mathbf{H}|^{1/2}$.

Covering Numbers. Case 1: K is Lipschitz. Let $\mathbb{Q}$ be a finite distribution on $(\mathscr{X}\times\mathbb{R},\mathscr{B}(\mathscr{X})\otimes\mathsf{Borel}(\mathbb{R}))$. Let $\mathbf{x},\mathbf{x}'\in\mathscr{X}$. In the proof of Lemma SA-2, we showed that $\sup_{\xi\in\mathscr{X}}\sup_{\mathbf{x},\mathbf{x}'\in\mathscr{X}}\frac{|g_n(\xi,\mathbf{x})-g_n(\xi,\mathbf{x}')|}{\|\mathbf{x}-\mathbf{x}'\|_\infty}\lesssim|\mathbf{H}|^{-1}\underline{h}^{-1}$. Hence,

$$\|g_n(\mathbf{X}_i,\mathbf{x})\varepsilon_i-g_n(\mathbf{X}_i,\mathbf{x}')\varepsilon_i\|_{\mathbb{Q},2}\leq\|g_n(\cdot,\mathbf{x})-g_n(\cdot,\mathbf{x}')\|_\infty\|\varepsilon_i\|_{\mathbb{Q},2}\lesssim\underline{h}^{-1}\|F\|_{\mathbb{Q},2}\|\mathbf{x}-\mathbf{x}'\|_\infty.$$

It follows that $\sup_Q N(\mathscr{F},\|\cdot\|_{Q,2},\epsilon\|F\|_{Q,2})\lesssim(\frac{\mathrm{diam}(\mathscr{X})}{\epsilon\underline{h}})^d$, where sup is over all finite probability distributions on $(\mathscr{X}\times\mathbb{R},\mathscr{B}(\mathscr{X})\otimes\mathsf{Borel}(\mathbb{R}))$. Letting $A_1=\frac{\mathrm{diam}(\mathscr{X})}{\underline{h}}$ and $A_2=d$, we conclude that

$$\sup_Q N(\mathscr{F},\|\cdot\|_{Q,2},\epsilon\|F\|_{Q,2})\lesssim(A_1/\epsilon)^{A_2},\qquad\epsilon\in(0,1].$$

Case 2: K is the uniform kernel. Let

$$m_n(\xi, \mathbf{x}) = (\mathbf{H}^{-1}(\xi - \mathbf{x}))^{\mathbf{v}} |\mathbf{H}|^{-1} \mathbb{1}(\xi \in \mathscr{A}_t), \qquad \xi, \mathbf{x} \in \mathscr{X},$$

with $\mathscr{M} = \{(\xi, u) \in \mathscr{X} \times \mathbb{R} \to m_n(\xi, \mathbf{x})u : \mathbf{x} \in \mathscr{B}\}$ and envelope function $M(\xi, u) = C_1 |\mathbf{H}|^{-1} \underline{h}^{-|\mathbf{v}|} |u|$, for a positive constant C_1 depending only on K. By similar arguments as Case 1 and the proof of Lemma SA-2, it follows that $\sup_Q N(\mathscr{M}, \|\cdot\|_{Q,2}, \varepsilon \|M\|_{Q,2}) \lesssim \left(\frac{\operatorname{diam}(\mathscr{X})}{\varepsilon \underline{h}}\right)^d$, where the supremum is taken over all finite discrete measures. Taking $\mathscr{G} = \{\mathbb{1}(\mathbf{H}^{-1}(\cdot - \mathbf{x}) \in [-1, 1]^d) : \mathbf{x} \in \mathscr{B}\}$, the proof of Lemma SA-2 shows that

$$\sup_Q N(\mathscr{G}, \|\cdot\|_{Q,2}, \varepsilon) \leq 2d(4e)^{2d} \varepsilon^{-4d}, \qquad \varepsilon \in (0, 1],$$

where the supremum is taken over all finite discrete measures. The class $\mathscr{F}$ is contained in $\mathscr{M} \cdot \mathscr{G}$. Since $\|F\|_{Q,2} / \|M\|_{Q,2} \gtrsim \underline{h}^{|\mathbf{v}|}$, the product-covering inequality gives, for a constant $c > 0$,

$$\begin{aligned} N(\mathscr{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) &\leq N(\mathscr{M}, \|\cdot\|_{Q,2}, c\varepsilon \underline{h}^{|\mathbf{v}|} \|M\|_{Q,2}) N(\mathscr{G}, \|\cdot\|_{Q,2}, c\varepsilon \underline{h}^{|\mathbf{v}|}) \\ &\lesssim \varepsilon^{-5d} \underline{h}^{-d(5|\mathbf{v}|+1)}. \end{aligned}$$

Taking $A_1 = C\underline{h}^{-(|\mathbf{v}|+1)}$ and $A_2 = 5d$ for a large enough constant C, we have

$$\sup_Q N(\mathscr{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \lesssim (A_1/\varepsilon)^{A_2}, \qquad \varepsilon \in (0, 1],$$

the supremum is over all finite discrete measures.

Maximal Inequality. By Corollary 5.1 in Chernozhukov et al. [2014b],

$$\begin{aligned} \mathbb{E}\Big[\sup_{\mathbf{x} \in \mathscr{B}} \big|\mathbb{E}_n[g_n(\mathbf{X}_i, \mathbf{x})\varepsilon_i]\big|\Big] &\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{A_2 \log(A_1/\delta)} + \frac{\|M\|_{\mathbb{P},2} A_2 \log(A_1/\delta)}{n} \\ &\lesssim \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}} |\mathbf{H}|}. \end{aligned}$$

Since $\mathbf{Q}_{t,\mathbf{x}}$ is finite-dimensional, entry-wise convergence implies convergence in norm with the same rate. Hence, $\sup_{\mathbf{x} \in \mathscr{B}} \|\mathbf{Q}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}} |\mathbf{H}|}$. By Lemma SA-2,

$$\begin{aligned} \sup_{\mathbf{x} \in \mathscr{B}} \big|\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x}) - \mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}] - \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}}\big| &= \sup_{\mathbf{x} \in \mathscr{B}} \big|\boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} (\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}) \mathbf{Q}_{t,\mathbf{x}}\big| \\ &\lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} \Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}} |\mathbf{H}|}\Big), \end{aligned}$$

and

$$\sup_{\mathbf{x} \in \mathscr{B}} \big|\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x}) - \mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}]\big| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}} \Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}} |\mathbf{H}|}\Big),$$

which completes the proof. $\square$

SA-8.4 Proof of Lemma SA-4

Let $\eta_i(\mathbf{x}) = \sum_{t\in\{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)(\mu_t(\mathbf{X}_i) - \widehat{\boldsymbol{\beta}}_t(\mathbf{x})^\top \mathbf{r}_p(\mathbf{X}_i - \mathbf{x}))$. Then, for all $\mathbf{x}, \mathbf{y} \in \mathscr{B}$, the difference between the estimated and true variance matrices is

$$\widehat{\boldsymbol{\Sigma}}_{t,\mathbf{x},\mathbf{y}} - \boldsymbol{\Sigma}_{t,\mathbf{x},\mathbf{y}} = \mathbf{M}_{1,\mathbf{x},\mathbf{y}} + \mathbf{M}_{2,\mathbf{x},\mathbf{y}} + \mathbf{M}_{3,\mathbf{x},\mathbf{y}} + \mathbf{M}_{4,\mathbf{x},\mathbf{y}}$$

where

$$\begin{aligned}
\mathbf{M}_{1,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y}) \eta_i(\mathbf{x})\eta_i(\mathbf{y}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big],\\
\mathbf{M}_{2,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y}) (\eta_i(\mathbf{x}) + \eta_i(\mathbf{y}))\varepsilon_i \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big],\\
\mathbf{M}_{3,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y}) (\varepsilon_i^2 - \sigma_t(\mathbf{X}_i)^2) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big],\\
\mathbf{M}_{4,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y}) \sigma_t(\mathbf{X}_i)^2 \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big]\\
&\quad - \mathbb{E}\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y}) \sigma_t(\mathbf{X}_i)^2 \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big].
\end{aligned}$$

For $\mathbf{u}$ and $\mathbf{v}$ multi-indices, let $g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y}) = |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{y})(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^{\mathbf{u}}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{y}))^{\mathbf{v}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)$. Set

$$\mathfrak{R}_{\mathtt{EP}} = \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}.$$

First, we present a bound on $\max_{1\le i\le n} |\eta_i(\mathbf{x})| \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K))$. By Lemma SA-5 and Lemma SA-3, for any multi-index $\boldsymbol{\alpha}$ such that $|\boldsymbol{\alpha}| \le p$,

$$\sup_{\mathbf{x}\in\mathscr{B}} |\mathbf{e}_{\boldsymbol{\alpha}}^\top(\widehat{\boldsymbol{\beta}}_t(\mathbf{x}) - \boldsymbol{\beta}_t(\mathbf{x}))| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\alpha}}(\bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}}).$$

Since K is compactly supported, we have

$$\max_{1\le i\le n}\Big|\sum_{t\in\{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)(\widehat{\boldsymbol{\beta}}_t(\mathbf{x}) - \boldsymbol{\beta}_t(\mathbf{x}))^\top \mathbf{r}_p(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K))\Big| \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}}.$$

Since μ_t is $p+1$ times continuously differentiable,

$$\max_{1\le i\le n}\Big|\sum_{t\in\{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)(\mu_t(\mathbf{X}_i) - \boldsymbol{\beta}_t(\mathbf{x})^\top \mathbf{r}_p(\mathbf{X}_i - \mathbf{x})) \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K))\Big| \lesssim \bar{h}^{p+1}.$$

It follows that

$$\sup_{\mathbf{x}\in\mathscr{B}} \max_{1\le i\le n} |\eta_i(\mathbf{x})| \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K)) \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}}.$$

Term $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$. From the proof for Lemma SA-2, applied to the absolute-value class generated by g_n, $\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}_n[|g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})|] \lesssim_{\mathbb{P}} 1$. Thus,

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \big|\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})\eta_i(\mathbf{x})\eta_i(\mathbf{y})]\big|$$

$$
\begin{aligned}
&\leq \Big(\sup_{\mathbf{x}\in\mathscr{B}} \max_{1\leq i\leq n} |\eta_i(\mathbf{x})| \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K))\Big)^2 \sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}_n[|g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})|] \\
&\lesssim_{\mathbb{P}} (\bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}})^2,
\end{aligned}
$$

where we have used the preceding local-support bound on $\eta_i(\mathbf{x})$. Finite dimensionality of $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$ then implies

$$
\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \|\mathbf{M}_{1,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} (\bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}})^2.
$$

Term $\mathbf{M}_{2,\mathbf{x},\mathbf{y}}$. The same maximal-inequality argument used for Lemma SA-3, applied to the absolute-value class $|g_n(\cdot;\mathbf{x},\mathbf{y})u|$, gives $\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}_n[|g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})\varepsilon_i|] \lesssim_{\mathbb{P}} 1$. Thus,

$$
\begin{aligned}
&\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \big|\mathbb{E}_n[g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})(\eta_i(\mathbf{x}) + \eta_i(\mathbf{y}))\varepsilon_i]\big| \\
&\leq 2 \sup_{\mathbf{x}\in\mathscr{B}} \max_{1\leq i\leq n} |\eta_i(\mathbf{x})| \mathbb{1}(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}) \in \mathrm{supp}(K)) \sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}_n[|g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})\varepsilon_i|] \\
&\lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}},
\end{aligned}
$$

which implies that

$$
\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \|\mathbf{M}_{2,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathtt{EP}}.
$$

Term $\mathbf{M}_{3,\mathbf{x},\mathbf{y}}$. Define $l_n(\cdot,\cdot;\mathbf{x},\mathbf{y}) : \mathscr{X} \times \mathbb{R} \to \mathbb{R}$ as

$$
l_n(\xi,\varepsilon;\mathbf{x},\mathbf{y}) = |\mathbf{H}| K_{\mathbf{H}}(\xi - \mathbf{x}) K_{\mathbf{H}}(\xi - \mathbf{y}) (\mathbf{H}^{-1}(\xi - \mathbf{x}))^{\mathbf{u}} (\mathbf{H}^{-1}(\xi - \mathbf{y}))^{\mathbf{v}} \mathbb{1}(\xi \in \mathscr{A}_t)(\varepsilon^2 - \sigma_t^2(\xi)),
$$

and consider the function class $\mathscr{L} = \{l_n(\cdot,\cdot;\mathbf{x},\mathbf{y}) : \mathbf{x},\mathbf{y} \in \mathscr{B}\}$. Let $L : \mathscr{X} \times \mathbb{R} \to \mathbb{R}$ be $L(\xi,\varepsilon) = c|\mathbf{H}|^{-1}|\varepsilon^2 - \sigma_t^2(\xi)|$ with $c = \sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \big|(\mathbf{H}^{-1}(\xi - \mathbf{x}))^{\mathbf{u}} (\mathbf{H}^{-1}(\xi - \mathbf{y}))^{\mathbf{v}} K(\mathbf{H}^{-1}(\xi - \mathbf{x})) K(\mathbf{H}^{-1}(\xi - \mathbf{y}))\big|$. By similar argument as in the proof for Lemma SA-3, we can show $\mathscr{L}$ is a VC-type class such that $\mathbb{E}[l_n(\mathbf{X}_i,\varepsilon_i;\mathbf{x},\mathbf{y})] = 0$, for all $\mathbf{x},\mathbf{y} \in \mathscr{B}$,

$$
\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}[l_n(\mathbf{X}_i,\varepsilon_i;\mathbf{x},\mathbf{y})^2]^{\frac{1}{2}} \lesssim \sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \mathbb{E}[g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})^2]^{\frac{1}{2}} \sup_{\xi\in\mathscr{X}} \mathbb{E}[(\varepsilon_i^2 - \sigma_t^2(\xi))^2 | \mathbf{X}_i = \xi]^{1/2} \lesssim |\mathbf{H}|^{-1/2}
$$

because Assumption SA-1(v), with $v \geq 2$, gives a uniformly bounded conditional fourth moment for ε_i. and

$$
\mathbb{E}\big[\max_{1\leq i\leq n} L(\mathbf{X}_i,\varepsilon_i)^2\big]^{\frac{1}{2}} \lesssim |\mathbf{H}|^{-1} \mathbb{E}\big[\max_{1\leq i\leq n} \varepsilon_i^4\big]^{1/2} \lesssim |\mathbf{H}|^{-1} \mathbb{E}\big[\max_{1\leq i\leq n} |\varepsilon_i|^{2+v}\big]^{\frac{2}{2+v}} \lesssim |\mathbf{H}|^{-1} n^{\frac{2}{2+v}}.
$$

Applying Corollary 5.1 in Chernozhukov et al. [2014b], we obtain

$$
\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \big|\mathbb{E}_n[l_n(\mathbf{X}_i,\varepsilon_i;\mathbf{x},\mathbf{y})]\big| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}} |\mathbf{H}|}
$$

and

$$
\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}} \|\mathbf{M}_{3,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}} |\mathbf{H}|}.
$$

Term $\mathbf{M}_{4,\mathbf{x},\mathbf{y}}$. Notice that $\{g_n(\cdot;\mathbf{x},\mathbf{y})\sigma_t^2(\cdot):\mathbf{x},\mathbf{y}\in\mathscr{B}\}$ is a VC-type class with constant envelope function $C|\mathbf{H}|^{-1}$ for some positive constant C, where

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\sup_{\xi\in\mathcal{X}}|g_n(\xi;\mathbf{x},\mathbf{y})\sigma_t^2(\xi)|\lesssim|\mathbf{H}|^{-1}$$

and

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\mathbb{E}[g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})^2\sigma_t(\mathbf{X}_i)^2]^{\frac{1}{2}}\lesssim|\mathbf{H}|^{-1/2}.$$

Then, similar to the proof of $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$, we conclude that

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\big|\mathbb{E}_n[g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})\sigma_t^2(\mathbf{X}_i)]-\mathbb{E}[g_n(\mathbf{X}_i;\mathbf{x},\mathbf{y})\sigma_t^2(\mathbf{X}_i)]\big|\lesssim_{\mathbb{P}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}$$

and

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\|\mathbf{M}_{4,\mathbf{x},\mathbf{y}}\|\lesssim_{\mathbb{P}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}.$$

Final result. Combining the upper bounds of the four terms,

$$\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\big\|\widehat{\mathbf{\Sigma}}_{t,\mathbf{x},\mathbf{y}}-\mathbf{\Sigma}_{t,\mathbf{x},\mathbf{y}}\big\|\lesssim_{\mathbb{P}}\bar{h}^{p+1}+\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|},$$

which implies $\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\|\widehat{\mathbf{\Sigma}}_{t,\mathbf{x},\mathbf{y}}\|\lesssim_{\mathbb{P}}1$. It follows that, for each $t\in\{0,1\}$,

$$\begin{aligned}\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}|\widehat{\Omega}^{(\boldsymbol{\nu})}_{t,\mathbf{x},\mathbf{y}}-\Omega^{(\boldsymbol{\nu})}_{t,\mathbf{x},\mathbf{y}}|&\leq\frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big(\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}-\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\|\|\widehat{\mathbf{\Sigma}}_{t,\mathbf{x},\mathbf{y}}\|\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{y}}^{-1}\|\\&\quad+\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\|\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\|\|\widehat{\mathbf{\Sigma}}_{t,\mathbf{x},\mathbf{y}}-\mathbf{\Sigma}_{t,\mathbf{x},\mathbf{y}}\|\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{y}}^{-1}\|\\&\quad+\sup_{\mathbf{x},\mathbf{y}\in\mathscr{B}}\|\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\|\|\mathbf{\Sigma}_{t,\mathbf{x},\mathbf{y}}\|\|\widehat{\mathbf{\Gamma}}_{t,\mathbf{y}}^{-1}-\mathbf{\Gamma}_{t,\mathbf{y}}^{-1}\|\Big)\\&\lesssim_{\mathbb{P}}\frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big(\bar{h}^{p+1}+\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|}\Big).\end{aligned}$$

Summing the last display over $t\in\{0,1\}$ gives the asserted bound for $\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{y}}-\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{y}}$. By Assumption SA-1(iv) and Assumption SA-2(ii), $\inf_{\mathbf{x}\in\mathscr{B}}\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}\gtrsim(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$. Therefore, $\inf_{\mathbf{x}\in\mathscr{B}}\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}\gtrsim_{\mathbb{P}}(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$. Furthermore,

$$\sup_{\mathbf{x}\in\mathscr{B}}\Big|\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}-\sqrt{\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}\Big|\lesssim_{\mathbb{P}}\sup_{\mathbf{x}\in\mathscr{B}}\sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big|\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}-\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}\Big|\lesssim_{\mathbb{P}}\frac{1}{\sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}}\Big(\bar{h}^{p+1}+\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|}\Big)$$

and

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|\frac{\mathbf{h}^{-\boldsymbol{\nu}}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}}-\frac{\mathbf{h}^{-\boldsymbol{\nu}}}{\sqrt{\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}}\right|=\mathbf{h}^{-\boldsymbol{\nu}}\sup_{\mathbf{x}\in\mathscr{B}}\left|\frac{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}-\sqrt{\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}}}}\right|\lesssim_{\mathbb{P}}\sqrt{n|\mathbf{H}|}\Big(\bar{h}^{p+1}+\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}+\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|}\Big),$$

which completes the proof. $\square$

SA-8.5 Proof of Lemma SA-5

Define

$$\boldsymbol{\chi}_{t,\mathbf{x}} = \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\mathfrak{r}_t(\mathbf{X}_i;\mathbf{x})\Big], \quad \mathfrak{r}_t(\xi;\mathbf{x}) = \mu_t(\xi) - \sum_{0\leq|\boldsymbol{\omega}|\leq p}\frac{\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!}(\xi-\mathbf{x})^{\boldsymbol{\omega}}.$$

Since μ_t is $(p+1)$-times continuously differentiable, Taylor's theorem and compact support of K imply

$$\sup_{\mathbf{x}\in\mathscr{B}}\max_{1\leq i\leq n}\big|\mathfrak{r}_t(\mathbf{X}_i;\mathbf{x})\big|\mathbb{1}\{K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\neq 0\}\lesssim \bar{h}^{p+1}.$$

Since $\frac{\log(1/\bar{h})}{n|\mathbf{H}|}=o(1)$, the same argument as the proof of Lemma SA-2 shows

$$\sup_{\mathbf{x}\in\mathscr{B}}\mathbb{E}_n\Big[\big\|\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{x}))\big\|K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\Big]\lesssim_{\mathbb{P}} 1.$$

Therefore, $\sup_{\mathbf{x}\in\mathscr{B}}\big\|\boldsymbol{\chi}_{t,\mathbf{x}}\big\|\lesssim_{\mathbb{P}}\bar{h}^{p+1}$. It then follows from Lemma SA-2 that

$$\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}]-\mu_t^{(\boldsymbol{\nu})}(\mathbf{x})\big| = \sup_{\mathbf{x}\in\mathscr{B}}\big|\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1}\boldsymbol{\chi}_{t,\mathbf{x}}\big|\lesssim_{\mathbb{P}}\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}.$$

Now also assume that $\bar{h}=o(1)$. By Assumption SA-1(iii), for all $\mathbf{x}\in\mathscr{B}$ and $\xi\in\mathscr{X}$,

$$\mathfrak{r}_t(\xi;\mathbf{x}) = \sum_{|\mathbf{v}|=p+1}\frac{\mu_t^{(\mathbf{v})}(\mathbf{x})}{\mathbf{v}!}(\xi-\mathbf{x})^{\mathbf{v}} + \sum_{|\mathbf{v}|=p+1} r_{t,\mathbf{v}}(\xi;\mathbf{x})(\xi-\mathbf{x})^{\mathbf{v}},$$

where

$$\max_{|\mathbf{v}|=p+1}\sup_{\mathbf{x}\in\mathscr{B},\xi\in\mathscr{X}}|r_{t,\mathbf{v}}(\xi;\mathbf{x})|\mathbb{1}\{K_{\mathbf{H}}(\xi-\mathbf{x})\neq 0\} = o(1).$$

Letting

$$\widetilde{\boldsymbol{\chi}}_{t,\mathbf{x}} = \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)(\sum_{|\mathbf{v}|=p+1}\frac{\mu_t^{(\mathbf{v})}(\mathbf{x})}{\mathbf{v}!}(\mathbf{X}_i-\mathbf{x})^{\mathbf{v}})\Big],$$

we conclude that

$$\begin{aligned}\sup_{\mathbf{x}\in\mathscr{B}}\big\|\boldsymbol{\chi}_{t,\mathbf{x}}-\widetilde{\boldsymbol{\chi}}_{t,\mathbf{x}}\big\| &\lesssim M_n\sup_{\mathbf{x}\in\mathscr{B}}\Big\|\mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\Big(\sum_{|\mathbf{v}|=p+1}\frac{|\mathbf{X}_i-\mathbf{x}|^{\mathbf{v}}}{\mathbf{v}!}\Big)\Big]\Big\|\\ &= o_{\mathbb{P}}(\bar{h}^{p+1}),\end{aligned}$$

where the last equality employs the same arguments as in the proof of Lemma SA-2. Hence,

$$\begin{aligned}\sup_{\mathbf{x}\in\mathscr{B}}\Big|\mathbb{E}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}]-\mu_t^{(\boldsymbol{\nu})}(\mathbf{x})-\widehat{B}_{t,\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\Big| &= \sup_{\mathbf{x}\in\mathscr{B}}\Big|\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1}\boldsymbol{\chi}_{t,\mathbf{x}}-\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1}\widetilde{\boldsymbol{\chi}}_{t,\mathbf{x}}\Big|\\ &= o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).\end{aligned}$$

Using Lemma SA-2 and the maximal inequality as in the proof of Lemma SA-2, we conclude that

$$\max_{t\in\{0,1\}}\sup_{\mathbf{x}\in\mathscr{B}}\big|\widehat{B}^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h})-B^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h})\big|\lesssim_{\mathbb{P}}\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}.$$

Since $\max_{t\in\{0,1\}}\sup_{\mathbf{x}\in\mathscr{B}}|B^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h})|\lesssim\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}$, it follows that $\max_{t\in\{0,1\}}\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{B}^{(\boldsymbol{\nu})}_{t,\mathbf{x}}(\mathbf{h})|\lesssim_{\mathbb{P}}\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}$. □

SA-8.6 Proof of Theorem SA-1

For a fixed $\mathbf{x}\in\mathscr{B}$, the pointwise versions of Lemma SA-3 and Lemma SA-5 decompose $\widehat{\mu}^{(\boldsymbol{\nu})}_t(\mathbf{x})-\mu^{(\boldsymbol{\nu})}_t(\mathbf{x})$ into a stochastic component and a conditional bias component of the stated pointwise orders. Applying this decomposition to both $t=0$ and $t=1$ and using the triangle inequality gives the pointwise rate for $\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})-\tau^{(\boldsymbol{\nu})}(\mathbf{x})$. Taking suprema over $\mathbf{x}\in\mathscr{B}$ and using the displayed uniform bounds in the two lemmas gives the uniform rate. □

SA-8.7 Proof of Theorem SA-2

For the conditional bias, by Lemma SA-5,

$$\begin{aligned}
&\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{E}\big[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})-\tau^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big]^2-(B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}))^2\big|\\
&\leq\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{E}\big[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})-\tau^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big]-B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h})\big|\cdot\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{E}\big[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})-\tau^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big]+B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h})\big|\\
&=o_{\mathbb{P}}(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}}).
\end{aligned}$$

For the conditional variance, by Lemma SA-4,

$$\sup_{\mathbf{x}\in\mathscr{B}}\big|\mathbb{V}\big[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big]-(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}V^{(\boldsymbol{\nu})}_{\mathbf{x}}\big|=o_{\mathbb{P}}((n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}).$$

The pointwise MSE expansion follows directly. For the IMSE expansion, notice that

$$\begin{aligned}
&\Big|\mathrm{IMSE}^{(\boldsymbol{\nu})}-\int_{\mathscr{B}}\big[(B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}))^2+(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}V^{(\boldsymbol{\nu})}_{\mathbf{x}}\big]w(\mathbf{x})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})\Big|\\
&\leq\int_{\mathscr{B}}|w(\mathbf{x})|\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})\cdot\sup_{\mathbf{x}\in\mathscr{B}}\big|\,\mathrm{MSE}^{(\boldsymbol{\nu})}(\mathbf{x})-(B^{(\boldsymbol{\nu})}_{\mathbf{x}}(\mathbf{h}))^2-(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}V^{(\boldsymbol{\nu})}_{\mathbf{x}}\big|\\
&=o_{\mathbb{P}}\big(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}}+(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\big),
\end{aligned}$$

which completes the proof. □

SA-8.8 Proof of Theorem SA-3

We have $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})=\sum_{i=1}^n\chi_i$ with

$$\chi_i=\sum_{t\in\{0,1\}}s_t n^{-1}(\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{x}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{x})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\varepsilon_i,$$

where $s_1 = 1$, $s_0 = -1$, $\mathbb{E}[\chi_i] = 0$, and $\mathbb{V}[\chi_i] = n^{-1}$. By the Berry-Esseen Theorem,

$$\sup_{u\in\mathbb{R}} \Big|\mathbb{P}\big(\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) \leq u\big) - \Phi(u)\Big| \lesssim B_n^{-1} \sum_{i=1}^{n} \mathbb{E}[|\chi_i|^3],$$

where $B_n = \sum_{i=1}^n \mathbb{V}[\chi_i] = 1$. Moreover,

$$\begin{aligned}
\sum_{i=1}^{n} \mathbb{E}[|\chi_i|^3] &= n^{-3}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-3/2} \sum_{i=1}^{n} \mathbb{E}\Big[\Big| \sum_{t\in\{0,1\}} s_t \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\varepsilon_i\Big|^3\Big] \\
&\lesssim n^{-3}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-3/2} \sum_{i=1}^{n} \mathbb{E}\Big[\Big| \sum_{t\in\{0,1\}} s_t \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big|^3\Big] \\
&\lesssim n^{-2}\mathbf{h}^{-\boldsymbol{\nu}}|\mathbf{H}|^{-1}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-3/2} \mathbb{E}\Big[\Big| \sum_{t\in\{0,1\}} s_t \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big|^2\Big] \\
&\lesssim n^{-1}\mathbf{h}^{-\boldsymbol{\nu}}|\mathbf{H}|^{-1}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2} \\
&\lesssim (n|\mathbf{H}|)^{-1/2},
\end{aligned}$$

where the second line uses Assumption SA-1(v): since $v \geq 2$, the conditional third moment of ε_i is uniformly bounded, so conditioning on $\mathbf{X}_i$ absorbs the factor $|\varepsilon_i|^3$ into a constant. The third line uses

$$\Big| \sum_{t\in\{0,1\}} s_t \boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big| \lesssim \mathbf{h}^{-\boldsymbol{\nu}}|\mathbf{H}|^{-1}$$

and the fourth line uses the definition of $\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}$.

Finally, although Lemma SA-2 through Lemma SA-4 establish uniform convergence over $\mathbf{x} \in \mathscr{B}$, the corresponding pointwise results for a fixed $\mathbf{x}$ follow by restricting the function classes in those proofs to singletons indexed by $\mathbf{x}$. In particular, when handling the variance terms in Lemma SA-4 (specifically $\mathbf{M}_{3,\mathbf{x},\mathbf{y}}$), we bound terms involving ε_i^2 using either the Fuk–Nagaev inequality (under $2+v$ moments) or Markov's inequality. Thus, for each fixed $\mathbf{x} \in \mathscr{B}$,

$$\begin{aligned}
\|\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\| &= O_{\mathbb{P}}\big((n|\mathbf{H}|)^{-1/2}\big), \\
\|\widehat{\boldsymbol{\Sigma}}_{t,\mathbf{x},\mathbf{x}} - \boldsymbol{\Sigma}_{t,\mathbf{x},\mathbf{x}}\| &= O_{\mathbb{P}}\big((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1}\big),
\end{aligned}$$

and therefore

$$\begin{aligned}
\left|\frac{\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}}{\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}} - 1\right| &= O_{\mathbb{P}}\big((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1}\big), \\
\Big|(\widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2} - (\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2}\Big| &\lesssim_{\mathbb{P}} \sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\big((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1}\big).
\end{aligned}$$

Together with

$$\big|\boldsymbol{\nu}! \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1} \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}}\big| = O_{\mathbb{P}}\big(\mathbf{h}^{-\boldsymbol{\nu}}(n|\mathbf{H}|)^{-1/2}\big)$$

and the bias bound

$$\left|\mathbb{E}[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}] - \tau^{(\boldsymbol{\nu})}(\mathbf{x})\right| = O_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}),$$

this yields

$$\left|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) - \overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\right| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \frac{1}{\sqrt{n|\mathbf{H}|}}, \tag{SA-10}$$

provided that $\bar{h} \to 0$ and $n|\mathbf{H}| \to \infty$.

The final results follow by weak convergence to a Gaussian distribution, and properties of the distribution function. □

SA-8.9 Proof of Theorem SA-4

For all $\mathbf{x} \in \mathscr{B}$, we have $\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) = \overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) + G_1^{(\boldsymbol{\nu})}(\mathbf{x}) + G_2^{(\boldsymbol{\nu})}(\mathbf{x})$, where

$$G_1^{(\boldsymbol{\nu})}(\mathbf{x}) = \Big(\mathbb{E}\big[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})\big|\mathbf{X}\big] - \tau^{(\boldsymbol{\nu})}(\mathbf{x})\Big)(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-1/2},$$

and

$$G_2^{(\boldsymbol{\nu})}(\mathbf{x}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^{\top}\mathbf{S}^{-1}\Big[\big(\widehat{\boldsymbol{\Gamma}}_{1,\mathbf{x}}^{-1}\mathbf{Q}_{1,\mathbf{x}} - \widehat{\boldsymbol{\Gamma}}_{0,\mathbf{x}}^{-1}\mathbf{Q}_{0,\mathbf{x}}\big)(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-\frac{1}{2}} - \big(\boldsymbol{\Gamma}_{1,\mathbf{x}}^{-1}\mathbf{Q}_{1,\mathbf{x}} - \boldsymbol{\Gamma}_{0,\mathbf{x}}^{-1}\mathbf{Q}_{0,\mathbf{x}}\big)(\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-\frac{1}{2}}\Big].$$

By Lemma SA-5 and Lemma SA-4,

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|G_1^{(\boldsymbol{\nu})}(\mathbf{x})\right| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{1/2} \lesssim \bar{h}^{p+1}\sqrt{n|\mathbf{H}|}.$$

By Lemma SA-2, Lemma SA-3 and Lemma SA-4,

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^{\top}\mathbf{S}^{-1}\big[\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{x}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\big]\mathbf{Q}_{t,\mathbf{x}}(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-1/2}\right| \lesssim_{\mathbb{P}} \sqrt{\log(1/\bar{h})}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}\right)$$

and

$$\begin{aligned}
&\sup_{\mathbf{x}\in\mathscr{B}}\left|\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^{\top}\mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\mathbf{Q}_{t,\mathbf{x}}\Big[(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-1/2} - (\Omega^{(\boldsymbol{\nu})}_{\mathbf{x},\mathbf{x}})^{-1/2}\Big]\right| \\
&\lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}}\cdot\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}\right)\cdot\sqrt{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\right) \\
&\lesssim \frac{\log(1/\bar{h})}{\sqrt{n|\mathbf{H}|}} + \frac{(\log(1/\bar{h}))^{3/2}}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\sqrt{n|\mathbf{H}|}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}\right) \\
&\lesssim \frac{\log(1/\bar{h})}{\sqrt{n|\mathbf{H}|}} + \frac{(\log(1/\bar{h}))^{3/2}}{n^{\frac{v}{2+v}}|\mathbf{H}|} + o(\bar{h}^{p+1}\sqrt{n|\mathbf{H}|}).
\end{aligned}$$

The last term is absorbed by the bound for $G_1^{(\boldsymbol{\nu})}$, and the two displayed logarithmic terms give the second component of the rate in the theorem. The result now follows from combining the bounds above. □

SA-8.10 Proof of Theorem SA-5

We verify the high-level conditions of Theorem SA-23. We will employ the following technical lemma.

Lemma SA-12 (VC Class to VC2 Class)**.** *Assume $\mathscr{F}$ is a VC class on a measure space $(\mathcal{X}, \mathscr{B})$: there exists an envelope function F and positive constants $c(\mathscr{F}), d(\mathscr{F})$ such that for all $\varepsilon \in (0,1)$,*

$$\sup_Q N(\mathscr{F}, \|\cdot\|_{Q,1}, \varepsilon \|F\|_{Q,1}) \leq c(\mathscr{F})\varepsilon^{-d(\mathscr{F})},$$

where the supremum is taken over all finite discrete measures. Then, $\mathscr{F}$ is also VC2 class: for all $\varepsilon \in (0,1)$,

$$\sup_Q N(\mathscr{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \leq c(\mathscr{F})(\varepsilon^2/2)^{-d(\mathscr{F})},$$

where the supremum is taken over all finite discrete measures.

Proof of Lemma SA-12. Let Q be a finite discrete probability measure. Let $f, g \in \mathscr{F}$. Then, $\int |f-g|^2 \,\mathrm{d}Q \leq 2\int |f-g||F| \,\mathrm{d}Q$. Define another probability measure $\tilde{Q}(c_k) = F(c_k)Q(c_k)/\|F\|_{Q,1}$ on the support of Q, denoted by $\{c_1, \ldots, c_k, \ldots\}$. Then,

$$\int |f-g|^2 \,\mathrm{d}Q \leq 2\|F\|_{Q,1} \int |f-g| \,\mathrm{d}\tilde{Q} \leq 2\|F\|_{Q,1} \|f-g\|_{\tilde{Q},1}.$$

Hence, if we take an $\varepsilon^2/2$-net in $(\mathscr{F}, \|\cdot\|_{\tilde{Q},1})$ with cardinality no greater than $c(\mathscr{F})(\varepsilon^2/2)^{-d(\mathscr{F})}$, then for any $f \in \mathscr{F}$, there exists a $g \in \mathscr{F}$ such that $\|f-g\|_{\tilde{Q},1} \leq \varepsilon^2/2\,\|F\|_{\tilde{Q},1}$, and hence

$$\|f-g\|_{Q,2}^2 \leq 2\varepsilon^2/2\,\|F\|_{Q,1}\|F\|_{\tilde{Q},1} \leq \varepsilon^2\|F\|_{Q,2}^2,$$

which gives the result. $\square$

By the rectangular-support condition imposed in Theorem SA-5, after a componentwise affine rescaling we can take $\mathcal{X} = [0,1]^d$. Since f_X is continuous and bounded away from zero on this compact rectangle, Cattaneo and Yu [2025, Section 3.1] implies that $\mathbb{Q}_{\mathscr{F}_t} = \mathbb{P}_X$ is a valid surrogate measure for $\mathbb{P}_X$ with respect to $\mathscr{F}_t$, and that the Rosenblatt transform $\phi_{\mathscr{F}_t} = T_{\mathbb{P}_X}$ is a valid normalizing transformation. The constants $\mathtt{c}_1$ and $\mathtt{c}_2$ from Theorem SA-23 are therefore finite and uniformly bounded by constants depending only on the density bounds and the support geometry.

Consider first the class of functions $\mathscr{F}_t = \{\mathscr{K}_t^{(\boldsymbol{\nu})}(\cdot;\mathbf{x}) : \mathbf{x} \in \mathscr{B}\}$, for $t \in \{0,1\}$.

Envelope Function. By Lemma SA-1, Assumption SA-1(iv), and Assumption SA-2, uniformly in $\mathbf{x} \in \mathscr{B}$,

$$\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})} \asymp (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}.$$

Together with the compact support of K and $\|\mathbf{S}^{-1}\mathbf{e}_{\boldsymbol{\nu}}\| \asymp \mathbf{h}^{-\boldsymbol{\nu}}$, this gives

$$\sup_{\mathbf{x}\in\mathscr{B}} \sup_{\xi\in\mathcal{X}} \left|\mathscr{K}_t^{(\boldsymbol{\nu})}(\xi;\mathbf{x})\right| \lesssim n^{-1/2}(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{1/2}\mathbf{h}^{-\boldsymbol{\nu}}|\mathbf{H}|^{-1} \lesssim |\mathbf{H}|^{-1/2}.$$

Hence, there exists a constant $C_1 > 0$ such that $\mathtt{M}_{\mathscr{F}_t} = C_1|\mathbf{H}|^{-1/2}$ is a constant envelope function.

L_1 Bound. Since $\mathscr{K}_t^{(\boldsymbol{\nu})}(\cdot;\mathbf{x})$ is supported on $\mathbf{x}+\mathbf{H}\,\mathrm{supp}(K)$, whose Lebesgue measure is of order $|\mathbf{H}|$,

$$\mathtt{E}_{\mathscr{F}_t} = \sup_{\mathbf{x}\in\mathscr{B}} \mathbb{E}[|\mathscr{K}_t^{(\boldsymbol{\nu})}(\mathbf{X}_i;\mathbf{x})|] \lesssim |\mathbf{H}|^{1/2}.$$

Uniform Variation. Case 1: K is Lipschitz. By Assumption SA-1(iv) and Assumption SA-2,

$$\mathtt{L}_{\mathscr{F}_t} = \sup_{\mathbf{x}\in\mathscr{B}} \sup_{\xi,\xi'\in\mathscr{X}} \frac{|\mathscr{K}_t^{(\boldsymbol{\nu})}(\xi;\mathbf{x}) - \mathscr{K}_t^{(\boldsymbol{\nu})}(\xi';\mathbf{x})|}{\|\xi-\xi'\|_\infty} \lesssim |\mathbf{H}|^{-1/2}\underline{h}^{-1}.$$

Each entry of $\boldsymbol{\Gamma}_{t,\mathbf{x}}$ and $\boldsymbol{\Sigma}_{t,\mathbf{x},\mathbf{x}}$ is a smooth local integral with argument $\mathbf{H}^{-1}(\xi-\mathbf{x})$ and is therefore $\underline{h}^{-1}$-Lipschitz in $\mathbf{x}$. Similarly,

$$\left|\Omega_{t,\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})} - \Omega_{t,\mathbf{x}',\mathbf{x}'}^{(\boldsymbol{\nu})}\right| \lesssim (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}\underline{h})^{-1}\|\mathbf{x}-\mathbf{x}'\|_\infty,$$

which, using $\Omega_{t,\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})} \asymp (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$, implies

$$\left|(\Omega_{t,\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2} - (\Omega_{t,\mathbf{x}',\mathbf{x}'}^{(\boldsymbol{\nu})})^{-1/2}\right| \lesssim \underline{h}^{-1}(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{1/2}\|\mathbf{x}-\mathbf{x}'\|_\infty.$$

It follows that the class is Lipschitz with respect to the point of evaluation:

$$\mathtt{l}_{\mathscr{F}_t} = \sup_{\xi\in\mathscr{X}} \sup_{\mathbf{x},\mathbf{x}'\in\mathscr{B}} \frac{\left|\mathscr{K}_t^{(\boldsymbol{\nu})}(\xi;\mathbf{x}) - \mathscr{K}_t^{(\boldsymbol{\nu})}(\xi;\mathbf{x}')\right|}{\|\mathbf{x}-\mathbf{x}'\|_\infty} \lesssim |\mathbf{H}|^{-1/2}\underline{h}^{-1}.$$

Since $\mathscr{K}_t^{(\boldsymbol{\nu})}(\cdot;\mathbf{x})$ is supported on $\mathbf{x}+\mathbf{H}\,\mathrm{supp}(K)$, we have

$$\mathtt{TV}_{\mathscr{F}_t} \lesssim |\mathbf{H}|\,|\mathbf{H}|^{-1/2}\underline{h}^{-1} \lesssim |\mathbf{H}|^{1/2}\underline{h}^{-1}.$$

Case 2: $K = \mathbb{1}(\cdot\in[-1,1]^d)$. Define the smooth part

$$\tilde{\mathscr{K}}_t^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) = n^{-1/2}(\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))|\mathbf{H}|^{-1}, \qquad \mathbf{u}\in\mathscr{X}, \quad t\in\{0,1\}.$$

Then, $\mathscr{K}_t^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) = \tilde{\mathscr{K}}_t^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x})\mathbb{1}(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x})\in[-1,1]^d)$, and the preceding argument gives $\mathtt{TV}_{\tilde{\mathscr{F}}_t} \lesssim |\mathbf{H}|^{1/2}\underline{h}^{-1}$. For $\mathscr{L} = \{\mathbb{1}(\mathbf{H}^{-1}(\cdot-\mathbf{x})\in[-1,1]^d) : \mathbf{x}\in\mathscr{B}\}$, the boundary of each rectangle has perimeter $O(|\mathbf{H}|\underline{h}^{-1})$, so the product rule yields

$$\mathtt{TV}_{\mathscr{F}_t} \le \mathtt{TV}_{\tilde{\mathscr{F}}_t}\mathtt{M}_{\mathscr{L}} + \mathtt{M}_{\tilde{\mathscr{F}}_t}\mathtt{TV}_{\mathscr{L}} \lesssim |\mathbf{H}|^{1/2}\underline{h}^{-1}.$$

VC-type Class. Case 1: K is Lipschitz. We apply Cattaneo et al. [2024a, Lemma 7] to the rescaled maps

$$f_{\mathbf{x}}(\cdot) = \frac{\boldsymbol{\nu}!}{\sqrt{n\Omega_{\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})}}}\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{x}}^{-1}\mathbf{r}_p(\cdot)K(\cdot), \qquad \mathbf{x}\in\mathscr{B},$$

indexed by $\mathscr{H} = \{f_{\mathbf{x}}(\mathbf{H}^{-1}(\cdot-\mathbf{x})) : \mathbf{x}\in\mathscr{B}\}$. The boundedness and compact support conditions are unchanged;

the Lipschitz constant in the index is now of order $\underline{h}^{-1}$. Therefore, for $0 < \varepsilon \leq 1$,

$$\sup_Q N\big(\mathscr{F}_t, \|\cdot\|_{Q,2}, (2c+1)^{d+1}\varepsilon \mathtt{M}_{\mathscr{F}_t}\big) \leq \mathbf{c}' 2^{2d+4}\varepsilon^{-2d-4} + 1,$$

where the supremum is taken over all finite discrete measures.

Case 2: Suppose $K = \mathbb{1}(\cdot \in [-1,1]^d)$. The smooth class $\tilde{\mathscr{F}}_t$ satisfies the same entropy bound. The indicator class $\mathscr{L}$ is a VC class of axis-aligned rectangles with side lengths $2h_j$, so

$$\sup_Q N(\mathscr{L}, \|\cdot\|_{Q,2}, \varepsilon) \leq C\varepsilon^{-4d}, \qquad 0 < \varepsilon \leq 1.$$

The product entropy bound then gives, for a constant $c > 0$,

$$\begin{aligned}\sup_Q N(\mathscr{F}_t, \|\cdot\|_{Q,2}, \varepsilon C_1 \mathtt{M}_{\tilde{\mathscr{F}}_t}) &\leq \sup_Q N(\tilde{\mathscr{F}}_t, \|\cdot\|_{Q,2}, c\varepsilon \mathtt{M}_{\tilde{\mathscr{F}}_t}) \sup_Q N(\mathscr{L}, \|\cdot\|_{Q,2}, c\varepsilon) \\ &\leq C_2 \varepsilon^{-(6d+4)},\end{aligned}$$

for constants depending only on d.

Consider next the class of functions $\mathscr{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathscr{B}\}$, where $g_{\mathbf{x}}(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in \mathscr{A}_1)\mathscr{K}_1^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) - \mathbb{1}(\mathbf{u} \in \mathscr{A}_0)\mathscr{K}_0^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x})$. We have immediately that $\mathtt{M}_{\mathscr{G}} \lesssim |\mathbf{H}|^{-1/2}$, $\mathtt{E}_{\mathscr{G}} \lesssim |\mathbf{H}|^{1/2}$, and

$$\sup_Q N(\mathscr{G}, \|\cdot\|_{Q,2}, \varepsilon(2c+1)^{d+1}\mathtt{M}_{\mathscr{G}}) \leq 2\mathbf{c}'\varepsilon^{-4d-4} + 2.$$

Total Variation. Observe that $\mathbb{1}(\mathbf{u} \in \mathscr{A}_t)\mathscr{K}_t^{(\boldsymbol{\nu})}(\mathbf{u};\mathbf{x}) \neq 0$ implies $\mathbf{u} \in E_{t,\mathbf{x}} = \{\mathbf{y} \in \mathscr{A}_t : \mathbf{H}^{-1}(\mathbf{y}-\mathbf{x}) \in \operatorname{supp}(K)\}$. By assumption, $\mathscr{L}(E_{t,\mathbf{x}}) \leq C\bar{h}^{d-1}$. Using the product rule and the bandwidth comparability condition,

$$\begin{aligned}\mathtt{TV}_{\mathscr{G}} &\leq \sup_{\mathbf{x}\in\mathscr{B}} \sum_{t\in\{0,1\}} \mathtt{TV}_{\{\mathscr{K}_t^{(\boldsymbol{\nu})}(\cdot;\mathbf{x})\}} + \mathtt{M}_{\mathscr{F}_t}\mathtt{TV}_{\{\mathbb{1}_{E_{t,\mathbf{x}}}\}} \\ &\lesssim |\mathbf{H}|^{1/2}\underline{h}^{-1} + |\mathbf{H}|^{-1/2}\bar{h}^{d-1} \lesssim |\mathbf{H}|^{1/2}\underline{h}^{-1}.\end{aligned}$$

The imposed singleton regularity condition verifies the remaining truncated conditional-mean requirement in Theorem SA-23. We have therefore verified all high-level sufficient conditions, which immediately give the result. □

SA-8.11 Proof of Lemma SA-6

Let $\mathscr{N}_{\mathbf{x},h} = \mathbf{x} + h \operatorname{supp}(K)$. Since $\operatorname{supp}(K)$ is bounded, there is a constant $R < \infty$ such that $\mathscr{N}_{\mathbf{x},h} \subseteq B(\mathbf{x}, Rh)$ for all $\mathbf{x} \in \mathscr{B}$ and all $h > 0$. Since $\mathscr{B}$ is a finite union of line segments, say $\mathscr{B} = \cup_{j=1}^J L_j$, uniformly over $\mathbf{x} \in \mathscr{B}$,

$$\mathfrak{H}^1(\mathscr{B} \cap \mathscr{N}_{\mathbf{x},h}) \leq \mathfrak{H}^1(\mathscr{B} \cap B(\mathbf{x}, Rh)) \leq \sum_{j=1}^J \mathfrak{H}^1(L_j \cap B(\mathbf{x}, Rh)) \leq 2JRh \lesssim h.$$

Because $\mathscr{B}$ lies in the interior of $\mathscr{X}$, for all small enough h the window $\mathscr{N}_{\mathbf{x},h}$ does not intersect $\mathrm{bd}(\mathscr{X})$ uniformly over $\mathbf{x} \in \mathscr{B}$. Write $\mathscr{L}(\mathscr{A}_t; \mathscr{N}_{\mathbf{x},h})$ for the perimeter of $\mathscr{A}_t$ relative to $\mathscr{N}_{\mathbf{x},h}$. The relative boundary of $\mathscr{A}_t$ inside $\mathscr{N}_{\mathbf{x},h}$ is contained in $\mathscr{B} \cap \mathscr{N}_{\mathbf{x},h}$, up to finitely many endpoints, and therefore

$$\mathscr{L}(\mathscr{A}_t; \mathscr{N}_{\mathbf{x},h}) \lesssim \mathfrak{H}^1(\mathscr{B} \cap \mathscr{N}_{\mathbf{x},h}) \lesssim h.$$

If $K(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in [-1,1]^2)$, then $\mathscr{N}_{\mathbf{x},h}$ is a square with perimeter $8h$. More generally, if $\mathrm{supp}(K)$ has finite De Giorgi perimeter, the scaling property of perimeter gives $\mathscr{L}(\mathscr{N}_{\mathbf{x},h}) = h\mathscr{L}(\mathrm{supp}(K)) \lesssim h$. By the standard subadditivity/product rule for finite-perimeter sets,

$$\mathscr{L}(\mathscr{A}_t \cap \mathscr{N}_{\mathbf{x},h}) \leq \mathscr{L}(\mathscr{A}_t; \mathscr{N}_{\mathbf{x},h}) + \mathscr{L}(\mathscr{N}_{\mathbf{x},h}) \lesssim h,$$

uniformly over $\mathbf{x} \in \mathscr{B}$ and $t \in \{0,1\}$. Since $d = 2$ and $\bar{h} = h$, this is the claimed localized perimeter bound. □

SA-8.12 Proof of Theorem SA-6

The proof is divided into three theorem-specific technical lemmas. We first record a generic comparison lemma that will be used to translate a uniform process approximation into a Kolmogorov distance bound for suprema.

Lemma SA-13 (Generic Suprema Comparison)**.** *Let $\mathbb{T}_j = (\mathbb{T}_j(\mathbf{x}) : \mathbf{x} \in \mathscr{B})$, $j = 1,2$, be real-valued processes defined on a common probability space. Write*

$$M_j = \sup_{\mathbf{x} \in \mathscr{B}} |\mathbb{T}_j(\mathbf{x})|, \quad j = 1,2, \qquad \Delta_{12} = \sup_{\mathbf{x} \in \mathscr{B}} |\mathbb{T}_1(\mathbf{x}) - \mathbb{T}_2(\mathbf{x})|.$$

For any $\eta > 0$,

$$\sup_{u \in \mathbb{R}} |\mathbb{P}(M_1 \leq u) - \mathbb{P}(M_2 \leq u)| \leq \mathsf{AC}_2(\eta) + \mathbb{P}(\Delta_{12} > \eta),$$

where

$$\mathsf{AC}_2(\eta) = \sup_{u \in \mathbb{R}} \mathbb{P}(|M_2 - u| \leq \eta).$$

If, in addition, $\mathbb{T}_2$ is a separable mean-zero Gaussian process and the Gaussian anti-concentration bound applies to M_2, then

$$\sup_{u \in \mathbb{R}} |\mathbb{P}(M_1 \leq u) - \mathbb{P}(M_2 \leq u)| \leq 4\eta\{\mathbb{E}[M_2] + 1\} + \mathbb{P}(\Delta_{12} > \eta).$$

Consequently, if $\mathbb{E}[\Delta_{12}] \leq r_n$, then the last probability can be further bounded by r_n/η.

Proof of Lemma SA-13. On the event $\{\Delta_{12} \leq \eta\}$, the inequalities

$$M_1 \leq u \implies M_2 \leq u + \eta, \qquad M_2 \leq u - \eta \implies M_1 \leq u$$

hold for every $u \in \mathbb{R}$. Therefore,

$$\mathbb{P}(M_1 \le u) - \mathbb{P}(M_2 \le u) \le \mathbb{P}(u < M_2 \le u + \eta) + \mathbb{P}(\Delta_{12} > \eta),$$
$$\mathbb{P}(M_2 \le u) - \mathbb{P}(M_1 \le u) \le \mathbb{P}(u - \eta < M_2 \le u) + \mathbb{P}(\Delta_{12} > \eta).$$

Taking absolute values and then the supremum over u gives the first result. The second display follows from the Gaussian anti-concentration inequality of Chernozhukov et al. [2014a, Theorem 2.1], applied to the symmetrized Gaussian process whose supremum is M_2. The final claim follows from Markov's inequality. $\square$

For the theorem-specific lemmas below, define the strong-approximation rate

$$\mathfrak{R}_{\mathtt{sa}} = (\log n)^{\frac{3}{2}} \Big(\frac{1}{n|\mathbf{H}|}\Big)^{\frac{1}{2d+2} \cdot \frac{v}{v+2}} + \log(n)\sqrt{\frac{1}{n^{\frac{v}{2+v}}|\mathbf{H}|}},$$

the stochastic-linearization remainder

$$\mathfrak{R}_{\mathtt{lin}} = \sqrt{\log(1/\bar{h})}\Big(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|}\Big) + \bar{h}^{p+1}\sqrt{n|\mathbf{H}|},$$

and the covariance-estimation rate

$$\mathfrak{R}_{\mathtt{cov}} = \sqrt{\frac{\log n}{n|\mathbf{H}|}} + \frac{\log n}{n^{\frac{v}{2+v}}|\mathbf{H}|}.$$

Also set the covariance-comparison rate $\mathfrak{R}_{\mathtt{cc}} = \mathfrak{R}_{\mathtt{cov}} + \bar{h}^{p+1}$.

Lemma SA-14 (KS Distance Between $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$ and $Z^{(\boldsymbol{\nu})}$)**.** *Suppose the conditions of Theorem SA-6 hold. Then,*

$$\sup_{u\in\mathbb{R}} \Big| \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u\Big) - \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|Z^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u\Big) \Big| \lesssim (\log n)^{1/4}\mathfrak{R}_{\mathtt{sa}}^{1/2}.$$

Proof of Lemma SA-14. Let a_n be a positive sequence to be determined below. Applying Lemma SA-13 with $\mathbb{T}_1 = \overline{\mathrm{T}}^{(\boldsymbol{\nu})}$, $\mathbb{T}_2 = Z^{(\boldsymbol{\nu})}$, and $\eta = a_n$, and then using the coupling tail bound in Theorem SA-5, gives

$$\begin{aligned}
&\sup_{u\in\mathbb{R}} \Big| \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u\Big) - \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|Z^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le u\Big) \Big| \\
&\quad \le 4a_n \left(\mathbb{E}\left[\sup_{\mathbf{x}\in\mathscr{B}} |Z^{(\boldsymbol{\nu})}(\mathbf{x})|\right] + 1\right) + \frac{C\mathfrak{R}_{\mathtt{sa}}}{a_n}.
\end{aligned}$$

For the next displays, write $\mathscr{K}(\mathbf{u},\mathbf{x}) = g_{\mathbf{x}}(\mathbf{u})$ and $\sigma^2(\mathbf{u}) = \mathbb{V}(Y_i|\mathbf{X}_i = \mathbf{u})$. Notice that $Z^{(\boldsymbol{\nu})}(\mathbf{x}), \mathbf{x} \in \mathscr{B}$ is a mean-zero Gaussian process satisfying

$$\begin{aligned}
\mathbb{E}\big[\big(Z^{(\boldsymbol{\nu})}(\mathbf{x}) - Z^{(\boldsymbol{\nu})}(\mathbf{y})\big)^2\big]^{\frac{1}{2}} &= \mathbb{E}\big[(\mathscr{K}(\mathbf{X}_i,\mathbf{x}) - \mathscr{K}(\mathbf{X}_i,\mathbf{y}))^2\sigma(\mathbf{X}_i)^2\big]^{\frac{1}{2}} \\
&\le C' l_{n,2}\|\mathbf{x} - \mathbf{y}\|_\infty,
\end{aligned}$$

where C' is a constant and $l_{n,2} \asymp \underline{h}^{-1}$. Hence the canonical metric

$$\rho(\mathbf{x},\mathbf{y}) := \mathbb{E}\big[\big(Z^{(\boldsymbol{\nu})}(\mathbf{x}) - Z^{(\boldsymbol{\nu})}(\mathbf{y})\big)^2\big]^{1/2} \lesssim \underline{h}^{-1}\|\mathbf{x} - \mathbf{y}\|_\infty.$$

Moreover,

$$\sup_{\mathbf{x}\in\mathscr{B}} \mathbb{E}\big[Z^{(\boldsymbol{\nu})}(\mathbf{x})^2\big] = \sup_{\mathbf{x}\in\mathscr{B}} \mathbb{E}\big[\mathscr{K}(\mathbf{X}_i,\mathbf{x})^2\sigma^2(\mathbf{X}_i)\big] \lesssim 1.$$

Let $N(\varepsilon, \mathscr{B}, \rho)$ denote the covering number of $\mathscr{B}$ under the metric ρ. Since $\rho(\mathbf{x}, \mathbf{y}) \lesssim \underline{h}^{-1}\|\mathbf{x} - \mathbf{y}\|_\infty$ and $\mathscr{B}$ is compact in $\mathscr{R}^d$,

$$N(\varepsilon, \mathscr{B}, \rho) \lesssim N(\underline{h}\varepsilon, \mathscr{B}, \|\cdot\|_\infty) \lesssim \Big(\frac{1}{\underline{h}\varepsilon}\Big)^d, \qquad 0 < \varepsilon \leq 1,$$

and therefore

$$\log N(\varepsilon, \mathscr{B}, \rho) \lesssim d \log\Big(\frac{1}{\underline{h}\varepsilon}\Big).$$

Applying Corollary 2.2.8 in van der Vaart and Wellner [1996] yields

$$\begin{aligned}
\mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}} \big|Z^{(\boldsymbol{\nu})}(\mathbf{x})\big|\Big] &\lesssim \int_0^1 \sqrt{\log N(\varepsilon, \mathscr{B}, \rho)}\,\mathrm{d}\varepsilon \\
&\lesssim \int_0^1 \sqrt{\log\Big(\frac{1}{\underline{h}\varepsilon}\Big)}\,\mathrm{d}\varepsilon \lesssim \sqrt{\log(1/\underline{h})} \lesssim \sqrt{\log n}.
\end{aligned}$$

Choosing $a_n \asymp (\mathfrak{R}_{\mathtt{sa}}/\sqrt{\log n})^{1/2}$, the result now follows. $\square$

Lemma SA-15 (KS Distance Between $\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}$ and $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$)**.** *Suppose the conditions in Theorem SA-6 hold. Then, for any multi-index $|\boldsymbol{\nu}| \leq p$,*

$$\sup_{u\in\mathbb{R}} \Big|\mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} |\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})| \leq u\Big) - \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} |\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})| \leq u\Big)\Big| = o(1).$$

Proof of Lemma SA-15. Choose a deterministic sequence a_n such that $\mathfrak{R}_{\mathtt{lin}} = o(a_n)$ and $a_n\sqrt{\log n} = o(1)$; such a sequence exists under the side conditions of Theorem SA-6. Then,

$$\Delta_{\widehat{T},\overline{T}} := \sup_{\mathbf{x}\in\mathscr{B}} \big|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x}) - \widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| = o_{\mathbb{P}}(a_n).$$

Define

$$M_{\widehat{T}} = \sup_{\mathbf{x}\in\mathscr{B}} |\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|, \qquad M_{\overline{T}} = \sup_{\mathbf{x}\in\mathscr{B}} |\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|, \qquad M_Z = \sup_{\mathbf{x}\in\mathscr{B}} |Z^{(\boldsymbol{\nu})}(\mathbf{x})|.$$

Applying the first part of Lemma SA-13 with $\mathbb{T}_1 = \widehat{\mathrm{T}}^{(\boldsymbol{\nu})}$ and $\mathbb{T}_2 = \overline{\mathrm{T}}^{(\boldsymbol{\nu})}$ gives

$$\sup_{u\in\mathbb{R}} \big|\mathbb{P}(M_{\widehat{T}} \leq u) - \mathbb{P}(M_{\overline{T}} \leq u)\big| \leq \sup_{u\in\mathbb{R}} \mathbb{P}(|M_{\overline{T}} - u| \leq a_n) + \mathbb{P}(\Delta_{\widehat{T},\overline{T}} > a_n).$$

The anti-concentration term for $M_{\overline{T}}$ can be bounded through the Gaussian approximation in Lemma SA-14: for every $u \in \mathbb{R}$,

$$\begin{aligned}
\mathbb{P}(|M_{\overline{T}} - u| \leq a_n) &\leq \mathbb{P}(|M_Z - u| \leq a_n) + 2\sup_{v\in\mathbb{R}} \big|\mathbb{P}(M_{\overline{T}} \leq v) - \mathbb{P}(M_Z \leq v)\big| \\
&\leq 4a_n\{\mathbb{E}[M_Z] + 1\} + 2\sup_{v\in\mathbb{R}} \big|\mathbb{P}(M_{\overline{T}} \leq v) - \mathbb{P}(M_Z \leq v)\big|.
\end{aligned}$$

By Lemma SA-14, the last Kolmogorov distance is $O((\log n)^{1/4}\mathfrak{R}_{\mathtt{sa}}^{1/2}) = o(1)$. From the proof of that lemma, $\mathbb{E}[M_Z] \lesssim \sqrt{\log n}$. Since $a_n\sqrt{\log n} = o(1)$ and $\mathbb{P}(\Delta_{\widehat{T},\overline{T}} > a_n) = o(1)$, the result follows. $\square$

Lemma SA-16 (KS Distance Between $Z^{(\boldsymbol{\nu})}$ and $\widehat{Z}^{(\boldsymbol{\nu})}$)**.** *Suppose the conditions for Theorem SA-6 hold.*

Then, for any multi-index $|\boldsymbol{\nu}| \leq p$,

$$\sup_{u \in \mathbb{R}} \Big| \mathbb{P}\Big(\sup_{\mathbf{x} \in \mathscr{B}} \big| Z^{(\boldsymbol{\nu})}(\mathbf{x}) \big| \leq u \Big) - \mathbb{P}\Big(\sup_{\mathbf{x} \in \mathscr{B}} \big| \widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}) \big| \leq u \big| \mathbf{V} \Big) \Big| \lesssim_{\mathbb{P}} \log(n) \mathfrak{R}_{\mathsf{cc}}^{1/2}.$$

Proof of Lemma SA-16. In this proof, we suppress the superscript $(\boldsymbol{\nu})$ on Ω and $\widehat{\Omega}$. First, using Lemma SA-4, we provide an upper bound between covariance functions of the feasible Gaussian process and the infeasible Gaussian process. Letting $\mathbf{\Pi}_{\mathbf{x},\mathbf{y}} = \Omega_{\mathbf{x},\mathbf{y}} / \sqrt{\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}}}$ and $\widehat{\mathbf{\Pi}}_{\mathbf{x},\mathbf{y}} = \widehat{\Omega}_{\mathbf{x},\mathbf{y}} / \sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}}}$,

$$\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \big| \mathbf{\Pi}_{\mathbf{x},\mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x},\mathbf{y}} \big| = \sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \Big| \frac{\Omega_{\mathbf{x},\mathbf{y}} - \widehat{\Omega}_{\mathbf{x},\mathbf{y}}}{\sqrt{\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}}}} + \frac{\widehat{\Omega}_{\mathbf{x},\mathbf{y}}}{\sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}}}} \Big(\sqrt{\frac{\widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}}}{\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}}}} - 1 \Big) \Big|$$

From Lemma SA-4 and the fact that $|\sqrt{x} - \sqrt{y}| \leq (x \wedge y)^{-1/2} |x - y|/2$ for $x, y > 0$,

$$\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \frac{\big| \big(\widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}} \big)^{1/2} - \big(\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}} \big)^{1/2} \big|}{\big(\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}} \big)^{1/2}} \lesssim \frac{\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \big| \widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}} - \Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}} \big|}{\inf_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y},\mathbf{y}} \wedge \inf_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}}} \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathsf{cov}}$$

and

$$\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \frac{\big| \Omega_{\mathbf{x},\mathbf{y}} - \widehat{\Omega}_{\mathbf{x},\mathbf{y}} \big|}{\sqrt{\Omega_{\mathbf{x},\mathbf{x}} \Omega_{\mathbf{y},\mathbf{y}}}} \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \mathfrak{R}_{\mathsf{cov}}.$$

Therefore, $\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} |\mathbf{\Pi}_{\mathbf{x},\mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x},\mathbf{y}}| \lesssim_{\mathbb{P}} \mathfrak{R}_{\mathsf{cc}}$. Then, we bound the KS distance between the maximum of $Z^{(\boldsymbol{\nu})}$ and $\widehat{Z}^{(\boldsymbol{\nu})}$ on a δ_n-net of $\mathscr{B}$, denoted by $\mathscr{B}_{\delta_n}$: for all $\mathbf{x} \in \mathscr{B}$, there exists $\mathbf{z} \in \mathscr{B}_{\delta_n}$ such that $\|\mathbf{x} - \mathbf{z}\|_\infty \leq \delta_n$. Since $\mathscr{B}$ is compact, we can assume $M := \mathrm{Card}\,(\mathscr{B}_{\delta_n}) \lesssim \delta_n^{-d}$. Denote $\mathbf{Z}^{\delta_n}$ and $\widehat{\mathbf{Z}}^{\delta_n}$ as the restrictions of $Z^{(\boldsymbol{\nu})}$ and $\widehat{Z}^{(\boldsymbol{\nu})}$ to $\mathscr{B}_{\delta_n}$, respectively. Then, by Chernozhukov et al. [2022, Theorem 2.1],

$$\sup_{\mathbf{y} \in \mathbb{R}^M} \big| \mathbb{P}(\mathbf{Z}^{\delta_n} \leq \mathbf{y}) - \mathbb{P}(\widehat{\mathbf{Z}}^{\delta_n} \leq \mathbf{y} | \mathbf{V}) \big| \lesssim \log(M) \sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \big| \mathbf{\Pi}_{\mathbf{x},\mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x},\mathbf{y}} \big|^{\frac{1}{2}} \lesssim_{\mathbb{P}} \log(M) \mathfrak{R}_{\mathsf{cc}}^{\frac{1}{2}},$$

and hence

$$\begin{aligned} \sup_{x \in \mathbb{R}} \big| \mathbb{P}(\|\mathbf{Z}^{\delta_n}\|_\infty \leq x) - \mathbb{P}(\|\widehat{\mathbf{Z}}^{\delta_n}\|_\infty \leq x | \mathbf{V}) \big| &\leq \sup_{x \in \mathbb{R}} \big| \mathbb{P}(-x\mathbf{1} \leq \mathbf{Z}^{\delta_n} \leq x\mathbf{1}) - \mathbb{P}(-x\mathbf{1} \leq \widehat{\mathbf{Z}}^{\delta_n} \leq x\mathbf{1} | \mathbf{V}) \big| \\ &\lesssim_{\mathbb{P}} \log(M) \mathfrak{R}_{\mathsf{cc}}^{\frac{1}{2}} =: \mathfrak{R}_M. \end{aligned}$$

Finally, we extend the comparison from the finite net to the full boundary. Let $\delta_n = n^{-s}$ for a fixed constant $s > 0$ chosen large enough that

$$\sqrt{\log n}\, |\mathbf{H}|^{-1/2} \underline{h}^{-1} \delta_n = o(\mathfrak{R}_{\mathsf{cc}}), \qquad \sqrt{\log n}\, \underline{h}^{-1} \delta_n = o(\mathfrak{R}_{\mathsf{cc}}).$$

Such an s exists under the polynomial lower bound on $\bar{h}$ and the bounded ratio $\bar{h}/\underline{h}$. Put $b_n = \mathfrak{R}_{\mathsf{cc}}^{1/2}$ and define the modulus probabilities

$$\Psi_{\delta_n}(b_n) = \mathbb{P}\Big(\sup_{\substack{\mathbf{x},\mathbf{y} \in \mathscr{B} \\ \|\mathbf{x} - \mathbf{y}\|_\infty \leq \delta_n}} \big| Z^{(\boldsymbol{\nu})}(\mathbf{x}) - Z^{(\boldsymbol{\nu})}(\mathbf{y}) \big| \geq b_n \Big),$$

$$\widehat{\Psi}_{\delta_n}(b_n) = \mathbb{P}\Big(\sup_{\substack{\mathbf{x},\mathbf{y}\in\mathscr{B} \\ \|\mathbf{x}-\mathbf{y}\|_\infty \le \delta_n}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}) - \widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{y})\big| \ge b_n \Big| \mathbf{V}\Big).$$

The usual chaining argument for the local-polynomial Gaussian metrics gives

$$\mathbb{E}\Big[\sup_{\substack{\mathbf{x},\mathbf{y}\in\mathscr{B} \\ \|\mathbf{x}-\mathbf{y}\|_\infty \le \delta_n}} \big|Z^{(\boldsymbol{\nu})}(\mathbf{x}) - Z^{(\boldsymbol{\nu})}(\mathbf{y})\big|\Big] \lesssim \sqrt{\log n}\, \underline{h}^{-1}\delta_n,$$
$$\mathbb{E}\Big[\sup_{\substack{\mathbf{x},\mathbf{y}\in\mathscr{B} \\ \|\mathbf{x}-\mathbf{y}\|_\infty \le \delta_n}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x}) - \widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{y})\big|\Big|\mathbf{V}\Big] \lesssim_{\mathbb{P}} \sqrt{\log n}\, |\mathbf{H}|^{-1/2}\underline{h}^{-1}\delta_n.$$

The second display follows from Lemma SA-4, Lemma SA-2, and the Lipschitz bound for the feasible local-polynomial weights; the first display follows from the corresponding population bound. Therefore Markov's inequality implies

$$\Psi_{\delta_n}(b_n) + \widehat{\Psi}_{\delta_n}(b_n) = o_{\mathbb{P}}(\mathfrak{R}_{\mathsf{cc}}^{1/2}).$$

Moreover, the same entropy calculation gives

$$\mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}} |Z^{(\boldsymbol{\nu})}(\mathbf{x})|\Big] \lesssim \sqrt{\log n}, \qquad \mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}} |\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})|\Big|\mathbf{V}\Big] \lesssim_{\mathbb{P}} \sqrt{\log n}.$$

For every $t > 0$, the net approximation, the finite-dimensional comparison, and Gaussian anti-concentration give

$$\begin{aligned}
&\mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|Z^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le t\Big) \\
&\le \mathbb{P}\Big(\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le t\Big|\mathbf{V}\Big) + 4b_n\Big(\mathbb{E}\Big[\sup_{\mathbf{x}\in\mathscr{B}} |\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})|\Big|\mathbf{V}\Big] + 1\Big) + \Psi_{\delta_n}(b_n) + \widehat{\Psi}_{\delta_n}(b_n) + \mathfrak{R}_M,
\end{aligned}$$

and the same argument with the roles of $Z^{(\boldsymbol{\nu})}$ and $\widehat{Z}^{(\boldsymbol{\nu})}$ reversed gives the corresponding lower bound. Since $\log M \lesssim \log n$, $\mathfrak{R}_M \lesssim_{\mathbb{P}} \log(n)\mathfrak{R}_{\mathsf{cc}}^{1/2}$; the anti-concentration term is $O_{\mathbb{P}}(\sqrt{\log n}\mathfrak{R}_{\mathsf{cc}}^{1/2})$; and the two modulus probabilities are $o_{\mathbb{P}}(\mathfrak{R}_{\mathsf{cc}}^{1/2})$. Combining the upper and lower bounds gives the stated result. □

The proof of Theorem SA-6 now follows directly from Lemma SA-14, Lemma SA-15 and Lemma SA-16. Furthermore, by definition of $\widehat{\mathrm{I}}_\alpha^{(\boldsymbol{\nu})}(\mathbf{x})$,

$$\begin{aligned}
\mathbb{P}\Big[\tau^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}_\alpha^{(\boldsymbol{\nu})}(\mathbf{x}),\ \text{for all } \mathbf{x}\in\mathscr{B}\Big] &= \mathbb{P}\Big[\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le \mathfrak{q}_\alpha\Big] \\
&= \mathbb{P}\Big[\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le \mathfrak{q}_\alpha\Big] + o(1) \\
&= \mathbb{E}\Big[\mathbb{P}\Big[\sup_{\mathbf{x}\in\mathscr{B}} \big|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})\big| \le \mathfrak{q}_\alpha\Big|\mathbf{V}\Big]\Big] + o(1) \\
&= 1 - \alpha + o(1),
\end{aligned}$$

which completes the proof of the theorem. □

SA-8.13 Proof of Theorem SA-7

We employ the following result established by Chernozhukov et al. [2014b, Corollary 2.2].

Lemma SA-17 (Gaussian Approximation of Suprema)**.** *Let $X_1, \ldots, X_n$ be i.i.d. random variables in a measurable space $(S, \mathcal{S})$ with law P, where $n \geq 3$. Let $\mathscr{F}$ be a class of measurable functions $f : S \to \mathbb{R}$ that is P-centered, i.e., $Pf = 0$ for all $f \in \mathscr{F}$. Define the empirical process $\mathbb{G}_n$ indexed by $\mathscr{F}$ as $\mathbb{G}_n f = \frac{1}{\sqrt{n}} \sum_{i=1}^n f(X_i)$ for $f \in \mathscr{F}$. Suppose that:*

1. *The class $\mathscr{F}$ is pointwise measurable, that is, it contains a countable subset $\mathscr{G}$ such that for every $f \in \mathscr{F}$ there exists a sequence $g_m \in \mathscr{G}$ with $g_m(x) \to f(x)$ for every $x \in S$.*
2. *There exists a measurable envelope F such that $|f| \leq F$ for all $f \in \mathscr{F}$, and constants $A \geq e$ and $\upsilon \geq 1$ such that $\sup_Q N(\mathscr{F}, e_Q, \epsilon \|F\|_{Q,2}) \leq (A/\epsilon)^\upsilon$ for all $0 < \epsilon \leq 1$, where the supremum is taken over all finite discrete probability measures on $(S, \mathcal{S})$.*
3. *For some $b \geq \sigma > 0$ and $q \in [4, \infty]$, we have $\sup_{f \in \mathscr{F}} P|f|^k \leq \sigma^2 b^{k-2}$ for $k = 2, 3, 4$ and $\|F\|_{P,q} \leq b$.*

Let $Z = \sup_{f \in \mathscr{F}} \mathbb{G}_n f$. Then for every $\gamma \in (0,1)$, there exists, on a possibly extended probability space, a tight Gaussian process $(\mathbb{G}_P f)_{f \in \mathscr{F}}$ with mean zero, covariance function $\mathbb{E}[\mathbb{G}_P(f)\mathbb{G}_P(g)] = P(fg)$, and almost surely uniformly e_P-continuous trajectories, such that for $\tilde{Z} = \sup_{f \in \mathscr{F}} \mathbb{G}_P f$:

$$P\left\{|Z - \tilde{Z}| > \frac{bK_n}{\gamma^{1/q} n^{1/2-1/q}} + \frac{(b\sigma)^{1/2} K_n^{3/4}}{\gamma^{1/2} n^{1/4}} + \frac{(b\sigma^2 K_n^2)^{1/3}}{\gamma^{1/3} n^{1/6}}\right\} \leq C\left(\gamma + \frac{\log n}{n}\right)$$

where $K_n = \upsilon(\log n \vee \log(Ab/\sigma))$, and $C > 0$ is a constant that depends only on q.

Consider $\mathbf{z}_i = (\mathbf{X}_i, \varepsilon_i)$, $1 \leq i \leq n$, where $\varepsilon_i = Y_i - \mathbb{E}[Y_i|\mathbf{X}_i]$. For each $\mathbf{x} \in \mathscr{B}$, let $g_{\mathbf{x}}$ be the boundary-indexed local-polynomial weight function defined before Theorem SA-5, and consider the class

$$\mathscr{F} = \{(\mathbf{u}, u) \in \mathbb{R}^d \times \mathbb{R} \mapsto g_{\mathbf{x}}(\mathbf{u})u : \mathbf{x} \in \mathscr{B}\}, \qquad \mathscr{F}_\pm = \mathscr{F} \cup (-\mathscr{F}).$$

Then

$$\sup_{\mathbf{x} \in \mathscr{B}} |\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})| = \sup_{f \in \mathscr{F}_\pm} \frac{1}{\sqrt{n}} \sum_{i=1}^n \{f(\mathbf{z}_i) - \mathbb{E}[f(\mathbf{z}_i)]\}.$$

We apply Lemma SA-17 to this empirical process. First, the fixed indicators $\mathbb{1}(\mathbf{u} \in \mathscr{A}_t)$ are absorbed into $g_{\mathbf{x}}$ and do not increase $L_2(Q)$ covering numbers beyond a constant factor, because multiplication by a fixed indicator is a contraction in $L_2(Q)$. Therefore, using the same local-polynomial entropy calculation as in the proofs of Lemmas SA-2 and SA-3, but without the total-variation step, $\{g_{\mathbf{x}} : \mathbf{x} \in \mathscr{B}\}$ is pointwise measurable and has an envelope $\mathtt{M}_{\mathscr{G}} \lesssim |\mathbf{H}|^{-1/2}$ such that, for constants $\mathbf{c}, \mathbf{c}' > 0$ depending only on the compact set $\mathscr{X}$,

$$\sup_Q N(\mathscr{G}, \|\cdot\|_{Q,2}, \varepsilon(2\mathbf{c}+1)^{d+1}\mathtt{M}_{\mathscr{G}}) \leq 2\mathbf{c}'\varepsilon^{-4d-4} + 2, \qquad 0 < \varepsilon < 1,$$

where the supremum is taken over all finite discrete measures. Since the residual class is the singleton $\{\mathrm{Id}\}$, the class $\mathscr{F}_\pm$ satisfies the same VC-type entropy bound up to a universal constant; centering the functions changes only the envelope by a constant factor.

For the moment condition, take $q = 2 + v \geq 4$ in Lemma SA-17. Then, Assumption SA-1 implies $\sup_{\mathbf{x} \in \mathscr{X}} \mathbb{E}[|\varepsilon_i|^{2+v}|\mathbf{X}_i = \mathbf{x}] < \infty$. The envelope $F(\mathbf{u}, u) = \mathtt{M}_{\mathscr{G}}|u|$ satisfies $\|F\|_{\mathbb{P},2+v} \lesssim |\mathbf{H}|^{-1/2}$. The same local-polynomial moment calculation used in the proof of Lemma SA-3 gives, for $\sigma = 1$ and $b = |\mathbf{H}|^{-1/2}$, that $\sup_{f\in\mathscr{F}_\pm}\mathbb{P}|f|^k \le \sigma^2 b^{k-2}$ for $k=2,3,4$ and $\|F\|_{\mathbb{P},2+v}\le b$. Taking $\gamma=(\log n)^{-1}$ in Lemma SA-17 yields

$$\mathbb{P}\left\{\left|\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})| - \sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|\right| > \frac{\log n}{(n|\mathbf{H}|)^{1/6}} + \frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}}|\mathbf{H}|)^{1/2}}\right\} \le C\left(\frac{1}{\log n}+\frac{\log n}{n}\right),$$

for a constant $C>0$. This proves the theorem. □

SA-8.14 Proof of Theorem SA-8

Define

$$\mathfrak{R}_{\mathtt{sup}} = \frac{\log n}{(n|\mathbf{H}|)^{1/6}} + \frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}}|\mathbf{H}|)^{1/2}}.$$

In this proof, let $\mathfrak{R}_{\mathtt{lin}}$ denote the stochastic-linearization remainder defined at the beginning of the proof of Theorem SA-6. The entropy calculation in the proof of Theorem SA-7 gives

$$\mathbb{E}\left[\sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|\right] \lesssim \sqrt{\log n}.$$

The side condition $(\log n)^9/(n^{v/(2+v)}|\mathbf{H}|)=o(1)$ implies $\mathfrak{R}_{\mathtt{sup}}\sqrt{\log n}=o(1)$. Let

$$M_{\overline{T}} = \sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|, \qquad M_Z = \sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|.$$

Applying Lemma SA-13 with $\mathbb{T}_1 = M_{\overline{T}}$ and $\mathbb{T}_2 = M_Z$, viewed as singleton-indexed processes, and $\eta=\mathfrak{R}_{\mathtt{sup}}$, gives

$$\sup_{u\in\mathbb{R}}|\mathbb{P}(M_{\overline{T}}\le u)-\mathbb{P}(M_Z\le u)| \le \sup_{u\in\mathbb{R}}\mathbb{P}(|M_Z-u|\le\mathfrak{R}_{\mathtt{sup}}) + \mathbb{P}(|M_{\overline{T}}-M_Z|>\mathfrak{R}_{\mathtt{sup}}).$$

The first term is $O(\mathfrak{R}_{\mathtt{sup}}\sqrt{\log n})=o(1)$ by Gaussian anti-concentration, and the second term is $o(1)$ by Theorem SA-7. Therefore,

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|\le u\right)-\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|\le u\right)\right| = o(1). \tag{SA-11}$$

Next, by Theorem SA-4, the stochastic-linearization error is $O_{\mathbb{P}}(\mathfrak{R}_{\mathtt{lin}})$. The polynomial lower bound on $\bar{h}$, together with $(\log n)^9/(n^{v/(2+v)}|\mathbf{H}|)=o(1)$ and $\bar{h}^{p+1}\sqrt{n|\mathbf{H}|\log n}=o(1)$, implies $\mathfrak{R}_{\mathtt{lin}}\sqrt{\log n}=o(1)$. Choose a deterministic sequence a_n such that $\mathfrak{R}_{\mathtt{lin}}=o(a_n)$ and $a_n\sqrt{\log n}=o(1)$. Applying Lemma SA-13 with $\mathbb{T}_1=\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}$, $\mathbb{T}_2=\overline{\mathrm{T}}^{(\boldsymbol{\nu})}$, and $\eta=a_n$, and then bounding the anti-concentration of $M_{\overline{T}}$ through Equation (SA-11), gives

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|\le u\right)-\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|\le u\right)\right| = o(1). \tag{SA-12}$$

Finally, set

$$\mathfrak{R}_{\mathtt{cc}} = \sqrt{\frac{\log n}{n|\mathbf{H}|}} + \frac{\log n}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}.$$

This is the same covariance-comparison rate used in Lemma SA-16. The covariance-comparison and net-modulus arguments in that lemma do not use the De Giorgi perimeter condition. Under the present side conditions, they give

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|Z^{(\boldsymbol{\nu})}(\mathbf{x})|\leq u\right)-\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})|\leq u\,\middle|\,\mathbf{V}\right)\right|\lesssim_{\mathbb{P}}\log(n)\mathfrak{R}_{\mathtt{cc}}^{1/2}=o_{\mathbb{P}}(1). \tag{SA-13}$$

The first conclusion follows from the triangle inequality applied to Equations (SA-11), (SA-12), and (SA-13).

For the coverage statement, by definition of $\widehat{\mathrm{I}}_{\alpha}^{(\boldsymbol{\nu})}(\mathbf{x})$ and the first conclusion,

$$\begin{aligned}\mathbb{P}\Big[\tau^{(\boldsymbol{\nu})}(\mathbf{x})\in\widehat{\mathrm{I}}_{\alpha}^{(\boldsymbol{\nu})}(\mathbf{x}),\ \text{for all }\mathbf{x}\in\mathscr{B}\Big]&=\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}(\mathbf{x})|\leq\mathscr{q}_{\alpha}\right)\\&=\mathbb{E}\left[\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{Z}^{(\boldsymbol{\nu})}(\mathbf{x})|\leq\mathscr{q}_{\alpha}\,\middle|\,\mathbf{V}\right)\right]+o(1)\\&=1-\alpha+o(1),\end{aligned}$$

which completes the proof. □

SA-8.15 Proof of Lemma SA-7

By Lemma SA-5,

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|\mathbb{E}[\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x})|\mathbf{X}]-\tau^{(\boldsymbol{\nu})}(\mathbf{x})-B_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{h})\right|=o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).$$

Integrating this uniform expansion against $w(\mathbf{x})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})$ gives the displayed WBATE bias expansion because $\int_{\mathscr{B}}|w(\mathbf{x})|\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})<\infty$. When $\mathbf{h}=\mathbf{c}h$, the pointwise leading bias is $h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}_{\mathbf{x}}^{(\boldsymbol{\nu})}(\mathbf{c})$; integrating this leading term over $\mathscr{B}$ gives $\widetilde{B}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}(\mathbf{c})$, and the integrated remainder remains $o(h^{p+1-|\boldsymbol{\nu}|})$ by the same integrability argument. □

SA-8.16 Proof of Lemma SA-8

Since $\mathbb{V}[\widehat{\tau}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}|\mathbf{X}]=\mathbb{V}[\int_{\mathscr{B}}\widehat{\mu}_0^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})|\mathbf{X}]+\mathbb{V}[\int_{\mathscr{B}}\widehat{\mu}_1^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})|\mathbf{X}]$, it is enough to consider only one treatment assignment group $t\in\{0,1\}$. The conditional cross-covariance between the two side-specific integrals is zero because the side indicators are disjoint observation by observation, and observations are conditionally independent given $\mathbf{X}$. In addition,

$$\mathbb{V}\Big[\int_{\mathscr{B}}\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\Big|\mathbf{X}\Big]=\int_{\mathscr{B}}\int_{\mathscr{B}}\mathbb{C}\mathrm{ov}\big[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_1),\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_2)\big|\mathbf{X}\big]w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2)$$

and

$$\Omega_{t,\mathtt{WBATE}}^{(\boldsymbol{\nu})}=\int_{\mathscr{B}}\int_{\mathscr{B}}\Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2).$$

Proceeding as in the proof of Lemma SA-4, we have

$$\sup_{\mathbf{b}_1,\mathbf{b}_2\in\mathscr{B}}\left|\mathbb{C}\mathrm{ov}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_1),\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_2)|\mathbf{X}]-\Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}\right|\lesssim_{\mathbb{P}}\frac{1}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}}.$$

Since K is supported on a compact set, let $R\in(0,\infty)$ denote the diameter of the support, and define the "effective domain" $\mathscr{E}(\mathbf{H})=\{(\mathbf{x},\mathbf{y})\in\mathscr{B}\times\mathscr{B}:\|\mathbf{x}-\mathbf{y}\|\leq\bar{h}R\}$. Let $\nu_{\mathscr{B},2}=\mathfrak{H}^{d-1}\times\mathfrak{H}^{d-1}$ on $\mathscr{B}\times\mathscr{B}$. By

(SA-3), $\nu_{\mathscr{B},2}(\mathscr{E}(\mathbf{H})) \lesssim \bar{h}^{d-1}$. Therefore,

$$\begin{aligned}
&\Big|\mathbb{V}\Big[\int_{\mathscr{B}} \widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\Big|\mathbf{X}\Big] - \Omega_{t,\mathtt{WBATE}}^{(\boldsymbol{\nu})}\Big| \\
&= \Big|\int_{\mathscr{B}}\int_{\mathscr{B}} \big(\mathbb{C}\mathrm{ov}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_1), \widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_2)|\mathbf{X}] - \Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}\big)w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2)\Big| \\
&\lesssim \sup_{\mathbf{b}_1,\mathbf{b}_2\in\mathscr{B}}\big|\mathbb{C}\mathrm{ov}[\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_1), \widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}_2)|\mathbf{X}] - \Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}\big|\int_{\mathscr{B}}\int_{\mathscr{B}} \mathbb{1}((\mathbf{b}_1,\mathbf{b}_2)\in\mathscr{E}(\mathbf{H}))w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2) \\
&= o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big),
\end{aligned}$$

because $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$ and $\bar{h} = o(1)$. This proves the first claim. Next, for each $t \in \{0,1\}$,

$$\begin{aligned}
\Omega_{t,\mathtt{WBATE}}^{(\boldsymbol{\nu})} &= \int_{\mathscr{B}}\int_{\mathscr{B}} \Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2) \\
&\leq \sup_{\mathbf{b}_1,\mathbf{b}_2\in\mathscr{B}}\big|\Omega_{t,\mathbf{b}_1,\mathbf{b}_2}^{(\boldsymbol{\nu})}\big|\int_{\mathscr{B}}\int_{\mathscr{B}} \mathbb{1}((\mathbf{b}_1,\mathbf{b}_2)\in\mathscr{E}(\mathbf{H}))w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2) \\
&\lesssim (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\nu_{\mathscr{B},2}(\mathscr{E}(\mathbf{H})) \lesssim \frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}},
\end{aligned}$$

which verifies the upper bound after summing over t. The lower bound follows from the aggregate variance nondegeneracy condition (SA-2). This condition is needed because the covariance kernel can contain negative regions after local-polynomial weighting, so pointwise variance lower bounds alone do not rule out cancellation in the weighted boundary integral.

The third and final claim of the lemma follows from Lemma SA-4 and the same analysis as above. □

SA-8.17 Proof of Theorem SA-9

Lemma SA-7 gives

$$\mathbb{E}[\widehat{\tau}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}|\mathbf{X}] - \tau_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = B_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}(\mathbf{h}) + o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}),$$

so the squared conditional bias equals $(B_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}(\mathbf{h}))^2 + o_{\mathbb{P}}(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}})$. Lemma SA-8 gives the conditional variance expansion with remainder $o_{\mathbb{P}}(\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}))$. Adding the squared-bias and variance expansions gives the stated MSE expansion. □

SA-8.18 Proof of Theorem SA-10

Since $\widehat{\tau}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} - \tau_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = (\widehat{\mu}_{1,\mathtt{WBATE}}^{(\boldsymbol{\nu})} - \mu_{1,\mathtt{WBATE}}^{(\boldsymbol{\nu})}) - (\widehat{\mu}_{0,\mathtt{WBATE}}^{(\boldsymbol{\nu})} - \mu_{0,\mathtt{WBATE}}^{(\boldsymbol{\nu})})$, it is enough to start with only one treatment assignment group $t \in \{0,1\}$. The same argument applies to both side-specific components, and the two components combine by summing their variances because the treatment- and control-side kernel-weighted summands have disjoint side indicators observation by observation. Furthermore,

$$\begin{aligned}
\widehat{\mu}_{t,\mathtt{WBATE}}^{(\boldsymbol{\nu})} - \mu_{t,\mathtt{WBATE}}^{(\boldsymbol{\nu})} &= \int_{\mathscr{B}} (\widehat{\mu}_t^{(\boldsymbol{\nu})}(\mathbf{b}) - \mu_t^{(\boldsymbol{\nu})}(\mathbf{b}))w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) \\
&= \int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{Q}_{t,\mathbf{b}}w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})
\end{aligned}$$

$$+\int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}(\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1})\mathbf{Q}_{t,\mathbf{b}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) + O_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}})$$

using Lemma SA-5 to bound the approximation error. The side condition $n|\mathbf{H}|\bar{h}^{2p+2}/\bar{h}^{d-1} = o(1)$ implies that the bias term is $o_{\mathbb{P}}((\Omega_{\mathtt{WBATE}}^{(\boldsymbol{\nu})})^{1/2})$.

For the second integral, let

$$\overline{\boldsymbol{\Sigma}}_{t,\mathbf{x}_1,\mathbf{x}_2} = \mathbb{E}_n\Big[\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_1))\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}_2))^\top |\mathbf{H}| K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_1) K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}_2)\sigma_t^2(\mathbf{X}_i)\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\Big],$$

and since $\overline{\boldsymbol{\Sigma}}_{t,\mathbf{b}_1,\mathbf{b}_2} = 0$ if $\|\mathbf{b}_1 - \mathbf{b}_2\|$ is larger than a constant multiple of $\bar{h}$, let

$$\mathscr{E}(\mathbf{H}) = \{(\mathbf{b}_1, \mathbf{b}_2) \in \mathscr{B} \times \mathscr{B} : \|\mathbf{b}_1 - \mathbf{b}_2\| \lesssim \bar{h}\}.$$

By (SA-3), $\nu_{\mathscr{B},2}(\mathscr{E}(\mathbf{H})) \lesssim \bar{h}^{d-1}$. Then

$$\begin{aligned}
&\mathbb{E}\Big[\Big(\int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}(\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1})\mathbf{Q}_{t,\mathbf{b}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\Big)^2 \Big| \mathbf{X}\Big] \\
&= \int_{\mathscr{B}}\int_{\mathscr{B}} (\boldsymbol{\nu}!)^2 \mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}(\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}_1}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}_1}^{-1})(n|\mathbf{H}|)^{-1}\overline{\boldsymbol{\Sigma}}_{t,\mathbf{b}_1,\mathbf{b}_2}(\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}_2}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}_2}^{-1})\mathbf{S}^{-1}\mathbf{e}_{\boldsymbol{\nu}} w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2), \\
&\leq \mathbf{h}^{-2\boldsymbol{\nu}} \sup_{\mathbf{b}\in\mathscr{B}} \|\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\|^2 \sup_{\mathbf{b}_1,\mathbf{b}_2\in\mathscr{B}} \|\overline{\boldsymbol{\Sigma}}_{t,\mathbf{b}_1,\mathbf{b}_2}\| \sup_{\mathbf{b}\in\mathscr{B}} |w(\mathbf{b})|(n|\mathbf{H}|)^{-1}\nu_{\mathscr{B},2}(\mathscr{E}(\mathbf{H})) \\
&= o_{\mathbb{P}}\Big(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}}\Big),
\end{aligned}$$

and hence $\int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}(\widehat{\boldsymbol{\Gamma}}_{t,\mathbf{b}}^{-1} - \boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1})\mathbf{Q}_{t,\mathbf{b}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) = o_{\mathbb{P}}\big((\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}))^{1/2}\big)$.

Next, combine the two side-specific expansions with signs $s_1 = 1$ and $s_0 = -1$. Define the signed linearized statistic

$$\overline{\mathrm{T}}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = (\Omega_{\mathtt{WBATE}}^{(\boldsymbol{\nu})})^{-1/2} \sum_{t\in\{0,1\}} s_t \int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{Q}_{t,\mathbf{b}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}).$$

Using Lemma SA-8 and the previous side-specific bounds,

$$\widehat{\mathrm{T}}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} - \overline{\mathrm{T}}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = \big((\widehat{\Omega}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})})^{-1/2} - (\Omega_{\mathtt{WBATE}}^{(\boldsymbol{\nu})})^{-1/2}\big) \sum_{t\in\{0,1\}} s_t \int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{Q}_{t,\mathbf{b}} w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}) + o_{\mathbb{P}}(1) = o_{\mathbb{P}}(1),$$

Finally, apply the Berry–Esseen lemma to this signed treatment-control statistic. Write $\overline{\mathrm{T}}_{\mathtt{WBATE}}^{(\boldsymbol{\nu})} = \sum_{i=1}^n \chi_i$, where

$$\begin{aligned}
\chi_i = n^{-1}(\Omega_{\mathtt{WBATE}}^{(\boldsymbol{\nu})})^{-1/2} \sum_{t\in\{0,1\}} s_t \int_{\mathscr{B}} &\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{b}))K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{b}) \\
&\cdot \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)\varepsilon_i w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}).
\end{aligned}$$

This satisfies $\mathbb{E}[\chi_i] = 0$. The definition of $\Omega_{\mathtt{WBATE}}^{(\boldsymbol{\nu})}$ implies that $\sum_{i=1}^n \mathbb{V}[\chi_i] = 1$. Since the indicators $\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_t)$ are disjoint observation by observation, the Lyapunov bound is the same as the side-specific bound up to a fixed constant. Hence, it remains to bound

$$\sum_{i=1}^{n} \mathbb{E}[|\chi_i|^3]$$
$$\lesssim n^{-3}(\Omega^{(\boldsymbol{\nu})}_{\texttt{WBATE}})^{-3/2} \sum_{i=1}^{n} \sum_{t\in\{0,1\}} \mathbb{E}\left[\left|\int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{b}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{b})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\varepsilon_i w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\right|^3\right].$$

Let R denote the diameter of the (compact) support of K, and define $\mathscr{E}_3(\mathbf{H}) = \{(\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3) \in \mathscr{B}^3 : \|\mathbf{b}_i - \mathbf{b}_j\| \leq R\bar{h}, i,j=1,2,3\}$. Let $\nu_{\mathscr{B},3}$ denote the product measure $\mathfrak{H}^{d-1}\times\mathfrak{H}^{d-1}\times\mathfrak{H}^{d-1}$ on $\mathscr{B}^3$. Iterating (SA-3) gives $\nu_{\mathscr{B},3}(\mathscr{E}_3(\mathbf{H})) \lesssim \bar{h}^{2d-2}$. Then,

$$\begin{aligned}
&\mathbb{E}\left[\left|\int_{\mathscr{B}} \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{b}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{b})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\varepsilon_i w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\right|^3\right]\\
&\leq \mathbb{E}\left[\int_{\mathbf{b}_1\in\mathscr{B}}\int_{\mathbf{b}_2\in\mathscr{B}}\int_{\mathbf{b}_3\in\mathscr{B}} |G(\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3)|w(\mathbf{b}_1)w(\mathbf{b}_2)w(\mathbf{b}_3)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_3)\right]\\
&\lesssim \nu_{\mathscr{B},3}(\mathscr{E}_3(\mathbf{H})) \sup_{\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3\in\mathscr{B}} \mathbb{E}[|G(\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3)|],
\end{aligned}$$

where $G(\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3) = g(\mathbf{X}_i,\varepsilon_i,\mathbf{b}_1)g(\mathbf{X}_i,\varepsilon_i,\mathbf{b}_2)g(\mathbf{X}_i,\varepsilon_i,\mathbf{b}_3)$ with

$$g(\mathbf{X}_i,\varepsilon_i,\mathbf{b}) = \boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\boldsymbol{\Gamma}_{t,\mathbf{b}}^{-1}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i-\mathbf{b}))K_{\mathbf{H}}(\mathbf{X}_i-\mathbf{b})\mathbb{1}(\mathbf{X}_i\in\mathscr{A}_t)\varepsilon_i.$$

Proceeding as in the proof of Lemma SA-2 and Lemma SA-8, it can be shown that

$$\sup_{\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3\in\mathscr{B}} \mathbb{E}[|G(\mathbf{b}_1,\mathbf{b}_2,\mathbf{b}_3)|] \lesssim \mathbf{h}^{-3\boldsymbol{\nu}}|\mathbf{H}|^{-2}$$

provided that $\frac{\log(1/\bar{h})}{n|\mathbf{H}|} = o(1)$. Therefore, together with the rate of $\Omega^{(\boldsymbol{\nu})}_{\texttt{WBATE}}$ from Lemma SA-8, we have $\sum_{i=1}^n \mathbb{E}[|\chi_i|^3] \lesssim (\bar{h}^{d-1}/(n|\mathbf{H}|))^{1/2} \asymp (n\bar{h})^{-1/2}$, and the result follows. □

SA-8.19 Proof of Theorem SA-11

Follows by Theorem SA-1 after noting that

$$|\widehat{\tau}^{(\boldsymbol{\nu})}_{\texttt{LBATE}} - \tau^{(\boldsymbol{\nu})}_{\texttt{LBATE}}| \leq \sup_{\mathbf{x}\in\mathscr{B}} |\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}(\mathbf{x})|.$$

□

SA-8.20 Proof of Theorem SA-12

Consider the event $E = \left\{\sup_{\mathbf{b}\in\mathscr{B}} \frac{|\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b})-\tau^{(\boldsymbol{\nu})}(\mathbf{b})|}{(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}})^{1/2}} \leq \mathscr{q}_\alpha\right\}$. Theorem SA-8 implies that $\mathbb{P}(E) = 1-\alpha+o(1)$. On the event E, we also have

$$\widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}})^{1/2} \leq \tau^{(\boldsymbol{\nu})}(\mathbf{b}) \leq \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}})^{1/2}, \qquad \forall\, \mathbf{b}\in\mathscr{B},$$

which implies

$$\sup_{\mathbf{b}\in\mathscr{B}} \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}})^{1/2} \leq \sup_{\mathbf{b}\in\mathscr{B}} \tau^{(\boldsymbol{\nu})}(\mathbf{b}) \leq \sup_{\mathbf{b}\in\mathscr{B}} \widehat{\tau}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha(\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b},\mathbf{b}})^{1/2}.$$

The stated result then follows. □

SA-8.21 Proof of Theorem SA-13

Theorem SA-1 applies directly to the reduced-form estimator $\widehat{\tau}_Y^{(\boldsymbol{\nu})}$. For the treatment-status estimator $\widehat{\tau}_W^{(\boldsymbol{\nu})}$, the same convergence-rate proof applies with W in place of Y: Assumption SA-4(ii) gives the required smoothness of $\mu_{W,t}$, and $W_i(t) \in \{0,1\}$ gives the required moment bounds. The variance lower bound in Assumption SA-1(iv) is not used for these consistency rates. The strengthened bandwidth condition makes the uniform reduced-form convergence rate vanish, so $\inf_{\mathbf{x}\in\mathscr{B}} |\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})|$ is bounded away from zero with probability approaching one. Since $\inf_{\mathbf{x}\in\mathscr{B}} |\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, the displayed rates follow from the exact expansion

$$\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) - \zeta^{(\boldsymbol{\nu})}(\mathbf{x}) = \frac{\widehat{\tau}_Y^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})} - \frac{\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})}{\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x})}\big(\widehat{\tau}_W^{(\boldsymbol{\nu})}(\mathbf{x}) - \tau_W^{(\boldsymbol{\nu})}(\mathbf{x})\big),$$

and the same argument uniformly over $\mathbf{x} \in \mathscr{B}$. □

SA-8.22 Proof of Theorem SA-14

Lemma SA-18 (Covariance for Fuzzy Designs)**.** *Suppose Assumptions SA-1, SA-2 and SA-4 hold. If* $\frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} = o(1)$ *and* $\bar{h} = o(1)$*, then for* $t = 0, 1$*, and* $A_1, A_2 \in \{Y, W\}$*, we have*

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \big\|\widehat{\mathbf{\Sigma}}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2} - \mathbf{\Sigma}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2}\big\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1},$$

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \big|\widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}_1,\mathbf{x}_2} - \Omega^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}_1,\mathbf{x}_2}\big| \lesssim_{\mathbb{P}} (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}\Bigg(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\Bigg),$$

Proof. The argument parallels the proof of Lemma SA-4. The only additional moment input is for the treatment-status residuals and cross-products: W_i is binary, hence $\varepsilon_{W,i}$ is bounded, while Assumption SA-1(v) gives the required conditional fourth-moment control for $\varepsilon_{Y,i}$; mixed terms with $A_1, A_2 \in \{Y, W\}$ then follow by Cauchy–Schwarz. We begin with the decomposition

$$\widehat{\varepsilon}_{A_1,i}\widehat{\varepsilon}_{A_2,i} = (\varepsilon_{A_1,i} + \widehat{\varepsilon}_{A_1,i} - \varepsilon_{A_1,i})(\varepsilon_{A_2,i} + \widehat{\varepsilon}_{A_2,i} - \varepsilon_{A_2,i}),$$

which allows us to control the difference between estimated and population covariances through uniform bounds on $\widehat{\varepsilon}_{A,i} - \varepsilon_{A,i}$ for $A \in \{Y, W\}$. Fix $A \in \{Y, W\}$. For each i,

$$\begin{aligned}
&|\widehat{\varepsilon}_{A,i} - \varepsilon_{A,i}| \\
&\leq \sup_{\mathbf{x}\in\mathscr{B}} \Big| \sum_{s\in\{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_s)\mu_{A,s}(\mathbf{X}_i)
\end{aligned}$$

$$
- \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top \mathbf{S}\big[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\widehat{\boldsymbol{\beta}}_{A,0}(\mathbf{x}) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\widehat{\boldsymbol{\beta}}_{A,1}(\mathbf{x})\big]\Big|\mathbb{1}(K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \neq 0).
$$

Taking the maximum over i and decomposing the difference into approximation and estimation components yields

$$
\begin{aligned}
&\max_{1\le i\le n} |\widehat{\varepsilon}_{A,i} - \varepsilon_{A,i}| \\
&\le \max_{1\le i\le n}\sup_{\mathbf{x}\in\mathscr{B}}\Big|\sum_{s\in\{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_s)\mu_{A,s}(\mathbf{X}_i) \\
&\qquad\qquad - \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x}))^\top \mathbf{S}\big[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\widehat{\boldsymbol{\beta}}_{A,0}(\mathbf{x}) + \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\widehat{\boldsymbol{\beta}}_{A,1}(\mathbf{x})\big]\Big|\mathbb{1}(K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x}) \neq 0) \\
&\le \sup_{\mathbf{x}\in\mathscr{B}}\sup_{\mathbf{y}\in\mathscr{X}}\Big|\sum_{s\in\{0,1\}} \mathbb{1}(\mathbf{y} \in \mathscr{A}_s)\mu_{A,s}(\mathbf{y}) \\
&\qquad\qquad - \mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{y} - \mathbf{x}))^\top \mathbf{S}\big[\mathbb{1}(\mathbf{y} \in \mathscr{A}_0)\boldsymbol{\beta}_{A,0}(\mathbf{x}) + \mathbb{1}(\mathbf{y} \in \mathscr{A}_1)\boldsymbol{\beta}_{A,1}(\mathbf{x})\big]\Big|\mathbb{1}(K_{\mathbf{H}}(\mathbf{y} - \mathbf{x}) \neq 0) \\
&\quad + \max_{t=0,1}\sup_{\mathbf{x}\in\mathscr{B}}\sup_{\mathbf{y}\in\mathscr{X}}\Big|\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{y} - \mathbf{x}))^\top \mathbf{S}\big(\widehat{\boldsymbol{\beta}}_{A,t}(\mathbf{x}) - \boldsymbol{\beta}_{A,t}(\mathbf{x})\big)\Big|\mathbb{1}(K_{\mathbf{H}}(\mathbf{y} - \mathbf{x}) \neq 0).
\end{aligned}
$$

The first term is the usual local polynomial approximation error and is of order $\bar{h}^{p+1}$ uniformly over $\mathbf{x}$. The second term is controlled by the uniform convergence rate of $\widehat{\boldsymbol{\beta}}_{A,t}(\mathbf{x})$, giving

$$
\max_{1\le i\le n} |\widehat{\varepsilon}_{A,i} - \varepsilon_{A,i}| \lesssim_{\mathbb{P}} \bar{h}^{p+1} + \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{1+v}{2+v}}|\mathbf{H}|}.
$$

Substituting this bound into the covariance decomposition and proceeding exactly as in Lemma SA-4 yields the stated uniform bounds for $\widehat{\mathbf{\Sigma}}_{A_1,A_2,t,\mathbf{x}_1,\mathbf{x}_2}$, $\widehat{\Omega}^{(\boldsymbol{\nu})}_{A_1,A_2,\mathbf{x}_1,\mathbf{x}_2}$. □

Completion of the AMSE proof. Applying Lemma SA-5 with $A \in \{Y, W\}$ yields

$$
\sup_{\mathbf{x}\in\mathscr{B}}\Big|\mathbb{E}[\widehat{\tau}^{(\boldsymbol{\nu})}_A(\mathbf{x}) \mid \mathbf{X}] - \tau^{(\boldsymbol{\nu})}_A(\mathbf{x}) - B^{(\boldsymbol{\nu})}_{A,\mathbf{x}}(\mathbf{h})\Big| = o_{\mathbb{P}}\big(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}\big).
$$

Premultiplying the reduced-form bias vector by $\mathfrak{w}(\mathbf{x})$ gives $B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h})$ in (SA-5). Combining this bias expansion with Lemma SA-18 gives the AMSE expansion for the leading linear component, and integrating the same pointwise expansion over $\mathscr{B}$ establishes the AIMSE result. □

SA-8.23 Proof of Lemma SA-9

Let

$$
a_n = \sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}.
$$

Lemma SA-18 gives

$$
\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}}\big\|\widehat{\mathbf{\Theta}}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} - \mathbf{\Theta}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\big\| \lesssim_{\mathbb{P}} (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}a_n.
$$

The reduced-form convergence rates, together with $\inf_{\mathbf{x}\in\mathscr{B}} |\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, imply

$$\sup_{\mathbf{x}\in\mathscr{B}} \|\widehat{\mathfrak{w}}(\mathbf{x}) - \mathfrak{w}(\mathbf{x})\| \lesssim_{\mathbb{P}} \mathbf{h}^{-\boldsymbol{\nu}} a_n.$$

The displayed rate condition in Lemma SA-9 makes the right-hand side $o_{\mathbb{P}}(1)$, so the feasible weights are uniformly bounded with probability approaching one. Because $\mathfrak{w}(\cdot)$ is bounded and $\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \|\boldsymbol{\Theta}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\| \lesssim (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$, the feasible plug-in covariance satisfies

$$\sup_{\mathbf{x}_1,\mathbf{x}_2\in\mathscr{B}} \left|\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2} - \Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}_1,\mathbf{x}_2}\right| \lesssim_{\mathbb{P}} \frac{\mathbf{h}^{-\boldsymbol{\nu}}}{n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}} a_n.$$

Finally, $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}} = V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})$, so $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$ gives the required lower bound on the population denominator. The inverse-square-root bound follows from the displayed covariance bound and a mean-value expansion of $x \mapsto x^{-1/2}$ on this bounded-away-from-zero neighborhood. $\square$

SA-8.24 Proof of Theorem SA-15

For the fixed boundary point $\mathbf{x}$ in the theorem, a similar argument as the proof for Theorem SA-3 gives the following stochastic linearization,

$$\left|\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) - \overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})\right| \lesssim_{\mathbb{P}} \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \frac{1}{\sqrt{n|\mathbf{H}|}},$$

provided that $\bar{h} \to 0$ and $n|\mathbf{H}| \to \infty$. To see this, the pointwise reduced-form rates give

$$|\widehat{\tau}^{(\boldsymbol{\nu})}_A(\mathbf{x}) - \tau^{(\boldsymbol{\nu})}_A(\mathbf{x})| = O_{\mathbb{P}}\big(\mathbf{h}^{-\boldsymbol{\nu}}((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1})\big), \qquad A \in \{Y, W\}.$$

Since $|\tau^{(\boldsymbol{\nu})}_W(\mathbf{x})| > 0$, this implies $|\widehat{\tau}^{(\boldsymbol{\nu})}_W(\mathbf{x})|^{-1} = O_{\mathbb{P}}(1)$ and

$$\|\widehat{\mathfrak{w}}(\mathbf{x}) - \mathfrak{w}(\mathbf{x})\| = O_{\mathbb{P}}\big(\mathbf{h}^{-\boldsymbol{\nu}}((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1})\big).$$

The exact second-order expansion in Section SA-6.1 gives

$$|\mathfrak{R}_{\mathtt{F}}(\mathbf{x})| = O_{\mathbb{P}}\big(\mathbf{h}^{-2\boldsymbol{\nu}}((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1})^2\big),$$

and hence

$$\frac{|\mathfrak{R}_{\mathtt{F}}(\mathbf{x})|}{\sqrt{\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}}} = O_{\mathbb{P}}\Big(\mathbf{h}^{-\boldsymbol{\nu}}\sqrt{n|\mathbf{H}|}((n|\mathbf{H}|)^{-1/2} + \bar{h}^{p+1})^2\Big) = o_{\mathbb{P}}(1)$$

under the theorem's bandwidth conditions. Combining these bounds with the pointwise Gram and covariance bounds displayed in the proof of Theorem SA-3 yields the displayed linearization. Here

$$\overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x}) = \sum_{i=1}^{n} \chi_i,$$

with $\mathfrak{w}(\mathbf{x}) = (\mathfrak{w}_1(\mathbf{x}), \mathfrak{w}_2(\mathbf{x}))$ and

$$\chi_i = n^{-1}(\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top \mathbf{S}^{-1}\Big[\mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)\boldsymbol{\Gamma}^{-1}_{1,\mathbf{x}} - \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_0)\boldsymbol{\Gamma}^{-1}_{0,\mathbf{x}}\Big]\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{X}_i - \mathbf{x})) \\ \cdot K_{\mathbf{H}}(\mathbf{X}_i - \mathbf{x})\big(\mathfrak{w}_1(\mathbf{x})\varepsilon_{Y,i} + \mathfrak{w}_2(\mathbf{x})\varepsilon_{W,i}\big).$$

In particular, we have $\mathbb{E}[\chi_i] = 0$ and $\mathbb{V}\big[\sum_{i=1}^n \chi_i\big] = 1$. Since $\mathfrak{w}(\mathbf{x})$ is fixed at the point $\mathbf{x}$, $\varepsilon_{W,i}$ is bounded, and Assumption SA-1(v) gives a uniformly bounded conditional third moment for $\varepsilon_{Y,i}$,

$$\sum_{i=1}^n \mathbb{E}[|\chi_i|^3] \lesssim (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1/2} = o(1).$$

Berry–Esseen gives the normal approximation for $\overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})$, and the stochastic linearization transfers it to $\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})$. □

SA-8.25 Proof of Theorem SA-16

Use the exact second-order expansion in Section SA-6.1. The leading linear component is a fixed linear combination, with weights $\mathfrak{w}(\mathbf{x})$, of the reduced-form stochastic expansions for Y and W. Applying Theorem SA-4 to these two reduced-form estimators and then using Lemma SA-9 to replace $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}$ by $\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}$ gives

$$\sup_{\mathbf{x}\in\mathscr{B}}\left|\frac{\mathfrak{w}(\mathbf{x})^\top\mathfrak{T}(\mathbf{x})}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}}}} - \overline{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})\right| \\ \lesssim_{\mathbb{P}} \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \mathbf{h}^{-\boldsymbol{\nu}}\sqrt{\log(1/\bar{h})}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\right).$$

Equation (SA-4) and the lower bound $\inf_{\mathbf{x}\in\mathscr{B}}\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}} \gtrsim (n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$ from $\inf_{\mathbf{x}\in\mathscr{B}} V^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}} > 0$ imply that the normalized second-order remainder is bounded by the same rate. Combining the two displays proves the theorem. □

SA-8.26 Proof of Theorem SA-17

Consider $\mathbf{z}_i = (\mathbf{x}_i, \varepsilon_{Y,i}, \varepsilon_{W,i})$, $1 \le i \le n$. For each $\mathbf{x}\in\mathscr{B}$, define

$$g_{\mathtt{F},\mathbf{x}}(\mathbf{u}) = \mathbb{1}(\mathbf{u}\in\mathscr{A}_1)\mathscr{K}^{(\boldsymbol{\nu})}_{\mathtt{F},1}(\mathbf{u};\mathbf{x}) - \mathbb{1}(\mathbf{u}\in\mathscr{A}_0)\mathscr{K}^{(\boldsymbol{\nu})}_{\mathtt{F},0}(\mathbf{u};\mathbf{x}), \qquad \mathbf{u}\in\mathscr{X},$$

where

$$\mathscr{K}^{(\boldsymbol{\nu})}_{\mathtt{F},t}(\mathbf{u};\mathbf{x}) = n^{-1/2}(\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x},\mathbf{x}})^{-1/2}\boldsymbol{\nu}!\mathbf{e}_{\boldsymbol{\nu}}^\top\mathbf{S}^{-1}\boldsymbol{\Gamma}^{-1}_{t,\mathbf{x}}\mathbf{r}_p(\mathbf{H}^{-1}(\mathbf{u}-\mathbf{x}))K_{\mathbf{H}}(\mathbf{u}-\mathbf{x}), \qquad \mathbf{u}\in\mathscr{X}, \quad t\in\{0,1\}.$$

Consider the class of functions

$$\mathscr{F} = \{(\mathbf{u},\varepsilon_Y,\varepsilon_W)\in\mathbb{R}^d\times\mathbb{R}\times\mathbb{R}\mapsto g_{\mathtt{F},\mathbf{x}}(\mathbf{u})\ (\mathfrak{w}_1(\mathbf{x})\varepsilon_Y + \mathfrak{w}_2(\mathbf{x})\varepsilon_W) : \mathbf{x}\in\mathscr{B}\},$$

and let $\mathscr{F}_{\pm} = \mathscr{F} \cup (-\mathscr{F})$. Also define

$$\mathscr{G}_{\mathtt{F}} = \{g_{\mathtt{F},\mathbf{x}} : \mathbf{x} \in \mathscr{B}\}, \qquad \mathscr{R}_{\mathtt{F}} = \{(\varepsilon_Y, \varepsilon_W) \mapsto \mathfrak{w}_1(\mathbf{x})\varepsilon_Y + \mathfrak{w}_2(\mathbf{x})\varepsilon_W : \mathbf{x} \in \mathscr{B}\}.$$

Then we have

$$\sup_{\mathbf{x}\in\mathscr{B}} |\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| = \sup_{f\in\mathscr{F}_{\pm}} \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \Big\{ f(\mathbf{z}_i) - \mathbb{E}[f(\mathbf{z}_i)] \Big\}.$$

We use Lemma SA-17 to approximate the distribution of the right hand side of the above equation. First, we verify the VC-type entropy and moment conditions. The fixed indicators $\mathbb{1}(\mathbf{u} \in \mathscr{A}_t)$ are absorbed into $\mathscr{G}_{\mathtt{F}}$ and do not increase $L_2(Q)$ covering numbers beyond a constant factor, because multiplication by a fixed indicator is a contraction in $L_2(Q)$. Therefore, using the same local-polynomial entropy calculation as in the proofs of Lemmas SA-2 and SA-3, but without the total-variation step, we obtain that: (i) $\mathscr{G}_{\mathtt{F}}$ is pointwise measurable; (ii) there exists a constant envelope function $\mathtt{M}_{\mathscr{G}_{\mathtt{F}}} \lesssim |\mathbf{H}|^{-1/2}$ such that for all $0 < \varepsilon < 1$,

$$\sup_{Q} N(\mathscr{G}_{\mathtt{F}}, \|\cdot\|_{Q,2}, \varepsilon(2\mathbf{c}+1)^{d+1}\mathtt{M}_{\mathscr{G}_{\mathtt{F}}}) \leq 2\mathbf{c}'\varepsilon^{-4d-4} + 2,$$

where the supremum is taken over all finite discrete measures and some constants $\mathbf{c}$ and $\mathbf{c}'$ only depending on the compact set $\mathscr{X}$. Moreover, Assumption SA-1 and SA-4 imply that $\mathscr{R}_{\mathtt{F}}$ is also pointwise measurable and for an envelope function $M_{\mathscr{R}_{\mathtt{F}}}(e_1, e_2) = |e_1| + |e_2|$, $(e_1, e_2) \in \mathbb{R}^2$, we have for all $0 < \varepsilon < 1$,

$$\sup_{Q} N(\mathscr{R}_{\mathtt{F}}, \|\cdot\|_{Q,2}, \varepsilon\mathbf{c}M_{\mathscr{R}_{\mathtt{F}}}) \leq \mathbf{c}'\varepsilon^{-2d},$$

where the supremum is taken over all finite discrete measures. The last entropy bound follows because Assumption SA-1(iii), Assumption SA-4(ii), and $|\boldsymbol{\nu}| \leq p$ imply that $\tau_Y^{(\boldsymbol{\nu})}(\mathbf{x})$ and $\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})$ are continuously differentiable on a neighborhood of $\mathscr{B}$; combined with $\inf_{\mathbf{x}\in\mathscr{B}} |\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, the map $\mathbf{x} \mapsto \mathfrak{w}(\mathbf{x})$ is bounded and Lipschitz on the compact set $\mathscr{B}$. Thus the coefficient set $\{\mathfrak{w}(\mathbf{x}) : \mathbf{x} \in \mathscr{B}\} \subset \mathbb{R}^2$ has polynomial covering numbers inherited from $\mathscr{B}$. Since $\mathscr{F} \subseteq \mathscr{G}_{\mathtt{F}} \cdot \mathscr{R}_{\mathtt{F}}$, the product-covering inequality gives, for a constant $c > 0$,

$$N(\mathscr{F}, \|\cdot\|_{Q,2}, \varepsilon c\mathtt{M}_{\mathscr{G}_{\mathtt{F}}}M_{\mathscr{R}_{\mathtt{F}}}) \leq N(\mathscr{G}_{\mathtt{F}}, \|\cdot\|_{Q,2}, c\varepsilon\mathtt{M}_{\mathscr{G}_{\mathtt{F}}})N(\mathscr{R}_{\mathtt{F}}, \|\cdot\|_{Q,2}, c\varepsilon M_{\mathscr{R}_{\mathtt{F}}}).$$

Hence $\mathscr{F}_{\pm}$ is also pointwise measurable and satisfies the same VC-type bound as $\mathscr{F}$ up to a universal constant: for all $0 < \varepsilon < 1$,

$$\sup_{Q} N(\mathscr{F}_{\pm}, \|\cdot\|_{Q,2}, \varepsilon\mathbf{c}\ \mathtt{M}_{\mathscr{G}_{\mathtt{F}}}M_{\mathscr{R}_{\mathtt{F}}}) \leq 64\mathbf{c}'^2\varepsilon^{-6d-4} + 4\mathbf{c}'\varepsilon^{-2d},$$

where the supremum is taken over all finite discrete measures. The denominator condition $\inf_{\mathbf{x}\in\mathscr{B}} |\tau_W^{(\boldsymbol{\nu})}(\mathbf{x})| > 0$, continuity, and compactness of $\mathscr{B}$ imply that the population Wald weights $\mathfrak{w}(\mathbf{x})$ are uniformly bounded. For the moment condition, take $q = 2 + v \geq 4$ in Lemma SA-17. Since $\varepsilon_{W,i} = W_i - \mathbb{E}[W_i|\mathbf{X}_i]$ is bounded and Assumption SA-1 gives $\sup_{\mathbf{x}\in\mathscr{X}} \mathbb{E}[|Y_i(t)|^{2+v}|\mathbf{X}_i = \mathbf{x}] < \infty$, the envelope

$$F(\mathbf{x}, \varepsilon_Y, \varepsilon_W) = \mathtt{M}_{\mathscr{G}_{\mathtt{F}}}M_{\mathscr{R}_{\mathtt{F}}}(\varepsilon_Y, \varepsilon_W)$$

satisfies $\|F\|_{\mathbb{P},2+v} \lesssim |\mathbf{H}|^{-1/2}$. Together with the local-polynomial moment calculation used in the proof of Lemma SA-3, for

$$\sigma = 1, \qquad b = |\mathbf{H}|^{-1/2},$$

we have $\sup_{f \in \mathscr{F}_\pm} \mathbb{P}|f|^k \leq \sigma^2 b^{k-2}$ for $k = 2, 3, 4$ and $\|F\|_{\mathbb{P},2+v} \leq b$. Taking $\gamma = (\log n)^{-1}$ in Lemma SA-17,

$$\mathbb{P}\left\{ \Big| \sup_{\mathbf{x} \in \mathscr{B}} |\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| - \sup_{\mathbf{x} \in \mathscr{B}} |Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \Big| > \frac{\log n}{(n|\mathbf{H}|)^{1/6}} + \frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{\frac{v}{2+v}}|\mathbf{H}|)^{1/2}} \right\} \leq C \left(\frac{1}{\log n} + \frac{\log n}{n} \right).$$

□

Lemma SA-19 (KS Distance Between $Z_{\mathtt{F}}^{(\boldsymbol{\nu})}$ and $\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}$)**.** *Suppose the conditions for Theorem SA-18 hold. Then, for any multi-index $|\boldsymbol{\nu}| \leq p$,*

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{\mathbf{x} \in \mathscr{B}} \left| \widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) \right| \leq u \,\middle|\, \mathbf{V} \right) - \mathbb{P}\left(\sup_{\mathbf{x} \in \mathscr{B}} \left| Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) \right| \leq u \right) \right| = o_{\mathbb{P}}(1).$$

Proof. Let

$$\mathfrak{R}_{\mathtt{F},\mathtt{cc}} = \mathbf{h}^{-\boldsymbol{\nu}} \left(\sqrt{\frac{\log n}{n|\mathbf{H}|}} + \frac{\log n}{n^{v/(2+v)}|\mathbf{H}|} + \bar{h}^{p+1} \right).$$

For $\mathbf{x}, \mathbf{y} \in \mathscr{B}$, define

$$\boldsymbol{\Pi}_{\mathtt{F},\mathbf{x},\mathbf{y}} = \frac{\Omega_{\mathtt{F},\mathbf{x},\mathbf{y}}^{(\boldsymbol{\nu})}}{\sqrt{\Omega_{\mathtt{F},\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})} \Omega_{\mathtt{F},\mathbf{y},\mathbf{y}}^{(\boldsymbol{\nu})}}}, \qquad \widehat{\boldsymbol{\Pi}}_{\mathtt{F},\mathbf{x},\mathbf{y}} = \frac{\widehat{\Omega}_{\mathtt{F},\mathbf{x},\mathbf{y}}^{(\boldsymbol{\nu})}}{\sqrt{\widehat{\Omega}_{\mathtt{F},\mathbf{x},\mathbf{x}}^{(\boldsymbol{\nu})} \widehat{\Omega}_{\mathtt{F},\mathbf{y},\mathbf{y}}^{(\boldsymbol{\nu})}}}.$$

Lemma SA-9, the lower bound $\inf_{\mathbf{x} \in \mathscr{B}} V_{\mathtt{F},\mathbf{x}}^{(\boldsymbol{\nu})} > 0$, and the mean-value expansion of $x \mapsto x^{-1/2}$ give

$$\sup_{\mathbf{x},\mathbf{y} \in \mathscr{B}} \left| \widehat{\boldsymbol{\Pi}}_{\mathtt{F},\mathbf{x},\mathbf{y}} - \boldsymbol{\Pi}_{\mathtt{F},\mathbf{x},\mathbf{y}} \right| \lesssim_{\mathbb{P}} \mathfrak{R}_{\mathtt{F},\mathtt{cc}}.$$

The Gaussian processes $Z_{\mathtt{F}}^{(\boldsymbol{\nu})}$ and $\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}$ have the same boundary-indexed local-polynomial structure as the sharp Gaussian processes, with the scalar residual replaced by the linear combination $\mathfrak{w}(\mathbf{x})^\top \boldsymbol{\varepsilon}_i$, whose coefficients are uniformly bounded by the denominator nondegeneracy condition. The entropy and modulus calculations in Lemma SA-16 therefore apply with the sharp covariance-comparison rate $\mathfrak{R}_{\mathtt{cc}}$ used there replaced by the present fuzzy rate $\mathfrak{R}_{\mathtt{F},\mathtt{cc}}$. In particular, on a $\delta_n = n^{-s}$ net with s large enough,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{\mathbf{x} \in \mathscr{B}} \left| \widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) \right| \leq u \,\middle|\, \mathbf{V} \right) - \mathbb{P}\left(\sup_{\mathbf{x} \in \mathscr{B}} \left| Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) \right| \leq u \right) \right| \lesssim_{\mathbb{P}} \log(n) \mathfrak{R}_{\mathtt{F},\mathtt{cc}}^{1/2}.$$

The second displayed rate condition in Theorem SA-18 gives $\log^2(n) \mathfrak{R}_{\mathtt{F},\mathtt{cc}} = o(1)$, and hence the right-hand side is $o_{\mathbb{P}}(1)$. □

SA-8.27 Proof of Theorem SA-18

Define the convergence rates

$$\mathfrak{R}_{\mathtt{sup}} = \frac{\log n}{(n|\mathbf{H}|)^{1/6}} + \frac{(\log n)^{5/4}}{(n|\mathbf{H}|)^{1/4}} + \frac{(\log n)^{3/2}}{(n^{v/(2+v)}|\mathbf{H}|)^{1/2}},$$

and

$$\mathfrak{R}_{\mathtt{F},\mathtt{lin}} = \bar{h}^{p+1}\sqrt{n|\mathbf{H}|} + \mathbf{h}^{-\boldsymbol{\nu}}\sqrt{\log(1/\bar{h})}\left(\sqrt{\frac{\log(1/\bar{h})}{n|\mathbf{H}|}} + \frac{\log(1/\bar{h})}{n^{\frac{v}{2+v}}|\mathbf{H}|} + \bar{h}^{p+1}\right).$$

First, we establish the coupling between the intermediate processes. Let

$$M_{\overline{T},\mathtt{F}} = \sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})|, \qquad M_{Z,\mathtt{F}} = \sup_{\mathbf{x}\in\mathscr{B}}|Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})|.$$

Applying Lemma SA-13 with $\mathbb{T}_1 = M_{\overline{T},\mathtt{F}}$ and $\mathbb{T}_2 = M_{Z,\mathtt{F}}$, viewed as singleton-indexed processes, and $\eta = \mathfrak{R}_{\mathtt{sup}}$, gives

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}(M_{\overline{T},\mathtt{F}} \le u) - \mathbb{P}(M_{Z,\mathtt{F}} \le u)\right| \le \sup_{u\in\mathbb{R}}\mathbb{P}(|M_{Z,\mathtt{F}} - u| \le \mathfrak{R}_{\mathtt{sup}}) + \mathbb{P}(|M_{\overline{T},\mathtt{F}} - M_{Z,\mathtt{F}}| > \mathfrak{R}_{\mathtt{sup}}).$$

The first term is $O(\mathfrak{R}_{\mathtt{sup}}\sqrt{\log n}) = o(1)$ by Gaussian anti-concentration and the entropy calculation in the proof of Theorem SA-17. The second term is $o(1)$ by Theorem SA-17. Therefore,

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u\right) - \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u\right)\right| = o(1). \tag{SA-14}$$

Furthermore, Theorem SA-16 gives

$$\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x}) - \overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| = O_{\mathbb{P}}(\mathfrak{R}_{\mathtt{F},\mathtt{lin}}).$$

The side conditions of the theorem imply $\mathfrak{R}_{\mathtt{F},\mathtt{lin}}\sqrt{\log n} = o(1)$. Choose a deterministic sequence a_n such that $\mathfrak{R}_{\mathtt{F},\mathtt{lin}} = o(a_n)$ and $a_n\sqrt{\log n} = o(1)$. Applying Lemma SA-13 with $\mathbb{T}_1 = \widehat{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}$, $\mathbb{T}_2 = \overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}$, and $\eta = a_n$, and then bounding the anti-concentration of $M_{\overline{T},\mathtt{F}}$ through Equation (SA-14), gives

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u\right) - \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u\right)\right| = o(1), \tag{SA-15}$$

and Lemma SA-19 gives

$$\sup_{u\in\mathbb{R}}\left|\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{Z}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u \,\middle|\, \mathbf{V}\right) - \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}}|Z_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le u\right)\right| = o_{\mathbb{P}}(1). \tag{SA-16}$$

The distributional approximation follows from the triangle inequality applied to Equations (SA-14), (SA-15), and (SA-16). For coverage of the feasible confidence band, let

$$E_{\mathtt{F},\alpha} = \left\{\zeta^{(\boldsymbol{\nu})}(\mathbf{x}) \in \widehat{\mathrm{I}}_{\mathtt{F},\alpha}^{(\boldsymbol{\nu})}(\mathbf{x}) \text{ for all } \mathbf{x}\in\mathscr{B}\right\}.$$

By definition of the band, $E_{\mathtt{F},\alpha} = \{\sup_{\mathbf{x}\in\mathscr{B}}|\widehat{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{x})| \le \mathscr{q}_\alpha\}$. Hence the preceding approximation and the conditional definition of $\mathscr{q}_\alpha$ give

$$\begin{aligned}\mathbb{P}(E_{\mathtt{F},\alpha}) &= \mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}} |\widehat{\mathrm{T}}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})| \le \mathscr{q}_\alpha\right)\\ &= \mathbb{E}\left[\mathbb{P}\left(\sup_{\mathbf{x}\in\mathscr{B}} |\widehat{Z}^{(\boldsymbol{\nu})}_{\mathtt{F}}(\mathbf{x})| \le \mathscr{q}_\alpha \,\middle|\, \mathbf{V}\right)\right] + o(1)\\ &= 1-\alpha+o(1),\end{aligned}$$

using continuity of the supremum distribution of the feasible Gaussian process. □

SA-8.28 Proof of Lemma SA-10

A similar argument to the proof of Lemma SA-5 shows that under the assumptions imposed,

$$\sup_{\mathbf{x}\in\mathscr{B}} \left|\mathbb{E}[\mathfrak{w}(\mathbf{x})^\top \mathfrak{T}(\mathbf{x})|\mathbf{X}] - B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h})\right| = o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).$$

Because $\mathfrak{w}(\cdot)$ is bounded under the denominator nondegeneracy condition and the reduced-form bias expansion is uniform in $\mathbf{x}$, the displayed remainder is uniform over $\mathscr{B}$. Integrating against $w(\mathbf{x})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x})$ therefore preserves the order by $\int_{\mathscr{B}} |w(\mathbf{x})|\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{x}) < \infty$, and the integral of $B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{x}}(\mathbf{h})$ is $B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathtt{WBATE}}(\mathbf{h})$ by definition. Under the common-bandwidth restriction $\mathbf{h} = \mathbf{c}h$, the same integration of the pointwise leading term gives the stated $h^{p+1-|\boldsymbol{\nu}|}\widetilde{B}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathtt{WBATE}}(\mathbf{c})$ form. □

SA-8.29 Proof of Lemma SA-11

The proof is the fuzzy analog of Lemma SA-8. By Fubini's theorem,

$$\mathbb{V}[\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] = \int_{\mathscr{B}}\int_{\mathscr{B}} \mathbb{C}\mathrm{ov}[\mathfrak{w}(\mathbf{b}_1)^\top\mathfrak{T}(\mathbf{b}_1), \mathfrak{w}(\mathbf{b}_2)^\top\mathfrak{T}(\mathbf{b}_2)|\mathbf{X}]w(\mathbf{b}_1)w(\mathbf{b}_2)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_1)\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b}_2).$$

Lemma SA-9 replaces this covariance uniformly by $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2}$ with an integrated error of order $o_{\mathbb{P}}(\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}))$. Compact support of K implies that $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2} = 0$ unless $\|\mathbf{b}_1 - \mathbf{b}_2\| \lesssim \bar{h}$; by (SA-3), the effective integration set has product Hausdorff measure $O(\bar{h}^{d-1})$. The pointwise covariance is of order $(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})^{-1}$, which gives the stated upper bound. The lower bound follows from the aggregate fuzzy variance nondegeneracy condition (SA-6), which rules out cancellation after applying the fuzzy Wald weights and integrating over $\mathscr{B}$. Replacing $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2}$ by $\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathtt{F},\mathbf{b}_1,\mathbf{b}_2}$ uses Lemma SA-9 once more. □

SA-8.30 Proof of Theorem SA-19

By Lemma SA-10,

$$\mathbb{E}[\check{\zeta}^{(\boldsymbol{\nu})}_{\mathtt{WBATE}}|\mathbf{X}] = B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathtt{WBATE}}(\mathbf{h}) + o_{\mathbb{P}}(\bar{h}^{p+1}\mathbf{h}^{-\boldsymbol{\nu}}).$$

Squaring the bias expansion gives $\left(B^{(\boldsymbol{\nu})}_{\mathtt{F},\mathtt{WBATE}}(\mathbf{h})\right)^2 + o_{\mathbb{P}}(\bar{h}^{2p+2}\mathbf{h}^{-2\boldsymbol{\nu}})$. Lemma SA-11 gives the variance term $\Omega^{(\boldsymbol{\nu})}_{\mathtt{F},\mathtt{WBATE}}$ with remainder $o_{\mathbb{P}}(\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}}))$. Adding these two components gives the displayed AMSE expansion. □

SA-8.31 Proof of Theorem SA-20

The proof follows the same Lyapunov argument as Theorem SA-10, applied to the linearized fuzzy statistic obtained by integrating $\overline{\mathrm{T}}_{\mathtt{F}}^{(\boldsymbol{\nu})}(\mathbf{b})$ against $w(\mathbf{b})$. Lemma SA-11 gives the variance rate $\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})$ and consistency of its plug-in estimator. Lemma SA-10 and the side condition $n|\mathbf{H}|\bar{h}^{2p+2}/\bar{h}^{d-1} = o(1)$ make the linearized bias negligible relative to the standard deviation. The final two displayed rate conditions imply the two stochastic pieces of the uniform reduced-form consistency condition, while $\mathbf{h}^{-\boldsymbol{\nu}}\bar{h}^{p+1} = o(1)$ follows from $|\boldsymbol{\nu}| \leq p$ and $\bar{h}/\underline{h} \lesssim 1$. Moreover, (SA-4), integrability of w, and the final displayed rate conditions in the theorem imply

$$\frac{\left|\int_{\mathscr{B}} \mathfrak{R}_{\mathtt{F}}(\mathbf{b})w(\mathbf{b})\,\mathrm{d}\mathfrak{H}^{d-1}(\mathbf{b})\right|}{\left(\bar{h}^{d-1}/(n|\mathbf{H}|\mathbf{h}^{2\boldsymbol{\nu}})\right)^{1/2}} = o_{\mathbb{P}}(1).$$

The Lyapunov condition follows from the same calculation as in the sharp WBATE proof, with the scalar residual replaced by the linearized fuzzy residual $\mathfrak{w}_1(\mathbf{b})\varepsilon_{Y,i} + \mathfrak{w}_2(\mathbf{b})\varepsilon_{W,i}$. In particular, if χ_i denotes the normalized integrated linearized fuzzy summand, then the boundedness of $\mathfrak{w}(\mathbf{b})$, the boundedness of $\varepsilon_{W,i}$, and Assumption SA-1(v) imply

$$\sum_{i=1}^{n} \mathbb{E}[|\chi_i|^3] \lesssim \left(\frac{\bar{h}^{d-1}}{n|\mathbf{H}|}\right)^{1/2} = o(1).$$

Therefore the feasible statistic $\widehat{\mathrm{T}}_{\mathtt{F,WBATE}}^{(\boldsymbol{\nu})}$ is asymptotically standard normal. □

SA-8.32 Proof of Theorem SA-21

Follows by Theorem SA-13 after noting that

$$|\widehat{\zeta}_{\mathtt{LBATE}}^{(\boldsymbol{\nu})} - \zeta_{\mathtt{LBATE}}^{(\boldsymbol{\nu})}| \leq \sup_{\mathbf{x}\in\mathscr{B}} |\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{x}) - \zeta^{(\boldsymbol{\nu})}(\mathbf{x})|.$$

□

SA-8.33 Proof of Theorem SA-22

Consider the event $E_{\mathtt{F}} = \left\{\sup_{\mathbf{b}\in\mathscr{B}} \frac{|\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) - \zeta^{(\boldsymbol{\nu})}(\mathbf{b})|}{(\widehat{\Omega}_{\mathtt{F},\mathbf{b},\mathbf{b}}^{(\boldsymbol{\nu})})^{1/2}} \leq \mathscr{q}_\alpha\right\}$. Theorem SA-18 implies that $\mathbb{P}(E_{\mathtt{F}}) = 1-\alpha+o(1)$. On the event $E_{\mathtt{F}}$, we have

$$\widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha(\widehat{\Omega}_{\mathtt{F},\mathbf{b},\mathbf{b}}^{(\boldsymbol{\nu})})^{1/2} \leq \zeta^{(\boldsymbol{\nu})}(\mathbf{b}) \leq \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha(\widehat{\Omega}_{\mathtt{F},\mathbf{b},\mathbf{b}}^{(\boldsymbol{\nu})})^{1/2}, \qquad \forall\, \mathbf{b} \in \mathscr{B},$$

which implies

$$\sup_{\mathbf{b}\in\mathscr{B}} \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) - \mathscr{q}_\alpha(\widehat{\Omega}_{\mathtt{F},\mathbf{b},\mathbf{b}}^{(\boldsymbol{\nu})})^{1/2} \leq \sup_{\mathbf{b}\in\mathscr{B}} \zeta^{(\boldsymbol{\nu})}(\mathbf{b}) \leq \sup_{\mathbf{b}\in\mathscr{B}} \widehat{\zeta}^{(\boldsymbol{\nu})}(\mathbf{b}) + \mathscr{q}_\alpha(\widehat{\Omega}_{\mathtt{F},\mathbf{b},\mathbf{b}}^{(\boldsymbol{\nu})})^{1/2}.$$

The stated result then follows. □

SA-9 Implementation Details

This section provides additional details on the practical implementation of the pointwise and uniform inference methods discussed in Section 3.3 of the paper. The main issue for uniform inference is that the target process is indexed by the continuum $\mathbf{x} \in \mathscr{B}$, whereas computation is necessarily carried out on a finite discretization of the boundary. The following steps describe the default implementation used by the companion software and recommended in applications.

1. *Represent the boundary and choose the target set.* When $\mathscr{B}$ is piecewise linear, as in the empirical application, it is convenient to parameterize the boundary by arc length. Let $\gamma : [0, \mathfrak{L}(\mathscr{B})] \to \mathscr{B}$ denote such a parameterization, where $\mathfrak{L}(\mathscr{B})$ is the total boundary length. If the researcher wants inference only on a subset of the assignment boundary, for example after trimming endpoints, corners, or regions with weak local support, the same construction can be applied to the trimmed target set $\mathscr{B}_0 \subseteq \mathscr{B}$. This distinction is important: uniform confidence bands cover the chosen target set, so any trimming changes the estimand from the full boundary to the trimmed boundary.

2. *Construct a quasi-uniform grid on the boundary.* Choose grid points $\mathbf{b}_1, \ldots, \mathbf{b}_J \in \mathscr{B}_0$ with approximately equal spacing in arc length. For a single connected component this can be done by setting $\mathbf{b}_j = \gamma(s_j)$ with $s_j = (j-1)\mathfrak{L}(\mathscr{B}_0)/(J-1)$, $j = 1, \ldots, J$. For a piecewise linear boundary with multiple segments, grid points can be allocated across segments in proportion to segment length, while forcing vertices or other substantively important boundary points to be included when desired. In higher-dimensional settings, the analogous requirement is a quasi-uniform mesh on the $(d-1)$-dimensional boundary with respect to Hausdorff measure.

3. *Choose the mesh size.* The grid should be fine enough that the discretized supremum is a good approximation to the continuous supremum. A useful practical rule is to make the maximum spacing between adjacent boundary grid points small relative to the bandwidth scale used for inference. Equivalently, the mesh should be refined until the simulated critical value and the visual shape of the confidence band are stable. In simple two-dimensional applications, values such as $J = 40$ or $J = 80$ often give similar results, but the appropriate choice depends on the length and geometry of $\mathscr{B}$, the bandwidth, and the degree of treatment effect heterogeneity. We recommend reporting a grid-sensitivity check, for example comparing J, $2J$, and, when computationally feasible, $4J$.

4. *Select bandwidths and polynomial orders.* The plug-in selectors described in Section 3.3 of the paper can be implemented pointwise, producing $\widehat{h}_{\mathtt{MSE},\mathbf{x}}$, or globally, producing $\widehat{h}_{\mathtt{IMSE}}$. For uniform inference and graphical display along the boundary, a global bandwidth is often easier to interpret and tends to produce smoother covariance estimates. Pointwise bandwidths can also be used, provided the covariance matrix of the resulting process is computed using the same bandwidth choices. Robust bias-corrected inference is implemented by using the bandwidth selected for the pth-order point estimator and constructing the associated t-statistic with a $(p+1)$th-order local polynomial, including the corresponding variance adjustment. When the coordinates of the score have different scales, one may either standardize the score before using a scalar bandwidth or use a diagonal bandwidth matrix $\mathbf{H} = \mathrm{diag}(h_1, \ldots, h_d)$ as allowed in this supplemental appendix.

5. *Compute point estimates and covariance estimates on the grid.* For each grid point $\mathbf{b}_j$, compute the local-polynomial estimates, the robust bias-corrected t-statistics, and the variance estimates. For uniform

inference, also compute the estimated cross-covariances

$$\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b}_j,\mathbf{b}_k}, \qquad 1 \leq j, k \leq J,$$

using the formulas in Sections SA-3 and SA-6. The simulated Gaussian process is then represented by a mean-zero Gaussian vector with correlation matrix

$$\widehat{\mathbf{\Pi}}_{j,k} = \frac{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b}_j,\mathbf{b}_k}}{\sqrt{\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b}_j,\mathbf{b}_j}\widehat{\Omega}^{(\boldsymbol{\nu})}_{\mathbf{b}_k,\mathbf{b}_k}}}, \qquad 1 \leq j, k \leq J.$$

In finite samples, $\widehat{\mathbf{\Pi}}$ should be symmetrized before simulation.

6. *Regularize the covariance matrix if needed.* The estimated matrix $\widehat{\mathbf{\Pi}}$ may fail to be positive semidefinite because of simulation error, numerical precision, or unstable local covariance estimates. This is a finite-sample numerical issue rather than a change in the target parameter. A simple regularization is to replace $\widehat{\mathbf{\Pi}}$ by its nearest positive semidefinite approximation, or equivalently to truncate negative eigenvalues at a small nonnegative tolerance and rescale the diagonal entries to one. Another common approach is to use a small ridge adjustment, $\widehat{\mathbf{\Pi}}_\lambda = (1-\lambda)\widehat{\mathbf{\Pi}} + \lambda \mathbf{I}_J$, with the smallest $\lambda \geq 0$ that makes the matrix positive semidefinite. The chosen adjustment should be reported when it is not negligible. See Cattaneo et al. [2024b] for related discussion and theoretical results on covariance regularization.

7. *Simulate the critical value.* Given the regularized correlation matrix, draw M independent vectors from $N(0, \widehat{\mathbf{\Pi}})$ and compute the empirical $(1-\alpha)$ quantile of

$$\max_{1\leq j\leq J} |Z_j|.$$

This quantile is the simulated critical value for the uniform confidence band over the grid. In practice, M should be large enough that simulation error is small relative to the desired reporting precision; increasing M and checking that the critical value is stable is a simple diagnostic.

8. *Check local support and numerical stability.* Uniform inference can be sensitive to boundary regions where one side has little local support or the local design matrix is poorly conditioned. We recommend inspecting, for each grid point, effective local sample sizes or kernel masses on the two sides of the boundary, together with condition numbers of the estimated Gram matrices. Grid points with very small one-sided kernel mass, extremely unbalanced mass across sides, or unstable Gram matrices indicate that the local design is weak at those locations. In such cases, researchers should report the issue, consider sensitivity to bandwidth choices, and, when substantively justified, trim the problematic portion of the boundary and redefine the target set accordingly.

9. *Report sensitivity analyses.* A transparent empirical implementation should report the main estimates together with basic sensitivity checks: alternative grid sizes, alternative bandwidth choices, and whether covariance regularization was needed. For applications with corners or visibly irregular boundaries, it is also useful to report results with and without small neighborhoods of those regions. Researchers should also examine whether the density of the score changes abruptly near the boundary or whether observations appear to bunch on one side of $\mathscr{B}$. In BD designs this type of manipulation or sorting diagnostic is necessarily multivariate and boundary-local: useful checks include scatterplots near $\mathscr{B}$, side-

specific kernel masses along the boundary grid, and sensitivity to excluding small neighborhoods where the empirical distribution of $\mathbf{X}_i$ appears discontinuous or unusually concentrated. These checks help distinguish genuine treatment effect heterogeneity from artifacts of discretization, weak local support, numerical instability, or nonrandom sorting around the assignment boundary.

The companion software package `rd2d` and article [Cattaneo et al., 2025] discuss additional implementation details.

SA-10 Empirical Application

This section reports numerical results for the SPP empirical application that are omitted from the main paper. Replication codes are available at https://rdpackages.github.io/replication/. The main paper uses the ITT bandwidth selector throughout; this section reports the corresponding full boundary-grid results and then gives the analogous results using the fuzzy bandwidth selector.

SA-10.1 ITT Bandwidth Selector

Due to space limitations, the paper reports selected boundary points and aggregate summaries in Table 1. Here we provide the corresponding full boundary-grid results for all 40 evaluation points.

- Table SA-1 reports the reduced-form, or intention-to-treat (ITT), estimates for college enrollment.
- Table SA-2 reports the first-stage estimates for treatment take-up.
- Table SA-3 reports the fuzzy estimates for college enrollment.
- Table SA-4 reports the covariate-balance reduced-form estimates using mother's education as the outcome.

The figures in this subsection complement the confidence intervals and bands plots presented in the paper. For each quantity, we report heat maps along the boundary for the point estimates and associated pointwise robust bias-corrected p-values. Specifically:

- Figure SA-1 reports heatmaps for the reduced-form analysis of college enrollment.
- Figure SA-2 reports heatmaps for the first-stage estimates for treatment take-up.
- Figure SA-3 reports heatmaps for the fuzzy analysis of college enrollment.
- Figure SA-4 reports heatmaps for the covariate-balance analysis using mother's education as the outcome.

These plots make it easier to see where along the boundary the estimated effects are largest and where the evidence against the zero-effect null is strongest.

In addition, we report baseline estimates and treatment effects relative to control-side baselines.

- Table SA-5 reports the control-side baseline estimates $\widehat{\mu}_{Y,0}(\mathbf{x})$ for college enrollment.
- Table SA-6 reports ITT and Fuzzy treatment effects relative to control-side baseline estimates $\widehat{\mu}_{Y,0}(\mathbf{x})$ for college enrollment.

- Figure SA-5 reports baseline estimates and uncertainty quantification.
- Figure SA-6 reports treatment effects relative to the estimated control-side baseline.

The baseline estimates in Table SA-5 help interpret the magnitude of the treatment effects.

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	27.6	12.2	8921	3532	0.311	0.000	(0.204, 0.423)
$\mathbf{b}_2$	26.0	11.5	8998	3677	0.313	0.000	(0.217, 0.430)
$\mathbf{b}_3$	24.6	10.9	8510	3678	0.316	0.000	(0.218, 0.428)
$\mathbf{b}_4$	24.5	10.9	9236	3976	0.309	0.000	(0.197, 0.399)
$\mathbf{b}_5$	23.1	10.2	9044	3997	0.295	0.000	(0.164, 0.370)
$\mathbf{b}_6$	20.7	9.2	7176	3525	0.277	0.000	(0.128, 0.350)
$\mathbf{b}_7$	19.0	8.4	6089	3209	0.267	0.000	(0.118, 0.356)
$\mathbf{b}_8$	19.2	8.5	6725	3422	0.283	0.000	(0.137, 0.370)
$\mathbf{b}_9$	19.2	8.5	6803	3516	0.305	0.000	(0.159, 0.391)
$\mathbf{b}_{10}$	19.3	8.6	6925	3563	0.324	0.000	(0.189, 0.417)
$\mathbf{b}_{11}$	21.1	9.3	8665	4206	0.332	0.000	(0.219, 0.424)
$\mathbf{b}_{12}$	21.8	9.7	9017	4456	0.327	0.000	(0.223, 0.420)
$\mathbf{b}_{13}$	21.1	9.3	8683	4417	0.322	0.000	(0.209, 0.412)
$\mathbf{b}_{14}$	21.8	9.6	9069	4646	0.318	0.000	(0.194, 0.388)
$\mathbf{b}_{15}$	22.2	9.8	10063	4977	0.314	0.000	(0.190, 0.378)
$\mathbf{b}_{16}$	22.5	10.0	10210	5162	0.317	0.000	(0.208, 0.390)
$\mathbf{b}_{17}$	21.3	9.4	9409	4444	0.323	0.000	(0.251, 0.416)
$\mathbf{b}_{18}$	20.7	9.2	9182	3850	0.328	0.000	(0.264, 0.427)
$\mathbf{b}_{19}$	19.3	8.5	8468	3157	0.334	0.000	(0.263, 0.439)
$\mathbf{b}_{20}$	19.4	8.6	8945	2680	0.327	0.000	(0.231, 0.433)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.314	0.000	(0.157, 0.470)
$\mathbf{b}_{22}$	21.5	9.5	10221	2786	0.306	0.000	(0.183, 0.418)
$\mathbf{b}_{23}$	20.3	9.0	7743	2679	0.298	0.000	(0.186, 0.408)
$\mathbf{b}_{24}$	20.5	9.1	7097	2856	0.295	0.000	(0.198, 0.413)
$\mathbf{b}_{25}$	21.0	9.3	6188	3114	0.291	0.000	(0.211, 0.423)
$\mathbf{b}_{26}$	23.2	10.3	6948	3635	0.284	0.000	(0.211, 0.414)
$\mathbf{b}_{27}$	25.1	11.1	7232	4098	0.273	0.000	(0.195, 0.393)
$\mathbf{b}_{28}$	25.2	11.2	5984	4204	0.269	0.000	(0.182, 0.391)
$\mathbf{b}_{29}$	25.6	11.3	5319	4343	0.266	0.000	(0.171, 0.393)
$\mathbf{b}_{30}$	26.3	11.7	4698	4560	0.257	0.000	(0.157, 0.398)
$\mathbf{b}_{31}$	28.5	12.6	4759	4995	0.247	0.000	(0.139, 0.377)
$\mathbf{b}_{32}$	28.9	12.8	4482	4655	0.240	0.000	(0.123, 0.370)
$\mathbf{b}_{33}$	30.8	13.7	4559	4766	0.234	0.000	(0.115, 0.360)
$\mathbf{b}_{34}$	31.2	13.8	4048	4222	0.228	0.000	(0.103, 0.357)
$\mathbf{b}_{35}$	28.1	12.5	2875	2950	0.219	0.000	(0.077, 0.373)
$\mathbf{b}_{36}$	26.0	11.5	2214	2243	0.215	0.000	(0.064, 0.394)
$\mathbf{b}_{37}$	26.8	11.9	1987	2002	0.209	0.000	(0.059, 0.391)
$\mathbf{b}_{38}$	28.5	12.6	2049	2068	0.200	0.000	(0.052, 0.379)
$\mathbf{b}_{39}$	33.6	14.9	2758	2787	0.191	0.000	(0.053, 0.349)
$\mathbf{b}_{40}$	36.3	16.1	2916	2993	0.185	0.000	(0.044, 0.335)
WBATE					0.282	0.000	(0.248, 0.316)
LBATE					0.334		(0.263, 0.472)

Table SA-1: Intention-to-Treat Effects (SPP Application, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	27.6	12.2	8921	3532	0.551	0.000	(0.495, 0.655)
$\mathbf{b}_2$	26.0	11.5	8998	3677	0.546	0.000	(0.496, 0.651)
$\mathbf{b}_3$	24.6	10.9	8510	3678	0.544	0.000	(0.494, 0.646)
$\mathbf{b}_4$	24.5	10.9	9236	3976	0.536	0.000	(0.483, 0.629)
$\mathbf{b}_5$	23.1	10.2	9044	3997	0.528	0.000	(0.466, 0.612)
$\mathbf{b}_6$	20.7	9.2	7176	3525	0.517	0.000	(0.441, 0.595)
$\mathbf{b}_7$	19.0	8.4	6089	3209	0.507	0.000	(0.419, 0.582)
$\mathbf{b}_8$	19.2	8.5	6725	3422	0.507	0.000	(0.407, 0.570)
$\mathbf{b}_9$	19.2	8.5	6803	3516	0.512	0.000	(0.406, 0.571)
$\mathbf{b}_{10}$	19.3	8.6	6925	3563	0.524	0.000	(0.417, 0.582)
$\mathbf{b}_{11}$	21.1	9.3	8665	4206	0.538	0.000	(0.448, 0.599)
$\mathbf{b}_{12}$	21.8	9.7	9017	4456	0.542	0.000	(0.463, 0.607)
$\mathbf{b}_{13}$	21.1	9.3	8683	4417	0.552	0.000	(0.472, 0.616)
$\mathbf{b}_{14}$	21.8	9.6	9069	4646	0.562	0.000	(0.480, 0.617)
$\mathbf{b}_{15}$	22.2	9.8	10063	4977	0.564	0.000	(0.481, 0.614)
$\mathbf{b}_{16}$	22.5	10.0	10210	5162	0.568	0.000	(0.493, 0.622)
$\mathbf{b}_{17}$	21.3	9.4	9409	4444	0.576	0.000	(0.501, 0.636)
$\mathbf{b}_{18}$	20.7	9.2	9182	3850	0.577	0.000	(0.498, 0.642)
$\mathbf{b}_{19}$	19.3	8.5	8468	3157	0.575	0.000	(0.488, 0.654)
$\mathbf{b}_{20}$	19.4	8.6	8945	2680	0.574	0.000	(0.467, 0.674)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.573	0.000	(0.424, 0.762)
$\mathbf{b}_{22}$	21.5	9.5	10221	2786	0.576	0.000	(0.469, 0.715)
$\mathbf{b}_{23}$	20.3	9.0	7743	2679	0.574	0.000	(0.473, 0.696)
$\mathbf{b}_{24}$	20.5	9.1	7097	2856	0.574	0.000	(0.472, 0.683)
$\mathbf{b}_{25}$	21.0	9.3	6188	3114	0.579	0.000	(0.481, 0.684)
$\mathbf{b}_{26}$	23.2	10.3	6948	3635	0.587	0.000	(0.490, 0.679)
$\mathbf{b}_{27}$	25.1	11.1	7232	4098	0.593	0.000	(0.494, 0.674)
$\mathbf{b}_{28}$	25.2	11.2	5984	4204	0.600	0.000	(0.499, 0.682)
$\mathbf{b}_{29}$	25.6	11.3	5319	4343	0.611	0.000	(0.511, 0.697)
$\mathbf{b}_{30}$	26.3	11.7	4698	4560	0.623	0.000	(0.522, 0.711)
$\mathbf{b}_{31}$	28.5	12.6	4759	4995	0.636	0.000	(0.535, 0.720)
$\mathbf{b}_{32}$	28.9	12.8	4482	4655	0.649	0.000	(0.547, 0.738)
$\mathbf{b}_{33}$	30.8	13.7	4559	4766	0.661	0.000	(0.564, 0.751)
$\mathbf{b}_{34}$	31.2	13.8	4048	4222	0.674	0.000	(0.578, 0.772)
$\mathbf{b}_{35}$	28.1	12.5	2875	2950	0.691	0.000	(0.587, 0.811)
$\mathbf{b}_{36}$	26.0	11.5	2214	2243	0.710	0.000	(0.601, 0.849)
$\mathbf{b}_{37}$	26.8	11.9	1987	2002	0.719	0.000	(0.610, 0.861)
$\mathbf{b}_{38}$	28.5	12.6	2049	2068	0.722	0.000	(0.611, 0.863)
$\mathbf{b}_{39}$	33.6	14.9	2758	2787	0.721	0.000	(0.618, 0.852)
$\mathbf{b}_{40}$	36.3	16.1	2916	2993	0.723	0.000	(0.621, 0.855)
`WBATE`					0.592	0.000	(0.567, 0.620)
`LBATE`					0.723		(0.624, 0.860)

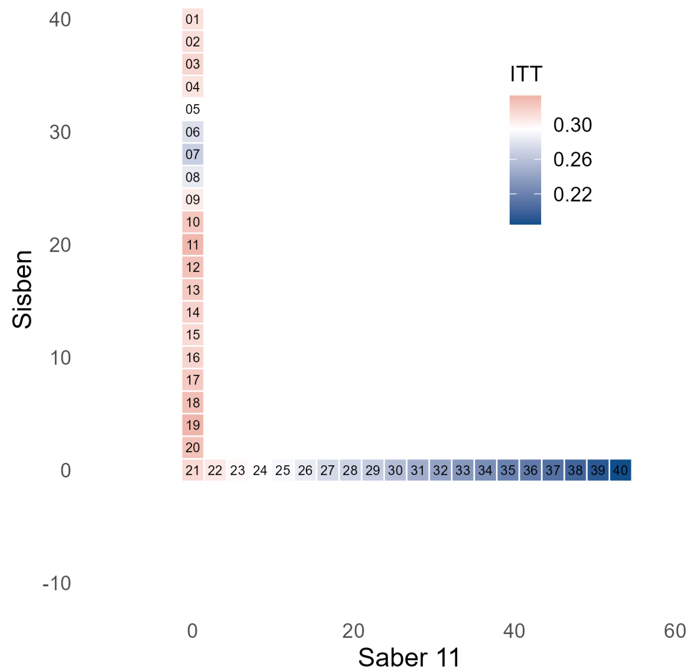
Table SA-2: First-Stage Effects (SPP Application, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	27.6	12.2	8921	3532	0.564	0.000	(0.374, 0.715)
$\mathbf{b}_2$	26.0	11.5	8998	3677	0.572	0.000	(0.398, 0.730)
$\mathbf{b}_3$	24.6	10.9	8510	3678	0.581	0.000	(0.403, 0.732)
$\mathbf{b}_4$	24.5	10.9	9236	3976	0.576	0.000	(0.374, 0.697)
$\mathbf{b}_5$	23.1	10.2	9044	3997	0.559	0.000	(0.323, 0.666)
$\mathbf{b}_6$	20.7	9.2	7176	3525	0.535	0.000	(0.268, 0.654)
$\mathbf{b}_7$	19.0	8.4	6089	3209	0.527	0.000	(0.259, 0.689)
$\mathbf{b}_8$	19.2	8.5	6725	3422	0.559	0.000	(0.304, 0.734)
$\mathbf{b}_9$	19.2	8.5	6803	3516	0.595	0.000	(0.350, 0.776)
$\mathbf{b}_{10}$	19.3	8.6	6925	3563	0.619	0.000	(0.401, 0.812)
$\mathbf{b}_{11}$	21.1	9.3	8665	4206	0.618	0.000	(0.437, 0.791)
$\mathbf{b}_{12}$	21.8	9.7	9017	4456	0.602	0.000	(0.433, 0.767)
$\mathbf{b}_{13}$	21.1	9.3	8683	4417	0.584	0.000	(0.401, 0.740)
$\mathbf{b}_{14}$	21.8	9.6	9069	4646	0.566	0.000	(0.371, 0.692)
$\mathbf{b}_{15}$	22.2	9.8	10063	4977	0.556	0.000	(0.362, 0.675)
$\mathbf{b}_{16}$	22.5	10.0	10210	5162	0.557	0.000	(0.388, 0.686)
$\mathbf{b}_{17}$	21.3	9.4	9409	4444	0.560	0.000	(0.458, 0.715)
$\mathbf{b}_{18}$	20.7	9.2	9182	3850	0.570	0.000	(0.484, 0.727)
$\mathbf{b}_{19}$	19.3	8.5	8468	3157	0.580	0.000	(0.484, 0.744)
$\mathbf{b}_{20}$	19.4	8.6	8945	2680	0.569	0.000	(0.440, 0.723)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.548	0.000	(0.338, 0.720)
$\mathbf{b}_{22}$	21.5	9.5	10221	2786	0.532	0.000	(0.359, 0.655)
$\mathbf{b}_{23}$	20.3	9.0	7743	2679	0.519	0.000	(0.357, 0.659)
$\mathbf{b}_{24}$	20.5	9.1	7097	2856	0.513	0.000	(0.373, 0.686)
$\mathbf{b}_{25}$	21.0	9.3	6188	3114	0.503	0.000	(0.385, 0.703)
$\mathbf{b}_{26}$	23.2	10.3	6948	3635	0.483	0.000	(0.382, 0.689)
$\mathbf{b}_{27}$	25.1	11.1	7232	4098	0.461	0.000	(0.352, 0.655)
$\mathbf{b}_{28}$	25.2	11.2	5984	4204	0.449	0.000	(0.326, 0.643)
$\mathbf{b}_{29}$	25.6	11.3	5319	4343	0.435	0.000	(0.302, 0.632)
$\mathbf{b}_{30}$	26.3	11.7	4698	4560	0.412	0.000	(0.272, 0.627)
$\mathbf{b}_{31}$	28.5	12.6	4759	4995	0.388	0.000	(0.238, 0.584)
$\mathbf{b}_{32}$	28.9	12.8	4482	4655	0.370	0.000	(0.206, 0.561)
$\mathbf{b}_{33}$	30.8	13.7	4559	4766	0.354	0.000	(0.188, 0.534)
$\mathbf{b}_{34}$	31.2	13.8	4048	4222	0.338	0.000	(0.164, 0.517)
$\mathbf{b}_{35}$	28.1	12.5	2875	2950	0.317	0.000	(0.123, 0.521)
$\mathbf{b}_{36}$	26.0	11.5	2214	2243	0.303	0.000	(0.100, 0.532)
$\mathbf{b}_{37}$	26.8	11.9	1987	2002	0.291	0.000	(0.090, 0.522)
$\mathbf{b}_{38}$	28.5	12.6	2049	2068	0.277	0.000	(0.079, 0.506)
$\mathbf{b}_{39}$	33.6	14.9	2758	2787	0.265	0.000	(0.079, 0.468)
$\mathbf{b}_{40}$	36.3	16.1	2916	2993	0.255	0.000	(0.067, 0.447)
WBATE					0.487	0.000	(0.435, 0.535)
LBATE					0.619		(0.487, 0.808)

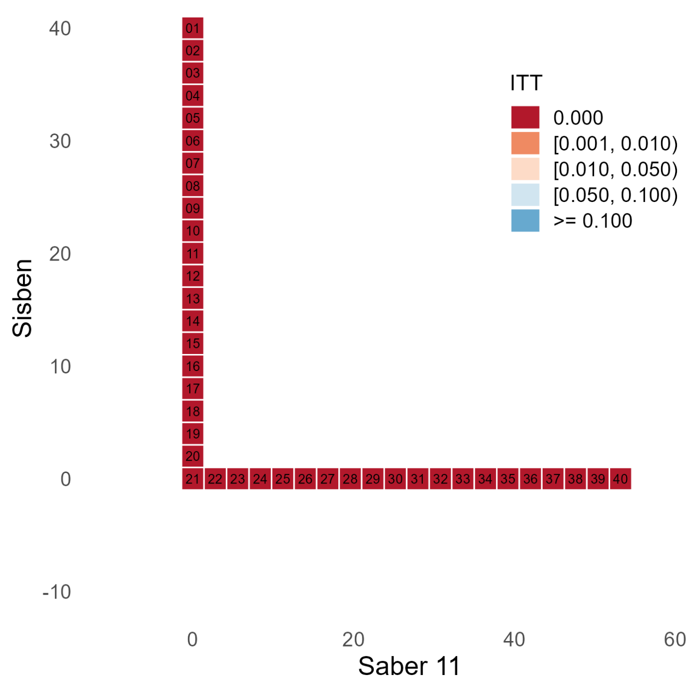
Table SA-3: Fuzzy Effects (SPP Application, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_{1}$	27.4	12.2	8750	3469	-0.027	0.170	(-0.153, 0.058)
$\mathbf{b}_{2}$	25.9	11.5	8454	3560	-0.032	0.115	(-0.158, 0.050)
$\mathbf{b}_{3}$	24.8	11.0	8447	3647	-0.031	0.115	(-0.157, 0.050)
$\mathbf{b}_{4}$	24.7	11.0	9184	3947	-0.022	0.213	(-0.139, 0.058)
$\mathbf{b}_{5}$	24.2	10.7	9767	4156	-0.015	0.398	(-0.123, 0.069)
$\mathbf{b}_{6}$	22.7	10.1	8801	3951	-0.008	0.739	(-0.108, 0.086)
$\mathbf{b}_{7}$	21.9	9.7	8282	3901	-0.001	0.984	(-0.099, 0.098)
$\mathbf{b}_{8}$	21.4	9.5	8372	3993	-0.002	0.678	(-0.115, 0.087)
$\mathbf{b}_{9}$	21.7	9.6	8680	4130	-0.010	0.367	(-0.130, 0.071)
$\mathbf{b}_{10}$	21.0	9.3	8085	3981	-0.011	0.272	(-0.139, 0.065)
$\mathbf{b}_{11}$	20.6	9.1	7894	3964	-0.011	0.444	(-0.129, 0.077)
$\mathbf{b}_{12}$	21.6	9.6	8832	4358	-0.009	0.931	(-0.100, 0.095)
$\mathbf{b}_{13}$	21.0	9.3	8098	4270	-0.001	0.851	(-0.094, 0.106)
$\mathbf{b}_{14}$	20.9	9.3	8189	4348	0.002	0.849	(-0.091, 0.104)
$\mathbf{b}_{15}$	20.3	9.0	8051	4329	0.000	0.909	(-0.093, 0.100)
$\mathbf{b}_{16}$	21.8	9.7	9040	4809	0.001	0.560	(-0.071, 0.105)
$\mathbf{b}_{17}$	22.7	10.0	10699	4678	0.002	0.516	(-0.057, 0.088)
$\mathbf{b}_{18}$	20.1	8.9	8788	3746	-0.002	0.472	(-0.058, 0.095)
$\mathbf{b}_{19}$	18.7	8.3	7730	3014	-0.005	0.815	(-0.085, 0.073)
$\mathbf{b}_{20}$	20.2	8.9	9828	2825	-0.009	0.613	(-0.103, 0.073)
$\mathbf{b}_{21}$	19.8	8.8	9415	2203	-0.004	0.503	(-0.118, 0.185)
$\mathbf{b}_{22}$	21.0	9.3	9846	2645	0.001	0.350	(-0.076, 0.144)
$\mathbf{b}_{23}$	17.9	7.9	5971	2207	0.012	0.065	(-0.043, 0.176)
$\mathbf{b}_{24}$	17.9	7.9	4992	2339	0.018	0.037	(-0.034, 0.180)
$\mathbf{b}_{25}$	18.7	8.3	4697	2598	0.020	0.058	(-0.039, 0.168)
$\mathbf{b}_{26}$	20.6	9.1	4929	3087	0.019	0.112	(-0.046, 0.148)
$\mathbf{b}_{27}$	22.3	9.9	5182	3532	0.015	0.233	(-0.057, 0.130)
$\mathbf{b}_{28}$	22.5	10.0	4222	3677	0.013	0.312	(-0.065, 0.130)
$\mathbf{b}_{29}$	23.0	10.2	3755	3831	0.009	0.412	(-0.077, 0.134)
$\mathbf{b}_{30}$	24.5	10.9	3842	3945	0.005	0.556	(-0.085, 0.126)
$\mathbf{b}_{31}$	29.4	13.0	5150	5113	0.004	0.678	(-0.079, 0.105)
$\mathbf{b}_{32}$	30.0	13.3	4746	4961	0.000	0.961	(-0.094, 0.097)
$\mathbf{b}_{33}$	27.7	12.3	3589	3723	-0.004	0.798	(-0.117, 0.099)
$\mathbf{b}_{34}$	26.1	11.6	2715	2794	-0.004	0.665	(-0.135, 0.101)
$\mathbf{b}_{35}$	23.1	10.3	1825	1842	-0.006	0.639	(-0.156, 0.114)
$\mathbf{b}_{36}$	21.2	9.4	1396	1393	-0.009	0.528	(-0.172, 0.113)
$\mathbf{b}_{37}$	21.7	9.6	1252	1223	-0.012	0.423	(-0.177, 0.103)
$\mathbf{b}_{38}$	22.7	10.1	1304	1274	-0.014	0.372	(-0.173, 0.094)
$\mathbf{b}_{39}$	29.3	13.0	1944	1969	-0.010	0.434	(-0.134, 0.079)
$\mathbf{b}_{40}$	27.8	12.3	1560	1511	-0.011	0.516	(-0.134, 0.087)
`WBATE`					-0.004	0.938	(-0.030, 0.028)
`LBATE`					0.020		(-0.031, 0.182)

Table SA-4: Covariate Balance for Mother's Education (SPP Application, ITT Bandwidth Selector).

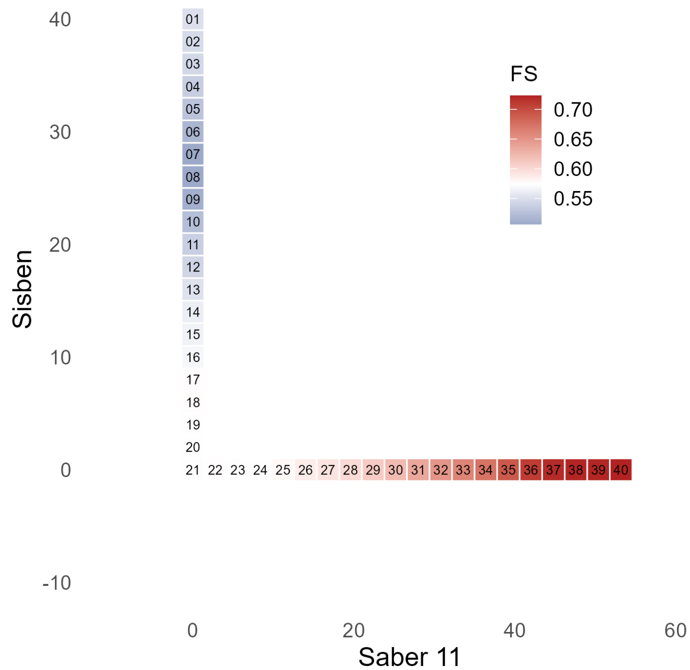


(a) Point estimates.

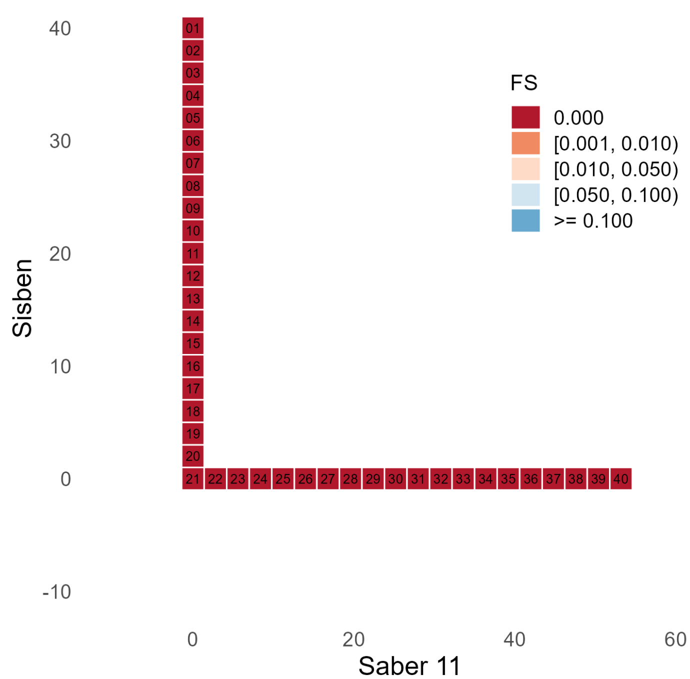


(b) Pointwise p-values.

Figure SA-1: Intention-to-Treat Effects (SPP Application, ITT Bandwidth Selector).

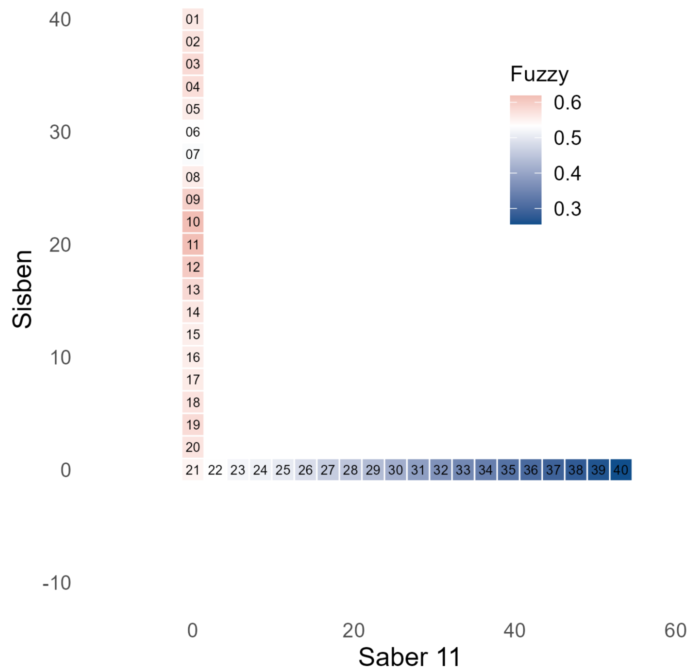


(a) Point estimates.

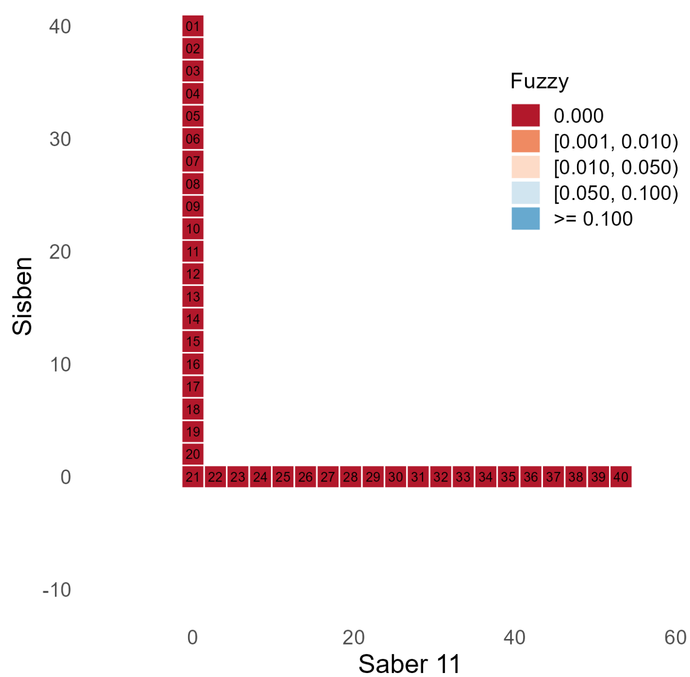


(b) Pointwise p-values.

Figure SA-2: First-Stage Effects (SPP Application, ITT Bandwidth Selector).

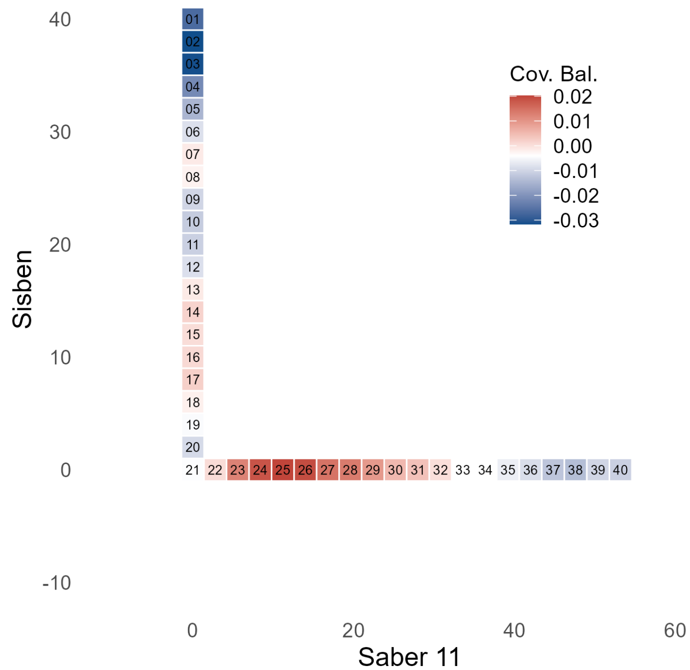


(a) Point estimates.

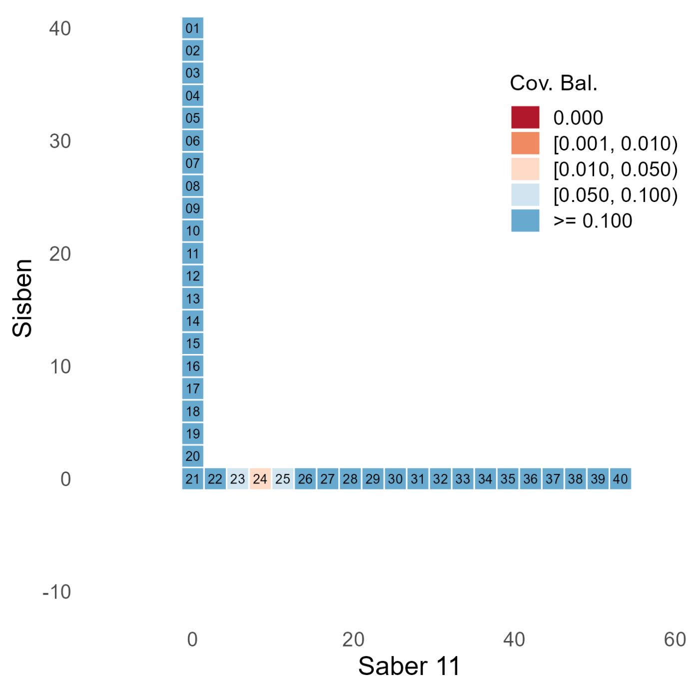


(b) Pointwise p-values.

Figure SA-3: Fuzzy Effects (SPP Application, ITT Bandwidth Selector).



(a) Point estimates.



(b) Pointwise p-values.

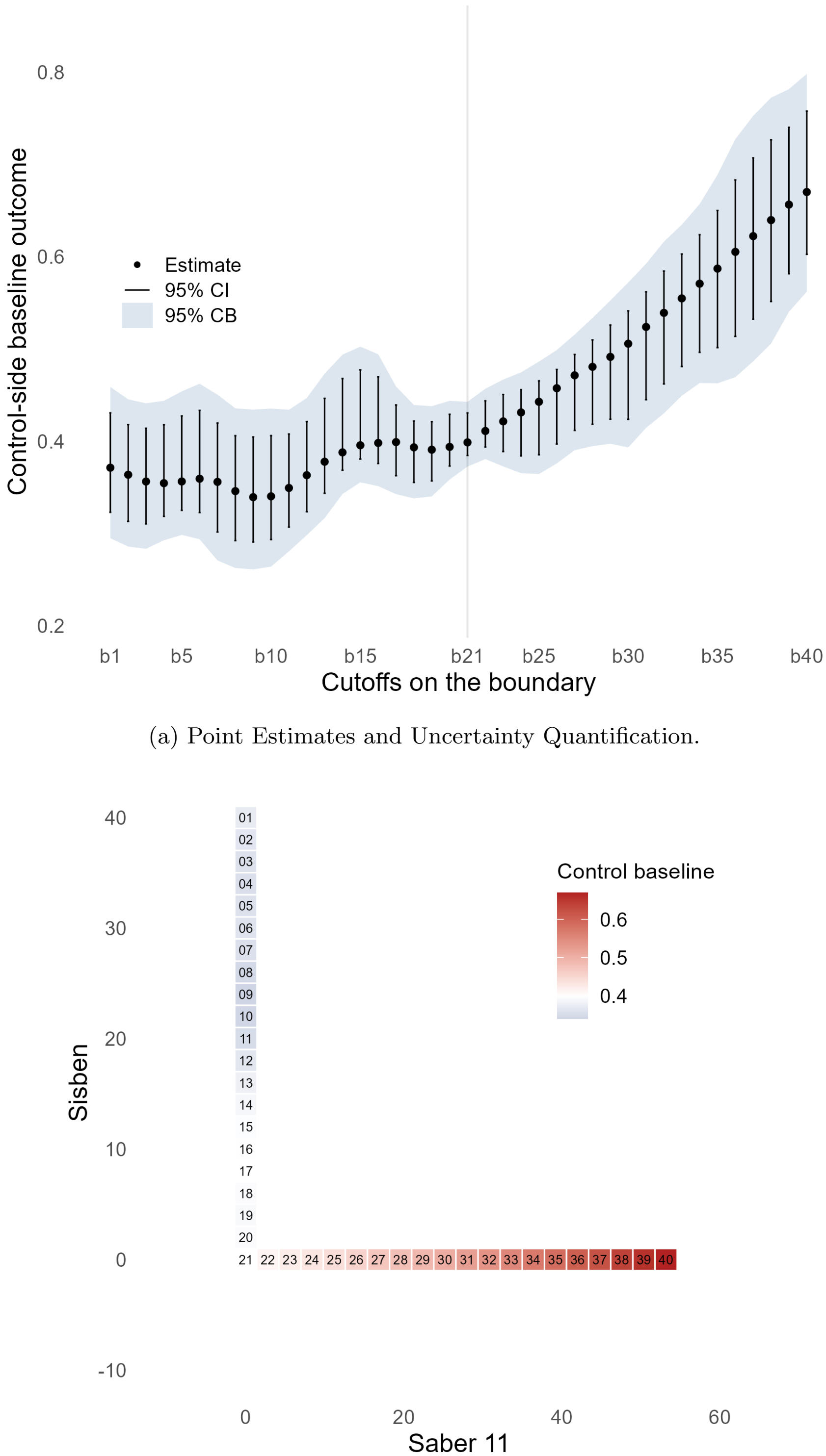
Figure SA-4: Covariate Balance for Mother's Education (SPP Application, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	27.6	12.2	8921		0.371	0.000	(0.295, 0.459)
$\mathbf{b}_2$	26.0	11.5	8998		0.364	0.000	(0.286, 0.445)
$\mathbf{b}_3$	24.6	10.9	8510		0.356	0.000	(0.283, 0.441)
$\mathbf{b}_4$	24.5	10.9	9236		0.355	0.000	(0.293, 0.444)
$\mathbf{b}_5$	23.1	10.2	9044		0.356	0.000	(0.298, 0.454)
$\mathbf{b}_6$	20.7	9.2	7176		0.359	0.000	(0.294, 0.462)
$\mathbf{b}_7$	19.0	8.4	6089		0.356	0.000	(0.271, 0.450)
$\mathbf{b}_8$	19.2	8.5	6725		0.346	0.000	(0.263, 0.436)
$\mathbf{b}_9$	19.2	8.5	6803		0.339	0.000	(0.261, 0.434)
$\mathbf{b}_{10}$	19.3	8.6	6925		0.340	0.000	(0.264, 0.435)
$\mathbf{b}_{11}$	21.1	9.3	8665		0.349	0.000	(0.281, 0.434)
$\mathbf{b}_{12}$	21.8	9.7	9017		0.363	0.000	(0.298, 0.447)
$\mathbf{b}_{13}$	21.1	9.3	8683		0.378	0.000	(0.317, 0.473)
$\mathbf{b}_{14}$	21.8	9.6	9069		0.388	0.000	(0.343, 0.494)
$\mathbf{b}_{15}$	22.2	9.8	10063		0.396	0.000	(0.355, 0.502)
$\mathbf{b}_{16}$	22.5	10.0	10210		0.398	0.000	(0.351, 0.494)
$\mathbf{b}_{17}$	21.3	9.4	9409		0.399	0.000	(0.343, 0.459)
$\mathbf{b}_{18}$	20.7	9.2	9182		0.393	0.000	(0.338, 0.439)
$\mathbf{b}_{19}$	19.3	8.5	8468		0.391	0.000	(0.340, 0.438)
$\mathbf{b}_{20}$	19.4	8.6	8945		0.394	0.000	(0.359, 0.444)
$\mathbf{b}_{21}$	20.9	9.3	10677		0.399	0.000	(0.373, 0.443)
$\mathbf{b}_{22}$	21.5	9.5	10221		0.411	0.000	(0.381, 0.457)
$\mathbf{b}_{23}$	20.3	9.0	7743		0.421	0.000	(0.373, 0.467)
$\mathbf{b}_{24}$	20.5	9.1	7097		0.431	0.000	(0.365, 0.475)
$\mathbf{b}_{25}$	21.0	9.3	6188		0.443	0.000	(0.364, 0.486)
$\mathbf{b}_{26}$	23.2	10.3	6948		0.457	0.000	(0.376, 0.499)
$\mathbf{b}_{27}$	25.1	11.1	7232		0.471	0.000	(0.390, 0.515)
$\mathbf{b}_{28}$	25.2	11.2	5984		0.481	0.000	(0.395, 0.534)
$\mathbf{b}_{29}$	25.6	11.3	5319		0.491	0.000	(0.397, 0.552)
$\mathbf{b}_{30}$	26.3	11.7	4698		0.506	0.000	(0.393, 0.572)
$\mathbf{b}_{31}$	28.5	12.6	4759		0.524	0.000	(0.415, 0.592)
$\mathbf{b}_{32}$	28.9	12.8	4482		0.539	0.000	(0.430, 0.616)
$\mathbf{b}_{33}$	30.8	13.7	4559		0.555	0.000	(0.449, 0.635)
$\mathbf{b}_{34}$	31.2	13.8	4048		0.571	0.000	(0.463, 0.657)
$\mathbf{b}_{35}$	28.1	12.5	2875		0.587	0.000	(0.463, 0.689)
$\mathbf{b}_{36}$	26.0	11.5	2214		0.605	0.000	(0.469, 0.727)
$\mathbf{b}_{37}$	26.8	11.9	1987		0.622	0.000	(0.487, 0.753)
$\mathbf{b}_{38}$	28.5	12.6	2049		0.640	0.000	(0.506, 0.772)
$\mathbf{b}_{39}$	33.6	14.9	2758		0.656	0.000	(0.540, 0.782)
$\mathbf{b}_{40}$	36.3	16.1	2916		0.670	0.000	(0.562, 0.798)
`WBATE`					0.447	0.000	(0.423, 0.473)
`LBATE`					0.670		(0.561, 0.800)

Table SA-5: Control-Side Baseline Outcome (SPP Application, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	ITT / ITT.0 (%)	Fuzzy / ITT.0 (%)
$\mathbf{b}_1$	27.6	12.2	8921	3532	83.631	151.785
$\mathbf{b}_2$	26.0	11.5	8998	3677	85.929	157.241
$\mathbf{b}_3$	24.6	10.9	8510	3678	88.773	163.114
$\mathbf{b}_4$	24.5	10.9	9236	3976	87.145	162.517
$\mathbf{b}_5$	23.1	10.2	9044	3997	82.811	156.772
$\mathbf{b}_6$	20.7	9.2	7176	3525	76.957	148.838
$\mathbf{b}_7$	19.0	8.4	6089	3209	75.122	148.193
$\mathbf{b}_8$	19.2	8.5	6725	3422	81.842	161.505
$\mathbf{b}_9$	19.2	8.5	6803	3516	89.783	175.477
$\mathbf{b}_{10}$	19.3	8.6	6925	3563	95.242	181.913
$\mathbf{b}_{11}$	21.1	9.3	8665	4206	95.023	176.783
$\mathbf{b}_{12}$	21.8	9.7	9017	4456	89.931	165.864
$\mathbf{b}_{13}$	21.1	9.3	8683	4417	85.359	154.546
$\mathbf{b}_{14}$	21.8	9.6	9069	4646	82.049	146.036
$\mathbf{b}_{15}$	22.2	9.8	10063	4977	79.337	140.579
$\mathbf{b}_{16}$	22.5	10.0	10210	5162	79.586	139.992
$\mathbf{b}_{17}$	21.3	9.4	9409	4444	80.823	140.364
$\mathbf{b}_{18}$	20.7	9.2	9182	3850	83.501	144.816
$\mathbf{b}_{19}$	19.3	8.5	8468	3157	85.366	148.486
$\mathbf{b}_{20}$	19.4	8.6	8945	2680	82.952	144.496
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	78.739	137.393
$\mathbf{b}_{22}$	21.5	9.5	10221	2786	74.537	129.406
$\mathbf{b}_{23}$	20.3	9.0	7743	2679	70.713	123.220
$\mathbf{b}_{24}$	20.5	9.1	7097	2856	68.327	119.007
$\mathbf{b}_{25}$	21.0	9.3	6188	3114	65.748	113.536
$\mathbf{b}_{26}$	23.2	10.3	6948	3635	62.000	105.621
$\mathbf{b}_{27}$	25.1	11.1	7232	4098	58.005	97.800
$\mathbf{b}_{28}$	25.2	11.2	5984	4204	56.030	93.353
$\mathbf{b}_{29}$	25.6	11.3	5319	4343	54.053	88.449
$\mathbf{b}_{30}$	26.3	11.7	4698	4560	50.756	81.532
$\mathbf{b}_{31}$	28.5	12.6	4759	4995	47.181	74.136
$\mathbf{b}_{32}$	28.9	12.8	4482	4655	44.580	68.662
$\mathbf{b}_{33}$	30.8	13.7	4559	4766	42.138	63.743
$\mathbf{b}_{34}$	31.2	13.8	4048	4222	39.864	59.187
$\mathbf{b}_{35}$	28.1	12.5	2875	2950	37.353	54.018
$\mathbf{b}_{36}$	26.0	11.5	2214	2243	35.513	50.042
$\mathbf{b}_{37}$	26.8	11.9	1987	2002	33.577	46.713
$\mathbf{b}_{38}$	28.5	12.6	2049	2068	31.279	43.332
$\mathbf{b}_{39}$	33.6	14.9	2758	2787	29.159	40.427
$\mathbf{b}_{40}$	36.3	16.1	2916	2993	27.554	38.108
WBATE					63.008	108.897
LBATE					49.787	92.377

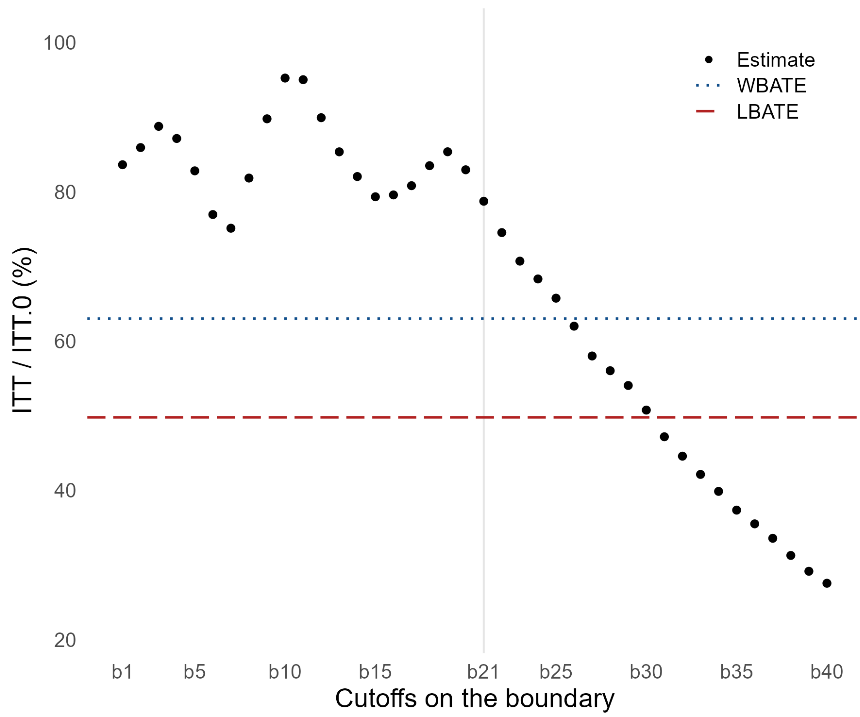
Table SA-6: Treatment Effect Relative to Control-Side Baseline (SPP Application, ITT Bandwidth Selector).



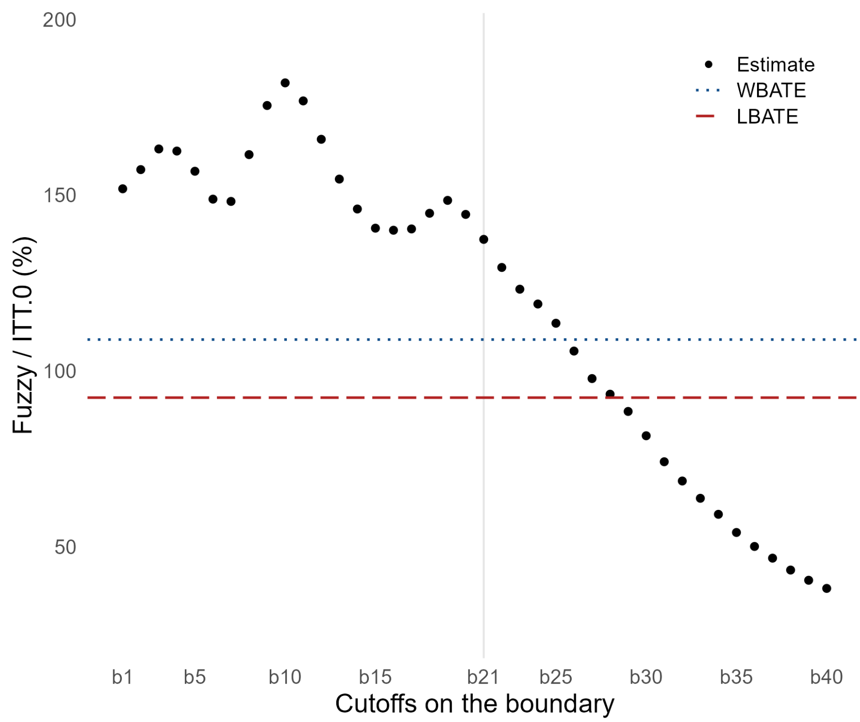
(a) Point Estimates and Uncertainty Quantification.

(b) Heatmap of Point Estimates.

Figure SA-5: Control-Side Baseline Outcome (SPP Application, ITT Bandwidth Selector).



(a) Intention-to-Treat Effect.



(b) Fuzzy Effect.

Figure SA-6: Treatment Effect Relative to Control-Side Baseline (SPP Application, ITT Bandwidth Selector).

SA-10.2 Fuzzy Bandwidth Selector

This subsection reports the same empirical quantities using the bandwidth selected for the fuzzy Wald estimand. These results provide a bandwidth-sensitivity check for the main ITT-bandwidth empirical analysis. We provide the corresponding full boundary-grid results for all 40 evaluation points.

- Table SA-7 reports the reduced-form, or intention-to-treat (ITT), estimates for college enrollment.
- Table SA-8 reports the first-stage estimates for treatment take-up.
- Table SA-9 reports the fuzzy estimates for college enrollment.
- Table SA-10 reports the covariate-balance reduced-form estimates using mother's education as the outcome.

The figures in this subsection complement the corresponding confidence intervals and bands plots for the fuzzy bandwidth selector. For each quantity, we report heat maps along the boundary for the point estimates and associated pointwise robust bias-corrected p-values. Specifically:

- Figure SA-7 reports the reduced-form, or ITT, estimates for college enrollment.
- Figure SA-8 reports the first-stage estimates for treatment take-up.
- Figure SA-9 reports the fuzzy estimates for college enrollment.
- Figure SA-10 reports the covariate-balance reduced-form estimates using mother's education as the outcome.

These plots make it easier to inspect how the estimated effects and pointwise significance patterns vary along the boundary.

In addition, we report baseline estimates and treatment effects relative to control-side baselines.

- Table SA-11 reports the control-side baseline estimates $\widehat{\mu}_{Y,0}(\mathbf{x})$ for college enrollment.
- Table SA-12 reports ITT and Fuzzy treatment effects relative to control-side baseline estimates $\widehat{\mu}_{Y,0}(\mathbf{x})$ for college enrollment.
- Figure SA-11 reports baseline estimates and uncertainty quantification.
- Figure SA-12 reports treatment effects relative to the estimated control-side baseline.

The baseline estimates in Table SA-11 help interpret the economic magnitude of the effects, while Table SA-12 reports the same treatment effects in percentage terms relative to the local control-side baseline.

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	28.2	12.5	9704	3687	0.311	0.000	(0.205, 0.418)
$\mathbf{b}_2$	26.0	11.5	8572	3612	0.313	0.000	(0.218, 0.429)
$\mathbf{b}_3$	24.9	11.0	8590	3720	0.316	0.000	(0.220, 0.425)
$\mathbf{b}_4$	23.2	10.3	8397	3724	0.308	0.000	(0.190, 0.401)
$\mathbf{b}_5$	21.8	9.6	7403	3568	0.291	0.000	(0.154, 0.371)
$\mathbf{b}_6$	20.0	8.9	6977	3427	0.273	0.000	(0.122, 0.351)
$\mathbf{b}_7$	19.7	8.7	6633	3404	0.271	0.000	(0.126, 0.352)
$\mathbf{b}_8$	21.3	9.4	8437	4032	0.289	0.000	(0.156, 0.362)
$\mathbf{b}_9$	21.9	9.7	8859	4218	0.306	0.000	(0.178, 0.378)
$\mathbf{b}_{10}$	22.2	9.8	9828	4469	0.321	0.000	(0.204, 0.399)
$\mathbf{b}_{11}$	22.3	9.9	9906	4584	0.330	0.000	(0.223, 0.415)
$\mathbf{b}_{12}$	22.3	9.9	9949	4710	0.327	0.000	(0.224, 0.414)
$\mathbf{b}_{13}$	20.6	9.1	7995	4229	0.322	0.000	(0.208, 0.412)
$\mathbf{b}_{14}$	20.7	9.2	8168	4350	0.315	0.000	(0.188, 0.390)
$\mathbf{b}_{15}$	21.0	9.3	8370	4493	0.309	0.000	(0.182, 0.380)
$\mathbf{b}_{16}$	20.8	9.2	8278	4560	0.312	0.000	(0.201, 0.398)
$\mathbf{b}_{17}$	19.7	8.7	7544	4057	0.319	0.000	(0.241, 0.428)
$\mathbf{b}_{18}$	19.6	8.7	7997	3623	0.326	0.000	(0.262, 0.431)
$\mathbf{b}_{19}$	20.4	9.0	9592	3367	0.333	0.000	(0.266, 0.432)
$\mathbf{b}_{20}$	20.0	8.9	9225	2755	0.327	0.000	(0.234, 0.430)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.314	0.000	(0.159, 0.469)
$\mathbf{b}_{22}$	22.0	9.8	10830	2843	0.307	0.000	(0.186, 0.414)
$\mathbf{b}_{23}$	21.3	9.4	8679	2871	0.298	0.000	(0.190, 0.401)
$\mathbf{b}_{24}$	21.7	9.6	8114	3140	0.294	0.000	(0.201, 0.403)
$\mathbf{b}_{25}$	21.8	9.6	6858	3223	0.289	0.000	(0.212, 0.416)
$\mathbf{b}_{26}$	24.3	10.8	7784	3867	0.282	0.000	(0.209, 0.401)
$\mathbf{b}_{27}$	27.4	12.1	8856	4564	0.271	0.000	(0.190, 0.370)
$\mathbf{b}_{28}$	26.9	11.9	7294	4540	0.267	0.000	(0.179, 0.373)
$\mathbf{b}_{29}$	26.7	11.8	5916	4593	0.264	0.000	(0.170, 0.377)
$\mathbf{b}_{30}$	27.3	12.1	5250	4759	0.256	0.000	(0.155, 0.378)
$\mathbf{b}_{31}$	30.1	13.3	5758	5285	0.247	0.000	(0.141, 0.356)
$\mathbf{b}_{32}$	31.8	14.1	5266	5607	0.240	0.000	(0.136, 0.359)
$\mathbf{b}_{33}$	32.1	14.2	4925	5218	0.234	0.000	(0.122, 0.355)
$\mathbf{b}_{34}$	31.7	14.0	4323	4517	0.228	0.000	(0.107, 0.355)
$\mathbf{b}_{35}$	28.8	12.7	3022	3135	0.220	0.000	(0.083, 0.368)
$\mathbf{b}_{36}$	27.0	12.0	2452	2467	0.215	0.000	(0.072, 0.386)
$\mathbf{b}_{37}$	28.1	12.4	2318	2326	0.208	0.000	(0.069, 0.383)
$\mathbf{b}_{38}$	29.7	13.1	2330	2353	0.200	0.000	(0.058, 0.370)
$\mathbf{b}_{39}$	35.2	15.6	2954	3036	0.191	0.000	(0.061, 0.341)
$\mathbf{b}_{40}$	39.6	17.5	3538	3725	0.185	0.000	(0.057, 0.322)
WBATE					0.281	0.000	(0.246, 0.313)
LBATE					0.333		(0.264, 0.473)

Table SA-7: Intention-to-Treat Effects (SPP Application, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	28.2	12.5	9704	3687	0.550	0.000	(0.496, 0.650)
$\mathbf{b}_2$	26.0	11.5	8572	3612	0.547	0.000	(0.498, 0.650)
$\mathbf{b}_3$	24.9	11.0	8590	3720	0.543	0.000	(0.495, 0.644)
$\mathbf{b}_4$	23.2	10.3	8397	3724	0.539	0.000	(0.481, 0.630)
$\mathbf{b}_5$	21.8	9.6	7403	3568	0.529	0.000	(0.463, 0.613)
$\mathbf{b}_6$	20.0	8.9	6977	3427	0.516	0.000	(0.440, 0.594)
$\mathbf{b}_7$	19.7	8.7	6633	3404	0.509	0.000	(0.424, 0.579)
$\mathbf{b}_8$	21.3	9.4	8437	4032	0.510	0.000	(0.422, 0.569)
$\mathbf{b}_9$	21.9	9.7	8859	4218	0.514	0.000	(0.419, 0.564)
$\mathbf{b}_{10}$	22.2	9.8	9828	4469	0.526	0.000	(0.431, 0.575)
$\mathbf{b}_{11}$	22.3	9.9	9906	4584	0.537	0.000	(0.452, 0.594)
$\mathbf{b}_{12}$	22.3	9.9	9949	4710	0.542	0.000	(0.465, 0.604)
$\mathbf{b}_{13}$	20.6	9.1	7995	4229	0.552	0.000	(0.471, 0.615)
$\mathbf{b}_{14}$	20.7	9.2	8168	4350	0.561	0.000	(0.477, 0.616)
$\mathbf{b}_{15}$	21.0	9.3	8370	4493	0.563	0.000	(0.478, 0.614)
$\mathbf{b}_{16}$	20.8	9.2	8278	4560	0.567	0.000	(0.489, 0.624)
$\mathbf{b}_{17}$	19.7	8.7	7544	4057	0.575	0.000	(0.499, 0.638)
$\mathbf{b}_{18}$	19.6	8.7	7997	3623	0.575	0.000	(0.497, 0.643)
$\mathbf{b}_{19}$	20.4	9.0	9592	3367	0.576	0.000	(0.493, 0.651)
$\mathbf{b}_{20}$	20.0	8.9	9225	2755	0.574	0.000	(0.471, 0.670)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.573	0.000	(0.428, 0.759)
$\mathbf{b}_{22}$	22.0	9.8	10830	2843	0.576	0.000	(0.470, 0.709)
$\mathbf{b}_{23}$	21.3	9.4	8679	2871	0.574	0.000	(0.477, 0.689)
$\mathbf{b}_{24}$	21.7	9.6	8114	3140	0.575	0.000	(0.478, 0.676)
$\mathbf{b}_{25}$	21.8	9.6	6858	3223	0.580	0.000	(0.486, 0.679)
$\mathbf{b}_{26}$	24.3	10.8	7784	3867	0.588	0.000	(0.494, 0.672)
$\mathbf{b}_{27}$	27.4	12.1	8856	4564	0.597	0.000	(0.498, 0.663)
$\mathbf{b}_{28}$	26.9	11.9	7294	4540	0.603	0.000	(0.504, 0.675)
$\mathbf{b}_{29}$	26.7	11.8	5916	4593	0.613	0.000	(0.515, 0.691)
$\mathbf{b}_{30}$	27.3	12.1	5250	4759	0.624	0.000	(0.526, 0.705)
$\mathbf{b}_{31}$	30.1	13.3	5758	5285	0.637	0.000	(0.541, 0.712)
$\mathbf{b}_{32}$	31.8	14.1	5266	5607	0.649	0.000	(0.557, 0.726)
$\mathbf{b}_{33}$	32.1	14.2	4925	5218	0.660	0.000	(0.568, 0.744)
$\mathbf{b}_{34}$	31.7	14.0	4323	4517	0.673	0.000	(0.581, 0.768)
$\mathbf{b}_{35}$	28.8	12.7	3022	3135	0.691	0.000	(0.591, 0.805)
$\mathbf{b}_{36}$	27.0	12.0	2452	2467	0.707	0.000	(0.606, 0.840)
$\mathbf{b}_{37}$	28.1	12.4	2318	2326	0.716	0.000	(0.617, 0.853)
$\mathbf{b}_{38}$	29.7	13.1	2330	2353	0.720	0.000	(0.618, 0.856)
$\mathbf{b}_{39}$	35.2	15.6	2954	3036	0.719	0.000	(0.626, 0.845)
$\mathbf{b}_{40}$	39.6	17.5	3538	3725	0.719	0.000	(0.635, 0.845)
WBATE					0.593	0.000	(0.567, 0.619)
LBATE					0.720		(0.636, 0.855)

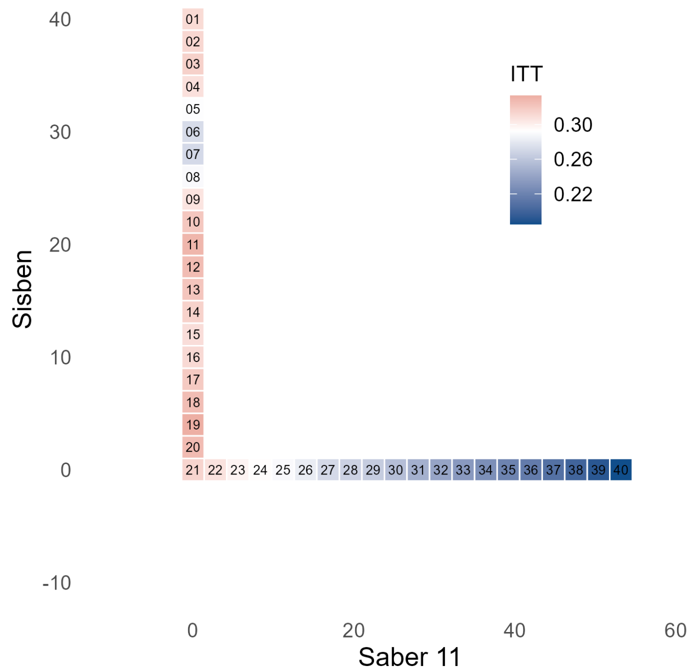
Table SA-8: First-Stage Effects (SPP Application, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	28.2	12.5	9704	3687	0.565	0.000	(0.381, 0.708)
$\mathbf{b}_2$	26.0	11.5	8572	3612	0.572	0.000	(0.401, 0.727)
$\mathbf{b}_3$	24.9	11.0	8590	3720	0.581	0.000	(0.407, 0.726)
$\mathbf{b}_4$	23.2	10.3	8397	3724	0.572	0.000	(0.365, 0.700)
$\mathbf{b}_5$	21.8	9.6	7403	3568	0.551	0.000	(0.308, 0.667)
$\mathbf{b}_6$	20.0	8.9	6977	3427	0.528	0.000	(0.261, 0.655)
$\mathbf{b}_7$	19.7	8.7	6633	3404	0.533	0.000	(0.275, 0.679)
$\mathbf{b}_8$	21.3	9.4	8437	4032	0.567	0.000	(0.338, 0.708)
$\mathbf{b}_9$	21.9	9.7	8859	4218	0.595	0.000	(0.387, 0.746)
$\mathbf{b}_{10}$	22.2	9.8	9828	4469	0.610	0.000	(0.428, 0.771)
$\mathbf{b}_{11}$	22.3	9.9	9906	4584	0.613	0.000	(0.446, 0.772)
$\mathbf{b}_{12}$	22.3	9.9	9949	4710	0.602	0.000	(0.438, 0.756)
$\mathbf{b}_{13}$	20.6	9.1	7995	4229	0.583	0.000	(0.401, 0.741)
$\mathbf{b}_{14}$	20.7	9.2	8168	4350	0.562	0.000	(0.363, 0.695)
$\mathbf{b}_{15}$	21.0	9.3	8370	4493	0.549	0.000	(0.351, 0.677)
$\mathbf{b}_{16}$	20.8	9.2	8278	4560	0.549	0.000	(0.378, 0.698)
$\mathbf{b}_{17}$	19.7	8.7	7544	4057	0.556	0.000	(0.441, 0.736)
$\mathbf{b}_{18}$	19.6	8.7	7997	3623	0.567	0.000	(0.482, 0.734)
$\mathbf{b}_{19}$	20.4	9.0	9592	3367	0.579	0.000	(0.490, 0.731)
$\mathbf{b}_{20}$	20.0	8.9	9225	2755	0.569	0.000	(0.446, 0.717)
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	0.548	0.000	(0.343, 0.715)
$\mathbf{b}_{22}$	22.0	9.8	10830	2843	0.533	0.000	(0.365, 0.652)
$\mathbf{b}_{23}$	21.3	9.4	8679	2871	0.519	0.000	(0.364, 0.649)
$\mathbf{b}_{24}$	21.7	9.6	8114	3140	0.510	0.000	(0.378, 0.667)
$\mathbf{b}_{25}$	21.8	9.6	6858	3223	0.499	0.000	(0.389, 0.689)
$\mathbf{b}_{26}$	24.3	10.8	7784	3867	0.479	0.000	(0.380, 0.666)
$\mathbf{b}_{27}$	27.4	12.1	8856	4564	0.455	0.000	(0.347, 0.618)
$\mathbf{b}_{28}$	26.9	11.9	7294	4540	0.443	0.000	(0.324, 0.613)
$\mathbf{b}_{29}$	26.7	11.8	5916	4593	0.430	0.000	(0.301, 0.606)
$\mathbf{b}_{30}$	27.3	12.1	5250	4759	0.410	0.000	(0.271, 0.596)
$\mathbf{b}_{31}$	30.1	13.3	5758	5285	0.388	0.000	(0.242, 0.551)
$\mathbf{b}_{32}$	31.8	14.1	5266	5607	0.370	0.000	(0.228, 0.544)
$\mathbf{b}_{33}$	32.1	14.2	4925	5218	0.354	0.000	(0.201, 0.526)
$\mathbf{b}_{34}$	31.7	14.0	4323	4517	0.338	0.000	(0.172, 0.512)
$\mathbf{b}_{35}$	28.8	12.7	3022	3135	0.319	0.000	(0.133, 0.513)
$\mathbf{b}_{36}$	27.0	12.0	2452	2467	0.304	0.000	(0.113, 0.520)
$\mathbf{b}_{37}$	28.1	12.4	2318	2326	0.291	0.000	(0.105, 0.510)
$\mathbf{b}_{38}$	29.7	13.1	2330	2353	0.278	0.000	(0.089, 0.492)
$\mathbf{b}_{39}$	35.2	15.6	2954	3036	0.266	0.000	(0.091, 0.455)
$\mathbf{b}_{40}$	39.6	17.5	3538	3725	0.257	0.000	(0.086, 0.426)
`WBATE`					0.485	0.000	(0.433, 0.531)
`LBATE`					0.613		(0.484, 0.781)

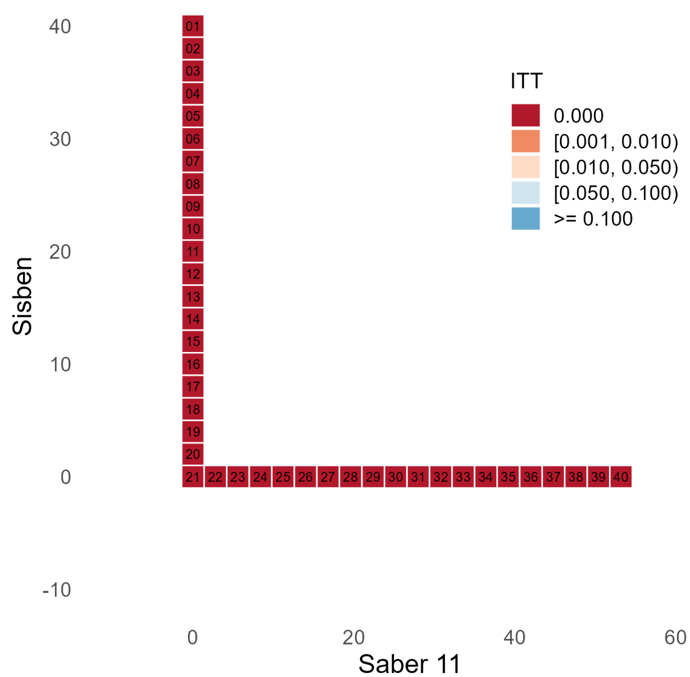
Table SA-9: Fuzzy Effects (SPP Application, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	27.2	12.1	8688	3446	-0.027	0.172	(-0.153, 0.058)
$\mathbf{b}_2$	25.8	11.5	8428	3552	-0.032	0.116	(-0.157, 0.049)
$\mathbf{b}_3$	25.4	11.3	9246	3841	-0.030	0.108	(-0.153, 0.046)
$\mathbf{b}_4$	25.4	11.3	10066	4154	-0.021	0.215	(-0.134, 0.056)
$\mathbf{b}_5$	24.4	10.8	9822	4176	-0.015	0.402	(-0.121, 0.068)
$\mathbf{b}_6$	22.7	10.1	8801	3951	-0.008	0.739	(-0.107, 0.086)
$\mathbf{b}_7$	21.9	9.7	8262	3891	-0.001	0.985	(-0.098, 0.097)
$\mathbf{b}_8$	21.4	9.5	8372	3993	-0.002	0.678	(-0.114, 0.086)
$\mathbf{b}_9$	21.6	9.6	8604	4098	-0.010	0.356	(-0.131, 0.070)
$\mathbf{b}_{10}$	20.9	9.3	8049	3966	-0.012	0.262	(-0.140, 0.064)
$\mathbf{b}_{11}$	20.7	9.2	7952	3986	-0.011	0.456	(-0.127, 0.076)
$\mathbf{b}_{12}$	21.5	9.5	8766	4328	-0.008	0.935	(-0.100, 0.095)
$\mathbf{b}_{13}$	21.0	9.3	8090	4265	-0.001	0.850	(-0.093, 0.106)
$\mathbf{b}_{14}$	21.0	9.3	8207	4358	0.002	0.848	(-0.090, 0.103)
$\mathbf{b}_{15}$	20.4	9.0	8059	4332	0.000	0.908	(-0.092, 0.099)
$\mathbf{b}_{16}$	21.6	9.6	8936	4750	0.001	0.550	(-0.071, 0.106)
$\mathbf{b}_{17}$	22.5	10.0	10627	4667	0.002	0.510	(-0.057, 0.088)
$\mathbf{b}_{18}$	20.1	8.9	8788	3746	-0.002	0.472	(-0.058, 0.094)
$\mathbf{b}_{19}$	18.7	8.3	7730	3014	-0.005	0.816	(-0.084, 0.072)
$\mathbf{b}_{20}$	20.1	8.9	9828	2825	-0.009	0.613	(-0.102, 0.073)
$\mathbf{b}_{21}$	19.9	8.8	9436	2206	-0.004	0.505	(-0.117, 0.184)
$\mathbf{b}_{22}$	20.8	9.2	9760	2625	0.001	0.337	(-0.075, 0.145)
$\mathbf{b}_{23}$	17.8	7.9	5890	2194	0.013	0.061	(-0.041, 0.177)
$\mathbf{b}_{24}$	17.7	7.9	4911	2316	0.019	0.035	(-0.032, 0.182)
$\mathbf{b}_{25}$	18.5	8.2	4641	2566	0.021	0.056	(-0.038, 0.169)
$\mathbf{b}_{26}$	20.2	9.0	4845	3018	0.020	0.114	(-0.046, 0.149)
$\mathbf{b}_{27}$	22.0	9.8	5110	3458	0.015	0.238	(-0.057, 0.130)
$\mathbf{b}_{28}$	22.5	10.0	4215	3666	0.013	0.312	(-0.065, 0.130)
$\mathbf{b}_{29}$	23.0	10.2	3754	3828	0.009	0.413	(-0.076, 0.133)
$\mathbf{b}_{30}$	24.4	10.8	3836	3936	0.005	0.556	(-0.084, 0.125)
$\mathbf{b}_{31}$	29.3	13.0	5139	5095	0.004	0.683	(-0.080, 0.104)
$\mathbf{b}_{32}$	30.0	13.3	4744	4958	0.000	0.963	(-0.093, 0.096)
$\mathbf{b}_{33}$	27.7	12.3	3587	3720	-0.004	0.799	(-0.116, 0.098)
$\mathbf{b}_{34}$	26.1	11.6	2714	2786	-0.004	0.667	(-0.134, 0.101)
$\mathbf{b}_{35}$	23.1	10.3	1825	1842	-0.006	0.639	(-0.155, 0.113)
$\mathbf{b}_{36}$	21.2	9.4	1396	1393	-0.009	0.528	(-0.171, 0.112)
$\mathbf{b}_{37}$	21.9	9.7	1350	1316	-0.012	0.429	(-0.173, 0.101)
$\mathbf{b}_{38}$	23.1	10.2	1313	1295	-0.013	0.382	(-0.168, 0.093)
$\mathbf{b}_{39}$	29.5	13.1	2021	2033	-0.010	0.423	(-0.133, 0.077)
$\mathbf{b}_{40}$	27.8	12.3	1556	1506	-0.011	0.516	(-0.133, 0.086)
`WBATE`					-0.004	0.951	(-0.030, 0.028)
`LBATE`					0.021		(-0.033, 0.185)

Table SA-10: Covariate Balance for Mother's Education (SPP Application, Fuzzy Bandwidth Selector).

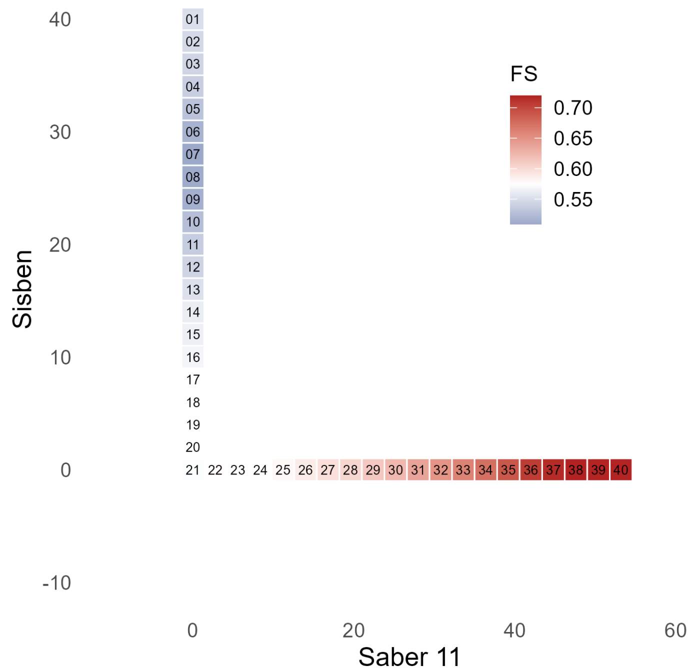


(a) Point estimates.

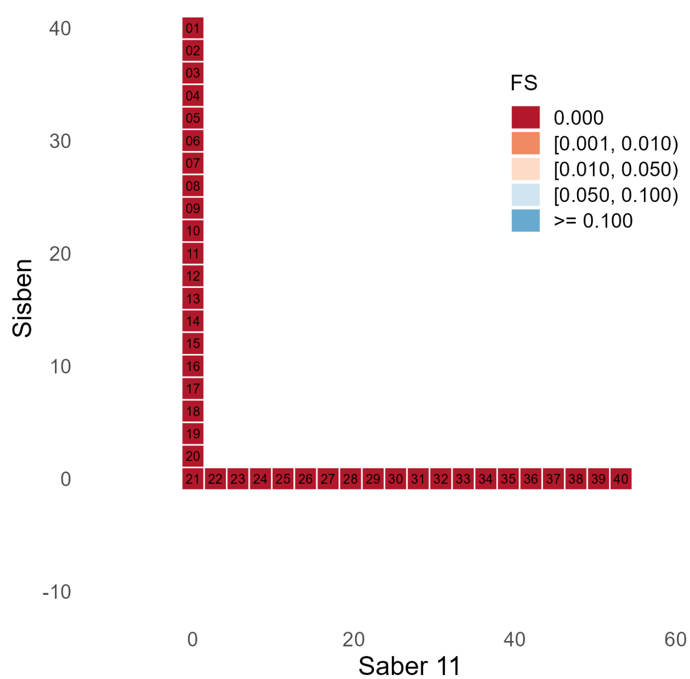


(b) Pointwise p-values.

Figure SA-7: Intention-to-Treat Effects (SPP Application, Fuzzy Bandwidth Selector).

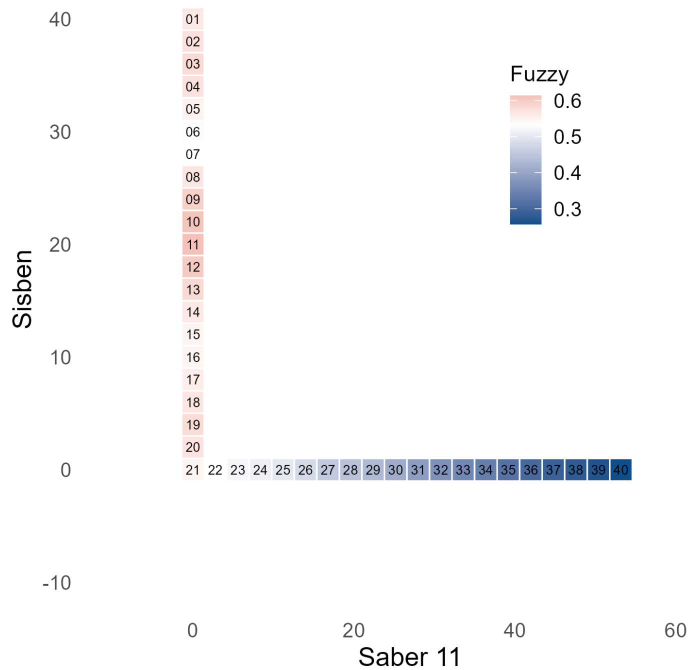


(a) Point estimates.

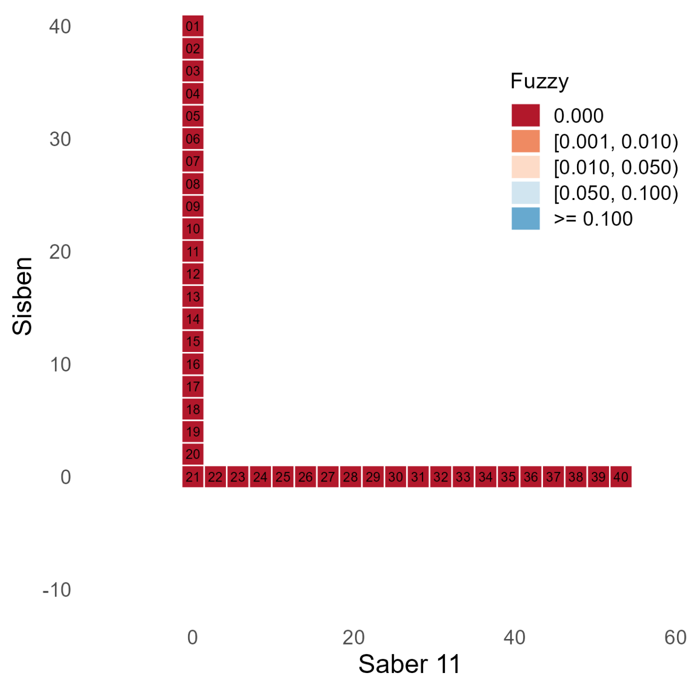


(b) Pointwise p-values.

Figure SA-8: First-Stage Effects (SPP Application, Fuzzy Bandwidth Selector).

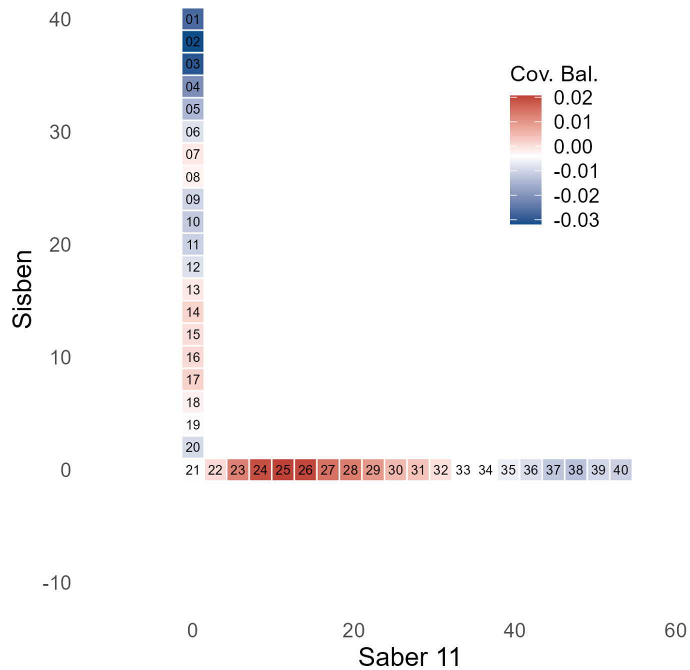


(a) Point estimates.

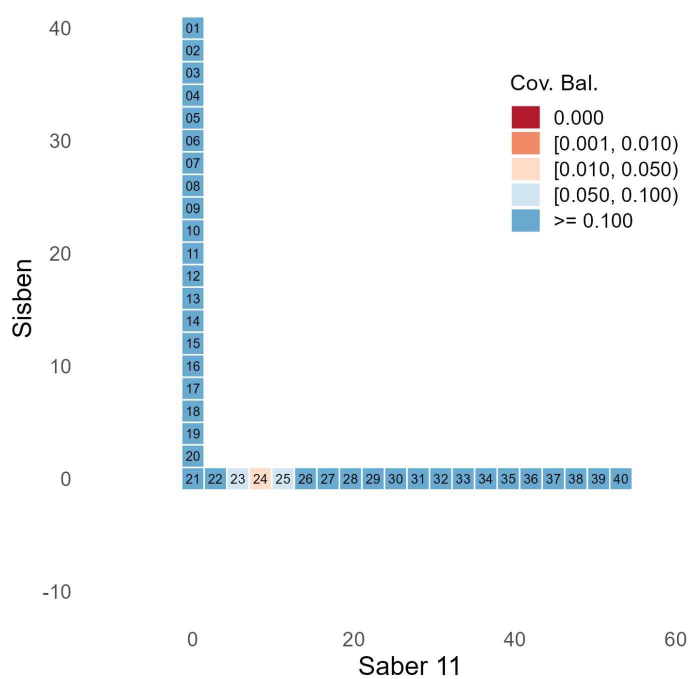


(b) Pointwise p-values.

Figure SA-9: Fuzzy Effects (SPP Application, Fuzzy Bandwidth Selector).



(a) Point estimates.



(b) Pointwise p-values.

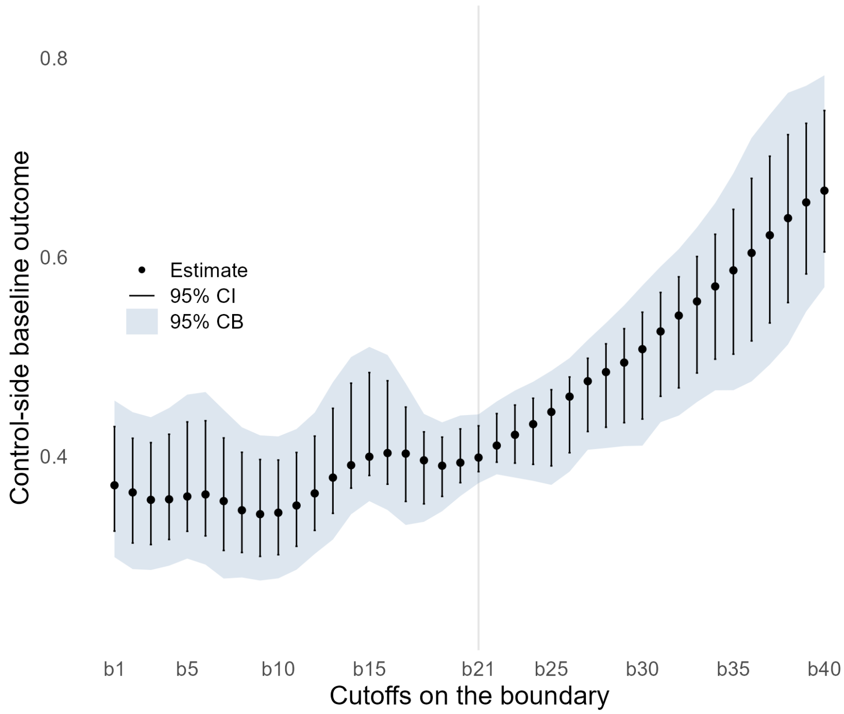
Figure SA-10: Covariate Balance for Mother's Education (SPP Application, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Estimate	p-value	95% CI
$\mathbf{b}_1$	28.2	12.5	9704		0.371	0.000	(0.299, 0.456)
$\mathbf{b}_2$	26.0	11.5	8572		0.364	0.000	(0.287, 0.444)
$\mathbf{b}_3$	24.9	11.0	8590		0.356	0.000	(0.286, 0.439)
$\mathbf{b}_4$	23.2	10.3	8397		0.357	0.000	(0.290, 0.448)
$\mathbf{b}_5$	21.8	9.6	7403		0.360	0.000	(0.297, 0.462)
$\mathbf{b}_6$	20.0	8.9	6977		0.362	0.000	(0.291, 0.465)
$\mathbf{b}_7$	19.7	8.7	6633		0.355	0.000	(0.277, 0.447)
$\mathbf{b}_8$	21.3	9.4	8437		0.346	0.000	(0.278, 0.429)
$\mathbf{b}_9$	21.9	9.7	8859		0.342	0.000	(0.275, 0.421)
$\mathbf{b}_{10}$	22.2	9.8	9828		0.343	0.000	(0.277, 0.420)
$\mathbf{b}_{11}$	22.3	9.9	9906		0.351	0.000	(0.286, 0.427)
$\mathbf{b}_{12}$	22.3	9.9	9949		0.363	0.000	(0.302, 0.444)
$\mathbf{b}_{13}$	20.6	9.1	7995		0.379	0.000	(0.316, 0.474)
$\mathbf{b}_{14}$	20.7	9.2	8168		0.391	0.000	(0.342, 0.500)
$\mathbf{b}_{15}$	21.0	9.3	8370		0.400	0.000	(0.355, 0.510)
$\mathbf{b}_{16}$	20.8	9.2	8278		0.403	0.000	(0.346, 0.502)
$\mathbf{b}_{17}$	19.7	8.7	7544		0.403	0.000	(0.331, 0.473)
$\mathbf{b}_{18}$	19.6	8.7	7997		0.396	0.000	(0.334, 0.443)
$\mathbf{b}_{19}$	20.4	9.0	9592		0.391	0.000	(0.345, 0.434)
$\mathbf{b}_{20}$	20.0	8.9	9225		0.394	0.000	(0.360, 0.441)
$\mathbf{b}_{21}$	20.9	9.3	10677		0.399	0.000	(0.373, 0.442)
$\mathbf{b}_{22}$	22.0	9.8	10830		0.411	0.000	(0.382, 0.455)
$\mathbf{b}_{23}$	21.3	9.4	8679		0.422	0.000	(0.379, 0.466)
$\mathbf{b}_{24}$	21.7	9.6	8114		0.432	0.000	(0.375, 0.475)
$\mathbf{b}_{25}$	21.8	9.6	6858		0.445	0.000	(0.371, 0.486)
$\mathbf{b}_{26}$	24.3	10.8	7784		0.460	0.000	(0.385, 0.499)
$\mathbf{b}_{27}$	27.4	12.1	8856		0.475	0.000	(0.407, 0.517)
$\mathbf{b}_{28}$	26.9	11.9	7294		0.485	0.000	(0.408, 0.534)
$\mathbf{b}_{29}$	26.7	11.8	5916		0.494	0.000	(0.410, 0.552)
$\mathbf{b}_{30}$	27.3	12.1	5250		0.508	0.000	(0.411, 0.572)
$\mathbf{b}_{31}$	30.1	13.3	5758		0.525	0.000	(0.434, 0.591)
$\mathbf{b}_{32}$	31.8	14.1	5266		0.541	0.000	(0.441, 0.608)
$\mathbf{b}_{33}$	32.1	14.2	4925		0.556	0.000	(0.455, 0.630)
$\mathbf{b}_{34}$	31.7	14.0	4323		0.571	0.000	(0.466, 0.654)
$\mathbf{b}_{35}$	28.8	12.7	3022		0.587	0.000	(0.466, 0.684)
$\mathbf{b}_{36}$	27.0	12.0	2452		0.604	0.000	(0.475, 0.720)
$\mathbf{b}_{37}$	28.1	12.4	2318		0.622	0.000	(0.492, 0.743)
$\mathbf{b}_{38}$	29.7	13.1	2330		0.639	0.000	(0.512, 0.765)
$\mathbf{b}_{39}$	35.2	15.6	2954		0.655	0.000	(0.546, 0.772)
$\mathbf{b}_{40}$	39.6	17.5	3538		0.667	0.000	(0.570, 0.783)
`WBATE`					0.448	0.000	(0.425, 0.474)
`LBATE`					0.667		(0.568, 0.784)

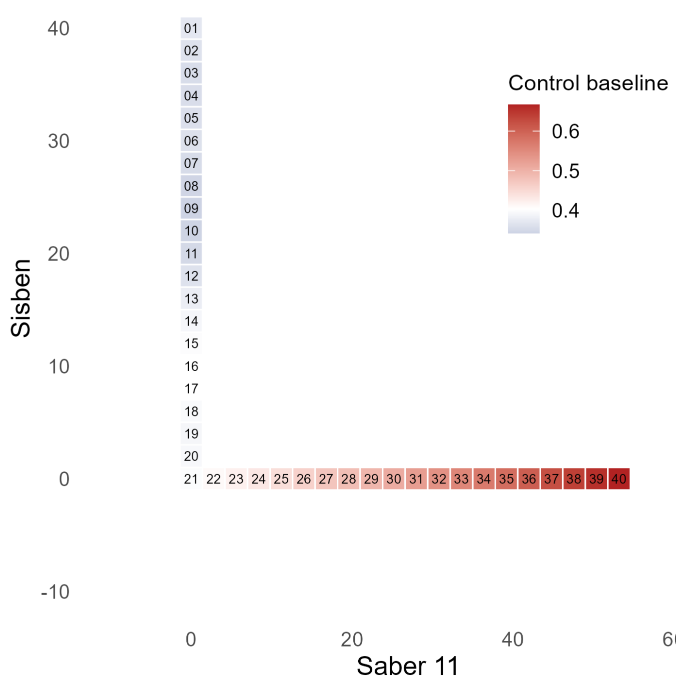
Table SA-11: Control-Side Baseline Outcome (SPP Application, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	ITT / ITT.0 (%)	Fuzzy / ITT.0 (%)
$\mathbf{b}_1$	28.2	12.5	9704	3687	83.801	152.231
$\mathbf{b}_2$	26.0	11.5	8572	3612	85.950	157.238
$\mathbf{b}_3$	24.9	11.0	8590	3720	88.686	163.182
$\mathbf{b}_4$	23.2	10.3	8397	3724	86.320	160.201
$\mathbf{b}_5$	21.8	9.6	7403	3568	80.977	153.111
$\mathbf{b}_6$	20.0	8.9	6977	3427	75.410	146.062
$\mathbf{b}_7$	19.7	8.7	6633	3404	76.445	150.264
$\mathbf{b}_8$	21.3	9.4	8437	4032	83.668	164.054
$\mathbf{b}_9$	21.9	9.7	8859	4218	89.489	174.020
$\mathbf{b}_{10}$	22.2	9.8	9828	4469	93.492	177.775
$\mathbf{b}_{11}$	22.3	9.9	9906	4584	94.019	174.994
$\mathbf{b}_{12}$	22.3	9.9	9949	4710	90.043	166.056
$\mathbf{b}_{13}$	20.6	9.1	7995	4229	85.004	153.958
$\mathbf{b}_{14}$	20.7	9.2	8168	4350	80.630	143.619
$\mathbf{b}_{15}$	21.0	9.3	8370	4493	77.426	137.434
$\mathbf{b}_{16}$	20.8	9.2	8278	4560	77.256	136.138
$\mathbf{b}_{17}$	19.7	8.7	7544	4057	79.328	138.034
$\mathbf{b}_{18}$	19.6	8.7	7997	3623	82.308	143.110
$\mathbf{b}_{19}$	20.4	9.0	9592	3367	85.368	148.263
$\mathbf{b}_{20}$	20.0	8.9	9225	2755	83.046	144.554
$\mathbf{b}_{21}$	20.9	9.3	10677	2442	78.747	137.406
$\mathbf{b}_{22}$	22.0	9.8	10830	2843	74.724	129.677
$\mathbf{b}_{23}$	21.3	9.4	8679	2871	70.735	123.194
$\mathbf{b}_{24}$	21.7	9.6	8114	3140	67.914	118.071
$\mathbf{b}_{25}$	21.8	9.6	6858	3223	65.063	112.204
$\mathbf{b}_{26}$	24.3	10.8	7784	3867	61.248	104.183
$\mathbf{b}_{27}$	27.4	12.1	8856	4564	57.067	95.654
$\mathbf{b}_{28}$	26.9	11.9	7294	4540	55.052	91.338
$\mathbf{b}_{29}$	26.7	11.8	5916	4593	53.365	87.115
$\mathbf{b}_{30}$	27.3	12.1	5250	4759	50.399	80.804
$\mathbf{b}_{31}$	30.1	13.3	5758	5285	47.095	73.880
$\mathbf{b}_{32}$	31.8	14.1	5266	5607	44.339	68.325
$\mathbf{b}_{33}$	32.1	14.2	4925	5218	42.053	63.694
$\mathbf{b}_{34}$	31.7	14.0	4323	4517	39.874	59.262
$\mathbf{b}_{35}$	28.8	12.7	3022	3135	37.509	54.321
$\mathbf{b}_{36}$	27.0	12.0	2452	2467	35.590	50.375
$\mathbf{b}_{37}$	28.1	12.4	2318	2326	33.480	46.767
$\mathbf{b}_{38}$	29.7	13.1	2330	2353	31.284	43.456
$\mathbf{b}_{39}$	35.2	15.6	2954	3036	29.205	40.612
$\mathbf{b}_{40}$	39.6	17.5	3538	3725	27.726	38.547
`WBATE`					62.655	108.222
`LBATE`					49.984	91.972

Table SA-12: Treatment Effect Relative to Control-Side Baseline (SPP Application, Fuzzy Bandwidth Selector).

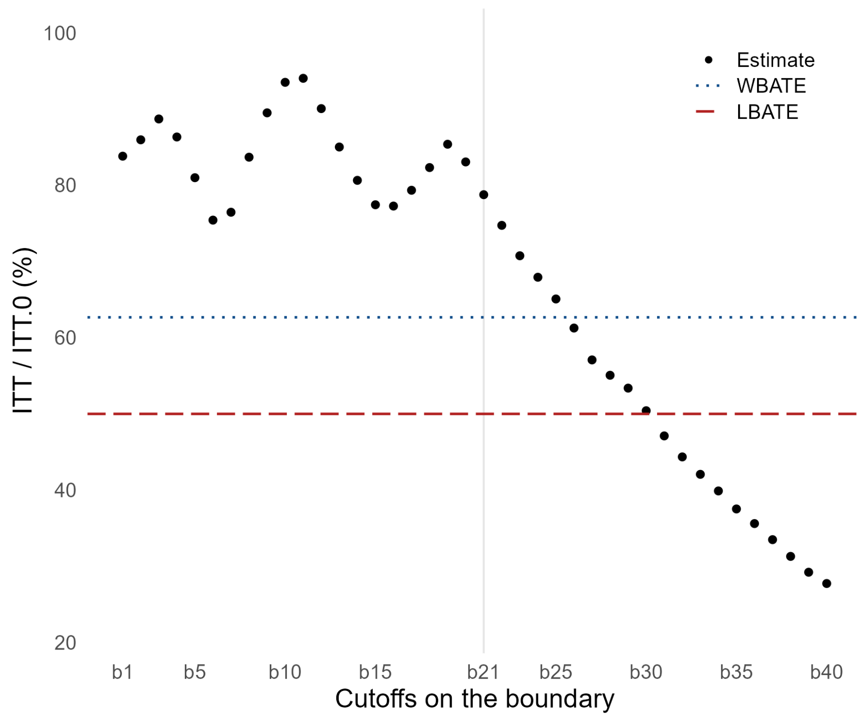


(a) Point Estimates and Uncertainty Quantification.

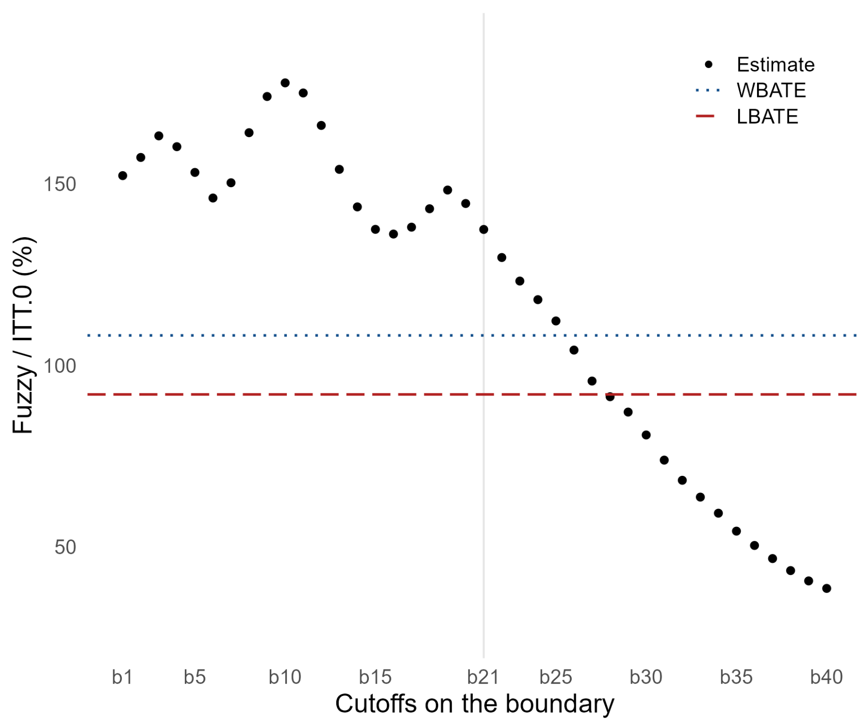


(b) Heatmap of Point Estimates.

Figure SA-11: Control-Side Baseline Outcome (SPP Application, Fuzzy Bandwidth Selector).



(a) Intention-to-Treat Effect.



(b) Fuzzy Effect.

Figure SA-12: Treatment Effect Relative to Control-Side Baseline (SPP Application, Fuzzy Bandwidth Selector).

SA-11 Simulation Study

The score $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$ is generated from the distribution $(100 \cdot \mathsf{Beta}(3,4) - 25,\ 100 \cdot \mathsf{Beta}(3,4) - 25)^\top$ with independent components, which approximately reproduces salient features of the SPP dataset. We employ the same L-shaped treatment assignment boundary as in the empirical application, featuring a kink at $\mathbf{x} = (0,0)^\top$. To reduce computation burden we consider 21 evaluation points with the kink corresponding to the evaluation grid point $\mathbf{b}_{11}$.

The simulation design sets $n = 10,000$ and uses $5,000$ Monte Carlo replications. We evaluate performance in terms of point estimation and uncertainty quantification, both pointwise and uniformly over the boundary $\mathscr{B}$. The Monte Carlo design jointly generates the reduced-form outcome and the treatment status, so that the sharp intention-to-treat (ITT), first-stage (FS), and fuzzy analyses can all be performed on the same simulated samples.

SA-11.1 Data-Generating Process

For each Monte Carlo replication, draw $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$ independently from the distribution described above and set $T_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)$. We consider DGP polynomial specifications of order $s \in \{1,2,3\}$, using the monomial vector $\mathbf{r}_s(\cdot)$, evaluated at the unscaled score. Thus, for $d = 2$, $\mathbf{r}_1(\mathbf{x}) = (1, x_1, x_2)^\top$ and $\mathbf{r}_2(\mathbf{x}) = (1, x_1, x_2, x_1^2, x_1x_2, x_2^2)^\top$, with $\mathbf{r}_3(\mathbf{x})$ defined analogously by adding all cubic monomials. For each $t \in \{0,1\}$ and $i = 1, \ldots, n$, define

$$\begin{aligned}
&\mu_{Y,t}(\mathbf{X}_i) = \mathbf{r}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}_{Y,t},\\
&\mu_{W,t}(\mathbf{X}_i) = \Lambda(\eta_{W,t}(\mathbf{X}_i)), \qquad \eta_{W,t}(\mathbf{X}_i) = \mathbf{r}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}_{W,t},
\end{aligned}$$

where $\Lambda(a) = 1/(1+\exp(-a))$. Given these conditional mean functions, draw $U_{W,t,i} \sim \mathsf{Uniform}(0,1)$ and $\varepsilon_{Y,t,i} \sim \mathsf{Normal}(0, \sigma^2_{Y,t})$ independently across i and t and independently of $\mathbf{X}_i$. The potential treatment statuses and treatment-indexed potential outcomes are

$$\begin{aligned}
W_i(t) &= \mathbb{1}\{U_{W,t,i} \le \mu_{W,t}(\mathbf{X}_i)\},\\
Y_i(t,r) &= \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{r - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}, \qquad r \in \{0,1\}.
\end{aligned}$$

The reduced-form potential outcome induced by assignment t is $Y_i(t, W_i(t))$. Equivalently,

$$Y_i(t, W_i(t)) = \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{W_i(t) - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}.$$

The observed outcomes and treatment status are

$$\begin{aligned}
Y_i &= Y_i(0, W_i(0))(1 - T_i) + Y_i(1, W_i(1))T_i,\\
W_i &= W_i(0)(1 - T_i) + W_i(1)T_i, \qquad T_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1),
\end{aligned}$$

for $i = 1, 2, \ldots, n$.

In the reduced-form notation used for sharp estimation and inference, $Y_i(t)$ is shorthand for $Y_i(t, W_i(t))$. This centered specification preserves the reduced-form outcome surface because $\mathbb{E}[Y_i(t, W_i(t))|\mathbf{X}_i = \mathbf{x}] =$

$\mu_{Y,t}(\mathbf{x})$, while allowing the potential outcome and potential treatment status equations to be correlated:

$$\mathbb{E}[W_i(t)|\mathbf{X}_i = \mathbf{x}] = \mu_{W,t}(\mathbf{x}) \qquad \text{and} \qquad \mathbb{C}\text{ov}(Y_i(t, W_i(t)), W_i(t)|\mathbf{X}_i = \mathbf{x}) = \lambda_t \mu_{W,t}(\mathbf{x})\{1 - \mu_{W,t}(\mathbf{x})\}.$$

Consequently, for each boundary point $\mathbf{b} \in \mathscr{B}$,

$$\begin{aligned}
\tau_W(\mathbf{b}) &= \mathbb{E}[W_i(1) - W_i(0)|\mathbf{X}_i = \mathbf{b}] = \mu_{W,1}(\mathbf{b}) - \mu_{W,0}(\mathbf{b}), \\
\tau_Y(\mathbf{b}) &= \mathbb{E}[Y_i(1, W_i(1)) - Y_i(0, W_i(0))|\mathbf{X}_i = \mathbf{b}] = \mu_{Y,1}(\mathbf{b}) - \mu_{Y,0}(\mathbf{b}), \\
\zeta(\mathbf{b}) &= \frac{\tau_Y(\mathbf{b})}{\tau_W(\mathbf{b})}.
\end{aligned}$$

In each Monte Carlo replication, the sharp analysis applied to Y_i targets the reduced-form or ITT curve $\tau_Y(\mathbf{b})$, the same analysis applied to W_i targets the first-stage curve $\tau_W(\mathbf{b})$, and the fuzzy analysis targets the Wald curve $\zeta(\mathbf{b})$. Thus, the ITT and fuzzy simulation results are different summaries of the same simulated samples rather than results from separate sharp and fuzzy DGPs.

SA-11.2 Calibration from the SPP data

The DGP parameters are calibrated once from the SPP data and then held fixed across Monte Carlo replications. We use the following steps separately for $s \in \{1, 2, 3\}$.

Step 1. Fix the DGP polynomial order $s \in \{1, 2, 3\}$ and define the side-specific calibration samples $\mathscr{I}_t = \{i : T_i = t\}$ and $N_t = |\mathscr{I}_t|$, for $t \in \{0, 1\}$.

Step 2. For each side t, estimate the outcome regression coefficients by side-specific least squares:

$$\widehat{\boldsymbol{\beta}}_{Y,t} = \operatorname*{arg\,min}_{\boldsymbol{\beta} \in \mathbb{R}^{\mathfrak{p}_s}} \sum_{i \in \mathscr{I}_t} \{Y_i - \mathbf{r}_s(\mathbf{X}_i)^\top \boldsymbol{\beta}\}^2.$$

This gives the fitted reduced-form outcome surface $\widehat{\mu}_{Y,t}(\mathbf{x}) = \mathbf{r}_s(\mathbf{x})^\top \widehat{\boldsymbol{\beta}}_{Y,t}$.

Step 3. For each side t, estimate the treatment-status regression coefficients by side-specific logit:

$$\widehat{\boldsymbol{\beta}}_{W,t} = \operatorname*{arg\,max}_{\boldsymbol{\beta} \in \mathbb{R}^{\mathfrak{p}_s}} \sum_{i \in \mathscr{I}_t} \left[W_i \mathbf{r}_s(\mathbf{X}_i)^\top \boldsymbol{\beta} - \log\{1 + \exp(\mathbf{r}_s(\mathbf{X}_i)^\top \boldsymbol{\beta})\} \right].$$

This gives the fitted first-stage surface $\widehat{\mu}_{W,t}(\mathbf{x}) = \Lambda(\mathbf{r}_s(\mathbf{x})^\top \widehat{\boldsymbol{\beta}}_{W,t})$.

Step 4. Construct side-specific residuals in the calibration sample:

$$\widehat{\varepsilon}_{Y,i} = Y_i - \widehat{\mu}_{Y,T_i}(\mathbf{X}_i), \qquad \widehat{\varepsilon}_{W,i} = W_i - \widehat{\mu}_{W,T_i}(\mathbf{X}_i).$$

Step 5. Estimate the residual dependence parameter and outcome-shock variance on each side:

$$\widehat{\lambda}_t = \frac{\sum_{i \in \mathscr{I}_t} \widehat{\varepsilon}_{Y,i} \widehat{\varepsilon}_{W,i}}{\sum_{i \in \mathscr{I}_t} \widehat{\varepsilon}^2_{W,i}}, \qquad \widehat{\sigma}^2_{Y,t} = \frac{1}{N_t} \sum_{i \in \mathscr{I}_t} \{\widehat{\varepsilon}_{Y,i} - \widehat{\lambda}_t \widehat{\varepsilon}_{W,i}\}^2,$$

Step 6. Set the Monte Carlo DGP parameters equal to the calibration estimates: $\boldsymbol{\beta}_{Y,t} = \widehat{\boldsymbol{\beta}}_{Y,t}$, $\boldsymbol{\beta}_{W,t} = \widehat{\boldsymbol{\beta}}_{W,t}$, $\lambda_t = \widehat{\lambda}_t$, and $\sigma^2_{Y,t} = \widehat{\sigma}^2_{Y,t}$ for $t \in \{0, 1\}$.

Step 7. Before running the Monte Carlo replications, compute the true target functions analytically from the calibrated conditional mean surfaces. For each evaluation point $\mathbf{b} \in \mathscr{B}$,

$$\tau_Y(\mathbf{b}) = \mathbf{r}_s(\mathbf{b})^\top(\boldsymbol{\beta}_{Y,1} - \boldsymbol{\beta}_{Y,0}), \qquad \tau_W(\mathbf{b}) = \Lambda(\mathbf{r}_s(\mathbf{b})^\top\boldsymbol{\beta}_{W,1}) - \Lambda(\mathbf{r}_s(\mathbf{b})^\top\boldsymbol{\beta}_{W,0}), \qquad \zeta(\mathbf{b}) = \frac{\tau_Y(\mathbf{b})}{\tau_W(\mathbf{b})}.$$

This calibration reproduces the empirical residual covariance between the outcome and take-up equations on each side of the boundary, while keeping the marginal reduced-form and first-stage surfaces fixed by the fitted SPP models.

SA-11.3 Monte Carlo Implementation

The following steps describe how to simulate from the calibrated model. They are repeated separately for each DGP polynomial order $s \in \{1, 2, 3\}$.

Step 1. Fix the Monte Carlo inputs: the sample size n, the number of replications M, the evaluation grid $\mathscr{B} = \{\mathbf{b}_1, \ldots, \mathbf{b}_{21}\}$ with $\mathbf{b}_{11}$ at the kink, the assignment regions $\mathscr{A}_0$ and $\mathscr{A}_1$, and the calibrated parameters

$$\{\boldsymbol{\beta}_{Y,t}, \boldsymbol{\beta}_{W,t}, \lambda_t, \sigma^2_{Y,t} : t \in \{0, 1\}\}.$$

Step 2. Before entering the Monte Carlo loop, evaluate the true target functions on the grid:

$$\tau_Y(\mathbf{b}_j), \qquad \tau_W(\mathbf{b}_j), \qquad \zeta(\mathbf{b}_j), \qquad j = 1, \ldots, 21,$$

using the analytic formulas in the calibration step above. These target values are fixed across Monte Carlo replications.

Step 3. For replication $m = 1, \ldots, M$, draw independent score vectors

$$\mathbf{X}_i = (X_{1i}, X_{2i})^\top, \qquad X_{\ell i} = 100B_{\ell i} - 25, \quad B_{\ell i} \sim \mathsf{Beta}(3, 4), \quad \ell \in \{1, 2\},$$

with B_{1i} and B_{2i} independent. Set $T_i = \mathbb{1}(\mathbf{X}_i \in \mathscr{A}_1)$.

Step 4. Form the design vectors $\mathbf{r}_s(\mathbf{X}_i)$ and compute the side-specific conditional means

$$\mu_{Y,t}(\mathbf{X}_i) = \mathbf{r}_s(\mathbf{X}_i)^\top\boldsymbol{\beta}_{Y,t}, \qquad \mu_{W,t}(\mathbf{X}_i) = \Lambda(\mathbf{r}_s(\mathbf{X}_i)^\top\boldsymbol{\beta}_{W,t}), \qquad t \in \{0, 1\}.$$

Step 5. Draw the treatment-status shocks and outcome shocks independently:

$$U_{W,t,i} \sim \mathsf{Uniform}(0, 1), \qquad \varepsilon_{Y,t,i} \sim \mathsf{Normal}(0, \sigma^2_{Y,t}), \qquad t \in \{0, 1\}.$$

Then construct the potential treatment statuses and treatment-indexed potential outcomes:

$$\begin{aligned} W_i(t) &= \mathbb{1}\{U_{W,t,i} \leq \mu_{W,t}(\mathbf{X}_i)\}, \\ Y_i(t, r) &= \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{r - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}, \qquad r \in \{0, 1\}. \end{aligned}$$

Evaluate the reduced-form potential outcome at the simulated treatment status:

$$Y_i(t, W_i(t)) = \mu_{Y,t}(\mathbf{X}_i) + \lambda_t\{W_i(t) - \mu_{W,t}(\mathbf{X}_i)\} + \varepsilon_{Y,t,i}.$$

Step 6. Select the observed variables using the assignment indicator:

$$\begin{aligned} W_i &= W_i(0)(1 - T_i) + W_i(1)T_i, \\ Y_i &= Y_i(0, W_i(0))(1 - T_i) + Y_i(1, W_i(1))T_i. \end{aligned}$$

The simulated dataset for replication m is therefore $\{Y_i, W_i, T_i, \mathbf{X}_i : i = 1, \ldots, n\}$.

Step 7. **ITT Analysis**. Apply the sharp estimator to Y_i using the simulated assignment rule and score vectors for the target $\tau_Y(\mathbf{b}_j)$.

Step 8. **FS Analysis**. Apply the sharp estimator to W_i using the simulated assignment rule and score vectors for the target $\tau_W(\mathbf{b}_j)$.

Step 9. **Fuzzy Analysis**. Apply the fuzzy estimator using Y_i as the outcome, W_i as the treatment status, and using the simulated assignment rule and score vectors for the Wald target $\zeta(\mathbf{b}_j)$.

Step 10. After all M replications, compute Monte Carlo summaries by comparing the stored estimates and confidence sets with the fixed analytic targets. For each target and each grid point, report bias, standard deviation, root mean squared error, pointwise coverage, and average interval length. For uniform inference, report simultaneous coverage and average band length over the grid $\mathscr{B}$. Finally, also recover WBATE and LBATE and report their associated performance.

We use the fuzzy simulations to study the finite-sample behavior of the reduced-form/ITT estimators, the first-stage estimators, and the fuzzy Wald estimator. A complier-average causal interpretation of $\zeta(\mathbf{b})$ would require additional assumptions that are not imposed in the simulations. In particular, one would need a local exclusion restriction ruling out a direct effect of assignment on the outcome except through treatment take-up, a monotonicity condition ruling out defiers near the boundary, and a nonzero first-stage effect. Under those conditions, the Wald curve $\zeta(\mathbf{b})$ can be interpreted as a local average treatment effect for compliers at the boundary point $\mathbf{b}$.

SA-11.4 Results

The following tables report Monte Carlo summaries separately for the reduced-form/ITT, first-stage, and fuzzy estimators, and separately by DGP polynomial order $s \in \{1, 2, 3\}$. We report two sets of results: one using the ITT bandwidth selector, matching the main empirical specification, and one using the fuzzy bandwidth selector.

The first set of simulation results uses the ITT bandwidth selector. Specifically:

- Table SA-13 reports the reduced-form (ITT) simulation results for $s = 1$.
- Table SA-14 reports the reduced-form (ITT) simulation results for $s = 2$.
- Table SA-15 reports the reduced-form (ITT) simulation results for $s = 3$.
- Table SA-16 reports the take-up (FS) simulation results for $s = 1$.

- Table SA-17 reports the take-up (FS) simulation results for $s = 2$.
- Table SA-18 reports the take-up (FS) simulation results for $s = 3$.
- Table SA-19 reports the fuzzy simulation results for $s = 1$.
- Table SA-20 reports the fuzzy simulation results for $s = 2$.
- Table SA-21 reports the fuzzy simulation results for $s = 3$.

The second set of simulation results uses the fuzzy bandwidth selector. Specifically:

- Table SA-22 reports the reduced-form (ITT) simulation results for $s = 1$.
- Table SA-23 reports the reduced-form (ITT) simulation results for $s = 2$.
- Table SA-24 reports the reduced-form (ITT) simulation results for $s = 3$.
- Table SA-25 reports the take-up (FS) simulation results for $s = 1$.
- Table SA-26 reports the take-up (FS) simulation results for $s = 2$.
- Table SA-27 reports the take-up (FS) simulation results for $s = 3$.
- Table SA-28 reports the fuzzy simulation results for $s = 1$.
- Table SA-29 reports the fuzzy simulation results for $s = 2$.
- Table SA-30 reports the fuzzy simulation results for $s = 3$.

The columns report the average bandwidths, average bias, standard deviation, root mean squared error, empirical coverage, and average interval or band length. For the Uniform row, empirical coverage and average length refer to uniform coverage and average uniform band length.

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.220	13.220	732	1267	0.001	0.045	0.045	0.950	0.263
$\mathbf{b}_2$	13.525	13.525	765	1343	0.001	0.044	0.044	0.949	0.255
$\mathbf{b}_3$	13.740	13.740	787	1393	0.001	0.043	0.043	0.950	0.251
$\mathbf{b}_4$	13.857	13.857	808	1400	0.001	0.042	0.042	0.949	0.246
$\mathbf{b}_5$	13.904	13.904	869	1329	0.001	0.041	0.041	0.953	0.234
$\mathbf{b}_6$	13.890	13.890	924	1229	0.001	0.040	0.040	0.955	0.232
$\mathbf{b}_7$	13.811	13.811	959	1116	0.000	0.040	0.040	0.950	0.239
$\mathbf{b}_8$	13.639	13.639	968	989	0.001	0.042	0.042	0.949	0.253
$\mathbf{b}_9$	13.993	13.993	1024	928	0.000	0.046	0.046	0.948	0.271
$\mathbf{b}_{10}$	15.131	15.131	1147	961	0.000	0.051	0.051	0.948	0.306
$\mathbf{b}_{11}$	14.782	14.782	1091	803	0.000	0.066	0.066	0.936	0.425
$\mathbf{b}_{12}$	14.679	14.679	1101	949	0.000	0.048	0.048	0.941	0.288
$\mathbf{b}_{13}$	13.614	13.614	969	965	0.000	0.043	0.043	0.949	0.257
$\mathbf{b}_{14}$	13.811	13.811	951	1137	0.000	0.040	0.040	0.953	0.237
$\mathbf{b}_{15}$	13.870	13.870	889	1286	0.001	0.041	0.041	0.954	0.232
$\mathbf{b}_{16}$	13.837	13.837	805	1397	0.001	0.043	0.043	0.949	0.246
$\mathbf{b}_{17}$	13.678	13.678	781	1379	0.001	0.044	0.044	0.945	0.251
$\mathbf{b}_{18}$	13.289	13.289	739	1284	0.001	0.045	0.045	0.945	0.260
$\mathbf{b}_{19}$	13.011	13.011	697	1195	0.001	0.047	0.047	0.947	0.269
$\mathbf{b}_{20}$	13.077	13.077	671	1154	0.001	0.048	0.048	0.949	0.274
$\mathbf{b}_{21}$	13.330	13.330	651	1130	0.000	0.049	0.049	0.944	0.278
`Uniform`								0.930	0.381
`WBATE`					0.001	0.026	0.026	0.943	0.138
`LBATE`					0.053	0.038	0.065	0.971	0.433

Table SA-13: Intention-to-Treat Effects (Simulations, $s = 1$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.191	13.191	728	1262	0.001	0.045	0.045	0.949	0.263
$\mathbf{b}_2$	13.494	13.494	762	1338	0.001	0.044	0.044	0.951	0.256
$\mathbf{b}_3$	13.717	13.717	784	1390	0.002	0.043	0.043	0.951	0.251
$\mathbf{b}_4$	13.830	13.830	804	1397	0.002	0.042	0.042	0.954	0.247
$\mathbf{b}_5$	13.870	13.870	864	1325	0.001	0.041	0.041	0.955	0.235
$\mathbf{b}_6$	13.846	13.846	919	1223	0.000	0.040	0.040	0.950	0.233
$\mathbf{b}_7$	13.777	13.777	954	1112	-0.001	0.040	0.040	0.947	0.239
$\mathbf{b}_8$	13.630	13.630	966	988	-0.001	0.043	0.043	0.949	0.253
$\mathbf{b}_9$	14.017	14.017	1026	932	-0.002	0.046	0.046	0.944	0.271
$\mathbf{b}_{10}$	15.156	15.156	1148	965	-0.003	0.051	0.051	0.942	0.305
$\mathbf{b}_{11}$	14.778	14.778	1090	803	-0.003	0.067	0.067	0.939	0.423
$\mathbf{b}_{12}$	14.646	14.646	1097	945	-0.003	0.049	0.049	0.939	0.287
$\mathbf{b}_{13}$	13.581	13.581	965	961	-0.002	0.044	0.044	0.946	0.256
$\mathbf{b}_{14}$	13.768	13.768	945	1131	-0.001	0.041	0.041	0.947	0.236
$\mathbf{b}_{15}$	13.834	13.834	883	1281	-0.001	0.042	0.042	0.949	0.231
$\mathbf{b}_{16}$	13.800	13.800	801	1391	0.000	0.043	0.043	0.945	0.246
$\mathbf{b}_{17}$	13.629	13.629	776	1370	0.000	0.044	0.044	0.945	0.251
$\mathbf{b}_{18}$	13.237	13.237	734	1275	0.000	0.045	0.045	0.951	0.260
$\mathbf{b}_{19}$	12.978	12.978	693	1189	-0.000	0.047	0.047	0.952	0.269
$\mathbf{b}_{20}$	13.065	13.065	669	1153	-0.001	0.047	0.047	0.947	0.273
$\mathbf{b}_{21}$	13.327	13.327	649	1131	-0.001	0.048	0.048	0.947	0.277
`Uniform`								0.932	0.381
`WBATE`					-0.001	0.027	0.027	0.944	0.138
`LBATE`					0.034	0.046	0.057	0.982	0.448

Table SA-14: Intention-to-Treat Effects (Simulations, $s = 2$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.216	13.216	731	1266	0.002	0.045	0.045	0.948	0.262
$\mathbf{b}_2$	13.505	13.505	763	1339	0.001	0.044	0.044	0.950	0.255
$\mathbf{b}_3$	13.716	13.716	784	1389	0.001	0.043	0.043	0.948	0.251
$\mathbf{b}_4$	13.816	13.816	803	1393	0.000	0.043	0.043	0.946	0.247
$\mathbf{b}_5$	13.843	13.843	861	1320	-0.001	0.042	0.042	0.951	0.235
$\mathbf{b}_6$	13.821	13.821	915	1219	-0.001	0.041	0.041	0.951	0.233
$\mathbf{b}_7$	13.740	13.740	950	1107	-0.002	0.041	0.041	0.951	0.239
$\mathbf{b}_8$	13.566	13.566	959	980	-0.002	0.043	0.043	0.950	0.253
$\mathbf{b}_9$	13.909	13.909	1014	919	-0.002	0.046	0.046	0.952	0.272
$\mathbf{b}_{10}$	15.049	15.049	1138	951	-0.002	0.051	0.051	0.949	0.306
$\mathbf{b}_{11}$	14.775	14.775	1090	803	-0.001	0.064	0.064	0.944	0.423
$\mathbf{b}_{12}$	14.642	14.642	1097	945	-0.000	0.048	0.048	0.949	0.288
$\mathbf{b}_{13}$	13.564	13.564	963	959	0.000	0.044	0.044	0.946	0.257
$\mathbf{b}_{14}$	13.758	13.758	944	1130	0.001	0.041	0.041	0.940	0.237
$\mathbf{b}_{15}$	13.839	13.839	885	1281	0.003	0.042	0.042	0.949	0.232
$\mathbf{b}_{16}$	13.799	13.799	802	1390	0.004	0.044	0.044	0.948	0.246
$\mathbf{b}_{17}$	13.634	13.634	777	1371	0.004	0.044	0.044	0.948	0.251
$\mathbf{b}_{18}$	13.260	13.260	737	1279	0.004	0.045	0.045	0.949	0.260
$\mathbf{b}_{19}$	13.014	13.014	697	1195	0.003	0.046	0.047	0.951	0.269
$\mathbf{b}_{20}$	13.095	13.095	672	1158	0.003	0.047	0.047	0.948	0.274
$\mathbf{b}_{21}$	13.341	13.341	651	1133	0.003	0.048	0.048	0.948	0.277
`Uniform`								0.932	0.381
`WBATE`					0.001	0.026	0.026	0.941	0.138
`LBATE`					0.035	0.044	0.056	0.980	0.444

Table SA-15: Intention-to-Treat Effects (Simulations, $s = 3$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.220	13.220	732	1267	0.000	0.038	0.038	0.950	0.230
$\mathbf{b}_2$	13.525	13.525	765	1343	0.000	0.037	0.037	0.947	0.223
$\mathbf{b}_3$	13.740	13.740	787	1393	0.000	0.036	0.036	0.945	0.219
$\mathbf{b}_4$	13.857	13.857	808	1400	0.000	0.036	0.036	0.946	0.217
$\mathbf{b}_5$	13.904	13.904	869	1329	0.000	0.036	0.036	0.944	0.219
$\mathbf{b}_6$	13.890	13.890	924	1229	0.000	0.037	0.037	0.946	0.226
$\mathbf{b}_7$	13.811	13.811	959	1116	0.001	0.040	0.040	0.944	0.239
$\mathbf{b}_8$	13.639	13.639	968	989	0.001	0.044	0.044	0.945	0.261
$\mathbf{b}_9$	13.993	13.993	1024	928	0.001	0.048	0.048	0.944	0.289
$\mathbf{b}_{10}$	15.131	15.131	1147	961	0.001	0.054	0.054	0.939	0.338
$\mathbf{b}_{11}$	14.782	14.782	1091	803	0.001	0.070	0.070	0.931	0.483
$\mathbf{b}_{12}$	14.679	14.679	1101	949	0.000	0.051	0.051	0.942	0.313
$\mathbf{b}_{13}$	13.614	13.614	969	965	0.000	0.044	0.044	0.947	0.266
$\mathbf{b}_{14}$	13.811	13.811	951	1137	0.000	0.039	0.039	0.950	0.235
$\mathbf{b}_{15}$	13.870	13.870	889	1286	0.000	0.036	0.036	0.948	0.220
$\mathbf{b}_{16}$	13.837	13.837	805	1397	0.000	0.036	0.036	0.951	0.215
$\mathbf{b}_{17}$	13.678	13.678	781	1379	0.000	0.036	0.036	0.949	0.217
$\mathbf{b}_{18}$	13.289	13.289	739	1284	0.001	0.037	0.037	0.947	0.224
$\mathbf{b}_{19}$	13.011	13.011	697	1195	0.001	0.038	0.038	0.947	0.232
$\mathbf{b}_{20}$	13.077	13.077	671	1154	0.001	0.039	0.039	0.947	0.236
$\mathbf{b}_{21}$	13.330	13.330	651	1130	0.001	0.039	0.039	0.945	0.238
`Uniform`								0.924	0.364
`WBATE`					0.001	0.025	0.025	0.944	0.134
`LBATE`					0.017	0.035	0.039	0.984	0.421

Table SA-16: First-Stage Effects (Simulations, $s = 1$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.191	13.191	728	1262	0.002	0.039	0.039	0.948	0.231
$\mathbf{b}_2$	13.494	13.494	762	1338	0.002	0.038	0.038	0.943	0.224
$\mathbf{b}_3$	13.717	13.717	784	1390	0.002	0.037	0.037	0.944	0.219
$\mathbf{b}_4$	13.830	13.830	804	1397	0.002	0.036	0.037	0.948	0.218
$\mathbf{b}_5$	13.870	13.870	864	1325	0.002	0.037	0.037	0.944	0.220
$\mathbf{b}_6$	13.846	13.846	919	1223	0.001	0.038	0.038	0.941	0.227
$\mathbf{b}_7$	13.777	13.777	954	1112	0.001	0.040	0.040	0.942	0.239
$\mathbf{b}_8$	13.630	13.630	966	988	0.000	0.043	0.043	0.936	0.259
$\mathbf{b}_9$	14.017	14.017	1026	932	-0.000	0.048	0.048	0.937	0.286
$\mathbf{b}_{10}$	15.156	15.156	1148	965	-0.002	0.054	0.054	0.936	0.333
$\mathbf{b}_{11}$	14.778	14.778	1090	803	-0.002	0.069	0.069	0.932	0.476
$\mathbf{b}_{12}$	14.646	14.646	1097	945	-0.002	0.051	0.051	0.942	0.309
$\mathbf{b}_{13}$	13.581	13.581	965	961	-0.001	0.044	0.044	0.944	0.262
$\mathbf{b}_{14}$	13.768	13.768	945	1131	-0.001	0.039	0.039	0.949	0.232
$\mathbf{b}_{15}$	13.834	13.834	883	1281	-0.001	0.036	0.036	0.945	0.218
$\mathbf{b}_{16}$	13.800	13.800	801	1391	-0.001	0.035	0.035	0.940	0.213
$\mathbf{b}_{17}$	13.629	13.629	776	1370	-0.001	0.036	0.036	0.943	0.215
$\mathbf{b}_{18}$	13.237	13.237	734	1275	-0.001	0.037	0.037	0.941	0.222
$\mathbf{b}_{19}$	12.978	12.978	693	1189	-0.002	0.038	0.038	0.940	0.229
$\mathbf{b}_{20}$	13.065	13.065	669	1153	-0.002	0.039	0.039	0.941	0.233
$\mathbf{b}_{21}$	13.327	13.327	649	1131	-0.002	0.039	0.039	0.939	0.235
`Uniform`								0.917	0.361
`WBATE`					-0.000	0.024	0.024	0.938	0.133
`LBATE`					0.021	0.038	0.043	0.978	0.427

Table SA-17: First-Stage Effects (Simulations, $s = 2$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.216	13.216	731	1266	0.001	0.038	0.038	0.951	0.230
$\mathbf{b}_2$	13.505	13.505	763	1339	0.001	0.037	0.037	0.949	0.223
$\mathbf{b}_3$	13.716	13.716	784	1389	0.001	0.036	0.036	0.950	0.219
$\mathbf{b}_4$	13.816	13.816	803	1393	0.000	0.036	0.036	0.950	0.217
$\mathbf{b}_5$	13.843	13.843	861	1320	-0.000	0.036	0.036	0.951	0.220
$\mathbf{b}_6$	13.821	13.821	915	1219	-0.000	0.037	0.037	0.949	0.226
$\mathbf{b}_7$	13.740	13.740	950	1107	-0.000	0.039	0.039	0.947	0.239
$\mathbf{b}_8$	13.566	13.566	959	980	0.000	0.043	0.043	0.942	0.260
$\mathbf{b}_9$	13.909	13.909	1014	919	0.001	0.048	0.048	0.941	0.288
$\mathbf{b}_{10}$	15.049	15.049	1138	951	0.002	0.053	0.053	0.940	0.336
$\mathbf{b}_{11}$	14.775	14.775	1090	803	0.003	0.068	0.068	0.930	0.478
$\mathbf{b}_{12}$	14.642	14.642	1097	945	0.002	0.051	0.051	0.939	0.311
$\mathbf{b}_{13}$	13.564	13.564	963	959	0.002	0.044	0.044	0.938	0.264
$\mathbf{b}_{14}$	13.758	13.758	944	1130	0.002	0.039	0.039	0.943	0.235
$\mathbf{b}_{15}$	13.839	13.839	885	1281	0.002	0.037	0.037	0.943	0.220
$\mathbf{b}_{16}$	13.799	13.799	802	1390	0.003	0.036	0.036	0.943	0.215
$\mathbf{b}_{17}$	13.634	13.634	777	1371	0.002	0.036	0.036	0.945	0.217
$\mathbf{b}_{18}$	13.260	13.260	737	1279	0.002	0.037	0.037	0.947	0.224
$\mathbf{b}_{19}$	13.014	13.014	697	1195	0.002	0.038	0.038	0.948	0.231
$\mathbf{b}_{20}$	13.095	13.095	672	1158	0.002	0.038	0.038	0.947	0.235
$\mathbf{b}_{21}$	13.341	13.341	651	1133	0.002	0.039	0.039	0.943	0.237
`Uniform`								0.920	0.363
`WBATE`					0.001	0.024	0.024	0.944	0.133
`LBATE`					0.024	0.036	0.043	0.980	0.431

Table SA-18: First-Stage Effects (Simulations, $s = 3$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.220	13.220	732	1267	0.001	0.072	0.072	0.954	0.433
$\mathbf{b}_2$	13.525	13.525	765	1343	0.001	0.069	0.069	0.958	0.419
$\mathbf{b}_3$	13.740	13.740	787	1393	0.001	0.067	0.067	0.958	0.411
$\mathbf{b}_4$	13.857	13.857	808	1400	0.001	0.066	0.066	0.958	0.402
$\mathbf{b}_5$	13.904	13.904	869	1329	0.001	0.064	0.064	0.957	0.373
$\mathbf{b}_6$	13.890	13.890	924	1229	0.001	0.061	0.061	0.959	0.364
$\mathbf{b}_7$	13.811	13.811	959	1116	0.001	0.060	0.060	0.955	0.369
$\mathbf{b}_8$	13.639	13.639	968	989	0.001	0.063	0.063	0.955	0.387
$\mathbf{b}_9$	13.993	13.993	1024	928	0.001	0.067	0.067	0.958	0.411
$\mathbf{b}_{10}$	15.131	15.131	1147	961	0.000	0.072	0.072	0.959	0.463
$\mathbf{b}_{11}$	14.782	14.782	1091	803	-0.000	0.092	0.092	0.964	0.719
$\mathbf{b}_{12}$	14.679	14.679	1101	949	0.000	0.068	0.068	0.951	0.425
$\mathbf{b}_{13}$	13.614	13.614	969	965	0.000	0.061	0.061	0.954	0.378
$\mathbf{b}_{14}$	13.811	13.811	951	1137	0.000	0.057	0.057	0.959	0.348
$\mathbf{b}_{15}$	13.870	13.870	889	1286	0.001	0.059	0.059	0.955	0.342
$\mathbf{b}_{16}$	13.837	13.837	805	1397	0.001	0.062	0.062	0.954	0.369
$\mathbf{b}_{17}$	13.678	13.678	781	1379	0.001	0.063	0.063	0.951	0.374
$\mathbf{b}_{18}$	13.289	13.289	739	1284	0.001	0.065	0.065	0.951	0.382
$\mathbf{b}_{19}$	13.011	13.011	697	1195	0.001	0.066	0.066	0.952	0.390
$\mathbf{b}_{20}$	13.077	13.077	671	1154	0.000	0.066	0.066	0.954	0.393
$\mathbf{b}_{21}$	13.330	13.330	651	1130	-0.000	0.067	0.067	0.952	0.394
`Uniform`								0.962	0.586
`WBATE`					0.001	0.039	0.039	0.949	0.214
`LBATE`					0.053	0.058	0.078	0.997	0.753

Table SA-19: Fuzzy Effects (Simulations, $s = 1$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.191	13.191	728	1262	0.000	0.073	0.073	0.958	0.445
$\mathbf{b}_2$	13.494	13.494	762	1338	0.001	0.070	0.070	0.957	0.429
$\mathbf{b}_3$	13.717	13.717	784	1390	0.001	0.068	0.068	0.959	0.417
$\mathbf{b}_4$	13.830	13.830	804	1397	0.001	0.067	0.067	0.959	0.405
$\mathbf{b}_5$	13.870	13.870	864	1325	0.000	0.064	0.064	0.959	0.372
$\mathbf{b}_6$	13.846	13.846	919	1223	-0.001	0.061	0.061	0.954	0.359
$\mathbf{b}_7$	13.777	13.777	954	1112	-0.002	0.059	0.059	0.951	0.359
$\mathbf{b}_8$	13.630	13.630	966	988	-0.003	0.061	0.061	0.953	0.369
$\mathbf{b}_9$	14.017	14.017	1026	932	-0.003	0.064	0.064	0.952	0.383
$\mathbf{b}_{10}$	15.156	15.156	1148	965	-0.003	0.068	0.068	0.949	0.421
$\mathbf{b}_{11}$	14.778	14.778	1090	803	-0.004	0.085	0.085	0.961	1.058
$\mathbf{b}_{12}$	14.646	14.646	1097	945	-0.003	0.063	0.064	0.952	0.384
$\mathbf{b}_{13}$	13.581	13.581	965	961	-0.002	0.058	0.058	0.951	0.345
$\mathbf{b}_{14}$	13.768	13.768	945	1131	-0.001	0.054	0.054	0.953	0.321
$\mathbf{b}_{15}$	13.834	13.834	883	1281	-0.000	0.056	0.056	0.950	0.318
$\mathbf{b}_{16}$	13.800	13.800	801	1391	0.001	0.059	0.059	0.948	0.345
$\mathbf{b}_{17}$	13.629	13.629	776	1370	0.001	0.060	0.060	0.951	0.351
$\mathbf{b}_{18}$	13.237	13.237	734	1275	0.001	0.061	0.061	0.952	0.361
$\mathbf{b}_{19}$	12.978	12.978	693	1189	0.000	0.063	0.063	0.951	0.372
$\mathbf{b}_{20}$	13.065	13.065	669	1153	-0.001	0.064	0.064	0.951	0.376
$\mathbf{b}_{21}$	13.327	13.327	649	1131	-0.001	0.065	0.065	0.949	0.379
`Uniform`								0.961	0.588
`WBATE`					-0.001	0.038	0.038	0.947	0.224
`LBATE`					0.052	0.060	0.079	0.994	0.993

Table SA-20: Fuzzy Effects (Simulations, $s = 2$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.216	13.216	731	1266	0.001	0.071	0.071	0.958	0.428
$\mathbf{b}_2$	13.505	13.505	763	1339	0.001	0.069	0.069	0.954	0.412
$\mathbf{b}_3$	13.716	13.716	784	1389	0.001	0.067	0.067	0.955	0.400
$\mathbf{b}_4$	13.816	13.816	803	1393	0.000	0.066	0.066	0.952	0.388
$\mathbf{b}_5$	13.843	13.843	861	1320	-0.001	0.063	0.063	0.954	0.358
$\mathbf{b}_6$	13.821	13.821	915	1219	-0.002	0.059	0.059	0.953	0.346
$\mathbf{b}_7$	13.740	13.740	950	1107	-0.003	0.058	0.058	0.954	0.349
$\mathbf{b}_8$	13.566	13.566	959	980	-0.003	0.060	0.060	0.960	0.363
$\mathbf{b}_9$	13.909	13.909	1014	919	-0.004	0.063	0.063	0.961	0.383
$\mathbf{b}_{10}$	15.049	15.049	1138	951	-0.004	0.067	0.067	0.958	0.427
$\mathbf{b}_{11}$	14.775	14.775	1090	803	-0.005	0.084	0.084	0.964	0.636
$\mathbf{b}_{12}$	14.642	14.642	1097	945	-0.003	0.064	0.064	0.952	0.399
$\mathbf{b}_{13}$	13.564	13.564	963	959	-0.002	0.059	0.059	0.954	0.361
$\mathbf{b}_{14}$	13.758	13.758	944	1130	-0.000	0.056	0.056	0.950	0.336
$\mathbf{b}_{15}$	13.839	13.839	885	1281	0.002	0.059	0.059	0.951	0.332
$\mathbf{b}_{16}$	13.799	13.799	802	1390	0.003	0.062	0.062	0.949	0.360
$\mathbf{b}_{17}$	13.634	13.634	777	1371	0.003	0.062	0.062	0.949	0.366
$\mathbf{b}_{18}$	13.260	13.260	737	1279	0.004	0.064	0.064	0.956	0.376
$\mathbf{b}_{19}$	13.014	13.014	697	1195	0.003	0.065	0.065	0.954	0.386
$\mathbf{b}_{20}$	13.095	13.095	672	1158	0.003	0.066	0.066	0.951	0.390
$\mathbf{b}_{21}$	13.341	13.341	651	1133	0.002	0.066	0.066	0.954	0.392
`Uniform`								0.961	0.561
`WBATE`					-0.000	0.037	0.037	0.950	0.203
`LBATE`					0.050	0.059	0.078	0.995	0.687

Table SA-21: Fuzzy Effects (Simulations, $s = 3$, ITT Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.404	13.404	746	1303	0.000	0.044	0.044	0.951	0.260
$\mathbf{b}_2$	13.823	13.823	788	1402	0.000	0.042	0.042	0.951	0.251
$\mathbf{b}_3$	14.104	14.104	815	1466	0.000	0.041	0.041	0.949	0.245
$\mathbf{b}_4$	14.257	14.257	849	1470	0.001	0.040	0.040	0.951	0.238
$\mathbf{b}_5$	14.318	14.318	919	1394	0.001	0.039	0.039	0.951	0.227
$\mathbf{b}_6$	14.306	14.306	975	1291	0.001	0.038	0.038	0.951	0.226
$\mathbf{b}_7$	14.235	14.235	1009	1176	0.000	0.038	0.038	0.949	0.233
$\mathbf{b}_8$	13.991	13.991	1009	1036	0.000	0.041	0.041	0.948	0.248
$\mathbf{b}_9$	14.133	14.133	1040	946	0.000	0.045	0.045	0.948	0.270
$\mathbf{b}_{10}$	15.100	15.100	1144	957	-0.000	0.050	0.050	0.947	0.308
$\mathbf{b}_{11}$	14.722	14.722	1085	796	-0.001	0.063	0.063	0.935	0.431
$\mathbf{b}_{12}$	14.698	14.698	1103	951	-0.000	0.047	0.047	0.941	0.288
$\mathbf{b}_{13}$	13.915	13.915	1004	1004	-0.000	0.042	0.042	0.949	0.252
$\mathbf{b}_{14}$	14.252	14.252	1004	1200	0.000	0.038	0.038	0.952	0.231
$\mathbf{b}_{15}$	14.307	14.307	941	1352	0.000	0.039	0.039	0.954	0.225
$\mathbf{b}_{16}$	14.254	14.254	849	1469	0.001	0.041	0.041	0.948	0.237
$\mathbf{b}_{17}$	14.044	14.044	809	1453	0.001	0.042	0.042	0.945	0.245
$\mathbf{b}_{18}$	13.534	13.534	758	1332	0.001	0.044	0.044	0.943	0.256
$\mathbf{b}_{19}$	13.137	13.137	707	1218	0.001	0.046	0.046	0.947	0.267
$\mathbf{b}_{20}$	13.188	13.188	680	1174	0.000	0.047	0.047	0.947	0.272
$\mathbf{b}_{21}$	13.422	13.422	657	1146	-0.000	0.047	0.047	0.946	0.276
`Uniform`								0.934	0.375
`WBATE`					0.000	0.026	0.026	0.945	0.137
`LBATE`					0.049	0.034	0.060	0.976	0.430

Table SA-22: Intention-to-Treat Effects (Simulations, $s = 1$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.386	13.386	744	1300	0.001	0.044	0.044	0.949	0.260
$\mathbf{b}_2$	13.791	13.791	785	1397	0.002	0.042	0.042	0.951	0.251
$\mathbf{b}_3$	14.069	14.069	811	1460	0.002	0.041	0.041	0.949	0.245
$\mathbf{b}_4$	14.218	14.218	844	1464	0.002	0.040	0.040	0.953	0.239
$\mathbf{b}_5$	14.287	14.287	914	1389	0.001	0.039	0.039	0.953	0.227
$\mathbf{b}_6$	14.284	14.284	972	1288	0.000	0.038	0.038	0.949	0.226
$\mathbf{b}_7$	14.225	14.225	1008	1176	-0.000	0.039	0.039	0.947	0.233
$\mathbf{b}_8$	13.995	13.995	1009	1037	-0.001	0.041	0.041	0.951	0.248
$\mathbf{b}_9$	14.126	14.126	1038	946	-0.002	0.045	0.045	0.945	0.270
$\mathbf{b}_{10}$	15.108	15.108	1144	958	-0.003	0.050	0.050	0.942	0.307
$\mathbf{b}_{11}$	14.748	14.748	1087	799	-0.005	0.063	0.064	0.936	0.429
$\mathbf{b}_{12}$	14.726	14.726	1106	955	-0.004	0.048	0.048	0.940	0.287
$\mathbf{b}_{13}$	13.933	13.933	1006	1007	-0.003	0.043	0.043	0.946	0.251
$\mathbf{b}_{14}$	14.255	14.255	1003	1201	-0.002	0.039	0.039	0.946	0.229
$\mathbf{b}_{15}$	14.312	14.312	941	1353	-0.001	0.039	0.039	0.947	0.223
$\mathbf{b}_{16}$	14.250	14.250	847	1469	-0.001	0.041	0.041	0.944	0.236
$\mathbf{b}_{17}$	14.019	14.019	806	1449	-0.000	0.042	0.042	0.944	0.244
$\mathbf{b}_{18}$	13.498	13.498	754	1326	-0.000	0.044	0.044	0.949	0.255
$\mathbf{b}_{19}$	13.115	13.115	703	1214	-0.001	0.046	0.046	0.950	0.266
$\mathbf{b}_{20}$	13.180	13.180	678	1173	-0.001	0.046	0.047	0.946	0.271
$\mathbf{b}_{21}$	13.415	13.415	655	1146	-0.002	0.047	0.047	0.946	0.275
`Uniform`								0.934	0.374
`WBATE`					-0.001	0.026	0.026	0.947	0.136
`LBATE`					0.030	0.041	0.051	0.984	0.445

Table SA-23: Intention-to-Treat Effects (Simulations, $s = 2$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.416	13.416	746	1305	0.001	0.044	0.044	0.949	0.259
$\mathbf{b}_2$	13.823	13.823	788	1403	0.001	0.042	0.042	0.950	0.250
$\mathbf{b}_3$	14.093	14.093	813	1465	0.000	0.041	0.041	0.947	0.245
$\mathbf{b}_4$	14.229	14.229	845	1465	-0.000	0.041	0.041	0.945	0.238
$\mathbf{b}_5$	14.282	14.282	913	1388	-0.001	0.040	0.040	0.948	0.227
$\mathbf{b}_6$	14.281	14.281	971	1288	-0.002	0.038	0.039	0.950	0.226
$\mathbf{b}_7$	14.214	14.214	1006	1174	-0.003	0.039	0.039	0.952	0.232
$\mathbf{b}_8$	13.955	13.955	1005	1031	-0.003	0.041	0.041	0.948	0.248
$\mathbf{b}_9$	14.080	14.080	1034	940	-0.003	0.045	0.045	0.952	0.270
$\mathbf{b}_{10}$	15.038	15.038	1137	949	-0.002	0.049	0.049	0.949	0.308
$\mathbf{b}_{11}$	14.667	14.667	1079	790	-0.002	0.062	0.062	0.941	0.430
$\mathbf{b}_{12}$	14.687	14.687	1102	951	-0.001	0.047	0.047	0.949	0.288
$\mathbf{b}_{13}$	13.902	13.902	1003	1003	0.000	0.042	0.042	0.946	0.252
$\mathbf{b}_{14}$	14.233	14.233	1001	1197	0.001	0.039	0.039	0.941	0.230
$\mathbf{b}_{15}$	14.292	14.292	940	1350	0.003	0.040	0.040	0.948	0.225
$\mathbf{b}_{16}$	14.217	14.217	845	1463	0.004	0.042	0.042	0.946	0.237
$\mathbf{b}_{17}$	13.995	13.995	805	1443	0.004	0.042	0.043	0.945	0.245
$\mathbf{b}_{18}$	13.492	13.492	755	1324	0.004	0.044	0.044	0.947	0.256
$\mathbf{b}_{19}$	13.126	13.126	705	1216	0.003	0.046	0.046	0.952	0.267
$\mathbf{b}_{20}$	13.183	13.183	679	1173	0.003	0.046	0.046	0.947	0.272
$\mathbf{b}_{21}$	13.408	13.408	655	1145	0.003	0.047	0.047	0.951	0.276
`Uniform`								0.934	0.375
`WBATE`					0.000	0.026	0.026	0.943	0.136
`LBATE`					0.032	0.041	0.052	0.984	0.443

Table SA-24: Intention-to-Treat Effects (Simulations, $s = 3$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.404	13.404	746	1303	-0.000	0.037	0.037	0.949	0.227
$\mathbf{b}_2$	13.823	13.823	788	1402	-0.000	0.035	0.035	0.948	0.218
$\mathbf{b}_3$	14.104	14.104	815	1466	-0.000	0.035	0.035	0.949	0.213
$\mathbf{b}_4$	14.257	14.257	849	1470	-0.000	0.034	0.034	0.946	0.211
$\mathbf{b}_5$	14.318	14.318	919	1394	-0.000	0.035	0.035	0.945	0.214
$\mathbf{b}_6$	14.306	14.306	975	1291	-0.000	0.036	0.036	0.946	0.221
$\mathbf{b}_7$	14.235	14.235	1009	1176	0.000	0.038	0.038	0.945	0.234
$\mathbf{b}_8$	13.991	13.991	1009	1036	0.000	0.042	0.042	0.946	0.256
$\mathbf{b}_9$	14.133	14.133	1040	946	0.000	0.047	0.047	0.941	0.287
$\mathbf{b}_{10}$	15.100	15.100	1144	957	0.000	0.053	0.053	0.939	0.338
$\mathbf{b}_{11}$	14.722	14.722	1085	796	-0.001	0.069	0.069	0.930	0.486
$\mathbf{b}_{12}$	14.698	14.698	1103	951	0.000	0.050	0.050	0.944	0.313
$\mathbf{b}_{13}$	13.915	13.915	1004	1004	-0.000	0.043	0.043	0.948	0.261
$\mathbf{b}_{14}$	14.252	14.252	1004	1200	-0.000	0.037	0.037	0.949	0.229
$\mathbf{b}_{15}$	14.307	14.307	941	1352	-0.000	0.034	0.034	0.948	0.215
$\mathbf{b}_{16}$	14.254	14.254	849	1469	-0.000	0.034	0.034	0.948	0.209
$\mathbf{b}_{17}$	14.044	14.044	809	1453	-0.000	0.034	0.034	0.948	0.212
$\mathbf{b}_{18}$	13.534	13.534	758	1332	0.000	0.036	0.036	0.947	0.221
$\mathbf{b}_{19}$	13.137	13.137	707	1218	0.000	0.037	0.037	0.947	0.230
$\mathbf{b}_{20}$	13.188	13.188	680	1174	0.000	0.038	0.038	0.943	0.234
$\mathbf{b}_{21}$	13.422	13.422	657	1146	0.000	0.038	0.038	0.944	0.237
`Uniform`								0.929	0.359
`WBATE`					-0.000	0.024	0.024	0.946	0.133
`LBATE`					0.014	0.033	0.036	0.988	0.416

Table SA-25: First-Stage Effects (Simulations, $s = 1$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.386	13.386	744	1300	0.002	0.037	0.037	0.945	0.228
$\mathbf{b}_2$	13.791	13.791	785	1397	0.002	0.036	0.036	0.942	0.219
$\mathbf{b}_3$	14.069	14.069	811	1460	0.002	0.035	0.035	0.943	0.214
$\mathbf{b}_4$	14.218	14.218	844	1464	0.002	0.035	0.035	0.947	0.212
$\mathbf{b}_5$	14.287	14.287	914	1389	0.002	0.035	0.035	0.945	0.214
$\mathbf{b}_6$	14.284	14.284	972	1288	0.001	0.036	0.036	0.940	0.221
$\mathbf{b}_7$	14.225	14.225	1008	1176	0.001	0.038	0.038	0.941	0.233
$\mathbf{b}_8$	13.995	13.995	1009	1037	0.000	0.042	0.042	0.936	0.254
$\mathbf{b}_9$	14.126	14.126	1038	946	-0.001	0.047	0.047	0.938	0.285
$\mathbf{b}_{10}$	15.108	15.108	1144	958	-0.003	0.052	0.052	0.937	0.334
$\mathbf{b}_{11}$	14.748	14.748	1087	799	-0.005	0.067	0.067	0.932	0.479
$\mathbf{b}_{12}$	14.726	14.726	1106	955	-0.003	0.049	0.049	0.942	0.308
$\mathbf{b}_{13}$	13.933	13.933	1006	1007	-0.002	0.042	0.042	0.945	0.257
$\mathbf{b}_{14}$	14.255	14.255	1003	1201	-0.002	0.037	0.037	0.947	0.226
$\mathbf{b}_{15}$	14.312	14.312	941	1353	-0.002	0.034	0.034	0.943	0.212
$\mathbf{b}_{16}$	14.250	14.250	847	1469	-0.002	0.033	0.033	0.943	0.206
$\mathbf{b}_{17}$	14.019	14.019	806	1449	-0.002	0.034	0.034	0.944	0.209
$\mathbf{b}_{18}$	13.498	13.498	754	1326	-0.002	0.036	0.036	0.939	0.218
$\mathbf{b}_{19}$	13.115	13.115	703	1214	-0.002	0.037	0.037	0.939	0.227
$\mathbf{b}_{20}$	13.180	13.180	678	1173	-0.003	0.038	0.038	0.940	0.231
$\mathbf{b}_{21}$	13.415	13.415	655	1146	-0.003	0.038	0.038	0.938	0.234
`Uniform`								0.921	0.355
`WBATE`					-0.001	0.024	0.024	0.941	0.132
`LBATE`					0.017	0.034	0.038	0.982	0.422

Table SA-26: First-Stage Effects (Simulations, $s = 2$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.416	13.416	746	1305	0.001	0.037	0.037	0.949	0.227
$\mathbf{b}_2$	13.823	13.823	788	1403	0.000	0.035	0.035	0.947	0.219
$\mathbf{b}_3$	14.093	14.093	813	1465	0.000	0.034	0.034	0.949	0.213
$\mathbf{b}_4$	14.229	14.229	845	1465	-0.001	0.034	0.034	0.949	0.211
$\mathbf{b}_5$	14.282	14.282	913	1388	-0.001	0.034	0.034	0.951	0.214
$\mathbf{b}_6$	14.281	14.281	971	1288	-0.001	0.035	0.035	0.947	0.220
$\mathbf{b}_7$	14.214	14.214	1006	1174	-0.001	0.037	0.037	0.945	0.233
$\mathbf{b}_8$	13.955	13.955	1005	1031	-0.001	0.041	0.041	0.945	0.254
$\mathbf{b}_9$	14.080	14.080	1034	940	-0.000	0.047	0.047	0.940	0.286
$\mathbf{b}_{10}$	15.038	15.038	1137	949	-0.000	0.052	0.052	0.939	0.336
$\mathbf{b}_{11}$	14.667	14.667	1079	790	0.001	0.067	0.067	0.929	0.483
$\mathbf{b}_{12}$	14.687	14.687	1102	951	0.002	0.050	0.050	0.939	0.310
$\mathbf{b}_{13}$	13.902	13.902	1003	1003	0.002	0.043	0.043	0.938	0.260
$\mathbf{b}_{14}$	14.233	14.233	1001	1197	0.002	0.037	0.037	0.943	0.228
$\mathbf{b}_{15}$	14.292	14.292	940	1350	0.003	0.035	0.035	0.944	0.214
$\mathbf{b}_{16}$	14.217	14.217	845	1463	0.003	0.034	0.034	0.943	0.209
$\mathbf{b}_{17}$	13.995	13.995	805	1443	0.002	0.034	0.035	0.944	0.211
$\mathbf{b}_{18}$	13.492	13.492	755	1324	0.002	0.036	0.036	0.946	0.221
$\mathbf{b}_{19}$	13.126	13.126	705	1216	0.002	0.037	0.037	0.946	0.229
$\mathbf{b}_{20}$	13.183	13.183	679	1173	0.002	0.037	0.037	0.948	0.233
$\mathbf{b}_{21}$	13.408	13.408	655	1145	0.002	0.038	0.038	0.944	0.236
`Uniform`								0.925	0.357
`WBATE`					0.001	0.024	0.024	0.943	0.132
`LBATE`					0.021	0.033	0.040	0.984	0.429

Table SA-27: First-Stage Effects (Simulations, $s = 3$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.404	13.404	746	1303	0.001	0.073	0.073	0.955	0.427
$\mathbf{b}_2$	13.823	13.823	788	1402	0.001	0.069	0.069	0.958	0.411
$\mathbf{b}_3$	14.104	14.104	815	1466	0.002	0.067	0.067	0.958	0.401
$\mathbf{b}_4$	14.257	14.257	849	1470	0.002	0.066	0.066	0.958	0.387
$\mathbf{b}_5$	14.318	14.318	919	1394	0.002	0.064	0.064	0.959	0.360
$\mathbf{b}_6$	14.306	14.306	975	1291	0.002	0.061	0.061	0.959	0.353
$\mathbf{b}_7$	14.235	14.235	1009	1176	0.001	0.060	0.060	0.955	0.359
$\mathbf{b}_8$	13.991	13.991	1009	1036	0.001	0.062	0.062	0.955	0.378
$\mathbf{b}_9$	14.133	14.133	1040	946	0.001	0.067	0.067	0.958	0.407
$\mathbf{b}_{10}$	15.100	15.100	1144	957	0.001	0.074	0.074	0.959	0.462
$\mathbf{b}_{11}$	14.722	14.722	1085	796	0.001	0.098	0.098	0.965	0.964
$\mathbf{b}_{12}$	14.698	14.698	1103	951	0.000	0.070	0.070	0.953	0.424
$\mathbf{b}_{13}$	13.915	13.915	1004	1004	0.001	0.061	0.061	0.956	0.370
$\mathbf{b}_{14}$	14.252	14.252	1004	1200	0.001	0.057	0.057	0.959	0.338
$\mathbf{b}_{15}$	14.307	14.307	941	1352	0.001	0.058	0.058	0.953	0.331
$\mathbf{b}_{16}$	14.254	14.254	849	1469	0.001	0.061	0.061	0.956	0.355
$\mathbf{b}_{17}$	14.044	14.044	809	1453	0.001	0.062	0.062	0.950	0.365
$\mathbf{b}_{18}$	13.534	13.534	758	1332	0.001	0.064	0.064	0.950	0.375
$\mathbf{b}_{19}$	13.137	13.137	707	1218	0.001	0.066	0.066	0.952	0.387
$\mathbf{b}_{20}$	13.188	13.188	680	1174	0.000	0.067	0.067	0.953	0.389
$\mathbf{b}_{21}$	13.422	13.422	657	1146	-0.000	0.067	0.067	0.952	0.391
`Uniform`								0.964	0.591
`WBATE`					0.001	0.039	0.039	0.950	0.222
`LBATE`					0.055	0.065	0.085	0.998	0.931

Table SA-28: Fuzzy Effects (Simulations, $s = 1$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.386	13.386	744	1300	0.001	0.074	0.074	0.959	0.439
$\mathbf{b}_2$	13.791	13.791	785	1397	0.001	0.071	0.071	0.957	0.420
$\mathbf{b}_3$	14.069	14.069	811	1460	0.002	0.068	0.068	0.959	0.407
$\mathbf{b}_4$	14.218	14.218	844	1464	0.002	0.067	0.067	0.960	0.390
$\mathbf{b}_5$	14.287	14.287	914	1389	0.001	0.063	0.063	0.962	0.358
$\mathbf{b}_6$	14.284	14.284	972	1288	-0.001	0.060	0.060	0.956	0.347
$\mathbf{b}_7$	14.225	14.225	1008	1176	-0.001	0.059	0.059	0.953	0.348
$\mathbf{b}_8$	13.995	13.995	1009	1037	-0.002	0.061	0.061	0.952	0.360
$\mathbf{b}_9$	14.126	14.126	1038	946	-0.002	0.065	0.065	0.954	0.380
$\mathbf{b}_{10}$	15.108	15.108	1144	958	-0.003	0.070	0.070	0.952	0.422
$\mathbf{b}_{11}$	14.748	14.748	1087	799	-0.003	0.089	0.089	0.963	1.656
$\mathbf{b}_{12}$	14.726	14.726	1106	955	-0.003	0.064	0.064	0.951	0.382
$\mathbf{b}_{13}$	13.933	13.933	1006	1007	-0.002	0.057	0.057	0.950	0.337
$\mathbf{b}_{14}$	14.255	14.255	1003	1201	-0.002	0.053	0.053	0.954	0.311
$\mathbf{b}_{15}$	14.312	14.312	941	1353	-0.001	0.055	0.055	0.952	0.306
$\mathbf{b}_{16}$	14.250	14.250	847	1469	0.000	0.058	0.058	0.951	0.330
$\mathbf{b}_{17}$	14.019	14.019	806	1449	0.001	0.059	0.059	0.950	0.342
$\mathbf{b}_{18}$	13.498	13.498	754	1326	0.001	0.061	0.061	0.950	0.355
$\mathbf{b}_{19}$	13.115	13.115	703	1214	0.000	0.063	0.063	0.953	0.368
$\mathbf{b}_{20}$	13.180	13.180	678	1173	-0.001	0.064	0.064	0.953	0.373
$\mathbf{b}_{21}$	13.415	13.415	655	1146	-0.001	0.065	0.065	0.950	0.377
`Uniform`								0.961	0.619
`WBATE`					-0.001	0.038	0.038	0.943	0.249
`LBATE`					0.054	0.064	0.084	0.995	1.461

Table SA-29: Fuzzy Effects (Simulations, $s = 2$, Fuzzy Bandwidth Selector).

	h_1	h_2	N_{Co}	N_{Tr}	Bias	SD	RMSE	EC	IL
$\mathbf{b}_1$	13.416	13.416	746	1305	0.002	0.072	0.072	0.959	0.422
$\mathbf{b}_2$	13.823	13.823	788	1403	0.001	0.069	0.069	0.955	0.403
$\mathbf{b}_3$	14.093	14.093	813	1465	0.001	0.067	0.067	0.956	0.390
$\mathbf{b}_4$	14.229	14.229	845	1465	0.000	0.065	0.065	0.952	0.373
$\mathbf{b}_5$	14.282	14.282	913	1388	-0.001	0.062	0.062	0.954	0.344
$\mathbf{b}_6$	14.281	14.281	971	1288	-0.003	0.059	0.059	0.954	0.335
$\mathbf{b}_7$	14.214	14.214	1006	1174	-0.003	0.057	0.057	0.954	0.338
$\mathbf{b}_8$	13.955	13.955	1005	1031	-0.004	0.059	0.059	0.960	0.354
$\mathbf{b}_9$	14.080	14.080	1034	940	-0.004	0.063	0.063	0.962	0.379
$\mathbf{b}_{10}$	15.038	15.038	1137	949	-0.004	0.069	0.069	0.960	0.428
$\mathbf{b}_{11}$	14.667	14.667	1079	790	-0.003	0.090	0.090	0.966	0.675
$\mathbf{b}_{12}$	14.687	14.687	1102	951	-0.003	0.065	0.065	0.952	0.398
$\mathbf{b}_{13}$	13.902	13.902	1003	1003	-0.001	0.059	0.059	0.957	0.352
$\mathbf{b}_{14}$	14.233	14.233	1001	1197	-0.000	0.055	0.055	0.952	0.325
$\mathbf{b}_{15}$	14.292	14.292	940	1350	0.002	0.058	0.058	0.952	0.320
$\mathbf{b}_{16}$	14.217	14.217	845	1463	0.003	0.061	0.061	0.950	0.345
$\mathbf{b}_{17}$	13.995	13.995	805	1443	0.004	0.062	0.062	0.948	0.357
$\mathbf{b}_{18}$	13.492	13.492	755	1324	0.004	0.064	0.064	0.954	0.370
$\mathbf{b}_{19}$	13.126	13.126	705	1216	0.003	0.065	0.065	0.954	0.383
$\mathbf{b}_{20}$	13.183	13.183	679	1173	0.003	0.066	0.066	0.951	0.387
$\mathbf{b}_{21}$	13.408	13.408	655	1145	0.002	0.066	0.066	0.954	0.390
`Uniform`								0.962	0.552
`WBATE`					0.000	0.038	0.038	0.946	0.202
`LBATE`					0.053	0.065	0.083	0.994	0.717

Table SA-30: Fuzzy Effects (Simulations, $s = 3$, Fuzzy Bandwidth Selector).

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